

ARTIFICIAL INTELLIGENCE

FOUNDATIONS, APPLICATIONS
AND FUTURE DIRECTIONS

Editors
Ahmet Gürkan YÜKSEK
Serkan AKKOYUN



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Artificial Intelligence: Foundations, Applications and Future Directions

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PREFACE

Artificial intelligence and related technologies have become one of the fastest-growing and most transformative scientific fields today. The significant increase in data production, advancements in computational power, and developments in modeling techniques have moved artificial intelligence beyond a theoretical field to a practical application area offering effective solutions across various sectors, from engineering and healthcare to education and agriculture.

Within this scope, this book, prepared under the coordination of the Sivas Cumhuriyet University Artificial Intelligence and Data Science Application and Research Center, addresses the theoretical foundations, application areas, and future perspectives of artificial intelligence under the theme "Artificial Intelligence: Foundations, Applications, and Future Directions" in a comprehensive manner.

The chapters in this book cover the scientific foundations of artificial intelligence, including quantum algorithms, deep learning architectures, fuzzy logic systems, and ANFIS-based modeling. Fundamental topics such as signal processing, data augmentation, model comparisons, and class imbalance are evaluated within a theoretical framework.

The application sections built upon this foundation are supported with concrete examples, including disease diagnosis in healthcare informatics, personalized medicine, recommendation systems, software quality assessment, and smart agricultural systems. These examples demonstrate the functionality of artificial intelligence in fields such as engineering, healthcare, e-commerce, natural language processing, and control systems.

Additionally, the book covers future-oriented research areas such as large language models (LLM), transformer architectures, explainable AI, quantum AI, and edge/fog computing. The optimization power of metaheuristic algorithms and hybrid approaches for biomarker discovery further enrich this vision.

Consisting of 20 interdisciplinary chapters, this work serves as a comprehensive and up-to-date resource for researchers, academics, and industry practitioners. By combining the theoretical foundations of artificial intelligence with contemporary and future-oriented solutions, this study aims to contribute to the knowledge-based transformation of our country.

Editors

Prof. Dr. Serkan AKKOYUN

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CHAPTER I

QUANTUM ALGORITHMS AND APPLICATIONS

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1. Introduction

Quantum computing is a branch of science that aims to revolutionize information processing using the fundamental principles of quantum mechanics. It has the potential to solve specific problems much faster and more efficiently by overcoming the limitations of traditional computers. This potential can lead to significant advances in many fields, such as optimization (Aasim et al., 2023; Katircı and Oğuz, 2024), cryptography, materials science and artificial intelligence. Quantum computers can perform a large number of calculations at the same time using quantum phenomena such as superposition and entanglement. This allows specific algorithms to achieve exponential speedup compared to classical computers (Nielsen and Chuang, 2010). Quantum algorithms such as Shor's Algorithm have the potential to break today's common cryptographic systems (e.g. RSA) (Shor, 1994).

This necessitates the development of new approaches to data security. Quantum computers can accelerate the discovery of new materials and drugs by

simulating systems at the molecular and atomic levels (Lloyd, 1996). Complex optimization problems and big data analytics can be solved more efficiently with quantum algorithms, leading to significant advances in finance, logistics and artificial intelligence (Rebentrost et al., 2014).

The development of quantum mechanics began in the 1930s with the work of scientists such as Max Planck, Albert Einstein, Niels Bohr and Erwin Schrödinger (Jammer, 1966). Paul Benioff showed that quantum mechanics could be applied to the Turing machine (Benioff, 1980). Richard Feynman argued that quantum systems cannot be simulated effectively with classical computers, so quantum computers are necessary (Feynman, 1982). David Deutsch laid the theoretical foundation for quantum computing by introducing the concept of a universal quantum computer (Deutsch, 1985). Peter Shor developed an algorithm that can perform the prime factorization of integers at exponential speed (Shor, 1994). This demonstrated the potential impact of quantum computers on cryptography (1994). In 1996, Lov Grover introduced an algorithm that speeds up searching databases, pointing to different applications of quantum computers (Grover, 1996). In the 2000s, experimental progress accelerated. In these years, experimental work on ion traps and superconducting qubits intensified (Cirac and Zoller, 1995), and quantum computers of 5-7 qubits were used for the first time in a laboratory environment. In 2010 and later years, technology giants such as Google, IBM and Microsoft started to develop quantum computer prototypes. In 2019, Google announced that it had achieved “quantum supremacy” by completing a specific task faster than classical computers (Arute et al., 2019).

Quantum computing is recognized as heralding a new era in information processing. Its historical development shows a significant progression from theoretical foundations to practical applications. Despite the technical challenges faced, the importance and potential of quantum computing is supported by a growing interest in research and investment.

2. Fundamental Differences between Classical and Quantum Computing

Today’s computing technology is primarily based on classical principles. However, quantum computing, which uses the principles of quantum mechanics, promises to revolutionize computing. Understanding the fundamental differences between classical and quantum computing is critical to assessing the potential and limitations of this new technology.

Classical computers process information in units called bits, which can only take two states (0 or 1) (Tanenbaum and Austin, 2013). A bit can take the

value 0 or 1 at any given moment (Figure 1a). In quantum computers, the basic unit is called a qubit (quantum bit). Thanks to superposition, qubits can exist in both the 0 and 1 states at the same time (Figure 1b) (Nielsen and Chuang, 2010). Thanks to superposition, qubits can exist in a linear combination of two states. This means that a qubit can simultaneously be in more than one state (Schrödinger, 1935).

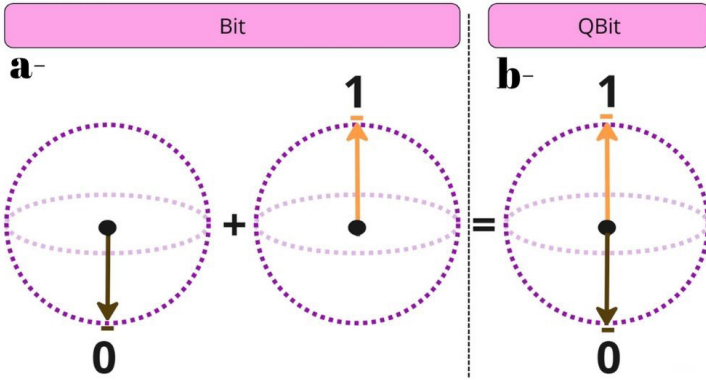


Figure 1. Classical and Quantum Units of Information.

A strong correlation between qubits, called entanglement, is possible (Figure 2). The state of entangled qubits cannot be defined independently of each other (Einstein et al., 1935). This property allows for remarkable parallelism and speedup in computations. Classical bits do not have this property; each bit is independent (Katırcı et al., 2024).



Figure 2. *Quantum Connectivity: Entangled Particles.* (OpenAI. (2024). DALL·E (Versiyon 3). 04.01.2025. <https://openai.com/dall-e>)

When the value of the classical bit is measured, the system's state does not change; the measurement process is passive. When the qubit's state is measured, the superposition state collapses and changes to a particular state. The measurement process directly affects the system's state (Dirac, 1981). In classical computing, multiple processors or cores are used for parallel processing. However, each processor processes a single state. Thanks to superposition and entanglement in quantum computing, quantum computers can simultaneously process an exponential number of states (Shor, 1995). This allows specific algorithms (for example, the Shor and Grover algorithms) to run much faster than classical computers. This offers significant advantages, especially in special fields such as factorization, search problems and simulation of quantum systems (Montanaro, 2016).

The main differences between classical and quantum computing stem from the fundamental principles of information processing. Quantum computing offers significant advantages over classical computing for certain problems by utilizing the unique properties of quantum mechanics. However, technical challenges need to be overcome, and more research is needed before quantum computing reaches its full potential.

3. Fundamentals of Quantum Algorithms

3.1. Quantum Circuit Model

The quantum circuit model is a fundamental framework in the design and analysis of quantum computations. This model represents quantum algorithms as a sequence of quantum gates applied to qubits, similar to how classical circuits (Figure 3) use logic gates on bits. In this model, qubits are initialized with a known state, a series of unitary transformations (quantum gates) are applied, and then measurement is performed to obtain classical results (Nielsen and Chuang, 2010).

A quantum circuit (Figure 4) consists of wires representing qubits and boxes representing quantum gates. Time moves from left to right, and the gates are applied in sequence. This model is powerful in that it can effectively represent complex quantum operations, and any quantum computation can be performed using a finite set of elementary gates (Deutsch, 1989).

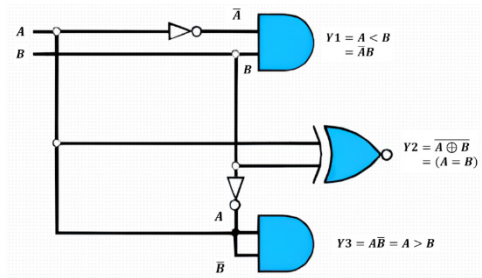


Figure 3. Classical Bit Circuit (Example).

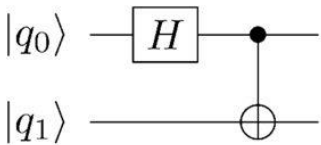


Figure 4. Quantum Bit Circuit (Example).

4. Quantum Gates and Operations

Quantum gates are the building blocks of quantum circuits and perform unitary transformations on qubits. Unlike classical logic gates, quantum gates are reversible and based on the principles of superposition and entanglement (Barenco et al., 1995).

4.1. Single Qubit Gates:

Pauli Gates (X, Y, Z): Rotates the state of the qubit around the respective axes on the Bloch sphere (Figure 5).

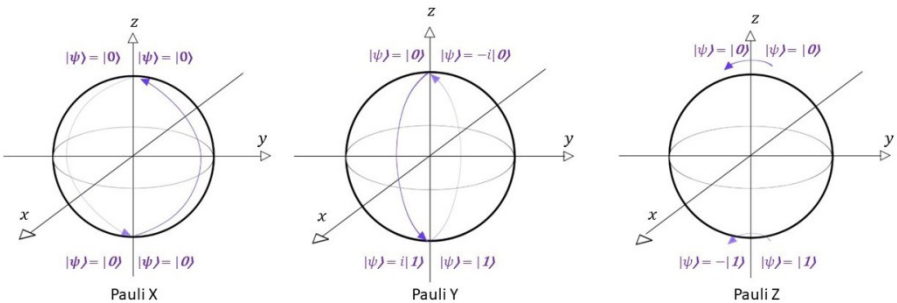


Figure 5. Representation of Pauli X, Pauli Y and Pauli Z gates on the Bloch Sphere.

X Gate: Functions as a quantum NOT gate, flipping the state $|0\rangle$ to $|1\rangle$ vice versa (Figure 5a).

Hadamard Gate (H): Converts a ground state into an equal superposition of $|0\rangle$ and $|1\rangle$, creating a superposition (Figure 6):

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

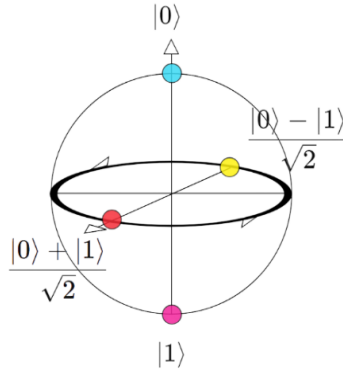


Figure 6. Illustration of hadamard gates on the Bloch sphere.

Phase Shift Gates (S, T): Affects interference patterns in quantum algorithms by adding a phase factor to the state of the qubit.

4.2. Multi-Qubit Gates:

- Controlled Gates: Operations where one qubit (control) determines the operation on another qubit (target).
- Controlled-NOT Gate (CNOT): Flip the target qubit if the control qubit is in the state $|1\rangle$.
- Swap Gate: Swaps the states of two qubits.
- Toffoli Gate: A three-qubit gate that translates the state of the target qubit if both control qubits are in state $|1\rangle$.
- These gates can be combined to create complex operations and are essential for entanglement between qubits, a vital resource in quantum computing (DiVincenzo, 1995).

5. Quantum Parallelism and Interferometry

5.1. *Quantum Parallelism:*

Quantum parallelism results from qubits' ability to exist in superposition states. When a quantum gate is applied to qubits in superposition, it performs computation on all possible states simultaneously. For example, a quantum gate applied to n qubits can process 2^n states at once, offering exponential scaling with respect to classical bits (Feynman, 1986). This parallelism is used in algorithms such as Shor's Algorithm for factoring large integers and Grover's Algorithm for database searching, providing significant speedups over their classical counterparts.

5.2. *Interferometry:*

Interferometry in quantum computing involves constructive and destructive interference of quantum amplitudes. By carefully designing quantum circuits, specific computational paths cancel each other out through destructive interference, while others reinforce the desired result through constructive interference (Grover, 1997).

Interference is critical in algorithms where the correct answer corresponds to the quantum state and is reinforced by interference effects. The Quantum Fourier Transform (QFT) is an essential example of this and plays a central role in period-finding algorithms, providing the speedup in Shor's Algorithm (Coppersmith, 2002).

6. Important Quantum Algorithms

6.1. *Shor's Algorithm: Prime Factorization and Cryptography Breakdown*

Shor's Algorithm is considered one of the most striking examples of quantum computing and can perform the prime factorization of large numbers with exponential speed. This is practically impossible on classical computers, especially for large numbers, and the security of existing cryptographic systems relies on this difficulty. RSA and similar encryption methods rely on the assumption that large integers cannot be factored. Shor's Algorithm makes it possible for quantum computers to factor these numbers quickly using the quantum Fourier transform. This threatens the security of existing cryptographic protocols and necessitates the development of quantum-resistant encryption methods. Shor's work is also of great importance in showing how quantum

computing can provide an exponential advantage over classical computing for specific problems (Shor, 1994).

6.2. Grover's Algorithm: Database Search and Quadratic Acceleration

Grover's Algorithm addresses the problem of searching for a given item in an extensive and disordered database. While classical algorithms perform such a search in $O(N)$ time on average, Grover's quantum algorithm reduces this time to $O(\sqrt{N})$. This quadratic speedup provides a significant advantage, especially for large data sets and optimization problems. The Algorithm iteratively increases the probability amplitude of the searched item using quantum superposition and interference principles. Grover's Algorithm is not limited to database search but can be applied to a wide range of problems and is considered one of the practical applications of quantum computing (Grover, 1996).

6.3. Deutsch-Jozsa Algorithm: Determining whether Functions are Balanced or Stationary

The Deutsch-Jozsa Algorithm is one of the first algorithms to demonstrate the decisive advantage of quantum computing over classical computing. The problem is determining whether a function is constant (giving the same output for all inputs) or balanced (giving an output of 0 for half the inputs and 1 for the other half). While a classical computer needs to try half the inputs to determine the function's properties in the worst case, the Deutsch-Jozsa Algorithm can do this in a single operation using quantum parallelism and interference. This Algorithm concretely demonstrates how quantum computers can outperform classical computers in certain situations and is considered a fundamental step in developing quantum algorithms (Deutsch and Jozsa, 1992).

6.4. Simon's Algorithm: Detection of Periodic Functions

Simon's Algorithm is a quantum algorithm that can find the period of a periodic function at exponential speed compared to classical methods. Simon's Algorithm can solve this problem in $O(2^{n/2})$ time, while classical algorithms solve it in $O(n)$ time. This algorithm also played an essential role in developing Shor's algorithm and is an example of a problem where quantum computing can provide exponential speedup. Simon's work shows how the quantum Fourier transform and quantum superposition can effectively solve problems. The Algorithm proves that quantum computers can quickly analyze functions

with certain structural properties and have applications in cryptography, error correction and information theory (Simon, 1994).

6.5. Quantum Walk Search Algorithm

The quantum walk search algorithm is an algorithm that solves search problems in quantum computing using a quantum version of random walks. Similar to classical random walks, quantum walks also traverse a graph or network but offer faster search capabilities thanks to quantum superposition and interference principles. This Algorithm is used to find a specific target, especially in large and complex graphs, and is considered a generalization of Grover's Algorithm. Quantum walks increase the probability amplitudes of certain states using quantum parallelism and interference, allowing the searched item to be found faster. This approach has applications in various fields, such as graph theory, optimization and simulation of physical systems (Ambainis, 2003).

6.6. Hidden Shift Problems

The hidden shift problem is an important problem where quantum algorithms have an advantage over classical algorithms. In this problem, there is a particular shift or translation between two given functions, and the objective is to find this shift. While classical algorithms have to evaluate many values of the function to find this shift, quantum algorithms can solve this problem more efficiently by using quantum Fourier transform and superposition. The hidden shift problem is closely related to Simon's Algorithm and other algorithms based on the quantum Fourier transform. This problem has critical applications in areas such as cryptography, coding theory, and the analysis of complex systems, and shows how quantum computing can take advantage of certain structural problems (van Dam and Hallgren, 2002).

6.7. QAOA (Quantum Approximate Optimization Algorithm)

QAOA, the Quantum Approximate Optimisation Algorithm, is an algorithm designed to solve the optimization problems of quantum computers. It offers a quantum approach to combinatorial optimization problems that are difficult on classical computers. QAOA is a combination of adiabatic quantum computing and variational quantum algorithms. Adiabatic quantum computation relies on the slow and controlled evolution of quantum systems to obtain the solution of the problem through the ground state. Variational quantum algorithms offer a hybrid

approach of quantum and classical computation. This algorithm aims to find the approximate best solution for the objective function using a parameterizable quantum circuit. By working with circuits of a certain depth, QAOA adapts to the limited capacity of existing quantum computers and aims to provide a viable quantum advantage in the near term. This Algorithm can be applied to graph problems such as Max-Cut as well as other NP-hard (Nondeterministic Polynomial time-hard) problems, allowing quantum computing to be used in practical optimization applications (Farhi et al., 2014).

7. Applications of Quantum Algorithms

By utilizing the unique principles of quantum mechanics, quantum computing has the potential to more quickly and efficiently address problems that classical computers have difficulty solving or can solve for a very long time (Nielsen and Chuang, 2010). This feature has led to the development of new and innovative applications in various disciplines. From cryptography to materials science, from financial optimization to machine learning, quantum algorithms provide the opportunity to improve existing methods and offer new solutions.

Quantum cryptography is a field that aims to maximize communication security using the fundamental principles of quantum mechanics (Bennett and Brassard, 1984; Nielsen and Chuang, 2010). While traditional cryptographic methods are based on the prime factorization of large numbers or the difficulty of certain mathematical problems, quantum cryptography is directly based on the laws of physics. In this way, the security of communication is guaranteed by the fundamental principles of quantum mechanics. Quantum key distribution protocol allows two parties to share a secure encryption key (Bennett and Brassard, 1984; Gisin et al., 2002). The main feature of QKD is that if a third party tries to eavesdrop on the communication, this eavesdropping attempt can be detected due to the principles of quantum mechanics. This security is based on two fundamental quantum phenomena:

1. Heisenberg's Uncertainty Principle: It is not possible to precisely measure certain properties of a quantum system at the same time (Dirac, 1981). When the state of a qubit is measured, this measurement process changes the state of the qubit. Thanks to this principle, detectable disturbances occur in the system if an eavesdropper eavesdrops on the communication.

2. Quantum Entanglement: It is a phenomenon that occurs between two or more quantum particles and ensures that the states of these particles are

interconnected (Einstein et al., 1935; Schrödinger, 1935). Measurements between entangled particles are instantaneously correlated regardless of distance. This property is used to share secure keys.

Quantum cryptography is robust in terms of the potential of quantum computers to break classical cryptographic systems (Gisin et al., 2002; Nielsen and Chuang, 2010). Quantum algorithms such as Shor's Algorithm can break common cryptographic systems such as RSA and ElGamal (Shor, 1994). QKD is a secure alternative to this threat. The security is not based on mathematical assumptions but directly on the fundamental principles of quantum mechanics. Thus, increases in computational power or algorithmic improvements do not affect security. Any interference with quantum systems changes the quantum states of the system and can be detected by it. This feature protects the integrity of the communication.

Today, quantum cryptography systems face physical and technological challenges due to the sensitive nature of qubits (Gisin et al., 2002). Signal loss in fibre optic cables and the lack of quantum repeaters make long-distance communication difficult. For the wide adoption of QKD systems, international standards need to be established, and different systems need to be harmonized. Current QKD systems are more expensive than classical cryptographic solutions. However, with the development of technology, costs are expected to decrease.

Quantum computers offer great advantages in the simulation of complex quantum systems and molecular structures (Cao et al., 2019; Feynman, 1982; Lloyd, 1996). With classical computers, it is possible to model properties such as DFT (Density functional theory) based analyses, material optimization, spectroscopy, and electronic structures of materials (Oğuz et al., 2022). However, classical computers lack the computational power required to fully model the behaviour of quantum systems because the number of possible states in quantum systems increases exponentially with the size of the system (Lloyd, 1996; Nielsen and Chuang, 2010). This makes the simulation of complex quantum systems, especially multi-electron atoms, molecules and solid-state systems, practically impossible for classical computers.

The importance of quantum simulations stems from the fact that quantum computers can simulate quantum systems in a natural way as they are based on the principles of quantum mechanics (Cao et al., 2019; Feynman, 1982). As Richard Feynman stated in 1982, "If you want to mimic nature, you have to do it quantum mechanically" (Feynman, 1982). This approach overcomes

the limitations of classical computers and enables more accurate and efficient simulations. Accurate calculation of the energy levels of molecules and materials is critical to understanding chemical reactions and material properties. Quantum computers can determine these energy levels more precisely, accelerating the discovery of new materials and drugs (Cao et al., 2019).

In the field of materials science, quantum simulations play an important role in the discovery of new superconductors. These materials can contribute to the development of more efficient energy infrastructures by reducing losses in energy transmission. In addition, the design of catalysts that accelerate chemical reactions increases the efficiency of industrial processes. Quantum computers enable the development of more effective catalysts by simulating the interactions of catalysts at the atomic level. Prediction of thermal, electrical and magnetic properties of materials also becomes possible thanks to quantum simulations, thus accelerating the design of materials with desired properties (Cao et al., 2019).

In the field of chemistry and drug design, quantum computers can more accurately model the interactions of drug molecules with target proteins. This can speed up the drug discovery process and reduce side effects. Understanding the mechanisms of complex chemical reactions helps to develop new synthetic routes, and quantum simulations can reveal how these reactions occur step by step (Cao et al., 2019).

In scientific research and industrial applications, quantum computers can be used in energy generation and storage. Quantum simulations play an important role in the development of materials to improve the efficiency of solar panels and better battery technologies. In environmentally friendly technologies, optimizing carbon capture and storage processes is critical in reducing greenhouse gas emissions, and quantum simulations can contribute to making these processes more efficient. In the field of nanotechnology and electronics, quantum computers perform critical simulations in the design of nanoscale electronic devices and quantum dots, leading to the development of faster and more efficient components in the electronics industry (Cao et al., 2019).

Quantum algorithms can lead to significant improvements in many sectors by providing significant speedups in optimization problems (Farhi et al., 2014; Montanaro, 2016; Orús et al., 2019). Especially in areas such as finance and logistics, where large and complex datasets and many variables are managed, quantum computing methods can offer more effective solutions than traditional methods.

In the financial sector, there are challenging tasks such as portfolio optimization, risk management and complex financial modelling (Orús et al., 2019). In portfolio optimization, a large amount of data and a large number of probabilities must be calculated to determine the optimal combination of investments according to investors' risk and return preferences. Quantum computers perform such calculations more quickly and efficiently, allowing investment strategies to be planned more effectively. In risk management, quantum algorithms make it possible to model the uncertainties and fluctuations of financial markets more accurately, enabling financial institutions to better prepare for potential risks (Orús et al., 2019).

In the field of logistics, complex optimization problems are central to daily operations (Neukart et al., 2017). Tasks such as route planning, supply chain optimization and resource allocation are often large-scale and multivariate problems that can take a long time to solve with classical computers. Thanks to the parallel processing capabilities of quantum computers, such problems can be solved in less time and at less cost. For example, determining the shortest and most efficient distribution routes for a distribution company is critical in terms of fuel saving and time management. Quantum algorithms can increase operational efficiency by quickly calculating these route optimizations (Neukart et al., 2017).

Quantum machine learning (Aasim et al., 2024) is an innovative field that aims to harness the power of quantum computing in the processing and analysis of large data sets (Biamonte et al., 2017; Rebentrost et al., 2014). Today, the exponential increase in the amount of data makes it difficult to process large data sets effectively and transform them into meaningful information (Katirci et al., n.d.). Classification of data using machine learning methods with classical computers is widely used (Kavalcı Yılmaz et al., 2023). However, the limited computational power of classical computers limits the performance of machine learning algorithms, especially in high-dimensional and complex data sets. Quantum computers, on the other hand, have the potential to perform such complex calculations faster and more efficiently by utilizing quantum phenomena such as superposition and entanglement (Biamonte et al., 2017).

Quantum-assisted algorithms can provide speed and performance advantages over classical algorithms in basic machine-learning tasks such as pattern recognition, classification, and clustering (Biamonte et al., 2017). For example, quantum support vector machines (QSVM) are quantum versions of traditional support vector machines and can perform similarity calculations

between data points faster thanks to the parallel processing capabilities of quantum computers. Similarly, quantum k-nearest neighbour (k-NN) algorithms can speed up classification processes on large data sets. This acceleration provides a great advantage, especially in real-time data analysis and applications where instant decisions need to be made.

Furthermore, quantum data processing techniques reduce computational complexity in high-dimensional data spaces (Biamonte et al., 2017), enabling the development of new approaches in artificial intelligence and deep learning. The exponential acceleration potential of quantum computers can accelerate the training processes of deep learning models and enable the implementation of more complex models (Biamonte et al., 2017). For example, quantum variational algorithms can be used in parameter optimization of deep learning models and can enable more efficient training of energy-based models (Preskill, 2018).

However, there are still many challenges that need to be solved in the field of quantum machine learning. The current hardware limitations of quantum computers, the small number of qubits and the lack of error correction mechanisms are obstacles to large-scale applications. Furthermore, more research is needed on the development of quantum algorithms and their integration with classical algorithms (Biamonte et al., 2017; Preskill, 2018).

8. Current Status of Quantum Computing

Significant progress has been made in the field of quantum computing in recent years. Hardware developments and quantum computer prototypes are undergoing a transition process from laboratory environments to commercial applications. Efforts are underway to develop more stable and scalable quantum computers using different physical systems such as superconducting qubits, trapped ions, topological qubits and photonic systems. For example, technology giants such as IBM and Google have produced prototype quantum computers with more than 50 qubits and made them available to researchers through cloud-based platforms. These prototypes provide an important platform for testing quantum algorithms and developing new applications.

Quantum error correction and stability issues are one of the biggest challenges of quantum computing. Qubits are highly sensitive to environmental effects and noise, resulting in high decoherence and error rates. Intensive research has been carried out on quantum error correction codes and stability techniques. Approaches such as topological qubits and surface codes aim to reduce error

rates and achieve more reliable quantum computation. Without error correction, practical and reliable operation of quantum computers is not possible; therefore, advances in this area are critical.

Industrial and academic collaborations contribute to the rapid development of the quantum computing field. Partnerships between universities, research institutes and technology companies facilitate the sharing of resources and know-how. The European Union's billion-euro investments in quantum technologies and initiatives such as the US Quantum National Initiative support research in this field. Furthermore, the development of quantum computing education programmes and curricula is essential for the training of future quantum scientists.

9. Future Perspectives and Research Areas

The extension of quantum superposition is a fundamental goal to increase the computational power of quantum computers. Controlling more qubits in entangled and superposition states will make it possible to solve complex problems. This requires the discovery of new materials, the development of technologies to increase the stability of qubits and the extension of the decoherence time. In addition, intensive studies on the scalability of quantum processors and the design of integrated quantum circuits are ongoing.

The development of new algorithms will increase the applicability of quantum computing in different fields. While existing algorithms provide advantages for specific problems, there is a need for general-purpose quantum algorithms that can be used in a wide range of applications. The development of innovative algorithms in areas such as quantum machine learning, artificial intelligence, cryptography and optimization will further unlock the potential of quantum computing. In addition, studies on the hybrid use of classical and quantum computing may also play an important role in the future.

Ethical and security issues are important areas to consider in terms of the societal impact of quantum technologies. The potential of quantum computers to break existing cryptographic systems raises serious concerns about data security and privacy. Therefore, the development of quantum-secure cryptographic methods and protocols is of great importance. In addition, it is necessary to establish policies and regulations on the ethical use of quantum technologies and their social and economic impacts. The adaptation and awareness raising of the society to these technologies is also an issue that should not be ignored.

10. Conclusion

The potential of quantum algorithms and their impact on society is seen as a harbinger of a new era in information processing. Quantum computing will lead to significant changes in scientific research, industrial applications and everyday life by providing exponential speedups over classical computers in solving specific problems. It will significantly contribute to economic growth and improved quality of life by stimulating innovation in areas such as materials science, drug discovery, finance, logistics and artificial intelligence.

The challenges and solutions are critical for the future of quantum computing. Technical challenges in hardware and software need to be overcome, quantum error correction techniques need to be developed, and scalable quantum computers need to be built. In addition, the development of education and human resources, multidisciplinary collaborations and the establishment of international standards are important for the successful adoption of quantum technologies. Addressing ethical and security issues and preparing society for these technologies are also among the keys to future success.

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CHAPTER II

METAHEURISTICS IN ACTION: UNLOCKING OPTIMAL SOLUTIONS FOR ENGINEERING APPLICATIONS

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1. Introduction

Classical optimization, one of the methods for problem solving, creates mathematical models based on inferred knowledge (Boussaid et al., 2013). However, with technological development, new problems with a large solution space and high complexity have emerged. Their methods are inadequate for solving these new generation problems consisting of nonlinear, discrete or discrete structures (Deb, 2001). To overcome the shortcomings, metaheuristics have been developed to model biological, physical or social processes in nature (Fister et al., 2023). Metaheuristics have the advantage of being able to efficiently search large and complex solution spaces. Moreover, thanks to their potential to find the global optimum, they provide more robust

and stable solutions without getting stuck in local minimum. This flexibility makes metaheuristics more effective than classical methods in solving both nonlinear and discrete problems (Karaboga, 2004).

In this context, various metaheuristic algorithms have been selected to comprehensively evaluate their performance and effectiveness in solving complex engineering applications. The selected algorithms include Puma Optimization Algorithm (PO), Secretary Bird Optimization Algorithm (SBOA), Walrus Optimization Algorithm (WO), Newton-Raphson Based Optimization Algorithm (NRBO), Chaotic Evolution Optimization Algorithm (CEO), Arctic Puffin Optimization Algorithm (APO), Black-Winged Kite Algorithm (BKA), and Starfish Optimization Algorithm (SFOA). These algorithms represent a diverse set of metaheuristic approaches, encompassing population-based methods, adaptive learning, and physics-inspired strategies, thereby providing a comprehensive evaluation of different search and convergence mechanisms. The algorithms are executed under identical experimental settings, ensuring a fair comparison of their performance across various real-world engineering applications.

The motivation and contributions of this study can be summarized as follows:

- While most studies in literature assess the performance of metaheuristic algorithms using theoretical test functions, this study provides a direct comparison of their performance on real-world engineering applications.
- A comprehensive analysis has been conducted to evaluate the performance of algorithms with different structures and characteristics.
- The algorithms have been directly compared under identical experimental conditions, enabling the identification of which algorithm performs best for specific types of applications. The convergence speed, minimum fitness value, and solution stability of the algorithms have been thoroughly examined.
- Based on the results, a general performance ranking of the algorithms has been established, and the most suitable algorithms for different application types have been identified.

This study is divided into six main sections. The Introduction section outlines the motivation, objectives, and scope of the study. The Related Work section provides an overview of existing metaheuristic algorithms and their

performance in engineering applications, as reported in the literature. The Methodology section details the structure, operating principles, and parameter settings of the eight algorithms evaluated in this study. The Experimental Studies section presents a comparative analysis of the performance of the algorithms on eight different engineering applications. The Results and Discussion section evaluates the overall ranking and stability of the algorithms based on the results obtained. Finally, the Conclusion section summarizes the main findings of the study and offers recommendations for future research.

2. Related Work

Metaheuristic algorithms that have emerged in recent years are recognized as effective methods for solving complex technical problems. The algorithms compared in this study and the evaluation criteria used for assessing their performance are listed in Table 1. An examination of the table reveals that the performance of these algorithms is generally evaluated using CEC test functions. Notably, accuracy is the most frequently used parameter among the evaluation criteria, highlighting its crucial role in determining algorithm performance. However, the new versions of the CEO and SFOA algorithms used in this study have not yet been developed, and therefore, they are not included in the table.

3. Metaheuristics

This section provides a general explanation of the structure and operating principles of metaheuristic algorithms.

Table 1. Reviewed papers on metaheuristics

Problem	Algorithm	Benchmark Algorithms	Criteria	Paper
Distributed generation planning in power grids	PO	CBO, GWO, ZOA, WO	Power loss reduction, Voltage profile enhancement, Annual savings	(Maurya vd., 2024)
Wheeled mobile robot control	Chaotic PO Optimizer Algorithm (CPOA)	CEC'2022 functions, POA, AOA, SCA, WOA, CAO, CAO	Response time, Stability, Accuracy, Optimization performance	(Kmich vd., 2025)
Structural and wind turbine blade optimization	Multi-Strategy Improvement Secretary Bird Optimization Algorithm (MISBOA)	CEC-2022 test suite, SBOA, GJO, PSO, PSA, QIO, NRBO, SCSO	Accuracy, Convergence speed, Stability, Energy efficiency improvement, Structural design optimization	(Qin vd., 2024)
Nonlinear constrained optimization	Crossover Strategy Integrated Secretary Bird Optimization Algorithm (CSBOA)	CEC-2017, CEC-2022, seven advanced metaheuristic algorithms	Accuracy, Convergence speed, Stability, Robustness, Optimization efficiency	(Mai vd., 2025)
Flexible job shop scheduling	Enhanced Walrus Optimization Algorithm (eWaOA)	30 test instances, WaOA variants	Makespan minimization, Solution quality, Optimization efficiency, Global search capability	(Lv vd., 2025)
Solar cell parameter estimation	Hybrid Walrus Optimization Algorithm (H-WaOA)	Standard solar cell models, classical and metaheuristic methods	Accuracy, Convergence speed, Robustness, Practical applicability	(Vujošević vd, 2024)

Table 1. Reviewed papers on metaheuristics (continued)

Problem	Algorithm	Benchmark Algorithms	Criteria	Paper
Solar cell parameter estimation	Hybrid Walrus Optimization Algorithm (H-WaOA)	Standard solar cell models, classical and metaheuristic methods	Accuracy, Convergence speed, Robustness, Practical applicability	(Vujošević vd, 2024)
Local optima entrapment, slow convergence, low optimization accuracy in BOA	Butterfly and Newton–Raphson Swarm Intelligence Algorithm (BOANRBO)	PSO, GWO, GA, ABC, DE, WOA	Accuracy, Convergence Speed, Stability	(Li vd., 2024)
Optimal integration of Fluid Antenna Systems (FAS) and Reconfigurable Intelligent Surfaces (RIS) in wireless systems	Newton–Raphson-Based Optimizer (NRBO)	BSDE, GWO, MGA, PSO	Energy Efficiency, Data Rate, Convergence Speed	(Alwakeel vd., 2025)
Aircraft fuel consumption prediction and optimization	CNN-LSTM Enhanced by CEEMDAN and Improved Arctic Puffin Optimization (IAPO)	WOA, PSO, GWO, SSA	Accuracy, Convergence Speed, Stability	(Tang vd., 2024)

Table 1. Reviewed papers on metaheuristics (continued)

Problem	Algorithm	Benchmark Algorithms	Criteria	Paper
Low convergence accuracy and local minimum entrapment in APO	Arctic Puffin Optimization with Multi-Strategy Blending (ETAAPO)	PSO, GWO, GA, WOA, ABC	Accuracy, Convergence Speed, Global Search Capability	(Sun vd., 2024)
Low accuracy, poor global search efficiency, and local optimum trap in BKA	Black-Winged Kite Optimization Enhanced by Osprey and Crossover (DKCBKA)	BKA, PSO, GWO, WOA, SSA	Solution accuracy, Convergence speed, Optimization capabilities in real-world problems	(Li, Y. vd., 2025)
Low population diversity, weak local search ability, and slow convergence in BKA	Complex-Valued Encoding Black-Winged Kite Algorithm (CBKA)	GTO, MGO, PO, AVOA, GCRA, HLOA, WO, SBOA, NRBO, APO, EHO, BKA	Population diversity, Global search efficiency, Convergence speed	(Du vd., 2025)
Engineering Applications	PO, SBOA, WO, NRBO, CEO, APO, BKA, SFOA		Accuracy, Convergence Speed, Global Search Capability	This Chapter

3.1. Puma Optimization Algorithm

PO is a metaheuristic inspired by the herd intelligence and hunting strategies of the PO (Abdollahzadeh et al., 2024). Pumas locate prey by scouting large areas and hunting at the optimal moment, which forms the basis for the algorithm’s balanced execution of the exploration and exploitation phases during the optimization process. In the exploration phase, the algorithm achieves diversity by systematically scanning the solution space. Subsequently, the algorithm shifts to a more goal-oriented optimization process by focusing on the best solutions.

The primary advantage of PO is its ability to dynamically optimize the balance between exploration and exploitation through an adaptive mechanism. This balance is automatically adjusted based on the problem structure via a phase transition mechanism. As a result, the algorithm can efficiently handle various problem types while reducing dependence on parametric settings (Jiang et al., 2024). The flowchart illustrating the algorithm’s operating principle is presented in Fig. (1).

3.2. Secretary Bird Optimization Algorithm

SBOA is an optimization algorithm inspired by the hunting and escape strategies of the African secretary bird (Fu et al., 2024). Each secretary bird is modeled as a candidate solution, and the algorithm conducts optimization through two main stages: exploration and exploitation. In the exploration phase, a large-scale search is performed by mimicking the bird's hunting behavior, enhancing diversity within the solution space and increasing the probability of reaching the global optimum. In the exploitation phase, local search and directed escape mechanisms inspired by the bird's escape strategies are applied. This approach enables a more efficient optimization process by reducing the risk of the algorithm getting trapped in local minimal. The performance of SBOA depends on parameters such as population size and the number of iterations. In some cases, the algorithm may face the risk of premature convergence. To mitigate this limitation, several variants have been proposed in the literature (Wang and Wang, 2025). The pseudo-code of the algorithm is presented in Algorithm 1 (Fu et al., 2024).

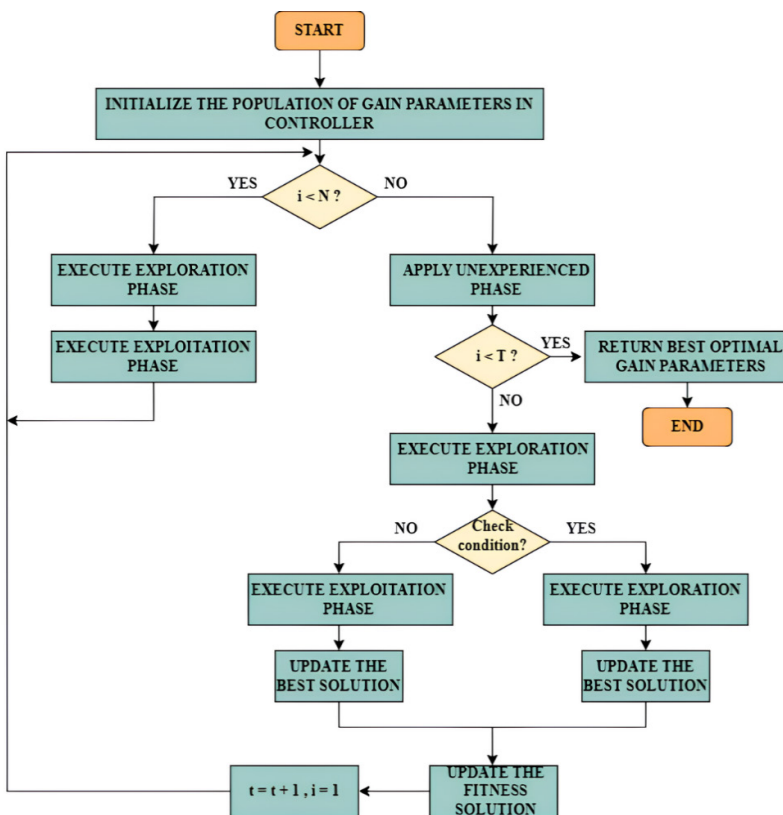


Figure 1. PO Flowchart (Naresh vd., 2024)

Algorithm 1: SBOA pseudo code
INPUT: Set maximum iterations (T), set population size (N), set problem dimension (Dim), set upper bound (ub) and lower bound (lb) and set weights for fitness function ($w1, w2, w3$). Initialize Parameters: Set max iterations (T), population size (N), problem dimension (Dim), upper and lower bounds (ub, lb), and fitness weights ($w1, w2, w3$). Generate Initial Population (S) of N solutions: For $i = 1$ to N: $s[i] = \text{GenerateRandomSolution}(Dim, ub, lb)$ Evaluate Fitness of Each Solution: For each $s[i]$ in S: $fitness[i] = \text{EvaluateFitness}(s[i], w1, w2, w3)$ Set Iteration Counter ($t = 0$) While $t < T$: For each solution $s[i]$: Exploration Phase: $s[i] = \text{ApplyExploration}(s[i], best_solution, t, T)$ Evaluate: $newFitness[i] = \text{EvaluateFitness}(s[i], w1, w2, w3)$ Adaptation: If $newFitness[i] < fitness[i]$, update $fitness[i]$. Else revert $s[i]$. Exploitation Phase: Fine-tune and select best solution using adaptive perturbation. Update population S with new solutions. Check Termination: ($t \geq T$ or Convergence Condition). Increment t. Output: the best solution obtained.

3.3. Walrus Optimization Algorithm

WO is an innovative metaheuristic based on swarm intelligence, inspired by the social interactions and behaviors of walruses within their community (Han et al., 2024). The algorithm operates through four main phases based on the walrus life cycle: migration, feeding and resting, gathering, and reproduction. The migration phase facilitates global exploration of the solution space, while the feeding and resting phase represents local optimization. In the gathering phase, individuals accelerate the optimization process by exchanging and sharing the

best solutions. The reproduction phase strengthens the algorithm's exploration capability by increasing the diversity of solutions.

An escape mechanism is triggered when a distress signal is detected, reducing the risk of the algorithm getting trapped in local minimal while enabling a more thorough search in safe regions. This versatile and adaptive structure allows WO to dynamically balance exploration and exploitation. Works have shown that WO performs well in engineering optimization problems and produces competitive results across various application domains (Said et al., 2024). The flowchart illustrating the operation of WO is presented in Fig. (2).

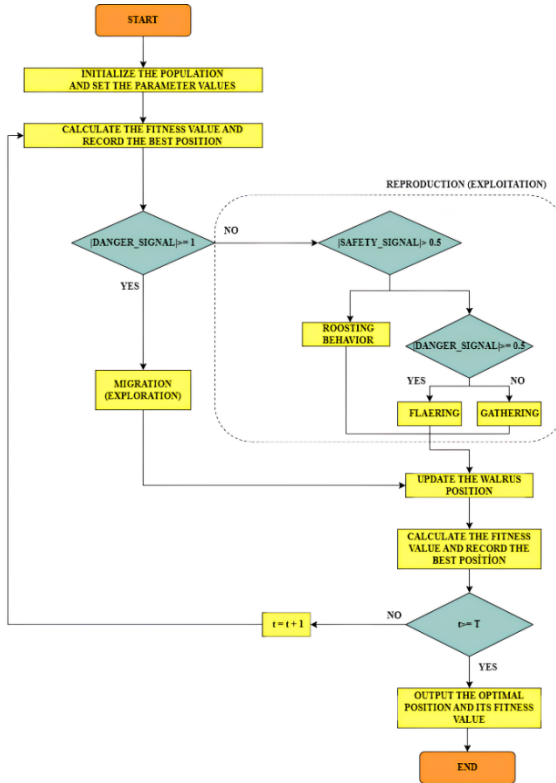


Figure 2. WO Flowchart (Li vd., 2025)

3.4. Newton-Raphson Based Optimization Algorithm

NRBO is a population-based metaheuristic inspired by the Newton-Raphson method (Sowmya et al., 2024). The algorithm employs the Newton-Raphson Search Rule (NRSR) and the Trap Avoidance Operator (TAO) to achieve an effective balance between exploration and exploitation in the solution

space. The NRSR leverages the second derivative information of the Newton-Raphson method to explore the solution space more precisely. TAO maintains population diversity by preventing the algorithm from getting trapped in local minimal.

The algorithm starts with an initially randomized population of solution vectors, which are dynamically updated through the combined application of NRSR and TAO. These updates guide the population toward the optimal solution by enhancing both the search depth and the convergence rate.

3.5. Chaotic Evolution Optimization Algorithm

CEO is an optimization algorithm inspired by chaotic dynamics (Dong et al., 2025). This algorithm is designed to overcome common challenges such as getting trapped in local minimal and lack of solution diversity. CEO is based on the chaotic evolution process of a two-dimensional memristor map, which leverages the randomness and unpredictability of chaotic systems to diversify search directions and enhance exploration efficiency.

CEO is built upon the framework of the classical Differential Evolution (DE) algorithm, enabling broader exploration of the solution space through the mutation process applied to the chaotic map. In the crossover phase, the control parameter (C_r) and the scaling factor (F) are randomly selected at each iteration, introducing dynamism into the search process and reducing the risk of being trapped in local minimal. The algorithm balances global exploration with local refinement by conducting a precise search around the best solution in the current population. CEO also successfully addresses the zero-bias problem, demonstrating its reliability and stability in solving complex optimization tasks.

3.6 Arctic Puffin Optimization Algorithm

APO is a metaheuristic inspired by the survival and hunting strategies of Arctic puffins (Wang et al., 2024). The algorithm operates through two main phases: aerial flight and underwater foraging. In the first phase, large-scale searches are performed using Levy flight and a velocity factor. Levy flight enables the algorithm to explore different solution spaces with long-distance hops, while the velocity factor regulates the direction and speed of the search process.

In the second phase, the underwater foraging process is employed, consisting of three key approaches: collective hunting, intensified search, and threat avoidance. In collective hunting, efficient hunting grounds are identified by harmonizing within the group through a synergy factor. In the intensified

search phase, more detailed exploration is conducted around the current solution. The threat avoidance strategy helps the algorithm escape local minimal by introducing random location changes.

The transition between these two phases is controlled by the behavior transformation factor, which is dynamically adjusted as the iteration progresses. This adaptive mechanism allows the algorithm to maintain a balanced trade-off between global exploration and local optimization, enhancing both the search depth and convergence speed.

3.7. BlackWinged Kite Algorithm

BKA is a metaheuristic inspired by the migration and hunting behavior of black-winged kites (Wang, J. et al., 2024). The algorithm operates through two main phases: attack and migration. In the attack phase, the hovering behavior of rookies is mimicked to detect prey and capture it through a sudden dive. During this process, Cauchy mutation is applied to broaden the search space, and a leader strategy based on the best available solution is employed. The leader strategy sets the search direction, facilitating a more efficient search for solutions.

The migration phase replicates the movement of rookies to new regions under the guidance of a leader, which adjusts according to environmental conditions. When the current solution is worse than a randomly selected solution, the leader is replaced, and the algorithm is redirected. In this phase, the repeated application of Cauchy mutation preserves solution diversity, improving the chances of reaching the global optimum.

The transition between the attack and migration phases is governed by dynamic parameters, enabling the algorithm to balance local and global search capabilities effectively. The pseudo-code of the algorithm is presented in Algorithm 2 (Wang, J. et al., 2024).

3.8. Starfish Optimization Algorithm

SFOA is an optimization algorithm inspired by the exploration, hunting, and regeneration behavior of sea stars (Zhong et al., 2025). The algorithm operates through two main phases: exploring the solution space and improving existing solutions. The first phase mimics the exploratory behavior of sea stars in their environment, facilitating efficient searches in large solution spaces. When the problem size is large, the algorithm employs a five-dimensional search pattern, whereas for smaller problem sizes, it switches to a one-dimensional strategy. This adaptive structure enhances the algorithm's ability to adjust to different problem sizes.

Algorithm 2: BKA pseudo code**Initialization phase**

Initialization of the position of Black-winged kites and evaluation of the objective function.

Calculate the fitness value of each Black-winged kite

while ($t < T$) **do**

/* **Attacking behavior** */

if $p < r$

$$y_{t+1}^{i,j} = y_t^{j,j} + n(1 + \sin(r)) \times y_t^{j,j}$$

else if do

$$y_{t+1}^{i,j} = y_t^{j,j} + n \times (2r - 1) \times y_t^{j,j}$$

end if

/* **Migration behavior** */

if $F1 < F_{ri}$ **do**

$$y_{t+1}^{\ell,J} = y_t^{\ell,J} + C(0,1) \times (y_t^{t,J} - L_t^J)$$

else if do

$$y_{t+1}'^J = y_t^J + C(0,1) \times (L_t^J - m \times y_t'^J)$$

end if

/***Select the best individual***/

if $y_{t+1}'^J < L_{jt}$

$$X_{best} = y_{t+1}'^J, F_{best} = f(y_{t+1}'^J)$$

else if do

$$X_{best} = L_{t+1}'^J, F_{best} = f(L_{t+1}'^J)$$

end if

end while

Return X_{best} and F_{best}

The second phase models the hunting and regeneration processes of sea stars. Hunting is performed using a bidirectional search strategy, where each solution moves toward more optimal locations based on the current best solution and its alternatives. In the regeneration process, the regeneration of a lost arm is simulated, enabling the algorithm to recover from poor search outcomes. This mechanism helps the algorithm escape local minimal and increases the likelihood of reaching the global optimum.

4. Experimental Design

In optimization processes, algorithm parameters play a crucial role in maintaining a balance between exploration and exploitation. These parameters

facilitate effective search within the solution space and enable the algorithm to explore larger solution areas, thereby improving the efficiency of the search process and leading to more optimal results. The specific parameters of the algorithms used in this study are presented in Table 2.

Table 2. Parameters

Algorithm	Symbol	Description	Value
PO	PF_1	First function parameter	0.5
	PF_2	Second function parameter	0.5
	PF_3	Third function parameter	0.3
	Mega_Explor	Exploration performance coefficient	0.99
	Mega_Exploit	Exploitation performance coefficient	0.99
SBOA	RL	Levy flight coefficient	0.5
WO	A	Danger factor	2
	R	Random factor	[-1, 1]
	P	Distress coefficient	0.4
NRBO	DF	Deciding factor for TAO (Trap Avoidance Operator)	0.6
CEO		Crossover control parameter	Random value from [0, 1]
		Scaling factor	Random value from [0, 1]
		Number of chaotic samples	5 (Ackley), 20 (Rastrigin)
		Hyperchaotic map control parameter	2.66
APO		Synergistic factor	0.5
		Behavioral conversion coefficient	0.5
BKA		Switching probability between two attack behaviors	0.9
		Adaptive coefficient for balancing exploration and exploitation	0.05
		Control factor for leader's influence during migration	2
SFOA		Probability of selecting exploration or exploitation phase	0.5

5. Engineering Applications

We comprehensively evaluate the performance of the selected algorithms on real-world engineering applications. Eight well-known benchmark applications are implemented, including the tension/compression spring design, pressure vessel design, welded beam design, speed reducer, gear train design, three bar truss, multiple disk clutch brake design, and step cone pulley. These applications are widely recognized in the field of engineering optimization due to their complex, multi-modal, and constrained nature, making them suitable for testing the convergence, exploration, and exploitation capabilities of optimization algorithms (Liu vd., 2024).

5.1. Tension/Compression Spring Design

Tension/compression springs are widely used components in mechanical systems across various fields, including energy storage and force damping. To ensure that the springs perform the desired elongation and compression under a given force, their design must be optimized. The objective is to minimize the stress/compression weight shown in Fig. (3) by optimizing key design parameters such as wire diameter (d), coil diameter (D), and number of coils (N) while satisfying the constraints related to minimum deflection, shear stress, and fluctuation frequency.

The mathematical model for the spring design is provided by Zhao et al. (2022).

$$\min_{x_1, x_2, x_3} f(x) + P(x)$$

subject to the following inequality constraints:

$$\begin{aligned} \frac{x_3 x_2^3}{71785 x_1^4} &\leq 1 \\ \frac{4x_2^2 - x_1 x_2}{12566(x_2 x_1^3 - x_1^4)} + \frac{1}{5108 x_1^2} &\leq 1 \\ \frac{140.45 x_1}{x_2^2 x_3} &\leq 1 \\ \frac{x_1 + x_2}{1.5} &\leq 1 \end{aligned}$$

where:

$$P(x) = \sum_{k=1}^4 \lambda \cdot g_k(x)^2 \cdot H(g_k)$$

with:

$$\begin{aligned} g_1(x) &= 1 - \frac{x_3 x_2^3}{71785 x_1^4} \\ g_2(x) &= \frac{4x_2^2 - x_1 x_2}{12566(x_2 x_1^3 - x_1^4)} + \frac{1}{5108 x_1^2} - 1 \\ g_3(x) &= 1 - \frac{140.45 x_1}{x_2^2 x_3} \\ g_4(x) &= \frac{x_1 + x_2}{1.5} - 1 \end{aligned}$$



Figure 3. Tension/compression spring design

The best solutions obtained by all algorithms for the tension/compression spring design are summarized in Table 3. As shown in the table, all algorithms have achieved the same optimal solutions by generating different values for the three design parameters of the application.

Table 3. Best solutions for tension/compression spring design

Algorithm	x_1	x_2	x_3	f_{min}
PO	0.05165	0.35589	11.33752	0.01267
SBOA	0.05167	0.35625	11.31646	0.01267
WO	0.05146	0.35118	11.62129	0.01267
NRBO	0.05167	0.35621	11.31890	0.01267
CEO	0.05169	0.35672	11.28897	0.01267
APO	0.05169	0.35671	11.28913	0.01267
BKA	0.05157	0.35381	11.46129	0.01267
SFOA	0.05167	0.35633	11.31156	0.01267

5.2. Pressure Vessel Design

Pressure vessels are structures that contain a liquid or gas at a certain pressure. The application of designing them to withstand high pressure is another common engineering challenge. The goal of the application shown in Fig. (4) is to reduce the overall cost of the pressure vessel, including expenses related to materials, forming, and welding. This is accomplished by adjusting the shell thickness (x_1), head thickness (x_2), inner radius (x_3), and the length of the cylindrical section (x_4) while ensuring that the design adheres to the specified limits on stress, internal volume, and manufacturing feasibility. Mathematical models are (Zamani vd., 2022).

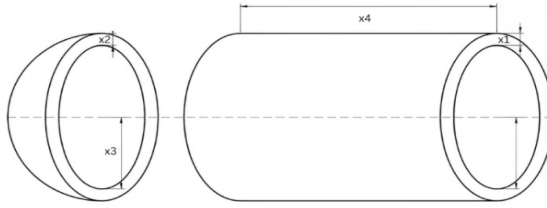


Figure 4. Pressure Vessel Design

$$\min_{x_1, x_2, x_3, x_4} f(x) + P(x)$$

where:

$$f(x) = 0.6224x_1x_3x_4 + 1.7781x_2x_3^2 + 3.1661x_1^2x_4 + 19.84x_1^2x_3$$

Stress constraints:

$$\begin{aligned} x_1 &\geq 0.0193x_3 \\ x_2 &\geq 0.00954x_3 \end{aligned}$$

Volume constraint:

$$\pi x_3^2 x_4 + \frac{4}{3} \pi x_3^3 \geq 1296000$$

Length constraint:

$$x_4 \leq 240$$

Penalty function:

$$P(x) = \sum_{k=1}^4 \lambda \cdot g_k(x)^2 \cdot H(g_k)$$

The results of the best solutions obtained by the algorithms are presented in Table 4. As shown in the table, SBOA and CEO achieved the best objective function values for the application, while SFOA produced the worst solution.

Table 4. Best solutions for pressure vessel design

Algorithm	x_1	x_2	x_3	x_4	f_{min}
PO	0.77817	0.38465	40.31968	199.99960	5885.34710
SBOA	0.77817	0.38465	40.31962	200.00000	5885.33277
WO	0.77822	0.38472	40.32227	199.96362	5885.55552
NRBO	0.77820	0.38466	40.32124	199.97772	5885.39281
CEO	0.77817	0.38465	40.31962	200.00000	5885.33277
APO	0.77817	0.38465	40.31962	199.99999	5885.33279
BKA	0.77817	0.38552	40.31964	199.99968	5887.86277
SFOA	0.77821	0.38474	40.32143	200.00000	5886.20773

5.3. Welded Beam Design

The welded beam design is a structural optimization task aimed at enabling the beam to carry a load without deformation and within safe working limits while minimizing the total cost, which includes material, fabrication, and welding expenses. To achieve this, the design optimizes key parameters, including weld thickness (h), beam width (b), beam height (l), and the length of the welded joint (t), while satisfying constraints on shear stress, bending stress, buckling load, and end deflection. The mathematical models for this application are provided by Zhong and Li (2022).

$$\min_{x_1, x_2, x_3, x_4} f(x) + P(x)$$

where:

$$f(x) = 1.10471x_1^2x_2 + 0.04811x_3x_4(14 + x_2)$$

Shear stress constraint:

$$\tau = \sqrt{\alpha^2 + 2\frac{\alpha\beta x_2}{2D} + \beta^2} \leq 13600$$

Bending stress constraint:

$$\sigma = \frac{504000}{x_4 x_3^2} \leq 30000$$

Geometry constraint:

$$x_1 \leq x_4$$

Fabrication cost constraint:

$$0.10471x_1^2 + 0.04811x_3x_4(14 + x_2) \leq 5$$

Minimum weld thickness constraint:

$$x_1 \geq 0.125$$

Deflection constraint:

$$\delta = \frac{65856000}{30 \cdot 10^6 \cdot x_4 \cdot x_3^3} \leq 0.25$$

Buckling load constraint:

$$F = \frac{4.013(30 \cdot 10^6)}{196} \sqrt{\frac{x_3^2 x_4^6}{36}} \left(1 - \frac{x_3 \sqrt{30/48}}{28} \right) \geq 6000$$

Penalty function:

$$P(x) = \sum_{k=1}^7 \lambda \cdot g_k(x)^2 \cdot H(g_k)$$

The experimental results are presented in Table 5. Four algorithms (PUMA, SBOA, CEO, and APO) achieved the best solutions for the welded beam design by generating the same position vectors. However, the best solutions obtained by all algorithms are very close to each other.

Table 5. Best solutions for welded beam design

Algorithm	x_1	x_2	x_3	x_4	f_{min}
PO	0.20573	3.47049	9.03662	0.20573	1.72485
SBOA	0.20573	3.47049	9.03662	0.20573	1.72485
WO	0.20570	3.47120	9.03674	0.20573	1.72492
NRBO	0.20598	3.46726	9.03105	0.20599	1.72578
CEO	0.20573	3.47049	9.03662	0.20573	1.72485
APO	0.20573	3.47049	9.03662	0.20573	1.72485
BKA	0.20574	3.47030	9.03629	0.20574	1.72491
SFOA	0.20569	3.47144	9.03692	0.20573	1.72498

5.4. Speed Reducer

The speed reducer is a well-known nonlinear engineering challenge in the fields of mechanical design and optimization. The objective is to minimize the overall weight of the speed reducer while maintaining its structural durability and operational efficiency, considering material strength and mechanical constraints. This is accomplished by optimizing key design parameters, including face width (x_1), module (x_2), number of teeth (x_3), length of the shaft between bearings (x_4), shaft diameter (x_5), bearing diameter (x_6), and gear center distance (x_7).

The design must also satisfy specific constraints related to bending stress, surface stress, shaft torsion, and geometric limitations to ensure that the speed reducer functions correctly and efficiently. The mathematical models for this application are provided by Abualigah et al. (2021).

$$\min_{x_1, x_2, x_3, x_4, x_5, x_6, x_7} f(x) + P(x)$$

where:

$$\begin{aligned} f(x) = & 0.7854x_1x_2^2(3.3333x_3^2 + 14.9334x_3 - 43.0934) \\ & - 1.508x_1(x_6^2 + x_7^2) + 7.4777(x_6^3 + x_7^3) \\ & + 0.7854(x_4x_6^2 + x_5x_7^2) \end{aligned}$$

In equation:

- x_1 : Ending stress constraint
- x_2 : Surface stress constraint
- x_3 : Torsional load constraint (first and second shafts)
- x_4 : Combined stress constraints
- x_5 : Gear size constraint
- x_6 : Face width and module ratio constraints
- x_7 : Shaft length to bearing diameter constraints

Penalty function:

$$P(x) = \sum_{k=1}^{11} \lambda \cdot g_k(x)^2 \cdot H(g_k)$$

The design variables are subject to the following practical and manufacturing limits:

$$\begin{aligned} 2.6 &\leq x_1 \leq 3.6 \\ 0.7 &\leq x_2 \leq 0.8 \\ 17 &\leq x_3 \leq 28 \\ 7.3 &\leq x_4, x_5 \leq 8.3 \\ 2.9 &\leq x_6, x_7 \leq 3.9 \end{aligned}$$

The optimization results for the speed reducer are presented in Table 6. As shown in Table 6, PO, SBOA, and CEO are among the algorithms that achieved the best solution. In contrast, NRBO did not perform well compared to the other algorithms for this application.

Table 6. Best solutions for speed reducer

Algorithm	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f_{min}
PO	3.50000	0.70000	17.00000	7.30000	7.71532	3.35021467	5.28665	2994.47107
SBOA	3.50000	0.70000	17.00000	7.30000	7.71532	3.35021467	5.28665	2994.47107
WO	3.50001	0.70000	17.00000	7.30031	7.71566	3.35023086	5.28666	2994.48858
NRBO	3.51318	0.70000	17.00000	7.30000	7.71711	3.35175763	5.28675	3000.14252
CEO	3.50000	0.70000	17.00000	7.30000	7.71532	3.35021467	5.28665	2994.47107
APO	3.50000	0.70000	17.00000	7.30000	7.71532	3.35021477	5.28665	2994.47116
BKA	3.50109	0.70000	17.00000	7.30000	7.77106	3.35021467	5.28823	2997.12008
SFOA	3.50000	0.70000	17.00000	7.30000	7.71532	3.35021467	5.28665	2994.47107

5.5. Gear Train Design

Gear trains are fundamental components of mechanical power transmission in the machinery and automotive industries. In design applications, the goal is to minimize the total size and weight of the gear train while ensuring efficient power transmission and mechanical durability. To achieve this, key design parameters, including gear module (x_1), number of teeth on the pinion (x_2), number of teeth on the gear (x_3), and face width (x_4) are optimized while satisfying the constraints on torque transmission, bending stress, surface durability, and gear ratio. The mathematical models for this application are provided by Qiu et al. (2022).

$$\min_{x_1, x_2, x_3, x_4} f(x)$$

where:

$$f(x) = \left(\frac{1}{6.931} - \frac{x_3 x_2}{x_1 x_4} \right)^2$$

Bending stress constraint:

$$\frac{T}{x_1 x_2 x_4} \leq \sigma_{\max}$$

Surface durability constraint:

$$\frac{T}{x_1^2 x_3 x_4} \leq \tau_{\max}$$

Gear ratio constraint:

$$\frac{x_3}{x_2} = R$$

Geometrical constraints:

$$x_1, x_2, x_3, x_4 > 0$$

Table 7 summarizes the best solution obtained of the gear train design solved by all algorithms. According to the table, all algorithms were able to find the optimal solution. PO has a very small error (E-34).

Table 7. Best solutions for gear train design

Algorithm	x_1	x_2	x_3	x_4	f_{min}
PO	35.63241	13.41589	13.15439	34.32741	7.70372E-34
SBOA	60.00000	43.28380	12.00000	60.00000	0.00000E+00
WO	59.70978	32.10792	13.13458	48.95296	0.00000E+00
NRBO	37.78417	13.55070	15.36845	38.20128	0.00000E+00
CEO	33.76850	24.28572	12.00000	59.81585	4.01146E-20
APO	56.26218	21.90135	16.93563	45.69320	6.36400E-19
BKA	54.15997	12.04481	19.51288	30.07729	0.00000E+00
SFOA	60.00000	25.67604	12.00000	35.59213	1.57419E-22

5.6. Three Bar Truss Design

For the three-bar truss design, analyzing the load-bearing capacity of each bar and the resulting deformations is critical for evaluating the overall performance of the structure. The objective is to minimize the total weight of the three-bar truss while ensuring structural stability and load-bearing capacity. To achieve this, key design parameters, including the cross-sectional area of the bars (x_1, x_2, x_3) and the length of the truss members (L), are optimized while satisfying constraints on stress limits, buckling stability, and displacement limits. The mathematical models for this application are provided by Coello (2000).

$$\min_{x_1, x_2} f(x) + P(x)$$

where:

$$f(x) = (2\sqrt{2}x_1 + x_2) \cdot 100$$

Stress constraint for diagonal bars:

$$\frac{\sqrt{2}x_1 + x_2}{\sqrt{2x_1^2 + 2x_1x_2}} \cdot 2 \leq 2$$

Stress constraint for vertical bar:

$$\frac{x_2}{\sqrt{2x_1^2 + 2x_1x_2}} \cdot 2 \leq 2$$

Stability constraint:

$$\frac{1}{\sqrt{2}x_2 + x_1} \cdot 2 \leq 2$$

Penalty function:

$$P(x) = \sum_{k=1}^3 \lambda \cdot g_k(x)^2 \cdot H(g_k)$$

The results of all algorithms for the three-bar truss design are provided in Table 8. Similar to Table 5, all algorithms in this table successfully found the optimal solution.

Table 8. Best solutions for three bar truss design

Algorithm	x_1	x_2	f_{min}
PO	0.78868	0.40825	263.89584
SBOA	0.78866	0.40829	263.89584
WO	0.78871	0.40816	263.89584
NRBO	0.78868	0.40825	263.89584
CEO	0.78868	0.40825	263.89584
APO	0.78868	0.40825	263.89584
BKA	0.78868	0.40824	263.89584
SFOA	0.78868	0.40825	263.89584

5.7. Multiple Disk Clutch Brake Design

The multiple disk clutch brake design is a significant engineering challenge in the development of efficient power transmission and safe braking systems for automotive and industrial applications. The objective is to minimize the total weight and size of a multiple disk clutch brake while ensuring efficient torque transmission and mechanical performance. To achieve this, the design optimizes key parameters, including inner radius (x_1), outer radius (x_2), thickness of the friction plate (x_3), applied force (x_4), and number of friction surfaces (x_5) while satisfying constraints on torque capacity, pressure distribution, material strength, and operational limits. The mathematical models for this application are provided by Zhao et al. (2022).

$$\min_{x_1, x_2, x_3, x_4, x_5} f(x) + P(x)$$

where:

$$f(x) = \pi x_3 (x_2^2 - x_1^2) (x_5 + 1) P_{\max}$$

Minimum thickness constraint:

$$x_2 - x_1 \geq 20$$

Maximum length constraint:

$$(x_5 + 1)(x_3 + 0.5) \leq 30$$

Maximum pressure constraint:

$$P_{\max} = \frac{x_4}{\pi(x_2^2 - x_1^2)} \leq 1$$

Pressure and sliding velocity compatibility:

$$P_{\max} \cdot V_{SR_{\max}} \geq \frac{x_4}{\pi(x_2^2 - x_1^2)} \cdot V_{rz}$$

Maximum sliding velocity constraint:

$$V_{rz} \leq 10000$$

Maximum torque transmission constraint:

$$T = \frac{I \pi n}{30(M_H + M_F)} \leq 15$$

Minimum torque capacity constraint:

$$M_H \geq 1.5 \cdot 40$$

Positive torque constraint:

$$T \geq 0$$

Penalty function:

$$P(x) = \sum_{k=1}^8 \lambda \cdot g_k(x)^2 \cdot H(g_k)$$

The design variables are subject to the following practical and manufacturing limits:

$$\begin{aligned} 50 &\leq x_1, x_2 \leq 150 \\ 1 &\leq x_3 \leq 10 \\ 1000 &\leq x_4 \leq 5000 \\ 1 &\leq x_5 \leq 10 \end{aligned}$$

The solutions obtained by all algorithms are presented in Table 9. As shown in the table, all algorithms produced the same optimal solution with similar position vectors. However, each algorithm generated a different value for x_4 .

Table 9. Best solutions for multiple disk clutch brake design

Algorithm	x_1	x_2	x_3	x_4	x_5	f_{min}
PO	70	90	1	600.000	2	30159.28943
SBOA	70	90	1	600.000	2	30159.28943
WO	70	90	1	914.656	2	30159.28943
NRBO	70	90	1	600.000	2	30159.28943
CEO	70	90	1	652.714	2	30159.28943
APO	70	90	1	961.531	2	30159.28943
BKA	70	90	1	1000.000	2	30159.28943
SFOA	70	90	1	703.354	2	30159.28943

5.8. Himmelblau's Problem

The Himmelblau Problem is widely used to evaluate the performance of optimization algorithms. Its significance lies in the presence of not only local minimal but also multiple global minimal, which plays a crucial role in determining the success rate of algorithms in complex solution spaces. The objective of the problem is to minimize the value of the Himmelblau function under these conditions. To achieve this, the problem optimizes the parameters (x_1) , (x_2) , (x_3) , (x_4) , and (x_5) while ensuring that the solution satisfies the constraints on smoothness and gradient properties. The mathematical models for this problem are provided by Zhong et al. (2025).

$$\min_{x_1, x_2, x_3, x_4, x_5} f(x) + P(x)$$

where:

$$f(x) = 5.3578547x_2^2 + 0.8356891x_1x_5 + 37.293239x_1 - 40792.141$$

First constraint:

$$\begin{aligned} 85.334407 + 0.0056858x_2x_5 + 0.0006262x_1x_4 - 0.0022053x_3x_5 &\leq 92 \\ 85.334407 + 0.0056858x_2x_5 + 0.0006262x_1x_4 - 0.0022053x_3x_5 &\geq 0 \end{aligned}$$

Second constraint:

$$\begin{aligned} 80.51249 + 0.0071317x_2x_5 + 0.0029955x_1x_2 + 0.0021813x_3^2 &\leq 110 \\ 80.51249 + 0.0071317x_2x_5 + 0.0029955x_1x_2 + 0.0021813x_3^2 &\geq 90 \end{aligned}$$

Third constraint:

$$\begin{aligned} 9.300961 + 0.0047026x_3x_5 + 0.0012547x_1x_3 + 0.0019085x_3x_4 &\leq 25 \\ 9.300961 + 0.0047026x_3x_5 + 0.0012547x_1x_3 + 0.0019085x_3x_4 &\geq 20 \end{aligned}$$

Penalty function:

$$P(x) = \sum_{k=1}^6 \lambda \cdot g_k(x)^2 \cdot H(g_k)$$

The design variables are subject to the following practical and manufacturing limits:

$$0 \leq x_1, x_2, x_3, x_4, x_5 \leq 100$$

Table 10 presents the global solutions obtained by each algorithm for the Himmelblau’s problem. The table shows that all algorithms produced the same value, except for WO, which exhibited a difference of approximately 0.002.

Table 10. Best solutions for Himmelblau’s problem

Algorithm	x_1	x_2	x_3	x_4	x_5	f_{min}
PO	78.000	33.000	29.995	45.000	36.776	-30665.539
SBOA	78.000	33.000	29.995	45.000	36.776	-30665.539
WO	78.000	33.000	29.995	45.000	36.776	-30665.537
NRBO	78.000	33.000	29.995	45.000	36.776	-30665.539
CEO	78.000	33.000	29.995	45.000	36.776	-30665.539
APO	78.000	33.000	29.995	45.000	36.776	-30665.539
BKA	78.000	33.000	29.995	45.000	36.776	-30665.539
SFOA	78.000	33.000	29.995	45.000	36.776	-30665.539

5. Performance Analysis of Optimization Algorithms

The statistical test results of the algorithms for each application are presented in Table 11. The table compares the performance of the algorithms based on the minimum value (min), mean value (mean), maximum value (max), and standard deviation (std) metrics. A detailed examination of the table leads to the following conclusions for each application:

In tension/compression spring design:

- The best result was obtained by PO, SBOA, WO, CEO, APO, BKA and SFOA algorithms with a value of 0.01.
- NRBO and produced a value of 0.012, which is consistent with the best performing algorithms.

In pressure vessel design:

- The lowest value was obtained by SBOA, CEO, and APO algorithms with a value of 5885.33.
- PO algorithm produced a very close result of 5885.35 (difference ≈ 0.01).
- WO produced a result of 5885.56, approximately 0.23 units higher than the best solution.
- NRBO produced a value of 5885.39, approximately 0.06 units higher than the best value.
- BKA produced a result of 5887.86, which is 2.53 units higher than the best result.
- SFOA algorithm produced a result of 5886.20, approximately 0.87 units higher than the best result.

In welded beam design:

- The best result was obtained by the PO and SBOA algorithms with a value of 1.72.
- WO produced a very close result of 1.724 (difference ≈ 0.00007).
- NRBO produced a value of approximately 1.73 (difference ≈ 0.001).
- CEO and APO algorithms produced identical results of 1.72485.
- BKA algorithm produced a result of 1.725, which is 0.00006 units higher than best result.
- SFOA produced a result of 1.72498, approximately 0.00013 units higher than the best result.

In speed reducer:

- The best result was obtained by PO, SBOA, and NRBO algorithms with a value of approximately 2994.48.
- WO produced a very close result of 2994.49 (difference ≈ 0.0176).
- CEO, APO, and SFOA algorithms produced identical results, consistent with the best solutions.

- BKA algorithm produced the worst result of 2997.12, which is approximately 2.65 units higher than the best value.

In gear train design:

- The best theoretical result is expected to be 0.0.
- SBOA, WO, NRBO, CEO, APO, BKA, and SFOA algorithms produced an ideal result of 0.0.

- PO algorithm produced a value of 7.70-34, which is practically equivalent to zero.

In three bar truss design:

- The best result was obtained by the SBOA algorithm with a value of 263.89.

- PO algorithm produced a result of 267.11 (difference ≈ 3.22).

- CEO, APO, and SFOA algorithms produced results in the range of 267 to 278.

Table 11. Performance Analysis of Algorithms

Applications	Algo.	PO	SBOA	WO	NRBO	CEO	APO	BKA	SFOA
Tension/ compression spring design	Min	0.01267	0.01267	0.01267	0.01267	0.01267	0.01267	0.01267	0.01267
	Mean	0.01269	0.01269	0.01392	0.01311	0.01267	0.01267	0.01284	0.01267
	Max	0.01302	0.01292	0.01777	0.01724	0.01267	0.01267	0.01575	0.01272
	Std	0.00004	0.00004	0.00173	0.00082	0.00000	0.00000	0.00048	0.00001
Pressure vessel design	Min	5885.34710	5885.33277	5885.55552	5885.39281	5885.33277	5885.33279	5887.86277	5886.20773
	Mean	6021.22543	5981.31019	6603.10998	6489.24745	5885.33277	5885.33291	7671.89935	5893.41026
	Max	7319.00080	6907.29481	7323.65046	7634.81729	5885.33277	5885.33360	55834.55720	5968.97952
	Std	219.52773	180.58886	562.92439	520.73674	0.00000	0.00012	6580.18918	8.77461
Welded beam design	Min	1.72485	1.72485	1.72492	1.72578	1.72485	1.72485	1.72491	1.72498
	Mean	1.72487	1.72487	1.81020	1.80152	1.72485	1.72485	1.77847	1.72538
	Max	1.72514	1.72557	2.52151	1.95879	1.72485	1.72485	2.66038	1.72626
	Std	0.00004	0.00008	0.16508	0.04806	0.00000	0.00000	0.18203	0.00029
Speed reducer	Min	2994.47107	2994.47107	2994.48858	3000.14252	2994.47107	2994.47116	2997.12008	2994.47107
	Mean	2994.47110	2994.47107	2996.27219	3041.96061	2994.47107	2994.47153	3010.90860	2994.47112
	Max	2994.47125	2994.47107	3005.35290	3145.40746	2994.47107	2994.47256	3144.25218	2994.47128
	Std	0.00003	0.00000	2.13458	37.60309	0.00000	0.00025	23.43123	0.00004
Gear train design	Min	7.70372E-34	0.00000E+00	0.00000E+00	0.00000E+00	4.01146E-20	6.36400E-19	0.00000E+00	1.57419E-22
	Mean	6.70458E-18	1.20679E-16	3.21118E-16	8.67506E-29	2.11016E-14	3.10958E-14	0.00000E+00	1.71862E-12
	Max	3.05254E-16	2.19489E-15	3.20555E-14	4.73428E-27	5.30670E-13	1.17079E-12	0.00000E+00	5.03792E-11
	Std	4.16822E-17	2.90079E-16	3.20549E-15	5.32442E-28	6.42846E-14	1.24899E-13	0.00000E+00	5.67205E-12

Table 11. Performance Analysis of Algorithms

Three bar truss design	Min	263.89584	263.89584	263.89584	263.89584	263.89584	263.89584	263.89584	263.89584	263.89584
	Mean	263.89584	263.89596	263.90959	263.89589	263.89584	263.89584	263.89584	263.90212	263.89584
	Max	263.89584	263.89672	264.00515	263.89800	263.89584	263.89584	263.89584	264.24590	263.89584
	Std	0.00000	0.00017	0.02097	0.00025	0.00000	0.00000	0.00000	0.03756	0.00000
Multiple Disk Clutch Brake Design	Min	30159.28943	30159.28943	30159.28943	30159.28943	30159.28943	30159.28943	30159.28943	30159.28943	30159.28943
	Mean	30159.28943	30159.28943	30159.52015	30159.28943	30159.28943	30159.28943	30159.28943	30526.85577	30159.28943
	Max	30159.28943	30159.28943	30176.05900	30159.28943	30159.28943	30159.28943	30159.28943	42411.50082	30159.28943
	Std	0.00000	0.00000	1.69465	0.00000	0.00000	0.00000	0.00000	2100.60008	0.00000
Himmelblau's Problem	Min	-30665.53869	-30665.53869	-30665.53650	-30665.53869	-30665.53869	-30665.53869	-30665.53869	-30665.53869	-30665.53869
	Mean	-30665.53869	-30665.53869	-30664.37693	-30636.00249	-30665.53869	-30665.53869	-30665.53849	-30655.87040	-30665.53814
	Max	-30665.53830	-30665.53869	-30650.65346	-30208.36939	-30665.53869	-30665.53710	-30665.53771	-30117.39271	-30665.53377
	Std	0.00004	0.00000	1.91792	68.79060	0.00000	0.00023	61.02052	0.00085	

In multiple disk clutch brake design:

- The best result was obtained by the PO, SBOA, and WO algorithms with a value of 30159.29.
- CEO, APO, and SFOA algorithms produced consistent results close to the best values.
- BKA produced the same result as the best values.

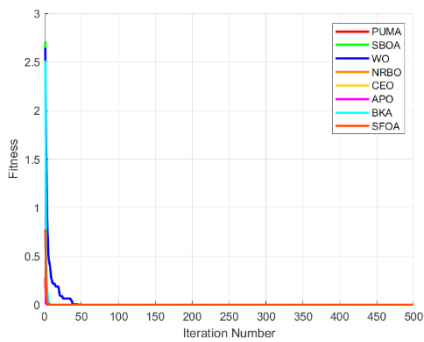
In Himmelblau's problem:

- The best result was obtained by the SBOA, PO, and WO algorithms with a value of -30665.53869.
- The NRBO and BKA algorithms produced a very close value of -30665.53650.

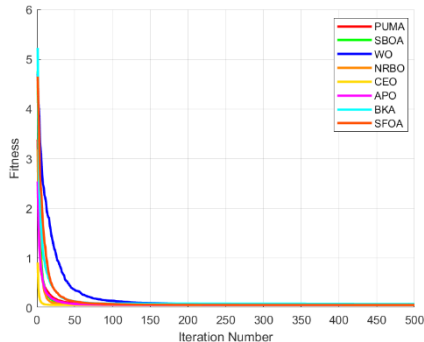
The convergence plots of the algorithms for all applications are shown in Fig. (5). The convergence behavior for the tension/compression spring illustrates the superior performance of the PO and SFOA algorithms. Both algorithms exhibit rapid convergence within the first 50 iterations, quickly stabilizing at the lowest fitness value. BKA and NRBO also display competitive performance, achieving similar fitness values after a slightly longer convergence period. In contrast, the WO algorithm shows a slower convergence rate and higher final fitness value, indicating its limitations in exploring the search space effectively. The rapid and stable convergence of PO and SFOA highlights their robustness.

The convergence plot for the pressure vessel design shows that PO, SFOA, and NRBO algorithms converge rapidly within the first 50 iterations, stabilizing at a low fitness value. BKA and CEO also demonstrate competitive performance, though their convergence rates are slower compared to PO and SFOA. WO exhibits the slowest convergence and higher final fitness value, indicating its inferior search capability for this application. The results highlight the ability of PO and SFOA to quickly locate high-quality solutions.

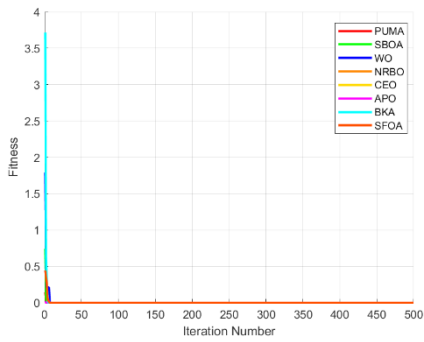
The convergence pattern for the welded beam design reveals that PO and SFOA provide the fastest and most stable convergence among all tested algorithms. Both algorithms reach the minimum fitness value within the first 50 iterations and remain stable thereafter. NRBO, BKA, and CEO also exhibit strong convergence behavior, but their final fitness values are slightly higher than those of PO and SFOA. On the other hand, WO shows slower convergence and a higher final fitness value, suggesting weaker performance.



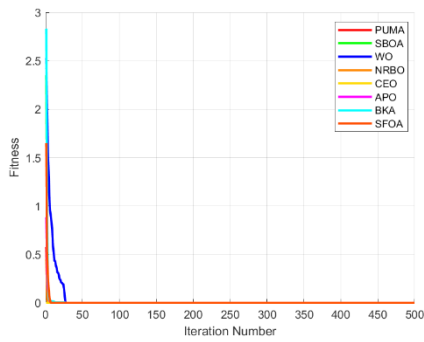
(a) Tension/Compression Spring Design



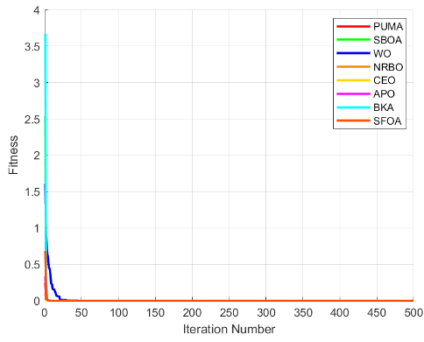
(b) Pressure Vessel Design



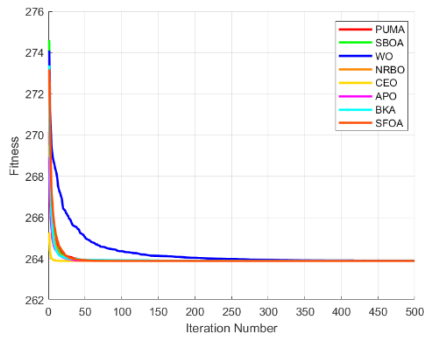
(c) Welded Beam Design



(d) Speed Reducer

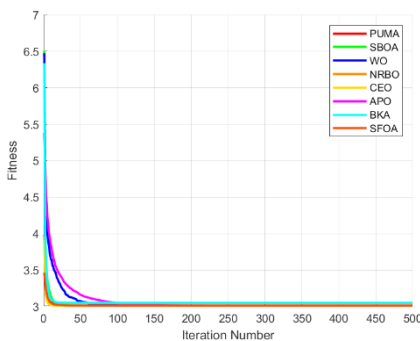


(e) Gear Train Design

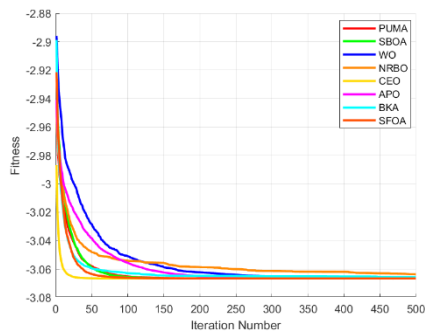


(f) Three Bar Truss Design

Figure 5. The convergence plots



(g) Multiple Disk Clutch Brake Design



(h) Himmelblau's problem

Figure 5. The convergence plots (continued)

The convergence plot for the speed reducer highlights the consistent performance of PO and SFOA, which achieve rapid and stable convergence within the initial 50 iterations. NRBO and BKA demonstrate competitive performance but require more iterations to reach the same fitness value. WO lags significantly, with slower convergence and a higher final fitness value. The strong early convergence and low final fitness values of PO and SFOA emphasize their effectiveness.

The convergence plot for the gear train design shows that PO, SFOA, and NRBO outperform other algorithms by achieving low fitness values within the first 50 iterations. BKA and CEO display competitive but slightly slower convergence. WO exhibits delayed convergence and higher final fitness values, reflecting poor search capability for this application. The results highlight the strength of PO and SFOA in efficiently solving this application through fast and consistent convergence.

The convergence plot for the three-bar truss design demonstrates the robust performance of PO and SFOA, which achieve rapid and stable convergence within the first 50 iterations. NRBO and BKA also exhibit competitive performance, converging to similar final fitness values after a longer search period. WO displays the weakest performance, with slower convergence and higher final fitness values. The superior early convergence and lower final values of PO and SFOA underline their capability.

The convergence plot for the multiple disk clutch brake design indicates that PO and SFOA achieve the lowest fitness values within the first 50 iterations, demonstrating rapid and stable convergence. NRBO and BKA also display

strong performance, converging at slightly higher fitness values. WO again shows slower convergence and higher final fitness values, highlighting its weaker search capability. The fast and consistent convergence of PO and SFOA underscores their effectiveness.

The convergence plot for Himmelblau's problem reveals that PO and SFOA exhibit the fastest and most stable convergence, reaching the lowest fitness values within the first 50 iterations. NRBO, BKA, and CEO also display competitive convergence patterns, though their final fitness values remain slightly higher. WO shows delayed convergence and higher fitness values, indicating less effective search performance. The rapid and stable convergence of PO and SFOA reflects their strength in handling this problem effectively.

7. Conclusion

In this study, eight different metaheuristic algorithms (PO, SBOA, WO, NRBO, CEO, APO, BKA and SFOA) were tested to solve eight different engineering optimization applications. The results obtained revealed that the PO and SFOA algorithms generally show the best performance. When considering the convergence speed, solution quality and solution stability of the algorithms, SFOA took first place and was characterized by its fast convergence and low fitness value. PO ranked second in terms of solution quality and fast convergence, while CEO ranked third with a low variance and high solution stability. On the other hand, the WO algorithm showed the lowest performance with slow convergence, high maximum and average fitness values. The BKA algorithm, on the other hand, performed well on some applications but was inadequate in terms of overall performance and stability. The results show that the SFOA, PO and CEO algorithms provide effective solutions for these applications and are competitive for general optimization problems.

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CHAPTER III

DATA AUGMENTATION METHODS IN ARTIFICIAL INTELLIGENCE

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1. Introduction

Artificial intelligence has significantly advanced in recent years and is increasingly used in various fields. Despite these developments, the success of an artificial intelligence model relies on large and labeled datasets. However, the size and diversity of these datasets are often limited. Therefore, data augmentation methods are employed to make artificial intelligence models work more efficiently and accurately (Shorten and Khoshgoftaar, 2019; Yang et al., 2022; Alomar et al., 2023).

Data augmentation is the process of generating more data by performing various operations on existing data. It is a crucial technique used to increase the size of datasets and improve the generalization ability of models, particularly in artificial intelligence applications (Mumuni and Mumuni, 2025). Data augmentation helps improve the performance of models trained with limited data, leading to more robust and accurate results. The goal of data augmentation is to enable the model to generalize better, prevent overfitting, and address data imbalances (Van Thang, 2019; Iwana and Uchida, 2021). The advantages of data augmentation are as follows:

- It improves model performance by providing more data.
- It prevents overfitting.
- It helps create a fair model by addressing data imbalances.
- It reduces the cost of collecting additional data.

Data augmentation plays a critical role in various types of data such as images, text, audio, and time-series data. With the evolving artificial intelligence techniques, data augmentation methods have become more diverse and sophisticated. This chapter will examine the data augmentation methods in artificial intelligence applications based on different data types, discussing their use cases and future developments.

2. Data Augmentation Methods

Data augmentation methods are specialized according to different data types (Kumar et al., 2024). The methods used for image, text, audio, and other types of data differ based on the characteristics of each data type. There are both basic and advanced data augmentation methods for each type of data, including image, text, and audio.

2.1. Data Augmentation Methods for Image Data

Image data is the most common data type in computer vision applications and deep learning methods (LeCun et al., 2015). Image data is used in object recognition, face recognition, segmentation, and other similar applications (Shin et al., 2016; Gürbüz and Yılmaz, 2023). The data augmentation techniques applied to image data help the model learn visual information more effectively (Krizhevsky et al., 2017). The basic and advanced data augmentation methods applied to image data can be listed as follows (Chlap et al., 2021; Khalifa et al., 2022; Yang et al., 2022; Lalitha and Latha, 2022; Lewy and Mańdziuk, 2023; Kumar et al., 2024; Bajaj et al., 2024; Rais et al., 2025; Jou et al., 2025):

2.1.1. Basic Methods

- **Rotation:** Rotating images at specific angles allows the model to recognize objects from different perspectives. For example, a 90-degree rotated image provides the model with more information about the orientation of the object.
- **Flipping:** Flipping images horizontally or vertically is especially useful when dealing with asymmetric objects. This operation increases the directional diversity of the image.
- **Scaling:** Changing the size of images enables the model to recognize objects at different scales. Shrinking or enlarging the image allows for a broader perspective of learning.

- **Cropping and Padding:** Randomly cropping images ensures that the model learns different visual sections of an image. Similarly, adding padding around the edges to keep the image at a fixed size is also a common method.
- **Color Adjustments:** Altering parameters like color saturation, brightness, and contrast helps the model make accurate predictions under different lighting conditions.

2.1.2. Advanced Methods

- **Generative Adversarial Networks (GANs):** GANs are models that take data augmentation techniques further by generating realistic images. This technique is highly effective in data augmentation. GANs are used to generate completely new and original images, especially when working with limited datasets.
- **Style Transfer:** This method applies a different artistic style or visual aesthetic to an image. For instance, transforming a photo into a painting in the style of a particular artist allows the model to work with various visual styles.
- **Autoencoders and Variational Autoencoders (VAE):** Autoencoders can be used to compress images and generate new and diverse variations. VAEs are an advanced approach that helps the model learn more features of the images.

2.2. Data Augmentation Methods for Text Data

Text data is widely used in natural language processing (NLP) applications (Kim and Kang, 2022). Augmenting text data allows for diversification without losing meaning, making it easier for the model to make accurate predictions when encountering linguistic diversity (Wei and Zou, 2019; Dai et al., 2025). The basic and advanced data augmentation methods for text data can be listed as follows (Liu et al., 2020; Corbeil and Ghadivel, 2020; Sawai et al., 2021; Beddiar et al., 2021; Bhattacharjee et al., 2022; Mi et al., 2022; Onan, 2023; Che et al., 2023; Zhao et al., 2024; Taheri et al., 2024; Al-Shameri and Al-Khalifa, 2024; ElSabagh et al., 2025):

2.2.1. Basic Methods

- **Synonym Replacement:** Replacing words in a sentence with their synonyms creates new text examples while maintaining the semantic structure of the text.
- **Word Shuffling:** Changing the order of words in a sentence increases linguistic diversity. This method helps the model understand different structures.

- **Word or Sentence Insertion/Deletion:** By adding or removing words or sentences within the text, new examples can be created while maintaining meaning.

2.2.2. Advanced Methods

- **Back Translation:** This method involves translating text from one language to another and then translating it back to the original language to express the text in different ways. This is used to generate new variations without altering the meaning of the text.

- **Paraphrasing:** Paraphrasing involves expressing the same message in different words without changing its meaning. This technique encourages the language model to use a wider vocabulary.

- **Transformer-Based Models (BERT, GPT):** Transformer architectures are powerful for text augmentation and generation. These models automate text data augmentation, generating a broader and more realistic set of text variations.

2.3. Data Augmentation Methods for Audio Data

Audio data plays a critical role in speech recognition, voice response systems, and other voice-interactive applications (Patil, 2024). Augmenting audio data enables the model to be trained on various audio examples (Salamon and Bello, 2017; Sugiura et al., 2021). The basic and advanced data augmentation methods for audio data can be listed as follows (Kumar et al., 2019; Mertes et al., 2020; Yella and Rajan, 2021; Ferreira-Paiva et al., 2022; Abayomi-Alli et al., 2022; Muthumari et al., 2022; Alex et al., 2023; Galić et al., 2024; Shi et al., 2024; Maguolo et al., 2025):

2.3.1. Basic Methods

- **Volume Adjustments:** Increasing or decreasing the volume of an audio recording allows the model to recognize speech at different sound levels.

- **Time Stretching:** Changing the speed of an audio recording helps the model understand speech at different speeds.

- **Adding Noise:** Adding background noise to audio data helps speech recognition systems perform better in noisy environments.

2.3.2. Advanced Methods

- **WaveGAN:** This model is an artificial intelligence model used to generate audio data. By producing realistic audio samples, it enhances existing audio datasets.

- **MelGAN and Vocoder-Based Method:** Audio data can be augmented by modifying the Mel frequency spectrum. It's possible to change the tone, speed, or other characteristics of the sounds.
- **Speech Enhancement and Augmentation (SESAR):** This method improves audio samples by generating cleaner and more recognizable speech examples for voice response systems.

2.4. Other Data Augmentation Methods

Data augmentation is not limited to image, text, and audio data. Different augmentation methods are also applied to specialized data types in artificial intelligence (Rashid and Louis, 2019; Liu et al., 2022; Volkova, 2023; Gao et al., 2023; Pérez et al., 2023; Onishi and Meguro, 2023; Shin et al., 2024; Su et al., 2024; Tang et al., 2024; Isomura et al., 2025).

- **Time Series Data Augmentation:** Time-series data such as financial, healthcare, and sensor data can be augmented. In particular, generating new examples from time-dependent data can improve model accuracy.
- **Tabular Data Augmentation:** For tables containing numerical and categorical data, new data samples can be generated by manipulating specific columns.
- **Multimodal Data Augmentation:** Augmenting a combination of image, text, and audio data helps create richer and more generalizable models.

3. Application Areas

Data augmentation is widely applied across various domains (Wang et al., 2024). The application areas of data augmentation are vast. Image processing (Shorten and Khoshgoftaar, 2019), natural language processing (Park and Ahn, 2019), speech recognition (Bakır et al., 2024), and time-series forecasting (Iwana and Uchida, 2021) are primary application fields. It is used in many industries to improve model performance.

- **Image Processing:** Data augmentation is commonly used in computer vision applications such as object recognition, face recognition, and segmentation.
- **Natural Language Processing (NLP):** In NLP tasks such as text classification, sentiment analysis, and summarization, data augmentation plays a significant role. Text data augmentation is an effective method to address the problem of insufficient labeled data.

- **Speech Recognition:** Voice response systems and speech recognition systems benefit from noise reduction and audio data augmentation techniques.
- **Time Series:** Augmentation techniques applied to time-series data can result in more accurate and reliable forecasts.

4. Conclusion and Future Directions

Data augmentation holds a significant place in artificial intelligence. The success of artificial intelligence applications is directly related to the quality and diversity of data. In the future, more sophisticated data augmentation methods and generative models are expected to become widespread. The use of GANs and transformer-based models will push the boundaries of data augmentation and make model training more efficient. Data augmentation is expected to contribute to the broader and more efficient use of artificial intelligence, especially in areas with limited data.

Additionally, ethical data use and security issues will gain more importance in the future. Along with the advantages of data augmentation, ensuring the ethical use and security of these processes will become critical. Therefore, attention must be paid to the ethical production and use of data in the data augmentation process.

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CHAPTER IV

VISION TRANSFORMER: ARCHITECTURE, VARIANTS, AND APPLICATIONS

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1. Introduction

The gap between machine and human capabilities is gradually closing thanks to artificial intelligence. Today, many researchers have made extraordinary progress using artificial intelligence techniques in many areas. One of these areas, computer vision, is rapidly developing with deep learning methods that help machines perceive the world with humans eyes. Most of the studies in this field use Convolutional Neural Networks (CNN) as the basic algorithm. CNN works effectively by using learnable biases and weights to detect and classify objects in images.(Bhatt et al., 2021). CNN-based architectures are an effective method for capturing local patterns in images. However, they have limitations in modeling long-range dependencies. Especially

in large and complex images, learning long-range relationships between specific regions is very important (Khan et al., 2023).

To overcome these limitations, Vaswani et al. (Vaswani et al., n.d.) proposed a model based on the attention mechanism. Initially developed to better learn long-range dependencies in texts, this model uses the self-attention mechanism to determine the importance of different parts of the input when generating predictions. Thanks to the attention mechanisms, the relationships between words can be interpreted independently of their positions (Kasneci et al., 2023)

These developments in the field of Natural Language Processing (NLP) have also inspired the field of computer vision. In this direction, Dosovitskiy et al. (Dosovitskiy et al., 2021) proposed the Vision Transformer (ViT) model in their study titled “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale”. Since the proposed ViT model needs to be trained with a large amount of data, it stands out as an architecture that requires large-scale datasets and computational power (He et al., 2023). ViT divides the input image into fixed-size pieces, then processes these pieces into a series of feature vectors with Transformer blocks. These blocks learn the relationship of each patch with the others thanks to the multi-head attention mechanism. In the last stage, the final prediction is performed with the classification head (Kang et al., 2024).

One of the most important advantages of ViT is its ability to learn long-range dependencies. ViT associates each region of an image with other regions using the self-attention mechanism. This ensures that ViT models perform well, especially when trained on large datasets (Touvron et al., 2021). The ViT model processes the image by dividing it into parts and leaves the important features entirely to the learning process. In this way, ViT can extract features with a more flexible and global perspective without relying on a specific filter structure. (Kumar, 2025).

The structure of the article is as follows. In Section 2, studies carried out with ViT in the literature are presented. In Section 3, the architecture, components, and variants of ViT are presented. In Section 4, the applications and impacts of ViT models in different fields are presented. Finally, in Section 5, the outputs, achievements, and limitations of the ViT model are presented.

2. Literature Review

Following the success of transformative models in natural language processing, ViTs have revolutionized computer vision and improved the processing of visual data. ViTs are used in various fields such as medical image

analysis, agricultural technologies, and exhibit superior performance, especially in high-resolution image classification, object detection, and segmentation tasks.

2.1. Health Science and Medicine

The use of ViTs in medical image analysis enables faster and more successful diagnosis of diseases. Thus, these artificial intelligence-supported systems provide doctors with a helpful decision system and help start the right treatment as soon as possible. It is seen that the ViT model is used in the classification of many diseases in different medical fields such as pneumonia (Murphy et al., 2022), diabetic retinopathy (Hagos & Kant, 2019), COPD (Y. Wu et al., 2021) Gheflati and Rivaz conducted a study using ViT and CNN models for the classification of 943 breast ultrasound images. This study showed that ViT models performed better than the best CNN models. Thus, it was shown that ViTs processed spatial information more efficiently and achieved high performance even with small data sets. This provides a great advantage in medical imaging operations where it is not always possible to access large data sets (Gheflati & Rivaz, 2022). ViT models are used in segmentation studies as well as classification operations. Cao et al. proposed the transformer-based Swin-Unet model. This proposed model was applied to segmentation tasks on multiple data sets. As a result of the study, it was observed that the Swin-Unet model exhibited superior performance by effectively learning local and global semantic features (Cao et al., 2021). The Swin UNETR (Swin-UNet TRansformer) model (Hatamizadeh et al., 2022) developed later was found to perform better than previous methods even in brain tumor segmentation operations that are difficult to segment (Baid et al., 2021). These studies show that ViTs are becoming widespread in the medical field and leading to more sensitive and reliable artificial intelligence solutions.

2.2. Agricultural Technologies

In the field of agriculture, many transformer models are used to provide early detection of plant diseases and pests and increase product productivity. Due to the superior performance of the ViT model in recent years, studies have started to focus on this direction. When the studies are examined, it is seen that ViT models perform better than traditional deep learning models in correctly classifying diseases in plants (Ul Abidin et al., 2023). In addition to disease detection, it also offers successful results in different study areas such as detection of pests in plants, crop productivity estimation, soil analysis, irrigation management. In addition to ViT models, hybrid models such as ConvViT have

been proposed. It is observed that performance is increased with these hybrid models (Dhanya et al., 2022). Similarly, successful segmentation operations have been carried out using the TransUNet model. Thus, operations such as agricultural area detection, crop and soil monitoring have been successfully analyzed and necessary operations have been applied early (Lei et al., 2024)

2.3. Finance

In recent years, the use of ViT architectures has become a hot topic, especially in the financial sector in the areas of risk management, automated trading, and intelligent customer service. Using ViT models, price change trends in the stock market can be predicted. Then, effective trading strategies can be developed using this predicted data (Liu et al., 2024) Using models such as ViT-DeepSort, fraud attempts in financial transactions and the identification of fake financial transactions can be performed (Verma, 2024) However, it is necessary to note that there are difficulties such as data privacy and computational cost when working with data in the financial field. As a result, ViT models are successfully used for many purposes in the financial field. In addition, this area needs further research and development (Raeini, 2024).

2.4. Security and Defense Technologies

Transformer models have also begun to be used frequently in the security field, which is one of the critical areas where accurate detection and classification are important. It is seen that they exhibit successful performance in the analysis of high-resolution images such as radar, satellite, or drone images in the process of correctly detecting and classifying the target. In addition to attack situations, transformer models have also begun to be used to detect and prevent smuggling and illegal crossings (Costa et al., 2024) In addition to these, transformer models are also used for security problems that may be encountered from both software and hardware perspectives. Thus, many dangerous situations such as aggressive software, watermarking, tourney horse, side-channel attacks can be detected correctly and attacks can be prevented (Latibari et al., 2024).

3. Methodology and Implementation

3.1. Vision Transformer

Vision Transformer (ViT), unlike CNNs, has a structure that is completely based on the attention mechanism to process visual data. The working principle

of ViT is based on dividing an image into patches of certain sizes and processing each patch as an array input. These patches are then transmitted to a Transformer model by adding position information. Thanks to the Self-attention mechanism in the Transformer process, global dependencies between different regions in the image are learned. Thus, long-range relationships are modeled more successfully and a more comprehensive visual representation is created (Saranya et al., 2024). In this section, the basic components of ViT will be detailed and how the model processes visual data will be examined.

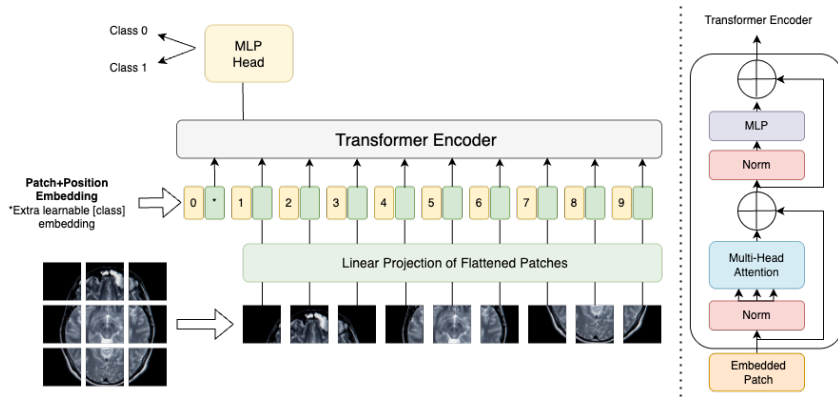


Figure 1. ViT Architecture

3.1.1. Patch Embedding

It is the component of the ViT model that divides images into patches and converts each of them into digital vectors. Each patch is reduced to a fixed vector size via the linear layer. The resulting vectors form digital representations of each part of the image. Then, location information is added to the patches and transmitted to the transformer model. Patch embedding improves the performance of ViT by enabling the efficient processing of visual information (Feng et al., 2023)

3.1.2. Positional Encoding

It is the component that helps understand the location information of each patch. In cases where the patches obtained from the images do not enter the model in a sequential manner, the location of each patch in the image must be understood. Therefore, location information is added to each patch. Location information is used for the attention mechanism to make sense of the location

of each patch and to establish global relationships (Yang et al., 2025) Positional Embedding helps the model make more accurate predictions by enabling it to learn the structure in the image (K. Jiang et al., 2022).

3.1.3. *Transformer Encoder*

A multi-layer Transformer Encoder is the basic building block of the ViT model. Transformer encoder effectively models the relationships between different regions in the image, enabling long-range dependencies to be captured. Transformer architecture consists of the following basic components (Dong et al., 2025).

- Multi-Head Self-Attention (MSA): The Multi-Head Self-Attention mechanism learns global dependencies by modeling the relationships of each patch with others. The biggest advantage of MSA is that it can learn the relationships of each region with others by considering the entire image context simultaneously (Y. Han et al., 2025).

The attention mechanism ensures that a query produces output by matching key-value pairs. In the multi-headed attention mechanism, each head evaluates the relationships in the data by performing calculations on the query, key and value vectors. The outputs of multiple heads are then combined to help the model make its final decision. Using multiple heads allows the model to learn from different perspectives, minimizing information loss and increasing the generalization ability of the model (Islam et al., 2024).

Query: In the attention mechanism, it is a vector used to determine the relationship of an input to other inputs. The model calculates how important each input is by comparing the query vector with the key vectors. This process helps establish context between the inputs and select the most relevant information.

Key: It is a vector that contains the representation of each input and allows the calculation of attention weights by comparing it with the query vectors. The model determines the importance level of the inputs by determining how much each query matches which key. Thus, it is determined which information will be taken into consideration and how much.

Value: It is the vector that contains the information used to create the final output of the model. The attention mechanism collects the value vectors according to the weights calculated by the query and key vectors. Thus, the model creates a better contextual representation by highlighting the most relevant information (Pacal et al., 2025).

The steps for calculating attention weights are as follows:

1. The relationship between query (Q) and key (K) is calculated. This determines the relationship of each query to each key.

$$Score = QK^T \quad (1)$$

2. These scores are made more stable by scaling them (Here $\sqrt{d_k}$, is the size of the key vectors. Square root scaling is used to prevent attention weights from being too large or too small for large vector sizes.).

$$Scaled\ Score = \frac{QK^T}{\sqrt{d_k}} \quad (2)$$

3. The scaled scores are normalized with the softmax function. The softmax function determines how much attention should be paid to which input by normalizing the sum of the weights to 1.

$$Attention\ Weights = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (3)$$

4. The resulting attention weights (A) are multiplied by value vectors to create outputs.

$$Attention\ Output = AV \quad (4)$$

- Feedforward Neural Network (FFN): After each attention layer, a two-layer fully connected feedforward neural network (FFN) is applied, which processes each patch representation through linear transformations and activation functions (Gong et al., 2024). This component enables the Transformer architecture to model more complex features. In particular, it helps to process the information content of each patch to produce more detailed and meaningful representations (Farzipour et al., 2023)

The harmony of these components enables ViT to provide a more flexible and powerful learning process compared to traditional CNN-based models. These advantages provided by Transformer Encoder have made ViT model highly competitive in large-scale image classification and computer vision applications. (Sharma & Vishwakarma, 2024)

3.1.4. Class Token

It is a special component used in the classification tasks in the ViT model. In addition to each patch in the image, a special “class token” is added so that the model can make the final classification prediction. This token is given as input to the transformer encoder. In the last layer of the encoder, this token is

used to determine which class the image belongs to. As a result, the class token makes the output of the model suitable for the classification problem (Huang et al., 2021)

3.1.5. Multi-Layer Perceptron (MLP) Head

Transformer is the component that makes the data processed by the encoder suitable for classification or other final tasks. The final features from the encoder are transferred to the MLP layer. This layer usually consists of several fully connected layers and makes the features learned by the model suitable for a particular class or task. The MLP head is designed to make final class predictions in classification tasks and usually results in a softmax activation function. As a result, the MLP head uses the learned features of the Transformer for various final tasks such as classification or regression problems (Abdallah et al., 2024).

3.2. Vision Transformer Model and Variants

With the emergence of the ViT model, great progress has been made in the processing of image data. ViT processes images with the attention mechanism of the Transformer architecture after dividing them into patches. Thanks to this approach, high performance is achieved when working with large data sets. However, it also brings some difficulties such as high computational cost (X. Jiang et al., 2024). Therefore, various ViT models have been proposed in order to increase the efficiency of ViT by improving its basic structure. In this section, ViT variants developed for different tasks will be discussed and the innovations offered by these models will be examined.

3.2.1. Data-Efficient Image Transformer (DeiT)

ViT exhibits high performance when working with large data sets. The DeiT model was developed to achieve high performance in cases where the data set is not large enough. DeiT uses CNN-based instructional models using a distillation token to support information transfer. In this way, high accuracy can be achieved even in smaller data sets. In this way, it has become a strong alternative for researchers and engineers working with limited data (Hossain et al., 2024) The architecture of the Deit model is shared in Figure 2.

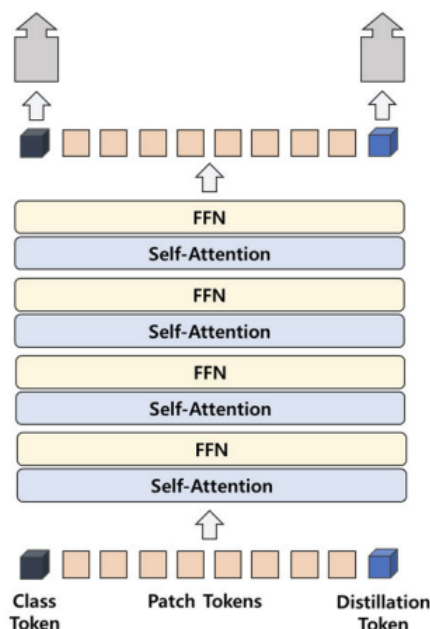


Figure 2. DeiT model architecture (Bang et al., 2023)

3.2.2. Pyramid Vision Transformer (PvT)

The PvT model was developed to increase the efficiency of ViT on high-resolution images. While traditional ViT models use fixed size patches, PvT uses progressively varying patch sizes and multi-level feature maps. This provides a better representation of images at different scales. Thanks to its hierarchical structure, it not only reduces the computational cost but also processes high-resolution details more effectively. It integrates the multi-scale feature learning advantage of CNN-based models into the Transformer architecture. Thanks to all these advantages, it stands out as a strong alternative in the field of computer vision (Wang et al., 2021). The architecture of the PvT model is shared in Figure 3.

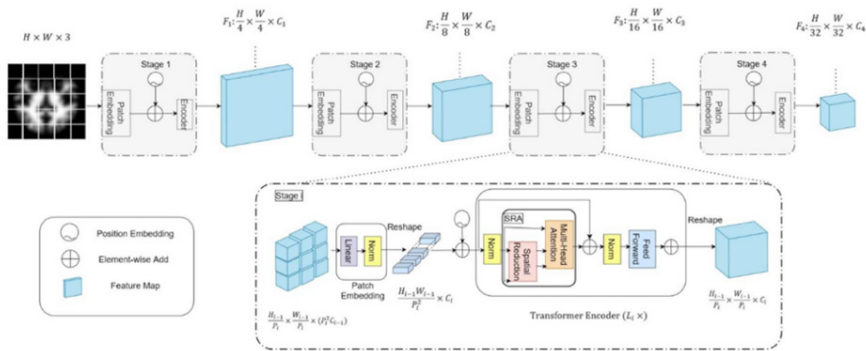


Figure 3. PvT model architecture (Aghdam et al., 2024)

3.2.3. Transformer in Transformer (TNT)

TNT works by dividing each patch into sub-parts to better represent its internal structure. This enables them to learn both local and global relations more effectively. While in traditional ViT models, each patch is transmitted directly to the Transformer layers, the TNT model uses a sub-Transformer structure within each patch. Thus, it enables more detailed feature extraction. Thanks to this structure, more successful results are achieved, especially when working on small objects or images containing fine details (K. Han et al., 2021). The architecture of the TNT model is shared in Figure 4.

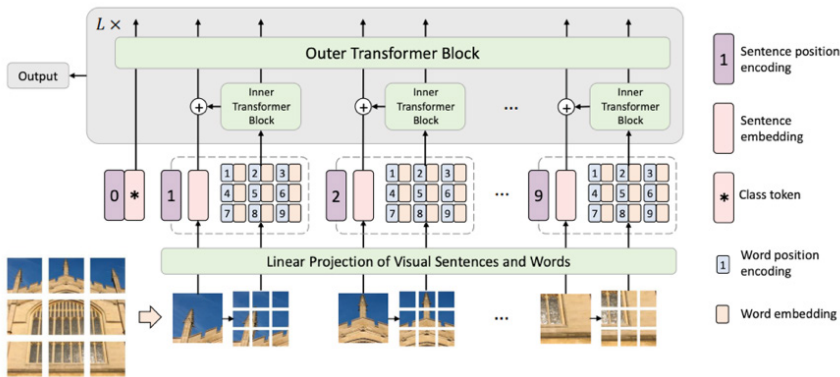


Figure 4. TNT model architecture (K. Han et al., 2021)

3.2.4. Swin Transformer

Unlike other models, Swin Transformer divides the image into small windows and applies the attention mechanism within each window. Thus, both the

computational cost is reduced and the performance is increased. Shifting these windows at a certain rate in each layer enables better learning of the global context and effective dissemination of regional information. Swin Transformer can learn features at different scales thanks to its hierarchical structure. In this way, it provides superior success especially in tasks such as object detection and segmentation. The efficient computational structure and scalability of this model make it a powerful alternative for large-scale computer vision tasks (Dümen et al., 2024) The architecture of the Swin Transformer model is shared in Figure 5.

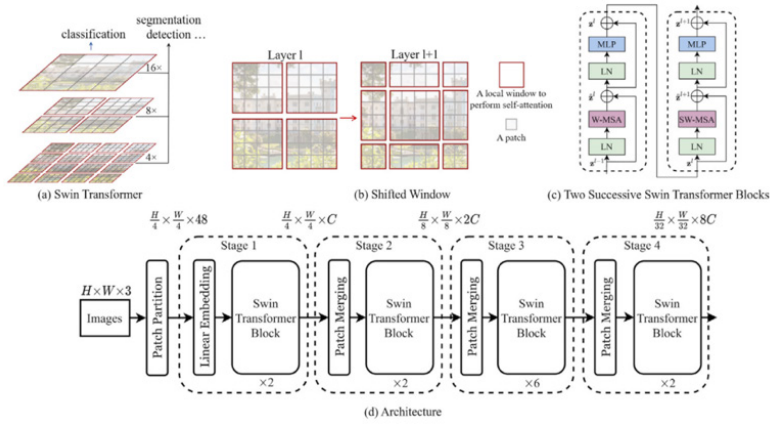


Figure 5. Swin transformer model architecture (Thisanke et al., 2023)

3.2.5. Convolutional Vision Transformer (CvT)

It is a hybrid model created by combining the traditional CNN and the ViT model. The ability of CNN to learn local information is combined with the ability of ViT to learn global information. Thus, the computational costs are reduced and the model is provided with better generalization. In this case, the CvT model can show higher performance even on small data sets (Fnu & Bansal, 2024) The architecture of the CvT model is shared in Figure 6.

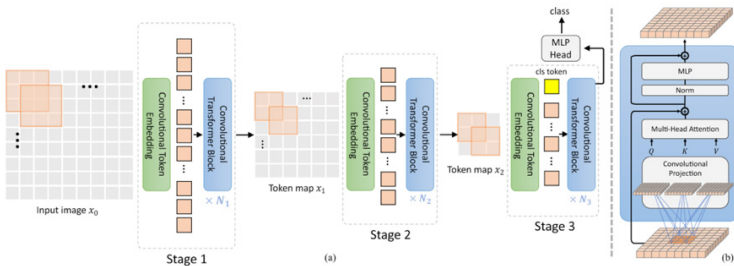


Figure 6. CvT model architecture (H. Wu et al., 2021)

3.2.6. *CrossViT*

It is a ViT model that aims to obtain higher performance by processing image data at different scales. This model processes image patches of different sizes in parallel. It also uses a cross-attention mechanism by combining two different-scale ViT components. Thanks to these features, it can learn small details and general structure at the same time. Compared to other models, CrossViT has the ability to generalize and can produce effective results in cases where the amount of data is limited (Abd Elaziz et al., 2024). The architecture of the CrossViT model is shared in Figure 7.

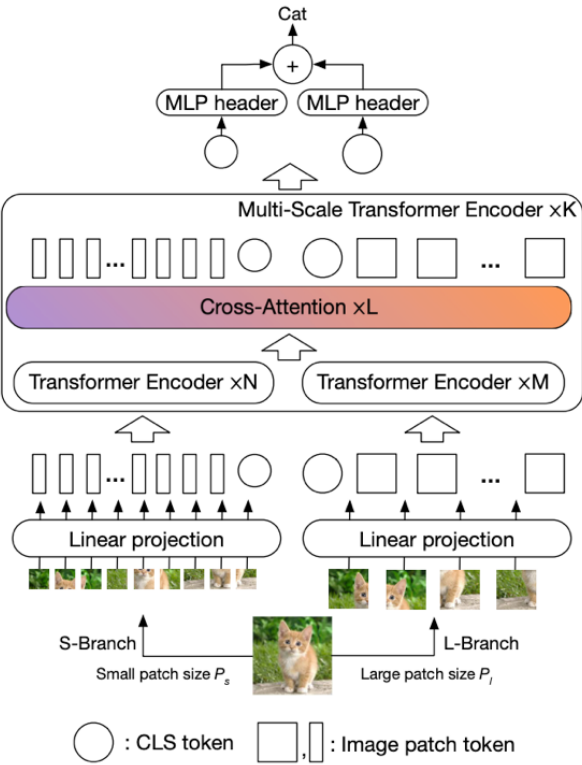


Figure 7. CrossViT model architecture (Chen et al., 2021)

In this section, different models derived from the ViT architecture are examined. In the studies, the advantages, disadvantages and the application areas in which the models are successful are important factors in selecting the most appropriate model. Table 1 presents a comparative analysis of the DeiT, PvT, TNT, Swin Transformer, CvT and CrossViT models.

Table 1: Comparison of ViT Variants

Model	Advantages	Disadvantages	Usage Areas
DeiT	It can be trained with less data, optimizes the attention mechanism, speeds up the training process.	Without large datasets, the ability to generalize is limited.	Image classification, model training with low data.
PvT	Produces multi-scale feature maps, low computational cost.	Its performance may degrade when trained with small datasets.	Object detection, segmentation, density mapping.
TNT	It has a high ability to learn finer details, capturing more detailed local features.	High computational cost, works better with large-scale data.	Medical imaging, detailed object recognition, facial analysis.
Swin Transformer	Scalable window mechanism reduces computational cost and is scalable.	The use of small windows can make it difficult to learn about long-range dependencies.	Computer vision, medical imaging, autonomous systems.
CvT	It combines the power of CNNs to learn local features with the advantages of Transformers to learn long-distance dependencies.	The computational cost is high for high-resolution images.	Image classification, object detection, video analysis.
CrossViT	Learn both local and global characteristics using a cross-attention mechanism with ViT branches at different scales.	It requires more computational resources, training time is longer.	Medical imaging, multi-scale object recognition.

4. Results and Discussions

Compared to traditional deep learning models, ViT and its variants appear to offer successful results in many areas such as natural language processing, object detection and segmentation, and image classification. ViT, which shows superior success in learning global knowledge when trained with large data

sets, offers a flexible architecture that can be adapted to different application scenarios. In addition, it is seen that it offers successful results in smaller sized data sets using hybrid models and optimization methods (Kavalcı Yılmaz et al., 2024; C. Wu & He, 2024). In this section, the application areas where Vision Transformer-based models are widely used and their impacts in these areas will be discussed.

4.1. Natural Language Processing

Transfer learning models were first introduced for studies in NLP. This method allows language models to develop a broad context understanding by training them with large amounts of text and can then be easily and successfully used in various tasks such as sentiment analysis, machine translation, text classification, and question-answer systems. In addition, when working on special fields such as medical and legal texts, fine-tuning can be done to ensure that language models produce more precise and domain-specific outputs. In this way, operations can be carried out at lower costs (Fields & Kennington, 2023).

4.2. Object Detection and Segmentation

ViT models are seen to exhibit strong performance in the field of object detection and image segmentation. ViT's multi-head self-attention (MSA) mechanism enables capturing more detailed and contextual features in segmentation processes by learning long-range dependencies (Ishibashi et al., 2024). ViT models offer significant advantages in overcoming challenges such as contrast changes, intra-class variations, and object overlaps when trained on large datasets. However, the high computational cost and large-scale data requirement remain one of the key factors limiting the widespread use of ViT in object detection and segmentation applications (Thisanke et al., 2023).

4.3. Image Classification

The ViT model shows high accuracy performance especially in image classification studies. It also attracts attention with its flexible architecture that can be adapted according to data sets. Thanks to its attention mechanism, it can extract deeper features by learning long-range dependencies. When pre-trained with large data sets, ViT models exhibit superior performance especially in recognizing fine details and complex visual patterns (Pantelaïos et al., 2024). Various versions such as DeiT have been optimized to achieve superior performance with small-sized data sets. The flexible structure of ViT offers

a wide range of use in many different fields of study. Thus, ViT has become an important alternative in image classification studies with its scalability and strong learning capacity (Singh et al., 2025).

4.4. Speech Recognition

The most important advantages of Transformer-based models in speech recognition systems are that they provide high-accuracy results after a rapid learning process. Unlike traditional methods, these models can better analyze audio data using the attention mechanism and produce more accurate speech texts. In addition, they can be used by integrating with language models. Thus, they help speech recognition systems produce more natural and fluent texts. However, the need for high computational power and a large amount of training data are among the main factors that limit the widespread use of these models (Orken et al., 2022).

4.5. Video Analysis

Transformer-based models provide comprehensive analysis in video analysis thanks to their ability to capture long-term dependencies. Transformer models can perform more detailed analysis because they can learn global features better. Models such as ViViT (Video Vision Transformer) can achieve successful results by processing spatial and temporal information together in tasks such as video classification, action recognition and anomaly detection (Arnab et al., 2021). It is widely used for abnormal event detection and behavior analysis, especially in areas such as security systems, smart cities, and driverless vehicles. However, high computational cost and the need for big data are important factors that limit its use in real-time applications (Guo et al., 2023).

5. Conclusion and Future Trends

Vision Transformer (ViT) models have shown remarkable success in various computer vision tasks, outperforming traditional CNN-based models. Thanks to this success, they have become widely used in many areas such as medicine, agriculture, finance, and security. However, there are concerns such as computational complexity and data efficiency. As a solution to this, hybrid models that integrate ViT models with convolutional or other attention-based architectures are being developed. Thanks to these hybrid models, a more comprehensive feature extraction is provided by combining the strengths of CNN in capturing local features with ViT's global attention mechanism.

This enables both the preservation of detailed spatial information and better modeling of long-range relationships. Thus, hybrid CNN-ViT models provide a strong balance by offering higher accuracy and computational efficiency in visual tasks.

ViT models exhibit superior performance when trained with large-scale datasets. However, they may exhibit lower performance compared to convolutional models when trained with small-scale datasets. New models such as DEiT have been developed and continue to be developed to overcome this limitation. Future research is expected to focus on optimizing ViT models for smaller datasets, reducing computational demands, and improving interpretability.

In conclusion, despite the challenges, the continued evolution of transformer-based architectures will lead to further advances in AI-driven vision systems, leading to the development of models that are more efficient and more approaching human-like perception capabilities.

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CHAPTER V

GENERATIVE ADVERSARIAL NETWORKS: ARCHITECTURE, VARIANTS AND APPLICATIONS

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1. Introduction

Generative Models have attracted considerable attention because of their capacity to execute tasks traditionally linked to human creativity, including writing (Iqbal & Qureshi, 2022), speaking (Y. A. Li et al., 2025), composing music (Hernandez-Olivan & Beltrán, 2023), creating art (E. Zhou & Lee, 2024) and programming (Fried et al., 2023). In the realm of deep learning, generative modeling represents a type of unsupervised learning task that focuses on identifying and understanding the overall patterns or structures within input data, enabling the model to generate or produce new instances that might be drawn from the entire dataset. Applying this concept can create examples or data distributions that closely resemble or convey the same semantic meaning as the original input data.

In the realm of deep neural networks and generative modeling, GANs have emerged as a groundbreaking method. Initially introduced by Goodfellow et al. (Goodfellow et al., 2014), GANs have gained significant interest from researchers, industry experts, and hobbyists alike. The core

idea behind GANs is a competitive interplay between the discriminator and generator neural networks, drawing inspiration from game theory. While the discriminator networks strive to distinguish real data from fake data, generator networks aim to create synthesized data instances that closely resemble real data. Through this adversarial mechanism, GANs progressively learn to generate data that increasingly mirrors real data, accurately reflecting the underlying data distribution. GANs have gained great interest and importance because they allow the production of useful data without the need for labeled examples through unsupervised learning and their adaptability in producing high-quality and realistic information has led to important applications in many different areas.

This work provides a comprehensive overview of GAN architectures, explores the evolution of their variants, and examines their practical applications. By understanding the theoretical foundations and practical implementations of GANs, researchers and practitioners can leverage these models effectively for a wide range of generative tasks.

The structure of the paper is as follows. In section 2, studies conducted with GAN in many different fields are presented. The architecture and components of GAN, GAN variants and evaluation criteria of GANs are presented in detail in section 3. In section 4, the applications and impacts of GANs in different fields are discussed. Finally, in section 5, the outputs and future trends of GANs are represented.

2. Literature Review

Generative Adversarial Networks (GANs) have emerged as the most prevalent type of generative models, renowned for their extraordinary capability to produce highly realistic data samples. This section will explore the diverse and expansive range of fields where GANs are making significant impacts. These applications span areas such as computer vision, where GANs are used to create lifelike images and enhance resolution and the realm of natural language processing, where they generate coherent and contextually relevant text. Additionally, GANs play a crucial role in the gaming industry for generating realistic environments and characters and in the medical field for synthesizing patient data that aids in research. Their versatility highlights the transformative potential of GANs across various domains, driving innovation and new possibilities in data generation.

2.1. Image Processing

Researchers at Nvidia created a particular kind of generative adversarial network called StyleGAN (Karras et al., 2019). StyleGAN's primary goal is to produce a large range of excellent facial photos. Adaptive instance normalization, noise mapping networks, and progressive growth techniques similar to those used in ProGAN were utilized to accomplish this. Given as an input of face image in any random posture, PosIXGAN (Bhattacharjee et al., 2018) was trained to produce high-quality face images with nine distinct pose variations. BeautyGAN (T. Li et al., 2018) can transfer makeup styles from a reference image of a made-up face to another without makeup, all while maintaining the person's identity. The goal of InfoGAN (X. Chen et al., 2016) was to train to acquire separated models without supervision, enabling modifications to facial features like spectacles and haircuts. Mohana et al. (Mohana et al., 2021) used DCGAN to generate high-quality pictures of human faces. The CelebFaces Attributes Dataset (CelebA) was used for training the DCGAN algorithm. The obtained results demonstrated that the generated photos' quality is about the same as that of the CelebA dataset's photos.

2.2. Video Processing

GANs were first used to create videos with VGAN (Vondrick et al., 2016). Two CNN architectures make up the Generator is composed of two CNN architectures: a 3D spatial-temporal convolutional network that records the motion of objects in the foreground and a 2D spatial convolutional model that records the static backdrop. The generated video is created by combining the distinct outputs from the Generator and the Discriminator then assesses it to decide whether it is authentic or not. A new video can be produced by Temporal Generative Adversarial Nets (TGAN) (Saito et al., 2017) after they have learned representations from an unlabeled video dataset. The temporal generator and the image generator are the two sub-generators that comprise the TGAN generator. The temporal generator generates numerous latent variables, every of them represents a distinct video frame, after receiving a single latent variable as input. This collection of latent variables is used by the picture generator to create the video. There are three-dimensional convolutional layers in the discriminator. To guarantee consistent training and satisfy the K-Lipschitz requirement, TGAN uses WGAN. FlowGAN and TextureGAN are the two GANs that make up FTGAN (Ohnishi et al., 2018). The TextureGAN model creates texture based

on the previous FlowGAN output, producing the required frames, whereas the FlowGAN network concentrates on motion. To generate videos, the Motion and Content decomposed GAN, often known as MoCoGAN (Tulyakov et al., 2018) uses a representation that breaks down motion and content. A recurrent neural network, an image generator and discriminators of image and video are the four sub-networks that make up MoCoGAN. To produce videos based on text input, Li et al. (Y. Li et al., 2018) combined the Variational Autoencoder (VAE) (Kingma & Welling, 2022) and GAN. The model consists of a video generator, a video discriminator and a conditional gist generator (conditional VAE). Depending on the decoded content, the conditional gist generator creates the original image or gist. The motion and substance of the video are then generated by cGAN, which is dependent on the input text as well as the gist. Temporal GANs conditioning on captions (TGANs-C) (Saito et al., 2017) encode and extract the representation from the input text using an LSTM-based encoder and a bidirectional LSTM. To produce and synthesize realistic films, this representation is then sent into the Generator, a 3D deconvolutional network, along with a random noise vector.

2.3. Medical and Healthcare

Similar to CycleGAN (J.-Y. Zhu et al., 2017), MR-GAN (Jin et al., 2019) is intended to convert 2D brain CT image slices into 2D brain MR image slices. However, in contrast to CycleGAN, MR-GAN is trained on both paired and unpaired data and contains dual cycle-consistent loss, adversarial loss and voxel-wise loss. In order to transform contrast CT scans into non-contrast pictures, Sandfort et al. (Sandfort et al., 2019) used CycleGAN (J.-Y. Zhu et al., 2017). A U-Net trained on the original dataset was compared to a U-Net trained on a dataset that included synthetic non-contrast images in order to assess the segmentation performance of the former. DermGAN generates artificial images that represent skin problems (Ghorbani et al., 2020). The model is trained to provide a realistic image that maintains the specified properties from a semantic map that describes a particular skin condition, including its size, location and underlying skin color. In order to counteract the checkerboard effect, the Generator of DermGAN alters a U-Net (Ronneberger et al., 2015) by replacing a nearest-neighbor resizing layer and a convolution layer for the deconvolution layers. A mixture of min-max GAN loss, l_1 reconstruction loss and feature matching loss for the entire picture and l_1 reconstruction loss for the diseased region is minimized by training the Generator and Discriminator. To create artificial medical images, Zhang et al. (Q. Zhang et al., 2018) used DCGAN (Radford et al., 2016), WGAN (Arjovsky et

al., 2017) and Boundary Equilibrium GANs (BEGANs) (Berthelot et al., 2017). They used the artificially created images to improve their datasets, which led to more accurate tissue recognition in the models. The three GAN models' total tissue recognition accuracy increased during training using enhanced datasets.

2.4. Material Science

A physics-aware GAN model was developed by Singh et al. (Singh et al., 2018) to produce binary microstructure images. To accomplish this, they used three distinct models. The WGAN-GP is the first model used (Gulrajani et al., 2017). The second approach uses an invariance checker that actively applies known physical invariances in place of the traditional discriminator in a GAN. The third model integrates the previous two to design microstructures that satisfy both explicit physical invariances and implicit constraints derived from picture data. CrystalGAN offers a GAN framework for constructing chemically stable crystallographic forms with a higher domain complexity (Nouira et al., 2019). A first-step GAN, a feature transfer process and a second-step GAN for synthesis are the three main parts of the CrystalGAN architecture. GAN produces pseudo-binary samples with mixed domains in the first step, much like a cross-domain GAN. The complexity of the data produced from the instances acquired in the previous step is increased by the feature transfer strategy. In the end, the second stage of GAN complies with geometric limitations and produces ternary stable chemical molecules. In order to create crystal structures with using a GAN, Kim et al. (S. Kim et al., 2020) proposed utilizing a coordinate-based crystal representation that was influenced by point clouds. Their Conditioned Composition Crystal GAN conditioning the network with a one-hot encoded composition vector, GAN may generate materials with the required chemical composition. GANs were successfully used by Mao et al. (Mao et al., 2020) to create intricately structured materials. A numerous randomly produced architected materials classified to train the model. The suggested approach is appropriate for a broad range of applications since it produces complex architected designs without the need for prior knowledge.

2.5. Finance

In order to precisely recreate three datasets from American Express, Efimov et al. (Efimov et al., 2020) combine Deep Regret Analytic Generate Adversarial Networks (DRAGANs) and cGAN. A regularization component is added to the discriminator's loss of DRAGANs to help stabilize convergence and avoid gradient

explosion or vanishing problems. For stock price prediction, Zhou et al. (X. Zhou et al., 2018) use the GAN-FD model, which attempts to minimize both forecasting error and direction prediction losses. Whereas the Discriminator uses CNN layers, its Generator uses LSTM layers. Stock-GAN is a conditional Wasserstein GAN (WGAN) developed by Li et al. (J. Li et al., 2020) to comprehend past dependencies in stock market order streams. The Generator network incorporates order-book properties as conditional information and models the double auction processes that are essential to stock exchanges. Quant GANs are introduced by Wiese et al. (Wiese et al., 2020) and use a Temporal Convolutional Networks (TCNs) architecture, also known as WaveNet (Oord et al., 2016), as the Generator. Long-range dependencies, such volatility clusters in stock data like the S&P 500 index, are well captured by this design. LSTM-GANs were created by Leangarun et al. (Leangarun et al., 2018) to identify anomalous trading activity brought on by stock price manipulation. The LSTM design serves as the foundation for both the discriminator and the generator. For evaluation, test scenarios with simulated alterations were used with trade data from the Stock Exchange of Thailand (SET).

2.6. Marketing

In order to streamline the logo development process, Sage et al. (Sage et al., 2018) presented iWGAN that can produce an endless number of logo variations by varying factors like shape and color. A grouped GAN model trained on multimodal data was proposed by the authors. Clustering delivered higher-quality samples from unlabeled datasets, prevented mode collapse and improved GAN training stability. The DCGAN (Radford et al., 2016) and WGAN-GP (Gulrajani et al., 2017) models served as the foundation for the GAN models. ACGAN (Odena et al., 2017) structure serves as the foundation for LoGAN (Mino & Spanakis, 2018), often referred to as the Auxiliary Classifier Wasserstein GAN with gradient penalty (AC-WGAN-GP). Twelve preset colors can be used by LoGAN to create logos. It is composed of a classification network that helps the discriminator classify logos, a generator and a discriminator. Instead of depending on the ACGAN loss, the WGAN-GP loss function (Gulrajani et al., 2017) was used to improve training stability. The stance Pose Guided Person Image Generation Network (PG2) was created by Ma et al. (Ma et al., 2017) to generate artificial representations of people in random positions using an input image and a new stance. The two stages of PG² operation are position integration and image refinement. Using the input image and target position, position integration produces a crude output that represents the general structure of the

human. In order to improve the original output through adversarial training and produce finer findings, image refinement uses the DCGAN (Radford et al., 2016) model. According to their appearance and posture, Deformable GANs (Siarohin et al., 2018) generate pictures of people. To handle large spatial deformations and correct misalignments between the real and generated pictures, the scientists used nearest neighbor loss and deformable skip connections. E2E, which uses unsupervised GANs, pose-guided person picture generation, was introduced by Song et al. (Song et al., 2019). In order to manage the complexity of the difficult task of creating a direct mapping across many postures, the authors divided it into appearance generation and semantic parsing transformation.

3. Methodology and Implementation

3.1. Generative Adversarial Networks

The Generative Adversarial Network (GAN) model, first presented in 2014 by Goodfellow et al. (Goodfellow et al., 2014), is a flexible approach commonly applied in image processing and computer vision fields. GANs operate based on the principle of unsupervised learning. Unlike traditional deep network structures, a GAN consists of two separate deep networks: generator and discriminator. These networks participate in a racing process that enhances their learning abilities. The generator hones its skill in creating remarkably realistic images and the discriminator enhances its ability to differentiate actual images from generated ones (Shah et al., 2022).

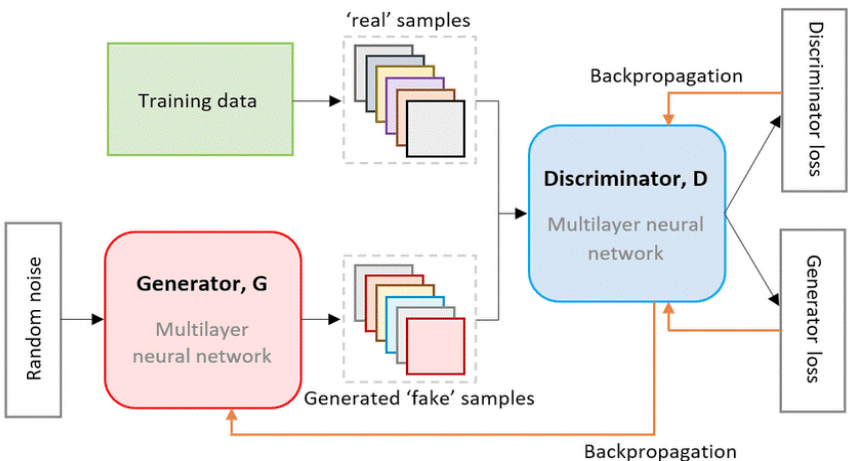


Figure 1. GAN Architecture

Architecture of GANs comprises two primary elements: the generator and the discriminator networks. The generator receives a random vector of noise and seeks to transform this noise into realistic data (Goodfellow et al., 2020). It generates synthetic data samples that mimic those found in the original train dataset. Conversely, the discriminator takes both real data and the synthetic data created by the generator. Its objective is to accurately distinguish between real and fake data. This setup fosters a competitive dynamic between the two networks. The generator aims to deceive the discriminator, while the discriminator strives to avoid being misled. This adversarial relationship enhances the learning process, resulting in improved outcomes for both models.

3.2. Different Generative Adversarial Networks Structures

3.2.1. Conditional GAN (cGAN)

Conditional Generative Adversarial Nets (cGANs) are a GAN variant for conditional sample generation (Mirza & Osindero, 2014). This allows for control over the data generation methods. In order to accomplish conditioning, cGANs concatenate additional information y with the input and feed it into both Generator and Discriminator. Examples of this additional information include class labels or other modalities.

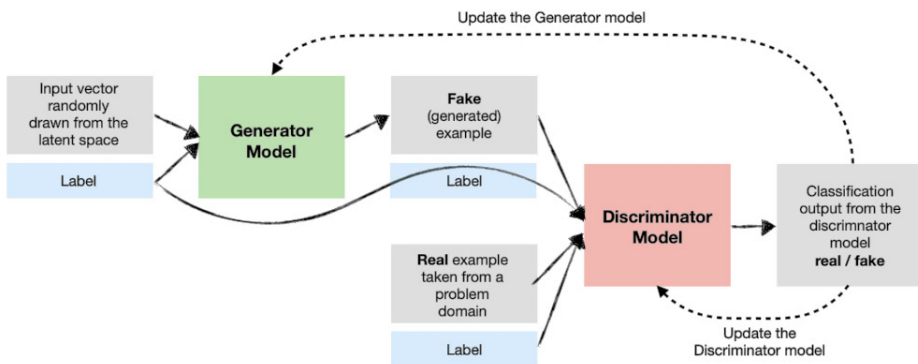


Figure 2. cGAN Architecture

3.2.2. Wasserstein GAN (WGAN)

An approach presented by the WGAN authors (Arjovsky et al., 2017) provided an alternative to conventional GAN training. They demonstrated how their novel approach avoided issues like mode collapse and increased the stability of model learning. Weight clipping, which WGAN employs for the

criticism model, guarantees that weight values remain within specified range values. It discovered that Jensen-Shannon divergence is not the best way to gauge how far apart the disjoint sections are distributed. Consequently, they employed the Wasserstein distance, which measures the difference between the generated and actual data distributions and attempts to preserve One-Lipschitz continuity throughout model training (Gulrajani et al., 2017). This distance is based on the idea of Earth mover's (EM) distance.

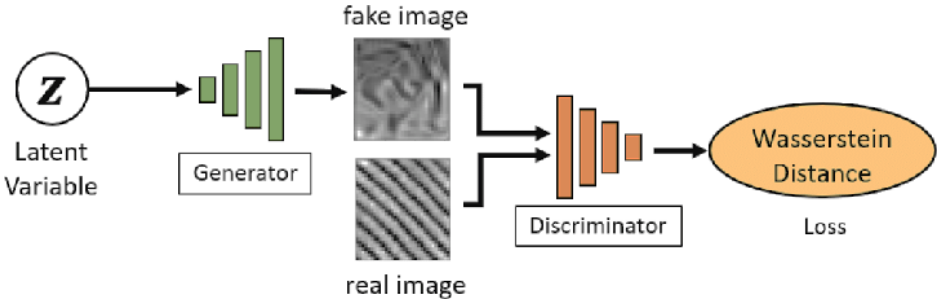


Figure 3. Wasserstein GAN Architecture

3.2.3. Deep Convolutional GAN (DCGAN)

Radford et al. (Radford et al., 2016) presented DCGANs. Deep Convolutional Neural Networks are used by DCGANs. The DCGAN's designers employed CNN into Generator and Discriminator because CNNs are better at pictures than multi-layer perceptrons, or MLPs, which were the only neural network architecture used in the original GAN architecture. The DCGAN neural network architecture includes transposed convolutions in the Generator, strided convolutions in the Discriminator, and fractional-strided convolutions in the Generator. Both the generators and the discriminator use batch normalization, ReLU is used in all layers except the output, and Adam optimizer is used instead of momentum-based SGD.

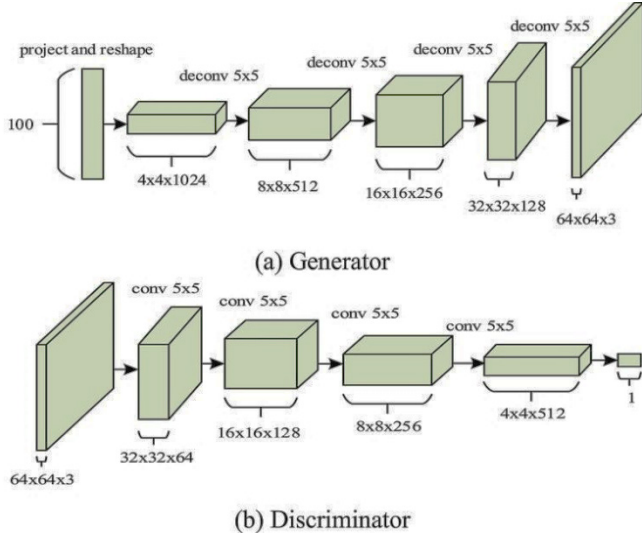


Figure 4. DCGAN Architecture

3.2.4. CycleGAN

The problem is addressed by CycleGAN (J.-Y. Zhu et al., 2017), which introduces a cycle consistency loss that attempts to maintain the original picture following a translation and reverse translation cycle. With this formulation, training no longer requires matching image pairings. Two discriminators (D_X , D_Y) and two generators (G , F) are used by CycleGAN. Images from the X domain are converted to the Y domain using G . In contrast, images from Y are converted to X via F . The Discriminator D_Y and D_X separate y from $G(x)$ and x from $F(y)$, respectively. Both mapping functions are subjected to the adversarial loss.

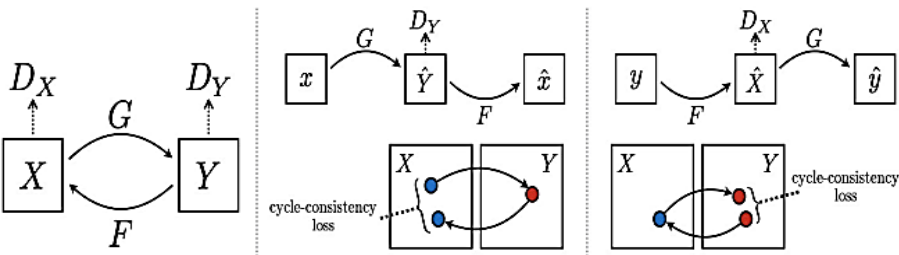


Figure 5. CycleGAN Architecture

3.2.5. StyleGAN

StyleGAN’s main objective is to generate diversified and high-resolution face images while giving users control over the synthetic images’ style that

are generated (Karras et al., 2020). The progressive growing approach is used by StyleGAN, a variant of the ProGAN (Karras et al., 2018), to create images with high quality and resolution. Changes made to StyleGAN only impact the Generator network, which implies that they only have an impact on the generative process. There has been no change to the discriminator or loss function, which are identical to those in a conventional GAN.

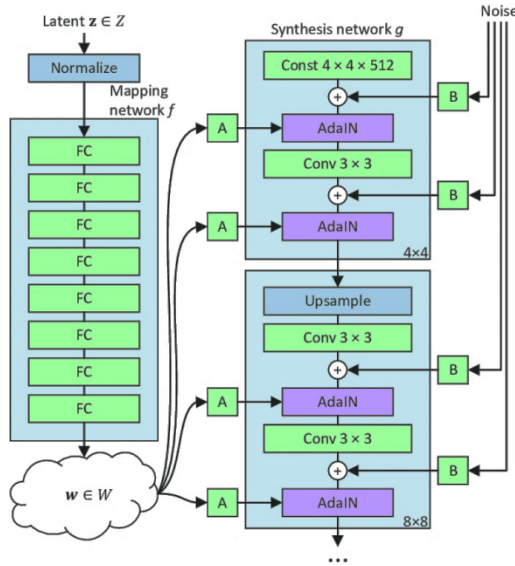


Figure 6. StyleGAN Architecture

3.2.6. Pix2Pix

A conditional generative adversarial network (cGAN) (Mirza & Osindero, 2014) called Pix2Pix (Isola et al., 2017) is used to solve general-purpose translation from image to image challenges. A PatchGAN (Isola et al., 2017) serves as the discriminator and the generator, which consist of U-Net (Ronneberger et al., 2015) structure, makes up the GAN. The pix2pix model builds a loss function to train the mapping in addition to learning the mapping from input to output image. It's interesting to note that the pix2pix Generator does not accept random noise vector input like standard GANs do. Instead, a mapping from the picture x to the output $G(x)$ is learned by the Generator. The classic function of adversarial loss serves as the discriminator's goal or loss function. To train the Generator, however, the adversarial loss is combined with the L_1 or pixel distance loss between the target or real image and the generated image. The generated image for a given input is encouraged by the L_1 loss to stay as close as possible. Training becomes more steady and converges more quickly as a result.

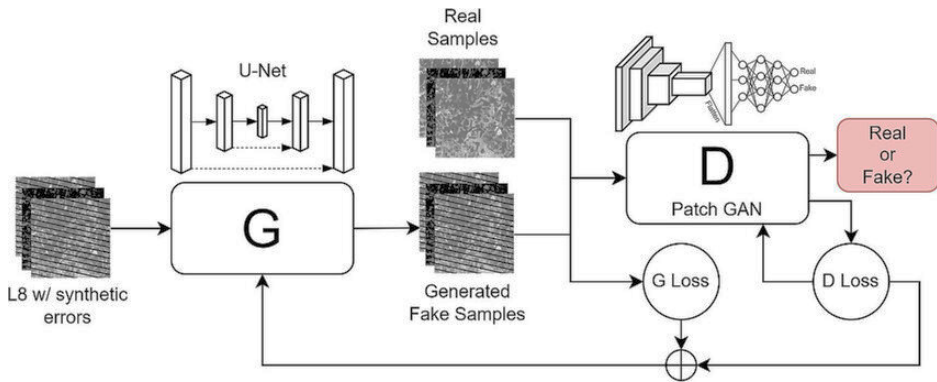


Figure 7. Pix2Pix Architecture

3.3. Different Evaluation Metrics of Generative Adversarial Networks

3.3.1. Inception Score (IS)

The IS (Salimans et al., 2016) is an instrument for assessing the level of quality of images generated by GANs. It employs a feature-extracting network (Szegedy et al., 2015) trained on a dataset similar to the generative model. An image is used as input and the result is a vector consisting of 1000 dimensions. The likelihood that an image belongs to a certain class is indicated by each dimension of this output vector. IS uses two measures to evaluate GAN performance: the diversity and quality of the images generated. The probability distribution from the Inception output should be as small as possible when assessing a single generated image. Higher image quality is indicated by a lower output, which increases the possibility that the generated image fits into a particular category. When generators produce a batch of images, the goal is for the average entropy of the probability distribution from the Inception output to be maximized, reflecting the variety in the output of the generator.

3.3.2. Fréchet Inception Distance (FID)

The FID (Dowson & Landau, 1982; Heusel et al., 2017) is a metric for calculating the separation between an actual image and the eigenvector of a produced image. This metric relies on Inception Net-V3, which alters the model's initial output layer such that the final pooling layer is the output. This creates a 2048-dimensional vector, meaning that each image is represented by 2048 features. FID calculates the distance between two distributions using the

mean and covariance matrix. A smaller FID value indicates that the generated distribution resembles the real image distribution.

3.3.3. Structural Similarity Index Measure (SSIM)

SSIM (Zhou Wang et al., 2004) evaluates the similarity between two images based on their brightness, contrast and structural elements. A higher SSIM value indicates better similarity, with a maximum possible value of 1. This computing model is based on human perception, allowing it to account for subtle variations in image structural information. Additionally, the model incorporates certain perceptual phenomena associated with shifts in perception, such as brightness and contrast masking.

3.3.4. Peak Signal-to-Noise Ratio (PSNR)

PSNR (Hore & Ziou, 2010) quantifies the ratio between the highest potential power of a signal and the noise power that negatively impacts its accuracy in representation. It serves as an objective criterion for evaluating an image's noise or distortion level. The two images are more strikingly similar when the PSNR value is higher. A commonly accepted threshold is 30dB, with noticeable image degradation occurring below this level.

4. Findings and Discussions

GANs are finding increasing applications in different areas and their ability to generate and alter data has changed a variety of industries, including autonomous cars, the creative arts, entertainment and healthcare. A few noteworthy uses for GANs are as follows: Using random noise vectors, GANs are widely used in image synthesis to create realistic images (Gadelha et al., 2017; Wu et al., 2016). They are frequently employed to create high-resolution images of objects, people, landscapes and artistic creations. GAN can be used for super-resolution (Ding et al., 2019; Guan et al., 2019; Ledig et al., 2017; X. Wang et al., 2018). They can enhance low-quality photos' resolution, resulting in crisper and more visually appealing photographs. They can be applied to the translation from image to image challenge (Isola et al., 2017; Lee et al., 2018; C. Wang et al., 2018). GANs make it easier to translate images between different visual domains. Applications include style transfer, innovative picture rendering and visual translation from day to night. GANs can replace or repair damaged or missing parts in photos by inpainting them

(Dolhansky & Ferrer, 2018; Iizuka et al., 2017). By training on photos with masked areas, the generator can predict appropriate material for the missing portions, making it appropriate for picture repair and alteration. By ageing or de-ageing faces in photos, GANs can mimic the effects of aging on a person's face (Antipov et al., 2017; P. Li et al., 2019; Liu et al., 2019; Z. Wang et al., 2018; Z. Zhang et al., 2017). These applications affect a wide range of fields, including entertainment, virtual character development and forensics. Pose estimation and human pose transfer are two applications for GANs (Ma et al., 2017; Siarohin et al., 2021). They could identify the positions of people's bodies in pictures. It also has the ability to change a person's position. This is important for virtual reality, animation and action recognition. High-quality images were successfully produced from text using GANs (Dash et al., 2017; Odena et al., 2017).

The application areas, advantages and disadvantages of different types of GAN are listed in Table 1.

Table 1. Applications, Advantages and Disadvantages of Different GAN Types

GAN Type	Application Areas	Advantages	Disadvantages
cGAN	• Image generation • Text-to-image generation • Image-to-image transformation • Super-resolution	• Data augmentation • Improved image quality • Conditional control • Versatility	• Mode collapse • Increased computational resources • Complexity • Training instability
WGAN	• High-quality image creation • Data augmentation • Domain modification • Data imputation and completion	• Improved stability • Meaningful loss function • Better training dynamics • Avoidance of mode collapse	• Slower training • Selecting the gradient penalty technique • Slower training • Hard to detect mode collapse
DCGAN	• Image generation • Translation from image to image • Data augmentation	• Stable training • High-quality image creation • Scale invariance • Disentangled representations	• Mode collapse • Hyperparameter sensitivity • Resource-intensive training • Difficulty in convergence
CycleGAN	• Style transfer • Domain adaptation • Transfiguring an object	• Less effort for data preparation • Domain generalization • Cycle consistency • Unpaired data usage	• Mode collapse • Hyperparameter tuning • Training complexity • Limited control
StyleGAN	• Face ageing and rejuvenation • Super-resolution • Data augmentation • Art and creativity • High-quality image generation	• Realistic and high-quality images • Control over styles • Smooth interpolation • Disentangled latent space	• Complexity • Resource-intensive • Data requirements • Mode collapse
Pix2Pix	• Data augmentation • Semantic segmentation • Style Transfer	• Adaptability to many fields • High-quality outputs • Consistent translations • Precise Control	• Complexity of implementation • Mode collapse • Data requirement • Overfitting

In generative models, generative adversarial networks are essential. GANs can effectively tackle the challenge of generating data. GANs offer several benefits. They are characterized by two distinct features that set them apart from other generative models. Firstly, GANs avoid making any assumptions. Many traditional methods rely on maximum likelihood to predict data distribution based on the presumption that data adheres to a specific distribution. The training process for GANs is more flexible and uncomplicated. Secondly, generating realistic samples is remarkably straightforward. Unlike conventional sampling methods, which can be complex, GANs use forward propagation through the generator to create lifelike samples. The emergence of GANs disrupts traditional artificial intelligence algorithms that restrict individual thought processes. They offer a successful approach for deep learning models that are not supervised. By means of ongoing adversarial training, GANs enable machine-to-machine communication.

GANs can generate new data that closely resembles the training set because they are designed for generative tasks. On the other hand, other neural networks are mostly used for classification or regression tasks. Whether the input is text, audio, photos, or other sorts, GANs are quite good at generating realistic and accurate data. Applications like style transfer, image synthesis and data augmentation gain from this. As an illustration of unsupervised learning, GANs may uncover hidden patterns and hierarchical structures in data without the use of ground truth labels. As a result, they are particularly helpful in applications where obtaining tagged data is costly or challenging. From a single input or latent vector, GANs can generate a variety of outputs. This unpredictability is useful in situations like improving data and creating a variety of simulations.

While GANs can perform effectively, they also come with certain limitations. A common problem that can arise during GAN training is mode collapse. This phenomenon happens when the GAN's generator network generates a relatively small range of outputs, which results in a high degree of homogeneity in the samples that are produced. Mode collapse makes it challenging for the GAN to adequately produce the diversity and complexity of the underlying data distribution. This issue is challenging to solve since the generator seeks to minimize its loss during training by producing outputs that closely resemble clearly identifiable examples in the training data. This could cause the generator to ignore other variations and duplicate outputs in order to make sure the discriminator is tricked by the simplest cases. Moreover, the discriminator and generator may both experience the problem of vanishing gradients. This occurs during training when the gradients of the loss function propagate backwards through multiple layers of the neural network until they reach a minimal value.

The application areas, application types and different types of GAN models are listed in Table 2.

Table 2. Application Areas and Applications of Different GAN Types

Application Area	Application Types	GAN Models
Image Processing	Enhancing images resolution	SRGAN (Ledig et al., 2017), ESRGAN (X. Wang et al., 2018), PESR (Vu et al., 2018), GMGAN (X. Zhu et al., 2020), TSRGAN (Jiang & Li, 2020), Best-BuddyGAN (W. Li et al., 2022)
	Editing and modifying images	IcGAN (Perarnau et al., 2016), ID-CGAN (H. Zhang et al., 2020)
	Generating synthetic faces	ProGAN (Karras et al., 2018), StyleGAN (Karras et al., 2020)
Video Processing	Unconditional video generation	DVD-GAN (Clark et al., 2019), TGAN (Saito et al., 2017), MoCoGAN (Tulyakov et al., 2018), FTGAN (Ohnishi et al., 2018), VGAN (Vondrick et al., 2016)
	Conditional video generation	TGANs-C (Saito et al., 2017), GAN with RNN (Vougioukas et al., 2018), VAE-GAN (Y. Li et al., 2018), TemporalGAN (H. Zhou et al., 2019), StoryGAN (Y. Li et al., 2019), TFGAN (Balaji et al., 2019), TiVGAN (D. Kim et al., 2020), Bo-GAN (Q. Chen et al., 2020), LSTM and cGAN (Jalalifar et al., 2018)
Medical and Healthcare	Multimodal translation from image to image	CycleGAN (J.-Y. Zhu et al., 2017), Pix2Pix (Isola et al., 2017), MCML-GANs (Yu et al., 2019), MR-GAN (Jin et al., 2019), DermGAN (Ghorbani et al., 2020), MedGAN (Armanious et al., 2020), Tub-GAN and Tub-sGAN (Zhao et al., 2018)
	Generating images for data enhancement	DCGAN (Radford et al., 2016), ACGAN (Odena et al., 2017), BEGAN (Berthelot et al., 2017), WGAN-GP (Gulrajani et al., 2017), ProGAN (Karras et al., 2018), fNIRS-GANs (Nagasawa et al., 2020), WGAN (Arjovsky et al., 2017)
	Reconstruction of images	VDSR (J. Kim et al., 2016a), DRCN (J. Kim et al., 2016b), SRCNN (Dong et al., 2016), mDCSRN-GAN (Y. Chen et al., 2018), ESRGAN (X. Wang et al., 2018), MedSRGAN (Gu et al., 2020)

Material Science	Generating and designing micro and crystal structure	Composition-Conditioned Crystal GAN (S. Kim et al., 2020), GAN+GP with Hedge Bayesian Optimization (Yang et al., 2018), CrystalGAN (Nouira et al., 2019)
	Creating structured material	GAN Based Model (Mao et al., 2020)
	Designing inorganic materials	WGAN Based Model (Hu et al., 2020), MatGAN (Dan et al., 2020)
Finance	Generating financial data	FIN-GAN (Takahashi et al., 2019), CGAN and DRAGANs (Efimov et al., 2020), Stock-GAN (J. Li et al., 2020)
	Forecasting stock market	GAN-FD (X. Zhou et al., 2018), Quant GAN (Wiese et al., 2020)
	Finding finance anomalies	LSTM-GAN (Leangarun et al., 2018)
Marketing	Creating logos	LoGAN (Mino & Spanakis, 2018), iWGAN (Sage et al., 2018)
	Generating pose and model	E2E (Song et al., 2019), PG ² (Ma et al., 2017), Deformable GAN (Siarohin et al., 2018)

Generative Adversarial Networks (GANs) have found extensive applications across various domains, demonstrating their versatility in tasks ranging from image and video processing to finance and healthcare. In image processing, GANs are widely utilized for tasks such as super-resolution, image editing, and synthetic face generation, with models like SRGAN, ESRGAN, and StyleGAN achieving state-of-the-art results. Similarly, GANs facilitate both unconditional and conditional video synthesis in video processing, leveraging models such as DVD-GAN and StoryGAN to generate high-quality and temporally coherent video sequences. These advancements highlight the capacity of GANs to enhance visual data representation and synthesis.

Beyond multimedia applications, GANs have proven valuable in scientific and commercial fields. They enable medical image translation, augmentation, and reconstruction in healthcare, improving diagnostic accuracy and data availability through models like CycleGAN, DCGAN, and MedSRGAN. In material science, GANs contribute to the design of novel materials by generating crystal structures and inorganic compounds. The financial sector benefits from GANs in data generation, stock market forecasting, and anomaly detection, as seen in models like Stock-GAN and Quant GANs. Furthermore, GANs play a role in marketing by generating synthetic product designs and advertisements.

As GAN research progresses, domain-specific adaptations and optimizations are expected to expand their impact across industries, reinforcing their significance in theoretical and applied machine learning.

5. Conclusion and Future Trends

Generative Adversarial Networks (GANs) have emerged as one of the most influential advancements in deep learning, enabling remarkable progress in image synthesis, data augmentation, and various other applications. Their ability to generate high-quality and diverse synthetic data has led to widespread adoption across industries, including healthcare, entertainment, and scientific research.

Future research in GANs is expected to address these challenges through more stable training techniques, improved loss functions, and enhanced architectures. Additionally, integrating GANs with other machine learning models, such as transformers and diffusion models, may further expand their capabilities. Another promising direction involves leveraging GANs for ethical AI applications, ensuring fairness, reducing biases, and maintaining data privacy in generative models.

Moreover, as GANs evolve, their applications will likely extend beyond their current scope, influencing areas such as personalized content generation, real-time simulation, and high-fidelity medical imaging. The intersection of GANs with quantum computing and edge AI also holds significant potential for revolutionizing real-time and resource-constrained generative tasks.

In conclusion, GANs have already profoundly impacted artificial intelligence and data science, and their future remains bright. Continued research and development will unlock even more possibilities, making generative modeling a cornerstone of intelligent systems in the years to come.

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CHAPTER VI

USING TRANSFORMER MODEL FOR RANDOM SIGNAL PREDICTION AND MODELSIM IMPLEMENTATION

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1. Introduction and Basic Concepts

The rapid evolution of signal processing technologies has fundamentally transformed numerous industries, including telecommunications, finance, and control systems. Signal prediction has emerged as a cornerstone in the development of intelligent systems, where accurate forecasting is pivotal for applications such as resource allocation, anomaly detection, and predictive maintenance. Traditional methodologies, including autoregressive integrated moving average (ARIMA) models and Kalman filters, have long dominated the field. While these approaches excel in handling linear patterns and moderately complex dynamics, they often falter when confronted with highly nonlinear, stochastic, or long-range dependencies. The advent of deep learning, particularly the transformer model, has introduced a paradigm shift, enabling unprecedented capabilities in signal prediction.

Transformers model, initially developed for natural language processing (NLP) tasks, have garnered widespread attention due to their innovative self-attention mechanism. This architecture eliminates the sequential nature of traditional recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, allowing for the efficient modeling of dependencies across vast temporal horizons. By leveraging parallel processing and positional encodings, transformers model capture intricate patterns and long-range

relationships in data, making them uniquely suited for the challenges posed by random signal prediction.

Random signals, characterized by their stochastic and often unpredictable nature, present unique challenges. They frequently exhibit a combination of trends, seasonal patterns, and high-frequency noise components, demanding sophisticated models that can disentangle these elements. Predicting such signals accurately is critical in applications ranging from stock market analysis to environmental monitoring and wireless communication. In these contexts, the transformer's ability to model complex temporal relationships without being hindered by sequence length constraints positions it as a transformative tool.

Applying ModelSim in random signal applications offers several advantages, particularly in fields such as digital signal processing (DSP), communications, and hardware design. ModelSim is a powerful simulation tool for verifying HDL (VHDL/Verilog) designs and is widely used in FPGA and ASIC development. Before committing to expensive FPGA or ASIC production, ModelSim allows designers to verify that their circuits handle random inputs correctly. Compared to hardware testing, ModelSim accelerates the debugging process by allowing quick design iterations without reprogramming FPGAs. Using ModelSim in random signal applications is crucial for testing and verifying the behavior of digital systems under real-world conditions. Its ability to simulate, analyze, and optimize designs before hardware implementation reduces costs, improves reliability, and enhances performance. Before programming in Modelsim, a preliminary study must be carried out by simulating with Python.

2. Literature and Field Studies

Random signal prediction is a critical area in signal processing, machine learning, and various engineering disciplines. The ability to predict random signals accurately has applications in telecommunications, financial forecasting, biomedical signal processing, and environmental monitoring. The aim of this study is to examine the basic methodologies, theoretical foundations, and recent advancements in random signal prediction. The Kalman filter, introduced by Kalman (1960), is a widely used recursive algorithm for estimating the state of a dynamic system from noisy observations. It operates optimally under the assumption of linearity and Gaussian noise. Applications include tracking systems, navigation, and control systems (Kalman, 1960). Wiener filtering, developed by Norbert Wiener, is another classical technique used for signal prediction. The Wiener filter is optimal for stationary processes and minimizes

the mean square error in signal estimation. It has been extensively applied in speech enhancement and radar signal processing (Wiener, 1949). Autoregressive (AR) and Moving Average (MA) models, along with their combined ARMA and ARIMA variants, have been fundamental in time-series forecasting. These models assume that future values of a signal are linear combinations of past observations and noise (Box & Jenkins, 1970). Particle filters (Gordon et al., 1993) extend the Kalman filter to nonlinear and non-Gaussian scenarios. These filters use a set of weighted samples (particles) to represent the posterior distribution of a system's state. Particle filters are commonly used in robotics and financial modeling. Singular Spectrum Analysis (SSA) is a data-driven technique used to decompose time-series data into interpretable components, such as trends and periodic signals. It has proven useful in climate science and biomedical signal analysis (Golyandina & Zhigljavsky, 2001). Stochastic resonance is a phenomenon where noise enhances the detectability of weak signals. It has applications in neuroscience, climate modeling, and mechanical systems (Gammaitoni et al., 1998). Artificial neural networks (ANNs), particularly Long Short-Term Memory (LSTM) networks, have revolutionized signal prediction. LSTMs, introduced by Hochreiter & Schmidhuber (1997), are well-suited for capturing long-range dependencies in time-series data. Convolutional neural networks (CNNs) have also been used for feature extraction in signal processing (LeCun et al., 2015). Support vector regression (SVR) has been successfully applied to random signal prediction, particularly in financial forecasting. Ridge regression and Lasso regression provide robust alternatives for high-dimensional signal prediction problems (Tibshirani, 1996). Reinforcement learning (RL) has been applied to adaptive filtering and predictive control problems. RL algorithms learn optimal prediction strategies through interactions with the environment (Sutton & Barto, 2018).

Random signal prediction is crucial in adaptive filtering for wireless communications, where signals must be extracted from noisy environments (Haykin, 2002). Machine learning models, including recurrent neural networks and probabilistic models, are used to predict stock prices and market trends (Fama, 1970). Electroencephalogram (EEG) and electrocardiogram (ECG) signal predictions benefit from deep learning and statistical models for diagnosing medical conditions (Liu et al., 2019). SSA and deep learning models aid in predicting weather patterns and climate change effects (Ghil et al., 2002).

Random signal prediction encompasses a diverse array of methodologies, from classical statistical techniques to advanced deep learning approaches. As

computational power increases, hybrid models that combine statistical and machine learning methods are likely to become more prominent. Future research should focus on improving model interpretability, computational efficiency, and adaptability to complex real-world scenarios.

The challenge of accurately predicting random signals holds both theoretical significance and practical value. In telecommunications, anticipating noisy signal behavior aids in efficient resource allocation for dynamic spectrum access. In finance, predicting stock price movements or market volatility supports informed investment strategies. Similarly, in industrial applications, forecasting sensor readings or equipment behavior enables predictive maintenance, reducing downtime and operational costs.

This study makes several key contributions to the field of signal processing and machine learning. It provides a detailed overview of transformer models and their application to random signal prediction, highlighting both theoretical and practical aspects. It introduces a comprehensive pipeline for integrating transformer-based predictions with ModelSim, bridging the gap between machine learning and hardware design. It presents experimental results demonstrating the effectiveness of transformers model in predicting synthetic random signals, including discussions on accuracy, computational efficiency, and hardware verification. By combining state-of-the-art machine learning techniques with practical hardware implementation, this study aims to provide a resource for researchers and engineers seeking to advance the field of random signal prediction.

3. Methodology and Technical Approaches

3.1 Transformers Model: A Paradigm Shift in Signal Processing

The Transformer model, initially introduced in the “Attention is All You Need” paper by Vaswani et al. in 2017, has become a cornerstone in modern machine learning, particularly in natural language processing (NLP). However, it can also be applied to other domains, such as signal processing. When working with random signals, the Transformer methodology can provide powerful tools for capturing long-range dependencies and patterns. The transformative impact of transformers model in machine learning cannot be overstated. Originally introduced by Vaswani et al. in their seminal paper “Attention Is All You Need,” transformers model have redefined state-of-the-art performance in NLP tasks such as machine translation, text summarization, and sentiment analysis. The core innovation of the transformer lies in its ability to process entire input

sequences simultaneously, leveraging self-attention to capture dependencies between elements, irrespective of their distance within the sequence.

In the context of signal processing, these attributes translate into several key advantages. Transformers models can handle long input sequences efficiently, making them suitable for time-series data with extended temporal horizons. The model's architecture can be adapted to various input modalities, including one-dimensional signals, multi-channel data, and even image-based time-series representations. Self-attention mechanisms enable transformers model to focus on the most relevant parts of the input, improving their resilience to noise and irrelevant features.

Transformers have revolutionized sequence modeling tasks, particularly in natural language processing. However, their application extends to time series forecasting and random signal prediction. This document provides a mathematical analysis of Transformers used in predicting random signals, exploring their architecture, self-attention mechanism, and optimization techniques.

A Transformer model consists of an encoder-decoder structure with multi-head self-attention and feed-forward networks. The encoder processes the input sequence, while the decoder generates predictions based on learned representations.

Mathematically, given an input sequence $X = \{x_1, x_2, \dots, x_n\}$, the Transformer encoder produces a contextual representation $Z = \{z_1, z_2, \dots, z_n\}$ via self-attention and feed-forward layers. The decoder then predicts future signals based on Z .

The core of the Transformer is the self-attention mechanism, which computes attention scores to weigh input tokens differently.

The scaled dot-product attention is formulated as:

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where:

$Q = XW_Q, K = XW_K, V = XW_V$ are the query, key, and value matrices, W_Q, W_K, W_V are learned weight matrices, d_k is the dimension of the key vector, *softmax* ensures that attention weights sum to one.

Multi-head attention allows the model to focus on different aspects of the input sequence:

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W_0 \quad (2)$$

where each head computes independent attention scores, and W_0 projects the concatenated output back to the model dimension.

Unlike recurrent networks, Transformers lack inherent sequential order. To encode position information, a sinusoidal function is used:

$$PE_{(pos, 2i)} = \sin(pos/1000^{2i/d}) \quad (3)$$

$$PE_{(pos, 2i+1)} = \cos(pos/1000^{2i/d}) \quad (4)$$

where pos is the position index and d is the embedding dimension.

Transformers are trained using gradient descent, commonly with the Adam optimizer. The loss function for time-series forecasting tasks is typically Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (5)$$

Where y_i is the actual value and $\hat{y}_i$ is the predicted value. To improve generalization, dropout regularization and layer normalization are applied:

$$LayerNorm(x) = \frac{x - \mu}{\sigma - \epsilon} \gamma + \beta \quad (6)$$

where μ and σ are mean and standard deviation, and γ, β are learned parameters.

For random signals, Transformers excel in learning temporal dependencies.

Given a stochastic process $\{X_t\}$, the model estimates future values $X_{t+1}, X_{t+2}, \dots$ based on past observations. Attention mechanisms help in identifying important patterns despite randomness.

Transformers model provide a powerful framework for random signal prediction by leveraging self-attention and parallel computation. Their ability to model dependencies without recurrence makes them efficient for large-scale time-series forecasting.

The transformer model's encoder-decoder structure is shown in Figure 1, emphasizing the feed forward neural networks (FNNs) in the encoder and decoder layers as well as the multi-head self-attention mechanism in the encoder.

The encoder’s examination of past weather data and the decoder’s incorporation of these findings to predict future wind power generation are crucial phases. The graphic also shows disguised self-attention in the decoder, highlighting the use of historical data to generate precise, future-oriented predictions.

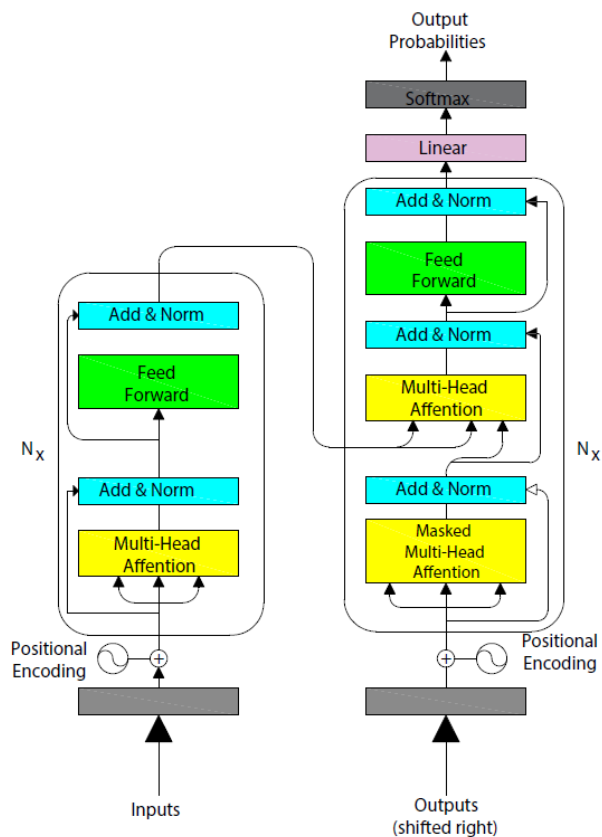


Figure 1. Transformer Model Architecture

These strengths make transformers model a natural choice for random signal prediction, where traditional models often struggle to balance complexity and accuracy. Furthermore, recent advancements such as the development of lightweight transformer variants and efficient training techniques have made the deployment of these models more practical, even in resource-constrained environments.

Predicting random signals is integral to applications such as financial market forecasting, radar signal processing, and biological signal analysis (Zhang et al., 2021). Traditional methods rely on domain-specific assumptions

and often struggle with complex, non-linear patterns. Deep learning, with its ability to learn data-driven representations, has emerged as a game-changer, offering robust solutions for such tasks. Transformer models, in particular, excel in sequence modeling by employing self-attention mechanisms that weigh the importance of each element in a sequence relative to others (Vaswani et al., 2017). This attribute is crucial for random signal prediction, where identifying underlying patterns amidst noise is paramount.

Despite their success, most studies on Transformers model focus on software-based implementations, leaving a gap in hardware realizations (Gupta & Kumar, 2020). ModelSim, a simulation tool for hardware description languages (HDLs) such as VHDL and Verilog, is instrumental in bringing this gap. Implementing Transformer models within ModelSim involves translating their mathematical operations into HDL-compatible formats, enabling their integration into digital systems (Bishop, 2006). This effort is motivated by the growing demand for deploying AI models in edge computing devices, where real-time processing and resource efficiency are critical (Silver et al., 2016).

3.2. Hardware Implementation and ModelSim Integration

ModelSim is a popular hardware simulation tool commonly used for verifying the functionality of digital circuits, especially in VHDL and Verilog designs. When working with random signals in the context of simulation and verification, ModelSim offers a range of techniques and methodologies for handling randomness in testbenches, signal generation, and analysis. It allows for functional verification, timing analysis, and debugging. ModelSim provides a powerful mechanism for simulating the behavior of digital circuits, which can be particularly useful when working with random signals. While much of the focus on transformers model has been on software-based applications, their integration with hardware systems represents a critical step toward real-world deployment. In applications such as autonomous vehicles, wireless communication, and industrial automation, signal prediction models must operate within embedded systems that demand low latency and high reliability. Hardware description languages (HDLs) like Verilog and simulation tools such as ModelSim play a crucial role in bridging this gap.

ModelSim, a widely used HDL simulation tool, allows engineers to verify the functionality and performance of digital designs before hardware implementation. By integrating transformer-based predictions with ModelSim, it becomes possible to Validate the accuracy of the predicted signals in a simulated

hardware environment, Optimize the implementation for specific hardware platforms, such as field-programmable gate arrays (FPGAs) or application-specific integrated circuits (ASICs) and ensure that the model's predictions align with the constraints and requirements of the target system.

While transformer models have been widely explored in software applications, their integration into hardware systems is a crucial step toward real-world usability. In fields such as autonomous vehicles, wireless communication, and industrial automation, signal prediction models must function within embedded systems that require low latency and high reliability. Hardware description languages (HDLs) like Verilog and simulation tools such as ModelSim facilitate this transition.

Using random signals in ModelSim is an essential technique for verifying and stress testing digital designs. By leveraging features such as random signal generation, System Verilog randomization, Monte Carlo simulations, and waveform analysis, ModelSim allows engineers to explore a wide range of possible scenarios, ensuring that the design can handle unpredictable inputs and complex conditions. Techniques like constraint random generation, stress testing, and code coverage can enhance the robustness of the verification process, making it easier to identify edge cases, vulnerabilities, and performance bottlenecks in the design.

This chapter outlines a practical approach to implementing transformer models for random signal prediction, including the generation of HDL code for hardware simulation. By combining the predictive capabilities of transformers model with the rigorous verification provided by ModelSim, the proposed methodology enables a seamless transition from algorithm development to hardware deployment.

4. Findings, Applications and Discussions

In ModelSim, random signal prediction is often used to simulate how a design reacts to unpredictable or stochastic inputs. The objective of a random signal prediction example is to observe the behavior of the design under random stimuli and to verify whether the system functions as expected. This process typically involves generating random inputs in a testbench, simulating the response of the design, and then analyzing the results to check for correctness, performance issues, or edge cases.

Before programming in ModelSim, it is highly beneficial to conduct a preliminary study using Python to simulate and analyze random signals. Python

offers powerful libraries such as NumPy, SciPy, and Matplotlib for signal processing, noise generation, and statistical analysis, making it an excellent tool for initial validation before implementing the design in VHDL/Verilog for ModelSim. Python allows for quick implementation and testing of signal processing algorithms without the complexity of HDL coding. Figure 2 shows the results obtained using the Transformers model in Python. Accordingly, similar results can be obtained by switching to the ModelSim application of the Transformers model.

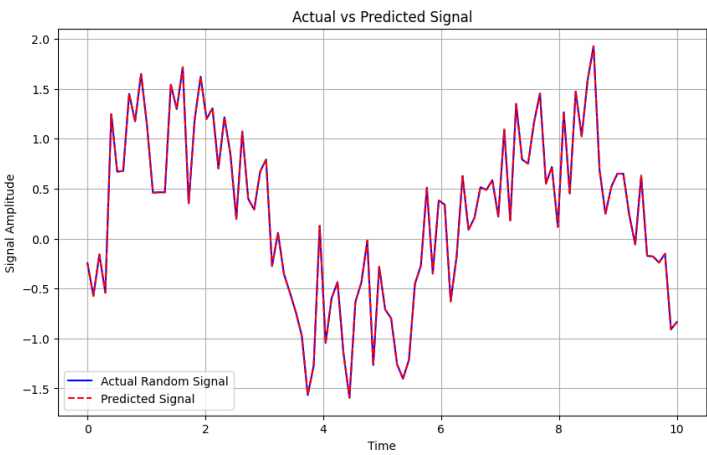


Figure 2. Comparison of Actual and Predicted Values with Python

In this example, let’s consider the simulation of a simple system where we apply random signals to the input of a design and observe the output. We will use ModelSim to generate random signals, predict the expected behavior, and discuss the results. In the testbench, we will generate two random 8-bit signals (a and b), apply them as inputs to the adder, and observe the sum. The \$random function will be used in Verilog to generate random values.

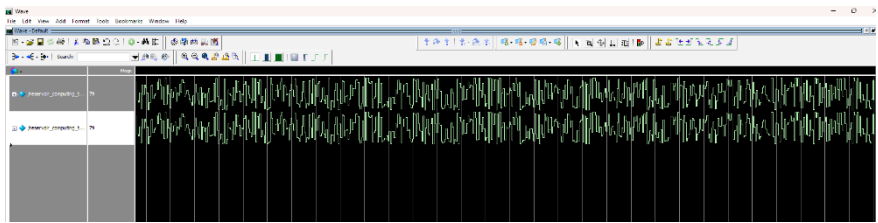


Figure 3. Comparison of Actual and Predicted Values with ModelSim as Analog Signal

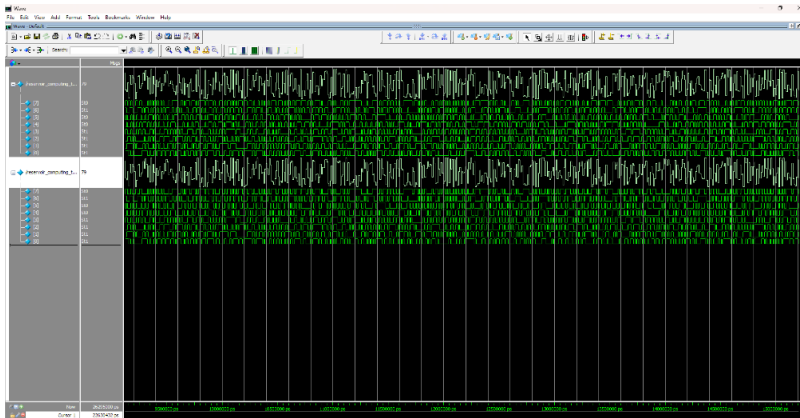


Figure 4. Comparison of Actual and Estimated Values as Analog and Digital Signals with ModelSim

After compiling the design and the testbench, we simulate the scenario in ModelSim. During the simulation, random values are applied to the inputs *a* and *b* every 10-time units. The ModelSim waveform viewer or console will display the results for each cycle of the simulation.

In this ModelSim random signal prediction example, we generated random signals for a simple 8-bit adder design, simulated the system, and observed the results. The simulation confirmed that the adder functioned correctly, generating the sum of the two random inputs. Random signal generation in verification allows for testing a wide range of input combinations, which can help ensure the design is robust and functioning as expected. However, there are also limitations, such as the potential for certain corner cases to remain uncovered. Expanding the testbench to include coverage metrics, overflow detection, and assertions would further enhance the reliability of the verification process.

5. Conclusion and Future Directions

Transformers model have emerged as a powerful tool for random signal prediction, offering unmatched capabilities in handling noise, capturing complex dependencies, and scaling large datasets. Their versatility, combined with ongoing advancements in architecture and training strategies, positions them as a cornerstone technology for tackling the challenges of stochastic and noisy data across various domains.

This study examines the use of transformer models for predicting random signals, a key aspect of contemporary signal processing. By highlighting the

self-attention mechanism within architecture, we illustrate its effectiveness in capturing and forecasting temporal patterns. Furthermore, we explore the implementation and validation of the predicted signal using ModelSim, a popular simulation tool for digital design. The chapter presents theoretical insights, practical applications, and experimental findings, offering a thorough resource for researchers and engineers.

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CHAPTER VII

NATURAL LANGUAGE PROCESSING IN THE AGE OF ARTIFICIAL INTELLIGENCE: TECHNICAL ADVANCES, OPPORTUNITIES AND CHALLENGES

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1. Introduction and Fundamental Concepts

Natural Language Processing (NLP) is a subfield of artificial intelligence that enables interaction between human languages and computers by aiming to understand and interpret human language. The primary goal of NLP is to transform complex, contextual, and inherently ambiguous human language into a form understandable by computers, thus facilitating natural communication between humans and machines. Tasks within the scope of NLP span a wide range, including text classification, machine translation, information extraction, sentiment analysis, and text generation.

Text processing, regarded as a subdomain of NLP, primarily focuses on the structural organization, preprocessing, and analysis of textual data. Essential preprocessing methods for making texts comprehensible include tokenization, stemming, lemmatization, stop word removal, and text normalization. These text processing techniques prepare raw text data for higher-level NLP tasks, typically forming the foundational step for advanced systems such as language models.

In this context, Large Language Models (LLMs) have recently brought about a paradigm shift in NLP and text processing. Large language models are deep learning architectures, typically based on the Transformer architecture and trained on vast amounts of textual data. Models such as GPT (Generative Pre-trained Transformer), BERT (Bidirectional Encoder Representations from

Transformers), and Mistral represent the most notable examples of this category. These models demonstrate human-like capabilities in language generation and comprehension, effectively addressing semantic ambiguities that are challenging to resolve with traditional methods.

Nevertheless, significant challenges still persist within NLP and text processing. Ambiguities inherent in natural language, noisy data, accurate context comprehension, multilingual and low-resource language limitations, and computational requirements represent prominent challenges for contemporary NLP research. Conversely, these difficulties also create opportunities for innovation. Developments in large language models, in particular, have opened up numerous avenues for novel solutions, such as enhancing automated decision-making systems with AI agents that leverage textual data and employing transfer learning techniques to facilitate NLP applications requiring fewer data resources.

2. Evolution of NLP

The historical development of the natural language processing field has followed a trajectory in which different approaches have gained prominence during different periods, from the earliest studies to the present day. These developments have profoundly influenced text processing techniques and have formed the foundation of modern NLP methods.

1950s – 1980s: Early Period and Rule-Based Approaches

The foundations of natural language processing date back to the 1950s. During this period, Alan Turing introduced the famous Turing Test, taking the first theoretical steps toward evaluating a machine's capability to interpret human language. One of the earliest practical attempts at enabling computers to understand human language was ELIZA, developed by Joseph Weizenbaum in the 1960s [1]. ELIZA was a chatbot capable of simulating psychotherapy conversations using basic rules and pattern-matching techniques.

However, contrary to the expectations of the time, rule-based systems encountered significant limitations due to the complexity and versatility of natural language. The 1966 ALPAC report further underscored these challenges, considerably slowing down research in areas such as machine translation.

1980s – 1990s: Rise of Statistical Approaches

Due to the limitations of rule-based methods, researchers began shifting toward data-driven statistical approaches in the late 1980s. Rabiner's seminal 1989 work provides a detailed theoretical foundation of Hidden Markov Models

(HMM) and their application, particularly in speech recognition, becoming a fundamental resource in the field of NLP [2]. Particularly, n-gram-based language models developed during the 1990s enabled text generation and language modeling by exploiting probabilistic relationships among words [3]. During this period, IBM's statistical machine translation systems, specifically IBM Models 1-5, provided substantial improvements over traditional rule-based translation methods [4].

Decision Trees and Maximum Entropy Models introduced in 1996 provided alternative statistical approaches, enhancing classification accuracy and flexibility [5]. Additionally, the first comprehensive release of WordNet in 1995 supplied a crucial linguistic resource, facilitating semantic understanding and lexical processing tasks [6]. In 1999, IBM's statistical machine translation systems further illustrated the potential of statistical methods, significantly outperforming earlier rule-based translation techniques. In the early 2000s, the integration of artificial neural networks into NLP gained momentum. The introduction of Long Short-Term Memory (LSTM) cells by Hochreiter and Schmidhuber in 1997 became a critical advancement, particularly for modeling the sequential nature of textual data [7].

2000s – 2010s: Beginning of the Deep Learning Revolution

In 2003, Bengio et al. demonstrated the capability of neural network-based language models to significantly outperform conventional n-gram-based statistical models, indicating the potential of neural approaches in NLP [8]. By the mid-to-late 2000s, deep neural networks, especially recurrent neural networks (RNNs), gained popularity due to their superior performance in various NLP tasks compared to traditional statistical methods. By the end of the 2000s, the effectiveness of deep learning became increasingly apparent within NLP research.

2010s: The Deep Learning Paradigm Shift

From the early 2010s, a significant paradigm shift occurred in text processing. Collobert et al. (2011) introduced a unified neural network architecture for NLP, demonstrating that deep learning approaches could simultaneously address multiple linguistic tasks such as part-of-speech tagging, named entity recognition, and semantic role labeling within a single framework [9]. In 2013, Mikolov et al. introduced the Word2Vec technique, successfully capturing semantic relationships between words by representing them as continuous vectors [10]. Similarly, the GloVe model developed by Pennington et al. in 2014 offered even more efficient and semantically rich word vectors [11].

Additionally, the Encoder-Decoder model proposed by Sutskever et al. in 2014 transformed many NLP tasks, notably machine translation [12]. The attention mechanism introduced by Bahdanau et al. in 2015 further improved this model, revolutionizing NLP by accurately capturing context even in long sequences of text [13].

2017: Transformer Architecture

One of the most critical milestones in NLP literature occurred in 2017 when Vaswani et al. introduced the Transformer architecture [14]. The Transformer eliminated the need for recurrent neural networks, relying exclusively on the attention mechanism to effectively capture dependencies between words. This innovation significantly accelerated model training by enabling parallel processing, paving the way for the development of larger language models.

Based on the Transformer architecture, large language models began with the ELMo model developed by Peters et al. in 2018, featuring context-dependent word representations [15]. The next year, Google's BERT (Bidirectional Encoder Representations from Transformers) model significantly advanced the usage of pre-trained language models, achieving state-of-the-art performance on multiple NLP tasks due to its bidirectional structure [16].

2020s: The Age of Large Language Models (LLMs)

The introduction of GPT-3 (Brown et al., 2020) in 2020 marked a new era in NLP [17]. With its immense size of 175 billion parameters, GPT-3 demonstrated unprecedented capabilities, performing complex tasks through few-shot learning and introducing the concept of "emergent capabilities" to the literature. Nonetheless, large language models also exhibited significant issues, reflecting biases from training data and occasionally generating erroneous content.

In 2021, Bommasani et al. from Stanford University introduced the concept of foundation models, comprehensively discussing the opportunities and risks posed by large language models, which resonated strongly within the NLP community [18]. A central issue identified was that the use of powerful but homogeneous foundation models could propagate biases and errors across downstream applications.

Similarly to **GPT-3**, Google's **T5** (Text-to-Text Transfer Transformer) approached NLP tasks uniformly as text-to-text problems, simplifying transfer learning and adaptation to multiple tasks [19]. Subsequently, Foundation Models emerged in 2021, offering versatile, large-scale pre-trained architectures

adaptable to various downstream applications. Techniques such as Instruction Tuning and Reinforcement Learning from Human Feedback (RLHF) in 2022 improved the controllability and alignment of large language models [20]. One notable model developed using RLHF is **Claude**, created by Anthropic. Claude leverages RLHF techniques to minimize harmful or misleading responses, ensuring more ethical and secure user interactions. The recent release of multimodal models like **GPT-4** and **Gemini** [21] expanded NLP into multimodal contexts, integrating visual and textual information.

Present Landscape

In 2024, models such as **Mistral** and **Phi-3** have become the focus of research, achieving high performance with lower computational costs through smaller and more efficient architectures. In parallel with these advancements, AI agent frameworks like ReAct (Reasoning and Acting), which incorporate reasoning capabilities, have enhanced the transparency and reliability of language models, making NLP systems more robust, explainable, and applicable to real-world scenarios [22]. The last model was introduced in 2024, **DeepSeek** is further refining model efficiency and multimodal capabilities, expanding beyond text to incorporate vision-based processing. DeepSeek models represent a growing trend in open-source AI development, providing powerful alternatives to commercial LLMs while maintaining transparency and accessibility in AI research.

Today, NLP research primarily focuses on scaling up large language models, reducing computational requirements, and ensuring the ethical, reliable, and transparent usage of these models. One critical concern is the issue of hallucination, where large models produce realistic yet inaccurate textual outputs. Additionally, research into multimodal models, transfer learning techniques, and NLP-based autonomous agents continues to expand rapidly.

3. Methodology and Techniques of NLP

Deep Learning Architectures in NLP

Deep learning architectures have played a crucial role in the evolution of NLP, with different models excelling in specific tasks. RNNs were initially designed to capture the sequential nature of language, making them effective in early applications like speech recognition and language modeling. Their enhanced variant, LSTM networks, overcame the limitations of standard RNNs by efficiently handling long-range dependencies in text sequences, enabling improvements in machine translation and text generation. Meanwhile, CNNs,

primarily known for computer vision, found applications in NLP tasks like text classification and sentiment analysis, where they identified local patterns in text using convolutional filters. However, the Transformer architecture, introduced in 2017, revolutionized NLP by eliminating the need for recurrence, leveraging self-attention mechanisms for superior efficiency and parallelism. Transformer-based models such as BERT, GPT, and T5 have since become the dominant approach, outperforming earlier architectures in diverse NLP tasks. Additionally, Transformers have extended beyond text to support multimodal processing, integrating vision, audio, and text, further reinforcing their role as a foundational AI architecture.

Pretraining and Fine-Tuning Techniques

The success of deep learning in NLP is largely driven by transfer learning, which allows a model to leverage knowledge gained from one task or dataset and apply it to another. In NLP, this typically involves pretraining a large language model on vast amounts of text data and then fine-tuning it for specific downstream tasks. This approach enables strong performance even when labeled data for the target task is limited. For example, ULMFiT demonstrated that a general-purpose pretrained language model could be effectively adapted to various text classification tasks [23]. Similarly, BERT and its variants undergo pretraining using masked language modeling and next sentence prediction before being fine-tuned for diverse tasks such as question answering, named entity recognition, and sentiment analysis. Transfer learning has made it possible to reuse models that have already learned linguistic structures, significantly reducing computational costs and data requirements while improving generalization across multiple NLP applications.

Various strategies are used to adapt pretrained models to specific tasks. The classic fine-tuning approach involves continuing training on a small amount of labeled data, updating all model parameters. However, in recent years, more efficient fine-tuning techniques have gained popularity. Instead of updating all parameters, methods like Low-Rank Adaptation (LoRA) introduce small modifications to specific layers, reducing computational cost while maintaining model effectiveness [24]. Another approach, prefix tuning, integrates new task-specific information into the model by learning prompt-based representations, eliminating the need for full retraining. These techniques allow fine-tuning of large-scale models with fewer resources while ensuring stability. Additionally, knowledge distillation and network compression are employed to transfer the

knowledge of a large, complex model to a smaller one, making deployment more efficient. These approaches enhance scalability, enabling the use of powerful NLP models in real-world applications with lower computational requirements.

Robustness and Security Approaches

Ensuring robustness and error tolerance in deep learning-based NLP systems is a critical challenge. Models often experience sharp performance drops when faced with adversarial examples or minor data distribution shifts. To address this, techniques such as data augmentation, adversarial training, uncertainty estimation, and contrastive learning have been developed. For example, training a model with synonym replacements or punctuation variations improves its adaptability. Additionally, enhancing model interpretability through attention visualization or rationale generation helps users understand its decision-making process, contributing to greater reliability and security in NLP applications.

Scalability and Computational Constraints

While scale (in terms of parameter count and data volume) is a key factor behind the success of LLMs, it alone is not sufficient. Research, such as the Chinchilla study, has shown that balanced growth in both model size and data is crucial for efficiency [25]. Larger models tend to exhibit emergent capabilities, enabling them to perform diverse tasks—such as translation, poetry generation, or code writing—without task specific training. Consequently, modern methodologies emphasize not only LLM training but also rigorous evaluation and regulation to ensure reliability and fairness.

Prompt Engineering and CoT Reasoning

A key approach in guiding LLMs is prompt engineering, where structured prompts enhance model reasoning. Chain-of-Thought (CoT) prompting improves logical consistency by instructing models to break problems into intermediate steps. For example, in math or logic puzzles, guiding the model with “Think step by step, then provide the answer” often yields more accurate results than directly requesting an answer. Such prompt-based techniques encourage models to externalize their reasoning process, reducing errors and improving reliability in complex tasks.

AI Agents

AI agent is an autonomous system designed to perceive its environment, process information, make decisions, and take actions to achieve specific

objectives. LLM-based agents use large language models as their “brain” to process text inputs, make decisions, and execute actions —such as performing web searches, making API calls, or controlling robotic systems. The advanced reasoning and language capabilities of modern LLMs have accelerated AI agent development. LLM-powered agents have surpassed traditional reinforcement learning methods in interactive tasks, such as game environments (e.g., ALFWorld) and web-based automation (e.g., WebShop) [22].

Multimodal Models in AI

Human communication often combines multiple sensory inputs such as text, images, and speech, leading to the development of multimodal AI models that integrate these different data modalities. Text-vision models, like **CLIP** (OpenAI, 2021), align visual and textual representations, enabling zero-shot classification and cross-modal understanding. Generative models such as **DALL-E** extend this capability by generating images from textual descriptions, opening new possibilities in creative content generation. In the text-audio domain, models like **wav2vec** have improved automatic speech recognition by learning rich speech representations before integrating them into language models. More recently, Multimodal Large Language Models (MLLMs) have expanded LLM capabilities to process and generate text, images, and even video-based content. These models leverage a LLM as a central reasoning unit, integrating external sensory modules to enhance comprehension across different modalities. Research suggests that MLLMs exhibit emergent abilities, such as mathematical reasoning or storytelling from images, which could mark a step toward Artificial General Intelligence (AGI).

4. Conclusions and Future Directions

Current State Summary

The integration of deep learning and LLMs has led to groundbreaking advancements in NLP. Machines can now understand and generate human-like text, even producing creative and versatile content. Pretrained models have enabled significant progress across various applications, from machine translation and text summarization to dialogue systems and code generation. Tasks that once required manual rule-based programming, such as grammar checking, are now automated with high accuracy through learned linguistic patterns.

Moreover, LLMs have reduced the need for task-specific models, demonstrating that a single general-purpose model can handle multiple tasks effectively. The emergence of LLM-based AI agents has further extended NLP

beyond text processing, incorporating decision-making and action execution in complex problem-solving scenarios. These developments indicate that AI is evolving into an interactive and multi-dimensional intelligence, moving beyond static textual understanding to dynamic, real-world interactions.

Limitations and Challenges

Despite significant progress, text processing technologies still face major challenges.

Noisy and Incomplete Data: Real-world text data, especially from social media and user-generated content, often contains spelling errors, slang, abbreviations, and missing context, reducing model accuracy. While preprocessing techniques help clean data, recent research suggests that preserving useful noise can improve certain NLP tasks.

Multilingual and Low-Resource: Most NLP models are trained primarily on high-resource languages like English, leaving thousands of other languages underrepresented. Methods such as transfer learning, multilingual embeddings (mBERT, XLM-R), and data augmentation aim to bridge this gap, though adapting to linguistically diverse structures remain an open challenge.

Semantic Ambiguity and Context Understanding: Words and phrases often have multiple meanings depending on context, making disambiguation difficult. While transformer-based models have improved context-aware understanding, long-range dependencies, cultural nuances, and implicit meaning interpretation remain challenging areas for NLP research.

Computational Cost and Scalability: Training and deploying large-scale models require immense computational resources, leading to long training times and high energy consumption, raising concerns about AI sustainability. The environmental impact of billion-parameter networks has sparked discussions on developing more efficient algorithms, model compression techniques, and specialized hardware to reduce costs while maintaining performance.

Bias and Ethical Issues: NLP models can inherit and amplify social biases from their training data, leading to stereotypical or discriminatory outputs. Additionally, risks related to data privacy, misinformation, and lack of transparency raise ethical concerns. Research in bias mitigation, fairness, and explainability aims to ensure more accountable AI systems.

Future Directions and Opportunities

The future of text processing will center on overcoming current challenges while expanding AI capabilities. Advancements in logical reasoning, causality,

and long-context coherence will drive models beyond statistical patterns toward genuine comprehension. Hybrid AI approaches (deep learning + symbolic AI) and memory-augmented architectures are expected to enhance text consistency and reliability. To maintain accuracy and relevance, LLMs will integrate external knowledge sources such as search engines and knowledge graphs. Improving interpretability and transparency will be a key focus, leveraging attention visualization, post-hoc explanations, and human-auditable components to build trust. Future models will be highly personalized, adapting to user-specific jargon, writing styles, and preferences through federated learning and dynamic fine-tuning. Addressing performance gaps in low-resource languages, AI research will emphasize data augmentation, cross-lingual transfer, and shared representation learning to make NLP technologies more inclusive. Additionally, ensuring responsible AI development will require measures against AI-generated misinformation, copyright violations, and data privacy risks through policies, regulations, and technical safeguards. Lastly, human-AI collaboration will be strengthened through Explainable AI (XAI), where AI assists but defers critical decisions to humans, fostering interactive and reliable NLP systems.

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CHAPTER VIII

USING LLM MODELS IN DIGITAL MARKETING

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1. Introduction

Language represents the capacity for communication and self-expression, which emerges during early childhood and evolves throughout an individual's lifespan (Hauser et al., 2002). Nevertheless, without sophisticated Artificial Intelligence (AI) algorithms, machines inherently struggle to grasp the nuances of human language comprehension and communication (Zhao et al., 2024). Pursuing this objective—specifically, the facilitation of machines to read, write, and engage in dialogue akin to human beings—has been a prominent area of academic study (Turing, 1950). Language models have emerged as pivotal technologies driving progress in this domain (Zhao et al., 2024). These models not only support unsupervised and multi-task learning to augment machine language proficiency but also function as essential elements within speech recognition frameworks, as well as in applications such as machine translation and information retrieval (Radford et al., 2019; Ghosh et al., 2017; F. Song & Croft, 1999). This swift advancement has precipitated the extensive

utilization of generative AI technologies by consumers and organizations across numerous industries (Bove, 2023). The introduction of ChatGPT on November 30, 2022, succeeded by Microsoft's unveiling of its AI-enhanced Bing search engine on February 7, 2023, and Google's launch of Bard on March 21, 2023, has notably propelled the prevalent discussion regarding AI across various societal segments (Huh et al., 2023).

The increasing prevalence of big data within enterprises, alongside the evolving configurations of the amassed data, has enabled the formulation of various models for data application. Organizations collect significant amounts of data from numerous sources, indispensable to their operational activities and workflows, from which meaningful insights can be extracted. Textual data is undeniably crucial when evaluating the different data categories. This data can be scrutinized by employing sophisticated language processing models to derive pertinent information for enterprises, thereby establishing a robust basis for informed decision-making processes (Arslan & Cruz, 2024).

Large Language Models (LLMs) represent a pivotal advancement in contemporary AI, relying on extensive datasets and exemplifying a significant application of the Generative Pre-Trained Transformer (GPT) framework (Zou, 2023). The escalating necessity for such models within organizational operations has catalyzed the emergence of LLM as a vast and international marketplace. As per the data regarding LLM, the market valuation, which stood at \$1,590 million in 2023, is anticipated to ascend to \$259.8 million by 2030, with an expected implementation of 750 million applications by 2025. Moreover, it is projected that by the same year, 50% of digital activities will undergo automation through applications leveraging these models (Uspenskyi, 2025).

The widespread adoption of AI and LLM technologies is driving significant transformations in key marketing domains, including strategic planning, content creation, customer engagement, and branding. These innovations offer distinctive advantages in comprehending and engaging with consumers, thereby allowing enterprises to formulate highly tailored marketing approaches that were once beyond reach (Ahmed, 2024). Heavily discussed LLMs, such as ChatGPT, have demonstrated remarkable capabilities in generating human-like texts and even providing creative and analytical insights (Borole, 2024; Gao et al., 2024; Sood & Pattinson, 2023). As the marketing realm persistently evolves through digital transformation, grasping the functionalities and constraints of LLMs is becoming increasingly critical. In this framework, the potential of LLMs holds particular significance for marketers aiming to comprehend the shifts in conventional marketing paradigms (Borole, 2024) and the impact of

this evolution in securing a competitive edge (Gao et al., 2024). Furthermore, the efficacy of LLMs in personalization facilitates the proficient application of digital marketing, which epitomizes a more individualized and bespoke iteration of traditional marketing methodologies (Ahmed, 2024).

In this framework, the present study delivers a literature-based evaluation of incorporating LLMs within the digital marketing domain and their deployment in marketing initiatives. In light of the expanding corpus of scholarly work concerning the application of AI and linguistic models in digital marketing (Anoop, 2021; Haleem et al., 2022; Shahid & Li, 2019) and the increasing importance of these innovations in digital marketing scholarship (Farseev et al., 2024; Robbert et al., 2023), this research aims to enhance academic discourse on the proficient application of language models in digital marketing and to practitioners providing LLM-enhanced marketing strategies.

2. The Role of Textual Data in Digital Marketing

Conventional information access mechanisms utilized in commercial operations, such as news articles, analytical reports, and competitor websites, possess intrinsic limitations (K. Xu et al., 2011). Initially, traditional mechanisms encounter considerable obstacles, including the absence of real-time data, exorbitant costs, and inadequate sample sizes (Kulviwat et al., 2004). Moreover, the intelligence disseminated by competitors is frequently skewed, limited in breadth, and devoid of impartiality (K. Xu et al., 2011). Thanks to the advent of Web 2.0, buyers have developed the power to take part in digital shopping and communicate their experiences through different e-commerce channels and social platforms, thereby affecting the purchasing choices of their contemporaries (Park & Lee, 2009). Furthermore, organizations have acknowledged the benefits presented by textual content from multiple sources in attaining sustainability and securing a competitive advantage (Wei et al., 2022). Through social media and online feedback, consumers produce extensive amounts of textual data encompassing insights regarding brands and businesses, which can be utilized to improve marketing efficacy (Kauffmann et al., 2019). Although the share of structured data leveraged in the business realm is approximated to be around 5% (Cukier, 2010), it is broadly recognized that unstructured data constitutes between 80% and 95% of all data, with a significant fraction being textual (Gandomi & Haider, 2015). Indeed, more than 80% of this data is unstructured and requires the implementation of information-supported text processing technologies (Rizkallah, 2017).

Textual data, an integral element of big data, significantly influences the configuration of modern market frameworks and the marketing strategies that

correspond to these dynamic frameworks (Balducci & Marinova, 2018). This data includes online feedback, email exchanges, digital advertisement copies, transcriptions from customer service interactions, open-ended survey replies, question-answer formats, marketing correspondences, and content disseminated through chatbots (Berger et al., 2020; Y. Chen et al., 2022; Grewal et al., 2021). The escalation of consumer exposure to advertisements and digital stimuli in environments where such content is shared and their capacity for interpersonal interaction fortifies the linkage between the data they generate and organizational processes and their personal relational memories (Venkatraman et al., 2021). Consequently, textual data, akin to all forms of digital data, harbors more profound informational richness and connectivity that can be exploited to craft bespoke messages for consumers (Grewal et al., 2021). The application of textual data within three-dimensional virtual environments, such as the metaverse (Lee et al., 2023), alongside the modeling of linguistic and paralinguistic features (e.g., emotions, sentiments, and speaking styles) in verbal exchanges featured in customer reviews on interactive digital platforms (Lin et al., 2024), further necessitates the utilization of LLMs. These environments, grounded in Web 3.0, are emerging as arenas where consumers can display purchasing behaviors and create substantial volumes of real-time textual data within the marketing context (Shen et al., 2021). Additionally, the processes of consumer decision-making, the rationale that supports these decisions, and even the biases embedded within decision-making can be discerned by examining textual data (Ross et al., 2024).

3. Large Language Models

LLMs have surfaced as sophisticated AI frameworks exhibiting extraordinary proficiency in the processing and generating human-like text across an extensive array of application areas (Naveed et al., 2023). These models can comprehend and analyze natural languages, enabling them to retrieve pertinent information and produce coherent, contextually relevant, intuitive responses (Shahab et al., 2024). By their text generation and reasoning skills, LLMs can mitigate the demands of repetitive activities and foster improved interactions between humans and data, thereby augmenting efficiency and quality across various contexts (Thirunavukarasu et al., 2023). The findings and interpretations rendered by LLMs are essential within specific tasks and for comprehending the wider societal ramifications of these technological advancements (Chang et al., 2024).

LLMs can engage with open-text inquiries without training in a particular domain (Y. Xu et al., 2024). The existing literature presents a dichotomy in

perspectives regarding LLMs; some researchers regard these models as a beneficial and revolutionary technology that allows individuals to concentrate on more intricate tasks by alleviating them from routine writing responsibilities, while others advocate for a prohibition on the utilization of these models, considering them to be inferior-quality plagiarism (Hughes & Heerden, 2024; Stokel-Walker & Noorden, 2023). Nevertheless, since the introduction of ChatGPT in 2022, these models, which initially provided limited query capabilities, have progressively advanced to answer more targeted questions across many academic domains and disciplines (Hughes & Heerden, 2024). The essential disciplines where LLMs are employed consist of forensic sciences (Gutiérrez, 2024), computer sciences (Kumar et al., 2023), data science (Hong et al., 2024), health sciences (Shahab et al., 2024; Thirunavukarasu et al., 2023), natural sciences (Latif et al., 2024), social sciences (Rossi et al., 2024), and economic sciences (Shahin et al., 2024).

Various categorizations of LLMs exist. One such categorization is predicated on the attributes of the employed language model. By this classification, types of AI are delineated as autoregressive language models, encoder-decoder architectures, transformer-based frameworks, pre-trained and fine-tuned iterations, sequential sequence models (Seq2Seq architectures), and multilingual frameworks (Kanade, 2023; Kniazieva, 2024).

While LLMs have achieved considerable acclaim through specific conversational frameworks such as OpenAI's GPT series, other linguistic models, including Google's BERT and PaLM, MetaAI's LLaMA, and the Claude models from Anthropic, are concurrently undergoing active development (Kniazieva, 2024; Najafi & Varol, 2024). Furthermore, the DeepSeek LLM, initiated by Lian Wengfeng and supported by the High-Flyer hedge fund, is anticipated to exhibit sophisticated encoding and learning proficiencies (Guo et al., 2024). This model formulates responses incrementally, utilizing a methodology akin to human cognitive processes. It enhances problem-solving capabilities in scientific domains relative to previous language models, thus establishing its significance as a research instrument (Gibney, 2025). In conjunction with ChatGPT, DeepSeek is also employed in generating impactful content across diverse fields, at times yielding superior software and linguistic results compared to ChatGPT (Koswara, 2025).

While ChatGPT is the preeminent and most frequently utilized model within the marketing domain among the diverse array of LLMs, it displays considerable distinctions when compared to alternative LLMs. Regarding

these distinctions, ChatGPT’s answer to the inquiry, “What characteristics set ChatGPT apart from LLMs?” is delineated in Table 1.

Table 1. Features That Set LLMs Apart from ChatGPT

Features	Large Language Models	ChatGPT
Purpose	Provides extensive language processing capabilities.	Optimized for chat and user interaction.
Scope	A very broad range of tasks.	Limited to conversation and dialogue-oriented missions.
User Audience	Researchers Developers Businesses	General users Customer support providers
Adaptation	It requires specific fine-tuning for a particular task.	It is pre-trained and ready for immediate use.
Output Format	Different formats (report, summary, analysis) for various tasks.	Provides written responses in a dialogue format.

An analysis of Table 1 reveals that LLMs exhibit distinctions from ChatGPT concerning their objectives, breadth, intended demographic, flexibility, and the nature of their outputs. These variances establish LLMs as adaptable instruments with extensive applicability that are proficient in being customized to tackle a diverse array of challenges in the field of digital marketing.

4. Applications of Large Language Models in Digital Marketing

LLMs are progressively employed in digital marketing to augment various dimensions, including consumer engagement, content generation, and market analysis. Their capacity to analyze and produce text that resembles human communication empowers marketers to refine campaigns and tailor customer interactions with enhanced effectiveness (Aghaei et al., 2025). AI assumes a crucial role in augmenting the performance of individuals and organizations and broadening the scope of human intelligence via information technologies (Anoop, 2021). Textual information, encompassing call center dialogues, press releases, promotional messages, content, and customer feedback, facilitates the collaboration of diverse marketing stakeholders (Berger et al., 2020). Furthermore, these technologies, with their evolving linguistic abilities, empower organizations to gain deeper insights into the cultures of prospective and current customers, thereby enabling the customization of content per the

target demographic (Lukose et al., 2025). In this setup, online influencers, like chatbots, are progressively used to facilitate electronic word-of-mouth (eWOM), allowing consumers to sway one another (Sands et al., 2022). Despite ongoing ethical dilemmas surrounding the deployment of AI-driven technologies and LLMs (Koswara, 2025), these tools provide substantial benefits from a marketing standpoint, including the enhancement of data diversity, the simplification of intricate tasks, the shaping of brand and product perceptions, and the reduction of human error (Lukose et al., 2025; Wels, 2024). When concentrating on particular marketing tactics, LLMs are utilized for the following objectives (Ahmed, 2024; Arora et al., 2024b; Borole, 2024; Bulut & Arslan, 2024; Koswara, 2025; Lukose et al., 2025; Ross et al., 2024; Wels, 2024):

- ***Improving the Efficiency of Marketing Research:*** LLMs can construct and analyze online surveys pertinent to marketing research with remarkable ease and efficiency. Furthermore, these sophisticated models can scrutinize textual data acquired from prospective consumers via diverse channels. Such capabilities promise to significantly improve marketing research's quality and efficacy (Arora et al., 2024a; Qiu et al., 2023).

- ***Customer Segmentation:*** LLMs enhance the examination of intricate consumer datasets and discerning behavioral trends. Furthermore, they function as AI-powered enablers by allowing for the swift and precise interpretation and evaluation of data for customer segmentation (Li et al., 2025).

- ***Effective Content Production:*** LLMs are adeptly employed for the swift and comparatively novel creation of visual and auditory materials, constituting a critical element of digital marketing (Jensen & Højmark, 2023).

- ***Customer Service and Value Creation:*** The incorporation of LLMs-driven automated response systems within e-commerce platforms and marketplaces facilitates swift and efficient consumer resolutions. Consequently, this bolsters customers' perceived value and satisfaction with the corresponding platform (Tomita, 2024).

- ***Providing Customer Interaction:*** LLMs scrutinize customer engagements with both the organization and their peers, thereby promoting more efficient communication. This improved interaction can markedly aid in sustaining ongoing customer involvement and, as a result, fortify their allegiance to the brand (Yu & Liu, 2024).

- ***Providing Insight by Analyzing Customer Behavior:*** LLMs facilitate the extraction of profound understandings by scrutinizing consumer behaviors

within extensive and intricate data frameworks. Such understandings permit a more precise recognition of consumer anticipations and emerging patterns (Shahin et al., 2024).

- ***Understanding Customer Trends and Biases:*** LLMs present effective methodologies for analyzing both overt and covert consumer biases in customer feedback. Utilizing these methodologies enables the formulation of marketing strategies that cater to consumers' perceptual and emotional requirements (Yazdi et al., 2024).

- ***Increasing Customer Relationship Management Effectiveness:*** LLMs can yield significant understanding regarding prospective domains for improving interpersonal connections through heightened customer engagement. Such insights can be utilized in promotional strategies to cultivate enduring relationships that promote elevated levels of customer contentment (Huang et al., 2024).

- ***Social Media Analysis:*** LLMs function on extensive, multilingual frameworks, with social media data constituting a crucial significant data source for these systems (Najafi & Varol, 2024).

- ***Improving Customer Experience:*** Enhancing customer experience can be achieved through individualized communication, the utilization of chatbots, and the implementation of recommendation systems that predominantly depend on LLMs and capitalize on users' perceptions as aggregated data (Shareef et al., 2024; Tajtit & Serrhini, 2024).

- ***Search Engine Optimization (SEO) and Search Engine Marketing (SEM):*** The capacity to perform keyword evaluation and discern competitive search phrases empowers LLMs to augment search engine optimization (SEO) efficacy, consequently enhancing the effectiveness of search engine marketing (SEM) (Nestaas et al., 2024).

- ***Managing Email Campaigns:*** LLMs can be adeptly utilized to formulate tailored email marketing initiatives and generate unique email content. Moreover, they augment the efficacy of pre-informed campaigns about the attributes of the targeted demographics. Automated email marketing strategies play a significant role in crafting methodologies designed to enhance customer engagement and improve conversion rates (Halder et al., 2024).

- ***Digital advertising Optimization:*** Implementing LLMs in digital advertising optimization offers novel avenues for augmenting the efficacy of advertising initiatives. LLMs can be adeptly utilized to facilitate audience segmentation, the creation of tailored content, and dynamic optimization by

examining extensive datasets about user interactions (Q. Yang et al., 2023). These models utilize advanced deep learning methodologies to forecast consumer behavior and refine advertising text, headlines, or image narratives (Skubis & Kołodziejczyk, 2024; X. Zhang et al., 2024).

- ***Creating Innovative Advertising Campaigns:*** In digital advertising, search marketers utilize many promotional messages, varying in intensity from overt to more nuanced expressions. By leveraging LLMs, online consumers can be precisely targeted using a spectrum of keywords, from broad matching techniques to exact matching strategies, facilitating the development of compelling advertisements aimed at individuals in pursuit of products or information (Bulut & Arslan, 2024; Sahbi et al., 2024).

- ***Influencer Marketing:*** Although conventional metrics hold significance in marketing, they frequently inadequately reflect the fluid characteristics of content generation driven by influencers, interactions grounded in trust, and evolving methodologies. This weakness has led to the implementation of elaborate computational solutions, featuring Machine Learning (ML) techniques and Natural Language Processing (NLP) applications, to capitalize on the significant and nuanced data generated in digital realms (Joshi et al., 2022). Consequently, consumer behavior is shaped, and enduring trust is fostered through AI- and LLM-enhanced virtual influencers, enhancing the congruence between products and influencers (Feng et al., 2024; Sands et al., 2022).

- ***Image-Supported Customer Response System:*** This framework, which amalgamates image processing with LLM technologies, allows customers to transmute their images into textual representations via LLM. Consequently, customer inquiries or grievances expressed through images can be converted into extensive textual datasets, which can be leveraged in the business's text analytics methodology (C. Song, 2024).

- ***Creation of Brand Concepts:*** LLMs are developed utilizing vast datasets and can produce innovative and persuasive slogans, brand narratives, and value propositions congruent with brand identity. As a result, marketing practitioners can devise more efficient and economically viable brand concepts by generating various ideas. Furthermore, by examining consumer behavior and emerging trends, LLMs can assist in formulating concepts that are more likely to resonate with the intended audience (Ghatora et al., 2024).

- ***Marketing Automation:*** LLMs exhibit the proficiency to scrutinize extensive datasets, facilitating automation that yields customized customer experiences. As a result, this promotes a more profound comprehension of

consumer behavior and the development of bespoke content, promotions, and marketing initiatives. By utilizing LLMs, automated and individualized interactions in domains such as email marketing, social media oversight, and customer support can diminish marketing expenditures while concurrently bolstering customer loyalty (Tawosi et al., 2024; Thukral et al., 2023).

- ***Personalization and Adaptation in Marketing Strategies:*** As digital marketing evolves, LLMs have solidified their position as key players. These models, having been developed using extensive datasets, can perform in-depth consumer data analyses and create tailored content for distinct customers. In turn, businesses are positioned to launch more impactful marketing efforts, raise customer contentment, and amplify their sales results. Furthermore, LLMs are utilized across many sectors, such as customized product suggestions, adaptive advertising, and sophisticated chatbots. This approach fosters the development of more robust customer relationships, thus allowing brands to secure a competitive edge within the marketplace (Earley & Mehta, 2024; Sun et al., 2024).

- ***Creating Marketing Plans:*** LLMs are revolutionizing the processes involved in marketing planning. These models, trained on vast datasets, are equipped with the capacity to conduct an in-depth analysis of historical marketing information. This proficiency allows marketers to formulate strategies that are more consistent and effective, firmly rooted in empirical data. By scrutinizing previous marketing information, LLMs undertake essential functions such as recognizing emerging trends, refining target audience definitions with greater precision, and executing competitive assessments. Consequently, marketing choices can be anchored in scientific findings. This methodology guarantees that marketing strategies are not merely reliant on intuition but are substantiated by evidence derived from data. Additionally, the efficacy of LLMs in analyzing target audiences enhances the customization of marketing communications, thus leading to increased conversion rates (Ding et al., 2024; Sood & Pattinson, 2023).

- ***A/B Testing and Optimization:*** A/B testing is a methodological approach that entails evaluating the efficacy of various iterations of a webpage or application to ascertain which iteration exhibits enhanced operational effectiveness. LLMs facilitate the enhancement of advertising efficacy by producing diverse iterations for A/B testing and scrutinizing user engagement. By leveraging NLP technologies, LLMs can assimilate user feedback, thus improving the capacity of advertisements to successfully connect with their intended demographic (Shankar et al., 2024).

ChatGPT, which is among the most prominent and conversationally adept language models within the spectrum of LLMs, in conjunction with its specialized AI applications for diverse functions, has garnered considerable attention in scholarly discourse both internationally (Jain et al., 2023; Rivas & Zhao, 2023; Y. Zhang & Prebensen, 2024; W. Zhou et al., 2023) and domestically (Akpur, 2023; Binbir, 2021; Erul & Işin, 2023; Gür, 2022; Sarioglu & Develi, 2022). ChatGPT represents a tailored adaptation of LLMs, initially conceived for general purposes, specifically modified for conversational scenarios (Mohammad et al., 2023). While numerous studies within the international literature have analyzed the relationship between LLMs and marketing, it has been noted that domestic literature tends to concentrate more specifically on ChatGPT and its specialized AI applications, in contrast to broader conversations concerning LLMs. In the course of preparing this study, a targeted inquiry via Google Scholar uncovered a limited array of works that investigate the intersection of LLMs and marketing (Özparlak & Çetin, 2023; Sadıkoğlu et al., 2023). The research by Özparlak and Çetin (2023) examines the legal dimensions of privacy and security in LLM-driven AI models. Still, it does not explore the particular application of LLMs within marketing. The research is situated within a legal rather than a marketing framework (Özparlak & Çetin, 2023). Conversely, Sadıkoğlu et al. (2023) concentrate on the development and application domains of LLM-based chatbots in the context of social media.

5. Challenges in Using Large Language Models in Marketing

LLMs offer promising benefits for the advancement of marketing; however, they also present certain potential challenges that necessitate careful consideration. Although these models exhibit remarkable proficiency, they also reveal considerable constraints that impede their capacity for comprehensive understanding and reasoning, resulting in various obstacles concerning the efficacy of their application (Williams & Huckle, 2024). The challenges and hazards linked to the implementation of LLMs in marketing can be delineated as follows (Eurostat, 2024; Williams & Huckle, 2024; Yin et al., 2024; X. Zhou et al., 2024):

- **User Biases:** LLMs continue to navigate the trajectory of acceptance as nascent technologies, representing innovations progressively being embraced. Moreover, as these technologies increase in prevalence, they are open to ongoing evaluation and susceptible to inaccuracies during their maturation. In

this framework, while users investigate the prospective applications of LLMs, they concurrently cultivate skepticism and bias regarding these technologies and their associated use cases (Ye et al., 2024).

- ***Domain Specify:*** LLMs frequently encounter difficulties providing precise answers within specialized markets, leading to potential discrepancies with the distinct requirements of targeted marketing strategies (X. Chen et al., 2023). This deficiency in accuracy may jeopardize the efficacy of campaigns that engage particular demographics.

- ***The Problematic of Contextual Interpretation:*** Despite the capacity of LLMs to produce linguistically precise outputs through the assimilation of patterns from extensive datasets, there exists a potential for misinterpretation or inadequacy in grasping the contextual nuances of the text during this operation (Bender & Koller, 2020). In particular, difficulties comprehending cultural, linguistic, and historical dimensions may result in LLMs generating a deceptive façade of accuracy (Marcus & Davis, 2020).

- ***Potential of Bias:*** LLMs can assimilate and disseminate biases present within the datasets utilized for their training, manifesting these biases in their generated outputs. Furthermore, the curation of training data alongside the methodologies employed in its modeling could inadvertently perpetuate systemic biases by these models (Taubenfeld et al., 2024). Mitigating these challenges necessitates enhancements in data preprocessing, model architecture, and evaluation protocols.

- ***Privacy and Security Problems:*** LLMs present significant challenges related to privacy and security owing to their capacity to extract information from extensive datasets utilized during their training. Specifically, the potential for these models to acquire sensitive or personal information throughout the training phase and subsequently produce such data inadvertently poses considerable security vulnerabilities. Furthermore, various security threats may arise, including malicious exploitation, the creation of deceptive content, and the alteration of model outputs (Carlini et al., 2021). These challenges can be alleviated by employing data anonymization techniques, instituting access controls, and establishing monitoring systems for model outputs (Yao et al., 2024).

- ***Ethical Issues:*** The potential for producing biased or detrimental content poses considerable ethical dilemmas, which can jeopardize brand integrity (Hadi et al., 2023). Marketers must guarantee that the outputs generated by LLMs adhere to ethical principles and reflect brand philosophy.

- **Knowledge Management Issues:** Phenomena such as the erosion of knowledge and the illusion of knowledge can result in LLMs delivering antiquated or superficial data, which may misguide marketing tactics (X. Chen et al., 2023). Consequently, marketers might face challenges depending on LLMs for instantaneous insights, potentially impairing their decision-making processes.

- **Potential of Misinformation:** LLMs possess the capacity to disseminate misinformation owing to intrinsic deficiencies or biases present in their training datasets. As these models lack the ability to differentiate between credible, validated information and misleading or incorrect data, they may yield persuasive yet fallacious outputs (Zellers et al., 2019). Moreover, this attribute of LLMs exacerbates the likelihood of nefarious applications, particularly in generating fabricated news or manipulative materials (Solaiman et al., 2019). To alleviate the threat of misinformation, it is essential to adopt strategies such as model evaluation, verification of sources, and educational initiatives for users.

- **Inadequacy of Legal Infrastructure:** The swift advancement of LLMs has rendered current legal frameworks inadequate in addressing the complexities introduced by these technologies. The obligations of LLMs concerning data privacy, intellectual property, misinformation, and ethical breaches remain vaguely articulated (Veale & Borgesius, 2021). Likewise, accountability for any harm resulting from the outputs generated by these models (like the developer, the end-user, or the data provider) lacks precision. This highlights the pressing need to create a comprehensive and internationally cohesive legal framework to facilitate LLMs' equitable and secure utilization (Yao et al., 2024).

- **Recency Problem:** LLMs face significant obstacles in assessing the pertinence of information, primarily owing to the restricted temporal boundaries of their training datasets. Given that these models are not exposed to the most contemporary information, the material provided to users may diverge from recent advancements (Brown et al., 2020). This predicament can produce erroneous or deceptive outputs, especially in rapidly changing domains such as commerce, healthcare, jurisprudence, and technology (Mousavi et al., 2024). To mitigate the relevance challenge, it is imperative that models undergo regular retraining or that mechanisms for real-time information retrieval be incorporated.

- **Overfitting Problems:** The generalization power of LLMs can suffer a setback due to an excessive fit to unique patterns in the datasets used for training (Raffel et al., 2023). This overfitting phenomenon can result in models generating outputs that are exclusively based on the training data, potentially

leading to suboptimal performance in novel or diverse contexts (M. Zhang et al., 2024). Moreover, such a situation may contribute to models perpetuating detrimental biases, producing excessively specific yet erroneous answers to user inquiries, and ultimately impairing their overall learning efficacy (Raffel et al., 2020). Addressing the overfitting dilemma requires that we utilize methods including regularization practices, data variation, and rigorous model testing.

- ***Linguistic Reasoning Problems:*** LLMs demonstrate inherent constraints in executing intricate tasks that necessitate sophisticated linguistic reasoning. Although these models adeptly imitate language patterns through the assimilation of extensive datasets, they frequently lack a thorough understanding of the logical architecture of language and the connections between meanings (Bender & Koller, 2020). Such a deficiency can lead to erroneous or superficial results, especially in marketing endeavors that require elevated reasoning capabilities, including inference, causal relationships, and semantic consistency across varying contexts (Arora et al., 2024a). To overcome these challenges in linguistic reasoning, it is imperative that models undergo training utilizing approaches that bolster logical structuring and contextual comprehension.

- ***Data Integration Problems:*** The heterogeneity and magnitude of datasets employed in the training of LLMs pose considerable obstacles regarding data integration. Achieving uniformity across datasets originating from various sources, with respect to format, linguistic attributes, and overall quality, has a direct correlation with the efficacy of the model's performance (Kayali et al., 2024). Instances of absent, erroneous, or contradictory information may lead the model to yield biased and deceptive outcomes, a concern that becomes particularly exacerbated in multilingual and multicultural contexts (Eurostat, 2024).

- ***Prediction Performance Problems:*** Evaluating how effectively LLMs function in distinct subtasks and formulating dependable forecasts poses a significant challenge in the field of NLP investigation. The performance-related concerns associated with LLMs can be generally classified into three categories: memorization, generalization, and the necessity for empirical validation in real-world contexts (Q. Zhang et al., 2024).

- ***Mathematical Reasoning Problems:*** LLMs might have hurdles in dealing with arithmetic challenges, grasping mathematical notations, and inferring results. Although these models are developed using extensive datasets comprising text and code, which enhances their capabilities in pattern recognition and the identification of statistical correlations (Yuan et al., 2023),

mathematical reasoning necessitates skills beyond mere pattern recognition. Consequently, these models may find it difficult to grasp abstract notions, make logical deductions, and implement effective problem-solving techniques (Ahn et al., 2024).

- ***Visual-Spatial Reasoning Problems:*** While LLMs are developed utilizing textual and coding datasets, they may experience difficulties in executing tasks necessitating visual-spatial reasoning. Visual-spatial reasoning involves the talent to understand and manipulate images and spatial information. This type of reasoning entails the ability to discern shapes, sizes, positions, and interrelations among objects (Nagar et al., 2024). Given that LLMs are predominantly engineered to process text-centric data, they may encounter obstacles such as the absence of visual stimuli, alongside challenges in abstraction, generalization, and the comprehension of spatial dynamics (Z. Chen et al., 2024).

6. Conclusions and Suggestions

The swift progression in AI and the emergence of LLMs have substantially reconfigured the digital marketing landscape. LLMs that employ NLP enable the execution of hyper-personalized marketing strategies, streamline content generation, enhance consumer engagement, and refine advertising methodologies. These models produce instantaneous consumer insights and assess sentiments, augmenting businesses' strategic decision-making processes. Nevertheless, ethical quandaries, intrinsic biases in AI-generated content, concerns related to data privacy, and the necessity for regulatory frameworks persist as considerable obstacles in adopting LLMs. Furthermore, issues such as contextual misinterpretation, the propagation of misinformation, and computational constraints continue to compromise the reliability of these systems within marketing environments. The research highlights that LLMs are still relatively nascent within the marketing sector and that a substantial gap exists in the scholarly literature.

The present study establishes a theoretical model for implementing LLMs in the sphere of digital marketing. Moreover, it examines the variables that obstruct the deployment of LLMs and curtail their efficacy. The challenges elucidated in this study will establish a robust basis for subsequent scholarly inquiries, proffering pragmatic resolutions that augment the prevailing academic discourse. LLMs confer substantial benefits in the realm of digital marketing, with pertinent applications in textual analysis, consumer segmentation, content

generation, customer support, personalization, and the formulation of marketing strategies, among other areas. Furthermore, the existing literature uncovers significant deficiencies, highlighting the necessity for additional exploration of LLMs in the context of digital marketing. In this light, LLMs emerge as a vital topic for studies across all research concerning textual material within the sphere of digital marketing.

LLMs facilitate an enhanced comprehension of consumer behavior through the examination of extensive datasets, thereby augmenting the efficacy of marketing initiatives and elevating customer experiences. In particular, the capacity of LLMs to produce tailored content and promote heightened customer engagement presents a significant competitive edge for digital marketing methodologies. Moreover, these models streamline marketing automation workflows, thereby contributing to a decrease in marketing expenditures and an enhancement in operational productivity.

While LLMs present a multitude of advantages within the realm of digital marketing, the inherent structural attributes of such models pose various obstacles. These obstacles encompass challenges associated with contextual understanding, the potential for bias, risks pertaining to privacy and security, ethical dilemmas, difficulties in information management, the threat of misinformation, inadequate legal frameworks, issues related to timeliness, overfitting, linguistic reasoning, data amalgamation, predictive efficacy, mathematical reasoning, and visual-spatial reasoning. Such challenges may limit the efficacy of LLMs in marketing contexts and necessitate thorough scrutiny. The quality, recency, and variety of datasets used in model training play a vital role in determining how well LLMs perform. Furthermore, an examination of existing literature indicates that themes such as hyper-personalization, content automation, the ethical deployment of AI, and improvements in efficiency and automation are likely to emerge as salient topics (Aghaei et al., 2025).

This study presents several limitations. Firstly, it is primarily centered on the domain of digital marketing. Subsequent inquiries could broaden the focus to encompass additional business functions, thereby adopting a more specialized approach. Furthermore, while illustrative instances of LLM applications within digital marketing are discussed, the challenges linked to these applications intersect with the more general issues faced by LLMs at large. Consequently, future research endeavors are anticipated to offer novel contributions to the academic discourse by refining the challenges of LLMs to the particular context of digital marketing.

The present study offers numerous implications for both scholarly researchers within the domain of digital marketing and practitioners engaged in digital marketing endeavors. Firstly, the research highlights substantial voids in the existing literature concerning language models and text analytics pertinent to digital marketing studies. In this scenario, it is likely that this examination will advance future inquiries that study the merging of LLMs and digital promotion. Moreover, industry professionals can utilize the groundbreaking solutions delineated in the literature and integrate them into their operational practices, while simultaneously reaping the benefits of the study's contributions towards accessing pertinent resources.

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CHAPTER IX

RECOMENDER SYSTEMS

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1. Introduction

We live in an era where billions of data points are generated every second. In our daily lives, we constantly experience the impact of this data bombardment. Whether following the news, watching movies, listening to music, or shopping, we make decisions to choose the most suitable products from thousands of options. Sometimes, we spend hours selecting a product to purchase or a movie to watch.

Although we have access to millions of choices in today's world, where the concept of big data has become a part of our lives, our time is limited and valuable. Given these circumstances, it is clear that we need a tool to simplify decision-making. Recommendation systems aim to present users with the most relevant products from an endless array of choices. These systems are indispensable for both users, who benefit from time savings, and companies which see significant contributions to their profit margins. As a result, recommendation systems have become a focal point of academic research.

Today, recommendation systems are used across all platforms, from tech giants like Amazon, Facebook, Netflix, and YouTube to mid-sized e-commerce sites. They are applied in various fields, from books and movies to grocery shopping and news. According to Forbes, recommendation systems account for 35% of Amazon's total sales [1]. As an information filtering method,

recommendation systems help users make decisions in all areas by narrowing down the overwhelming number of choices.

Recommendation systems accepted as a research field in the 1990s and gained widespread attention in the 2000s, particularly after Netflix launched the Netflix Prize competition in 2006. The competition announced a \$1,000,000 reward to any team that could develop a recommendation algorithm at least 10% more effective than Netflix's own system. In 2007, the BellKor team achieved an 8.43% improvement, earning a progress award. In 2009, BellKor's Pragmatic Chaos team won the grand prize with a hybrid algorithm that improved performance by 9.46% [Koren et al., 2009]. Over the past 20 years, interest in this field has surged, especially with the introduction of deep learning methods, which have significantly enhanced recommendation system performance.

User expectations from recommendation systems vary, but they generally include providing the most popular products, offering recommendation lists and product descriptions, delivering bundled recommendations, finding reliable recommenders, enabling self-expression, assisting other users, and influencing opinions (Herlocker et al., 1999).

Recommendation systems provide personalized suggestions, meaning each user receives unique recommendations. However, there are also non-personalized recommendations, such as the "Top 10" song lists in music stores or bestseller lists in bookstores.

2. Basic Concepts

Since there are many different data and information sources available for recommender systems (demographic, textual, numerical, contextual, visual, etc.), whether they are utilized depends on the recommendation technique. Although each technique uses different data sources, there are three basic sources of data used by recommender systems: items, users, and transactions (Ricci et al., 2011). Users are the audience that the recommendation system targets and the ones who interact with the system. Different characteristics of users can be used depending on the type of recommender system used. For example, a movie recommendation system can be based on the ratings users give to movies or whether they like the movies they have watched before. Items are the products or services that the system recommends to users. Items have different characteristics depending on the type of data used. For example, a movie item has characteristics such as director, actors, and genre. Transactions are the interactions between these two elements.

Data is obtained from the transaction between the user and the system. This data can be obtained either directly from the user-system interaction, which is called implicit data, or by asking the user for their opinions about the item. This is called explicit data. Implicit data is obtained by the system tracking the user's behavior. Behaviors such as which links the user visits, how much time they spend on which pages, which products they click on, which products they purchase are automatically tracked and recorded, and a user profile is constructed in this way. Explicit data usually consists of ratings. Ratings can be in different forms (Schafer et al., 2007):

- numerical ratings, which usually consist of a rating from 1 to 5
- ordinal ratings, which consist of ideas such as agree, disagree, undecided
- binary ratings, which directly classify the products as good or bad
- unary ratings, which evaluate the user's viewing or purchase of a product as positive feedback.

For example, a user's actions such as viewing a product, adding it to the cart, or purchasing it are considered positive interactions with that product. Similarly, clicking on a news article and the time spent on the page are examples of implicit data. Unlike explicit data, implicit data is collected without direct user involvement; it is derived from system logs that track user behavior. In contrast, obtaining explicit data requires active user participation. So users are asked to rate movies on a movie streaming platform. While doing this, various explanations are usually given to users. For example, it is stated that the more ratings are given, the better the recommendation quality will be. Rating behaviors are not the same for every user, and participation purposes are not the same either. While some users rate to increase the quality of their recommendations and improve their profiles, others may participate with the idea of helping or influencing other users (Ricci et al., 2011). In some cases, there may be users who rate for malicious purposes such as sabotaging the platform or misleading other users. This is one of the problems that the recommender system must overcome.

Recommender systems can be considered as either a prediction problem that aims to find the rating given by user u to item i , where u is the user and i is the item, or as a ranking problem that aims to present a recommendation set consisting of i items to user u . In other words, recommender systems are generated to predict whether a user prefers a particular item. A recommender system's basic elements are as follows (Zhang et al., 2021):

$$f:U \times I \rightarrow D$$

The function f , where $i \in I$ and $u \in U$, is the utility function that determines the fitness of a particular item for a particular user’s purpose. D is the list of recommendations ranked according to fitness for purpose. The recommendation is generated by maximizing the utility function (Zhang et al., 2021):

$$\forall u \in U, \mathop{\text{argmax}}_{i \in I} f(u, i)$$

3. Methods

Recommender systems are generally classified in three groups as shown in the figure below: collaborative filtering, content-based and hybrid recommender systems (Isinkaye et al., 2015).

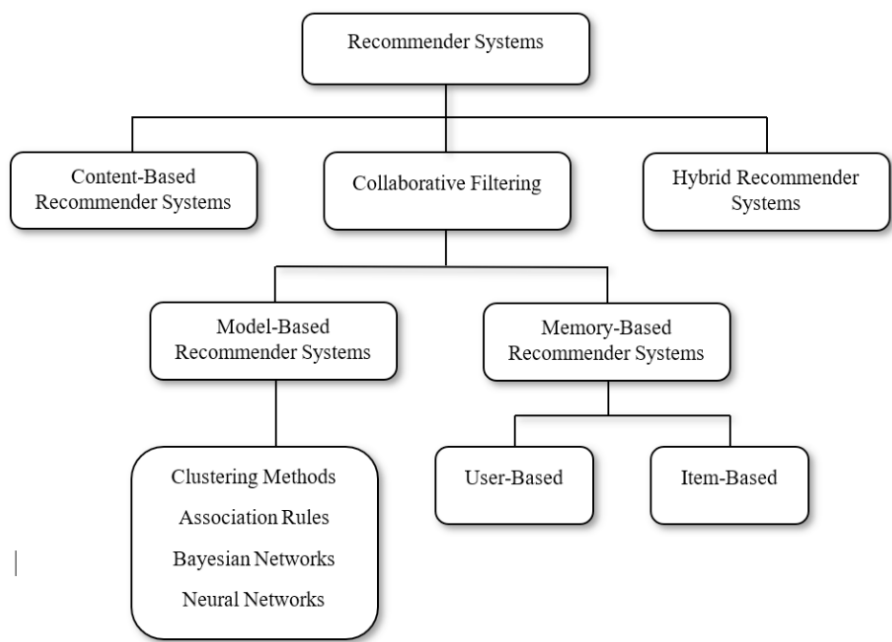


Fig. 1. Recommender Systems Classification

3.1. Collaborative Filtering Recommender Systems

Collaborative filtering recommendation systems aim to predict the ratings that users will give to items they have not rated before. In doing so, they use the

data called user-item or rating matrix as shown in the figure below (Moghaddam & Elahi, 2019). The rows in the matrix denote users, and the columns denote items. The intersection of the user and the item shows the rating that the user gives to the item. In collaborative filtering, the unknown ratings are predicted by the system and the items with the highest ratings are returned as the recommendation list.

	Items				
	5	?	5	?	2
	?	2	?	1	?
	?	5	?	?	3
	3	?	4	?	1
	?	?	?	?	?

Fig. 2. Rating Matrix

3.1.1. Memory-Based Recommender Systems

Collaborative filtering techniques, especially memory-based methods, have become one of the most widely used techniques in the field of recommender systems because they are simple, easy to apply and successful. Memory-based recommender systems are divided into two groups: user-based and item-based. The user-based method is based on the assumption that “users who had similar preferences in the past will have similar preferences in the future”. For this reason, similar users in the database are found using various similarity calculation methods. Memory-based collaborative filtering methods are also called neighborhood-based methods (Al-Shamri, 2019). For this, the k-nearest neighbor mechanism is usually used. The similarity of the user for whom recommendation is to be generated with other users is calculated and the weighted average of the items that are voted in common with the most similar user is found. The user-based collaborative filtering algorithm, where u is the active user, is as follows (Owen et al., 2012):

“for all other w users
calculate the similarity s between u and w
select the top n most similar users based on similarity rankings to
form the neighborhood
for all i items that some users in the neighborhood have rated but user
 u has not
for each user v in the neighborhood who has rated item i :
calculate the similarity s between user u and user v
weight the rating of item i given by user v using similarity and include
it in the averaging process.
Rank items based on the weighted averages and return the top n
recommended items”

In user-based collaborative filtering, when explaining the recommendations given to users, expressions such as “users who bought this item also bought these” or “users similar to you bought these items” are generally used.

In the item-based approach, the similarities between the items that users have previously rated and those that they have not, are calculated and the items having highest similarities are recommended. In other words, in the rating matrix, this time the similarities are calculated between the columns, not the rows.

Neighbor selection is a crucial process in collaborative filtering. Various methods can be used for selecting neighbors. The first approach is to calculate similarity with all neighbors. However, this method is computationally expensive. To reduce this cost, a limited number of neighbors can be selected randomly. However, a risk occurs in terms of prediction accuracy.

Another approach is to select neighbors whose similarity is above a certain threshold value (Threshold Neighborhood). Another similar method is the k -nearest neighbor method. K -nearest neighbor is one of the most common approaches used in collaborative filtering. Moreover, the system can also easily adapt any changes in the rating matrix. However, this requires recalculating the neighbors and the similarity matrix and this leads to an increase in the cost of computation. Although the k NN approach is simple and intuitive, it is seen that it gives accurate results and is suitable for development. However, it has been challenged as a general method in collaborative filtering applications only by dimensionality reduction-based approaches (Amatriain et al., 2011).

In addition to these methods, there are also hybrid approaches. For example, first selecting a random subset and then selecting the best n neighbors among them.

In addition to the neighbor selection method, the number of neighbors to be selected is also an important criterion. In applications, the number of neighbors is generally preferred between 25-100. The small number of neighbors increases the accuracy because it leads to the increase of similarity rate and decrease of noise, but finding neighbors who rated the same items is hard in small neighborhood and it raises the complexity. Additionally, there would be relatively a small number of items that can be recommended. Therefore, working with different neighbor sets for each item to be recommended can be considered as a solution because this increases the item coverage considerably. However, in this case, the user similarity and the prediction accuracy will decrease (Owen et al., 2012). Lathia et al. stated that an adaptive neighbor number should be determined for each user to minimize the error (Lathia et al., 2009).

In memory-based collaborative filtering, one of the most critical steps is similarity computation. The literature shows that various similarity measures are used. The most applied similarity metrics in memory-based collaborative filtering are as follows (Su & Khoshgoftaar, 2009):

- **Pearson and constrained Pearson correlation**
- **Spearman's rank correlation**
- **Kendall's Tau correlation**
- **Vector cosine-based similarity**
- **Adjusted vector cosine-based similarity**

In memory-based collaborative filtering, users or items are compared. Some of the commonly used distance metrics to calculate the similarity between items or users are Euclidean and Manhattan. Manhattan distance is calculated as below:

$$d = |x_1 - x_2| + |y_1 - y_2|$$

Euclidean distance:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

However, in order to use these methods, there should be no gaps in the data. Because in case of missing data, the number of items that some users rated together may be higher than others. To overcome this problem, the Minkowski measure, which is a generalization of these two distance measures, can be used:

$$d(x, y) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}$$

If $r=1$, Manhattan distance is obtained, if $r=2$, Euclidean distance is obtained, and if $r=\infty$, supremum distance is obtained.

Another problem is that each user has their own rating system. For example, a user may avoid extreme scores and only give 2, 3, 4 points. This user may give 4 points to his/her favorite movie, while another user may give 5 points to his/her favorite movie. To eliminate such scoring differences, Pearson correlation is used as a similarity measure. In this measure, -1 means complete opposition, and 1 means complete correlation (Herlocker et al., 1999):

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

$\bar{x}$ represents the average of the ratings given by user x ; and $\bar{y}$ shows the average of the ratings given by user y .

In the Spearman Rank Correlation similarity calculation, the items rated by the user are ranked according to their ratings, with the highest rating being the first. The prediction calculation is the same as the standard Pearson, but instead of the rating, the ranking value is used (Extrand et al., 2010). If the database is quite large and it is difficult to find items that most users rate in common, the Cosine similarity measure is used. Because in this method, 0.0 matches are ignored (Aggarwal, 2016).

$$\cos(x, y) = \frac{x \cdot y}{\|x\| \times \|y\|}$$

Here $x \cdot y$ is the dot product and $\|x\|$ is the length of the vector. The length of a vector is calculated as follows:

$$d = \sqrt{\sum_{i=1}^n x_i^2}$$

When making recommendations to a user, relying solely on the most similar user can lead to inaccurate results. To improve recommendation quality, the k-nearest neighbors (k-NN) approach is used, incorporating multiple similar users to provide more reliable and realistic recommendations.

However, memory-based CF techniques also have limitations. For example, similarity measurements are based on common items, and when the data is sparse, finding common items becomes quite difficult. For this reason, model-based CF methods have emerged. Model-based techniques use pure evaluation data for prediction.

User-based filtering is also called memory-based filtering because all evaluations need to be stored in order to produce recommendations.

Item-based filtering is also called model-based filtering because in this method, all evaluations do not need to be stored. A model is created that shows how similar each item is to all other items. If there are millions of users and items on the site, a scaling problem arises. In order to find the most similar neighbors of each user, millions of comparisons need to be made, which increases the computational cost considerably. The sparsity problem arises. Because if it is considered that each user evaluates only a handful of items among millions of products, finding similar neighbors becomes almost impossible.

Due to such limitations of memory-based methods, model-based methods are gaining importance. One of the methods used in model-based filtering is the adjusted Cosine similarity method (Sarwar et al., 2001):

$$s(i, j) = \frac{\sum_{u \in U} (R_{u,i} - \bar{R}_u)(R_{u,j} - \bar{R}_u)}{\sqrt{\sum_{u \in U} (R_{u,i} - \bar{R}_u)^2} \sqrt{\sum_{u \in U} (R_{u,j} - \bar{R}_u)^2}}$$

U is the set of users who have rated both i and j items. $R_{u,i}$ is the rating given by user u to item i and $\bar{R}_u$ is the average of the ratings given by user u to all items. This method scales better than memory-based methods. It is faster for large data sets and requires less memory. It also eliminates the negative effects of users' different rating preferences. Spertus et al. conducted an extensive study and stated that cosine measurement is the best (Spertus et al., 2005). Lathia et al. stated in their study that the similarity measurement technique has no effect on performance and can be selected randomly (Lathia et al., 2009).

Since the user and item counts are expressed in millions today, calculating similarity between all users causes scalability problems. To eliminate this

problem, systems calculate neighborhoods offline and produce recommendations in real time. In addition, since the item number is also very large, the number of items rated by users in the rating matrix is quite low. This is called the sparsity problem. The higher the sparsity, the more difficult it becomes to find users similar to the target user. This also negatively affects the system's prediction performance. On the other hand, in this method, since there are no items rated in the profiles of users who have newly joined the system, suggestions cannot be produced, which brings about the cold start problem. Therefore, with item-based collaborative filtering, similarities between items, not between users, are computed, and suggestions are tried to be produced even if the number of rated items is small. This method can be preferred in systems where the item count is small compared to the user count.

3.1.2. Slope One

Another popular algorithm for item-based collaborative filtering is the Slope One algorithm (Lemire & Maclachlan, 2005). The main advantage of this method is that it is simple and easy to implement. There are several different Slope One algorithms in literature. One of the most used is weighted Slope One. It consists of two steps. In the first step, deviations between objects are found:

$$dev_{i,j} = \sum_{u \in S_{i,j}(X)} \frac{u_i - u_j}{card(S_{i,j}(X))}$$

Here; $card(S_{i,j}(X))$ is the number of users who have rated both i and j items. In the second stage, the prediction is made using the calculated deviation values.

$$P^{wS1}(u)_j = \frac{\sum_{i \in S(u) - \{j\}} (dev_{j,i} + u_i) c_{j,i}}{\sum_{i \in S(u) - \{j\}} c_{j,i}}$$

$$c_{j,i} = card(S_{j,i}(\chi))$$

Here, $P^{wS1}(u)_j$ is the weighted Slope One prediction of the rating given by user u to item j . One of the biggest advantages of the Slope One algorithm is that when a new user enters the system, there is no need to recalculate the deviations of the item rated by that user and other items. A new deviation calculation can be easily made by adding the item of the new user to the existing deviation calculation.

3.1.3. Model-Based Recommender Systems

In model-based collaborative filtering, the ratings in the user–item matrix are used to build a model for making predictions. While the success of memory-based methods depends on neighborhood and similarity mechanisms, model-based methods depend on the performance of the employed learning algorithm. In Netflix’s competition, the matrix factorization method emerged as one of the successfully used approaches (Koren et al., 2009). Since it reduces the dimensionality of the rating matrix and makes it denser, the sparsity and scalability issues are mitigated (Zhang et al., 2021).

If user evaluations are categorical, classification algorithms are employed as collaborative filtering models; for numerical evaluations, regression models and SVD (Singular Value Decomposition) methods are used. Among the most used methods are Bayesian-based, clustering-based, regression-based, MDP (Markov Decision Process)-based, and Latent Semantic models. In addition, the literature also presents approaches such as ordinal learning, association rules, maximum entropy, dependency networks, decision trees, multiple multiplicative factor models, matrix factorization–based models, and probabilistic latent component analysis (Su & Khoshgoftaar, 2009).

In recommender systems, it is common not only to have feature-rich datasets that define a high-dimensional space but also to encounter spaces with very sparse data where each item is associated with a limited number of features. The concepts of density and inter-point distance which are critical for clustering and outlier detection, become meaningless in high-dimensional spaces. This is known as the dimensionality problem. Dimensionality reduction techniques aim to solve this problem by transforming the high-dimensional space into a lower-dimensional one. Even in the simplest applications, one often encounters matrices with thousands of rows and columns, most of which contain zeros. These dimensionality reduction techniques can be directly applied to the computation of predicted values and make a significant difference. Consequently, they are now regarded not merely as a preprocessing step but as an integral part of recommendation system design.

3.1.3.1. Matrix Factorization Methods

Matrix factorization is basically the mapping of the user item matrix to a much lower dimensional latent space. Thus, each user or item can be represented as a low dimensional vector in the latent space. In this way, a sparse matrix with many empty values is made dense and unnecessary parts are discarded, ensuring

that only the most important parts are kept. In other words, each element of the vectors indicates the degree of an important feature related to that item and user. Let $q_i \in R^f$ be the item and $p_u \in R^f$ be the user vectors, where i is the item and u is the user. The rating given by user u to item i is calculated as a point product (Koren et al., 2009).

$$\hat{r}_{u,i} = q_i^T p_u$$

A matrix factorization example can be seen in the figure below:

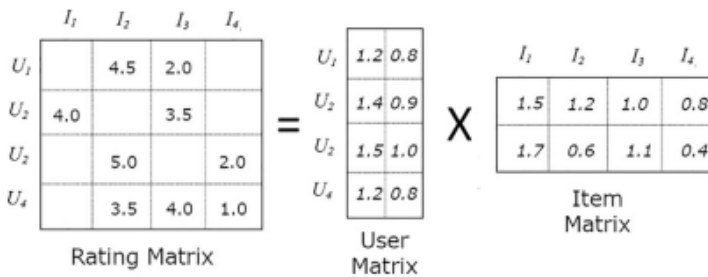


Fig. 3. A matrix factorization example

This model is also the basis of the Singular Value Decomposition (SVD) method and is one of the most widely used dimensionality reduction methods along with Principal Component Analysis (PCA).

It is a matrix factorization technique that is used as the fundamental of latent semantic analysis and is also related to PCA. SVD can be used to find the hidden connection between the customer and the product. Its biggest handicap is that there are too many missing values in the rating matrix. For this, the zero areas in the user-item matrix are filled with the item's average rating. The user's average is then normalized by subtracting it. This matrix is then decomposed with SVD, and the decomposed matrices are used directly to calculate the predictions. A major advantage of SVD is that incremental algorithms are used to find the approximate decomposition. In other words, there is no need to recalculate the model when a new user is added to the system. The validity of the incremental SVD method was accepted after its success in the Netflix competition (Koren et al., 2009).

Many other matrix factorization methods are used. These are similar methods to SVD. The basic idea is to divide the rating matrix into two separate matrices: one describing the user features and the other describing the object features. Matrix factorization methods outperform according to SVD in that

they handle missing values by adding a bias value to the model. SVD also uses missing values by replacing them with item means. However, both tend to overfit (Amatriain, 2011). In addition to dimensionality reduction methods, there are also classification approaches. Decision trees, rule-based classifiers, Bayesian classifiers, Artificial neural networks (ANNs) and Support vector machines (SVMs) can be used for applications where items are evaluated as binary classes of liked and disliked in recommendation systems.

3.1.3.2. Clustering Methods

Another approach used in recommendation systems is clustering. The aim of clustering is to reveal meaningful clusters in the data. In memory and content-based methods, a large number of comparisons are required when calculating similarity with methods such as k-nearest neighbor. The same applies to dimensionality reduction methods because even if the dimensions of the items are reduced, if the number of items to be compared is still high, the cost will still be high. With clustering, the distance within the cluster is aimed to be minimized and the distance between the clusters is maximized. In this way, the number of distance measurements to be made is significantly reduced. However, clustering methods do not always guarantee an increase in system performance. One of the most used clustering methods is the k-means method. Clustering is done by minimizing the sum of the distances of the center λ_i of each cluster to the other points in the cluster with k-means. The k-means algorithm can be defined as an iterative process that minimizes the following function (Koren et al., 2009):

$$E = \sum_1^k \sum_{n \in S_j} d(x_n, \lambda_j)$$

x_n is a vector representing the n th object, λ_j is the center of the S_j subset, and d is the distance measurement. The algorithm assigns items to closest clusters until E no longer decreases. Cluster centers are randomly selected. Therefore, created clusters change depending on these randomly selected cluster centers.

In the k-means method, the number of clusters k must first be determined. Then, observation values are assigned to each cluster, and the following operations are performed (Özkan, 2008):

- 1) The center of each cluster is determined ($M_1, M_2, \dots, M_k$).
- 2) The intra-cluster variations are calculated, and the value E , which is the sum of these variations, is determined.

3) The distances between the M_k center values and the observation values are calculated. An observation is assigned to the cluster corresponding to the center it is closest to.

4) Steps 2 and 3 above are continued until there is no change in the clusters.

Another one of the most used methods is the fuzzy c-means method, which is one of the fuzzy divisional clustering techniques. In the basic clustering logic, each element can belong to only one cluster, while in the fuzzy set theory, each element can belong to multiple clusters, but they have a membership degree between $[0, 1]$. The sum of these membership degrees must be 1. The element is included in the cluster to which it has the highest membership degree (Işık & Çamurcu, 2007).

The fuzzy c-means method also has its own advantages and disadvantages. Its ability to find overlapping clusters is higher than other divisional clustering techniques. The fact that each element has different levels of membership in different clusters increases the flexibility of the method. Similarly, having membership degrees at very different levels allows the elimination of outliers more easily because their membership degrees will be relatively very small. On the other hand, since the membership degree calculation will add extra load to the algorithm, the complexity increases and therefore it causes a time cost problem.

The fuzzy c-means method works by updating an objective function. It tries to minimize the function by shifting it, which is the generalized form of the least squares method given below (Kaynar et al., 2016):

$$Jm = \sum_{i=1}^N \sum_{j=1}^C u_{ij}^m \|x_i - c_j\|^2$$

First, the membership degrees are randomly determined to create the U membership matrix. Then, the center vectors are computed using the following formula (Moertini, 2002):

$$c_j = \frac{\sum_{i=1}^N u_{ij}^m x_i}{\sum_{i=1}^N u_{ij}^m}$$

According to the calculated cluster centers, the membership matrix U is recalculated using the following equation. The old membership matrix is compared with the new one. The same process is repeated until the difference between them is smaller than ϵ (Kaynar et al., 2016).

$$u_{ij} = \frac{1}{\sum_{k=1}^C \left(\frac{\|x_i - c_i\|}{\|x_i - c_k\|} \right)^{2/m-1}}$$

At the end of this process, the membership matrix U shows the result of the clustering process. The flow chart of the fuzzy c-means algorithm is as follows (Wang et al., 2015):

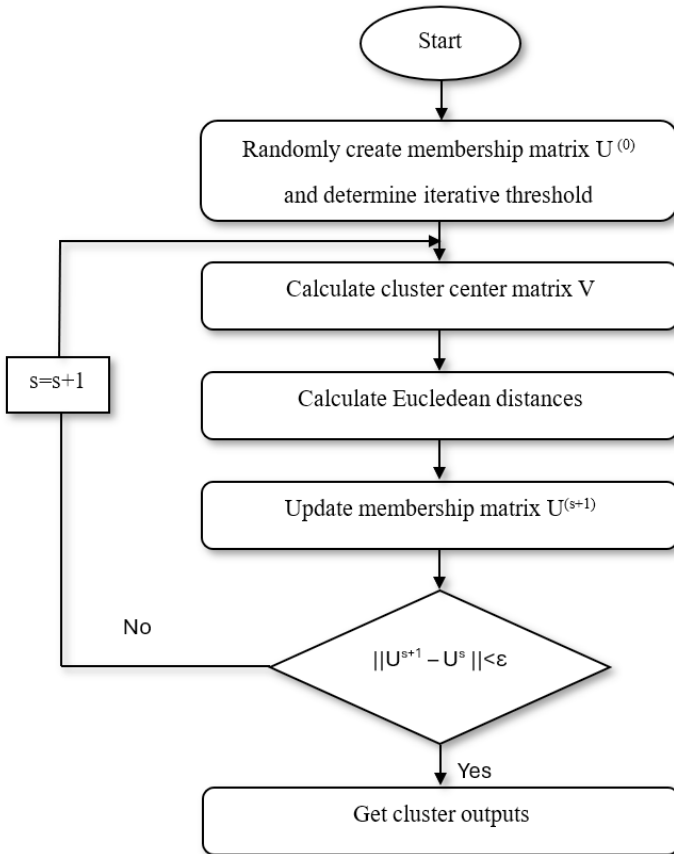


Fig. 4. The flow chart of the fuzzy c-means algorithm

Another method successfully used in clustering applications is Self-Organizing Maps (SOM), which map high-dimensional data into a one- or two-dimensional space. Initially proposed by Teuvo Kohonen, SOM networks are an unsupervised, topologically preserved, competitive learning method that consists

of two layers (Depren et al., 2004). The input layer is where the data—that is, the input vector—is fed into the network, and it has the same dimensions as the input vector. Between the input and output layers, there is a weight vector of the same size as the input vector. The final layer varies depending on the desired dimensionality reduction. Maps with different topologies, such as rectangular, hexagonal, or cubic, are generated from the output layer, with rectangular structures being among the most used. SOM networks utilize unsupervised training: by computing the distances between the input vector and the reference vectors in the output layer, the weights of the best-matching neuron and its neighboring neurons are updated. This causes the weight vectors to converge toward the cluster centers in the input layer. Because the weights of not only the best-matching neuron but also its neighbors are updated, clusters of adjacent neurons in the low-dimensional space are formed by the end of training. The structure of the SOM network is shown below (Kaynar et al., 2016):

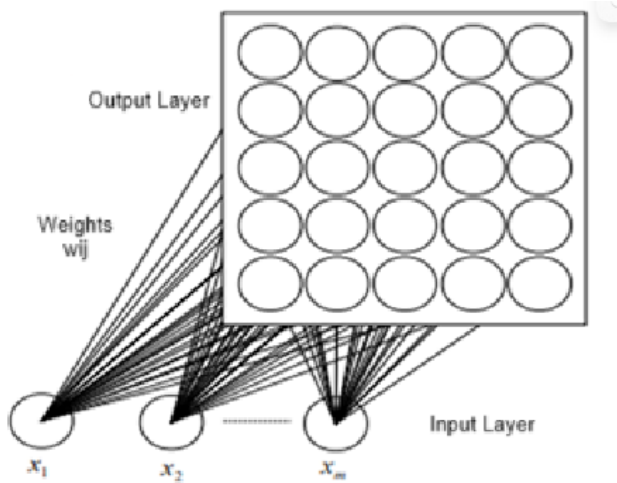


Fig. 5. SOM Network

In the SOM network training process, firstly the weight vectors are determined randomly. Then a random input vector is selected and given as input to the network. The distances between the reference vector of each neuron and the input vector are calculated and compared. The neuron with the smallest distance is called the best matching unit (BMU) (Haykin, 1998).

$$d_j = \|X - W_j\| = \sqrt{\sum_{i=1}^N (X_i - W_{ij})^2}$$

Here X is the input vector W and d is the distance between them. When neurons are updated, not only the best but also the neighboring neurons are updated. The new weight values are calculated as follows (Kaynar et al., 2016):

$$W_j(t+1) = W_j(t) + \eta(t)h_{j,i}(t)(X - W_j(t))$$

Here $W_j(t+1)$ is the weight vector in the next step, $W_j(t)$ is the current weight vector, X is the input vector, $\eta(t)$ is the learning rate that changes over time, $h_{i,j}(t)$ is the topological neighborhood function that changes depending on time and the distance between the winning neuron and the neighboring neurons. During the training process, the number of neurons around the winning neuron is reduced over time and the weights are changed so that only the winning neuron remains towards the end of the training. In this way, at the beginning of the training, a large part of the output space is interacted with, allowing the formation of larger clusters, while towards the end of the training, much smaller sized separations are formed (Alpdoğan & Bilge, 2009).

3.2. Content-Based Recommender Systems

In content-based recommender systems, a profile is created using the features of the items to be recommended and the users' past preferences. Recommendations are then generated based on this profile. To extract item features, keyword-based models such as TF-IDF (Term Frequency–Inverse Document Frequency) or the Vector Space Model (VSM) are commonly employed (Zhang et al., 2021).

If the items to be recommended can be expressed as features set, each item is defined by the same feature set, and the range of possible values for these features is known, then the item can be represented as structured data. In content-based applications, however, item explanations are usually textual features derived from news articles, emails, web pages, or product descriptions. Because these descriptions do not consist of predefined values, they are considered unstructured. In such cases, keyword-based approaches are used. If a text or formal variable appears in both the user profile and the document, that document is classified as relevant. Two significant challenges in text matching are homonyms and synonyms. The presence of synonyms may cause a relevant document to be overlooked, while homonyms may lead to an irrelevant document being mistakenly classified as relevant (Koren et al., 2009).

In the VSM, two major challenges in document representation are weighting the terms and computing the similarities between feature vectors. A term that

occurs in one document frequently but does not appear in the remainder of the corpus is more likely to be significant. Additionally, to avoid longer documents having an unfair advantage, weight vector is normalized. These concepts are encapsulated by the TF-IDF function (Koren et al., 2009).

$$TF - IDF(t_k, d_j) = \underbrace{TF(t_k, d_j)}_{TF} \cdot \underbrace{\log \frac{N}{N_k}}_{IDF}$$

Here N is document count in the corpus, N_k is document number in which the term t_k occurs at least once, the maximum of the frequencies $f_{z,j}$ of all terms (t_z) in the document d_j is calculated as follows (Kaynar et al., 2016):

$$TF(t_k, d_j) = \frac{f_{k,j}}{\max_{z,j} f_{z,j}}$$

In order to make the weights fall within the range $[0,1]$ and for the documents to be expressed as a vector of equal length, the weights obtained from the first equation are normalized with cosine normalization.

$$w_{k,j} = \frac{TF - IDF(t_k, d_j)}{\sqrt{\sum_{s=1}^{|T|} (TF - IDF(t_s, d_j))^2}}$$

A similarity measure is used to calculate how close two documents are. The most used is cosine similarity.

$$sim(d_i, d_j) = \frac{\sum_k w_{ki} \cdot w_{kj}}{\sqrt{\sum_k w_{ki}^2} \sqrt{\sum_k w_{kj}^2}}$$

In VSM-based content-based recommendation systems, user and item profiles are denoted as weighted term vectors. Whether a user will prefer a particular item is predicted by calculating the cosine similarity between their profile and the item.

Content-based recommendation systems offer many advantages over memory-based systems. Since item features and user history are used to generate the recommendation list, there is no need to find similar users—thus avoiding the sparsity problem that can occur when similar users cannot be identified. In memory-based systems, the only rationale provided for a recommended item is that similar users liked it. In contrast, because content-based systems compare item features

directly, it is much easier and more persuasive to explain the reasons behind a recommendation. This clarity helps increase user trust and loyalty to the system.

Furthermore, when a new item is included to the system, as long as its features are known, the cold start problem is avoided. The cold start problem only arises if a new user joins, since there is no existing data to build their profile.

Another disadvantage of content-based systems is that they tend to struggle with serendipity, that is, recommending items that a user might not discover on their own if not suggested by the system. This limitation exists because user profiles are built solely on past preferences, so without incorporating an element of randomness, the system cannot venture beyond that profile. However, if a user's tastes change over time, the system can adapt quite effectively.

3.3. Hybrid Recommender Systems

Collaborative filtering and content-based recommendation systems each have their own disadvantages. To handle these disadvantages, various hybrid systems have been proposed. Hybrid systems differ according to the methods of their creation. Some of them are “weighted, switching, cascaded, mixed hybridization, feature fusion, feature augmentation and meta-level” as shown in the table below (Isinkaye et al., 2015):

Table 1. Hybrid recommender systems according to their creation methods.

Method	Description
Weighted	The results of different recommendation systems are combined by weighing them. The strengths of the systems are highlighted by determining the extent to which each system will affect the recommendation.
Switching	A switching mechanism determines which recommendation system is active and which is not. The appropriate system is enabled to work according to the situation.
Cascaded	The output of one recommendation system is used as the input of another. The recommendation quality is improved with each system. It aims to present a solution to the cold start problem.
Feature fusion	Recommendations are created by combining the features of different recommendation systems into a single system.
Meta level	Two or more recommendation systems are combined in the form of layers. The output of the previous system is used as the input of the other system. The obtained data becomes more compact and summarized with each layer. Thus, less data is handled as the layers progress. Speed and efficiency are increased.
Mixed hybridization	The results of all of them are shown to the user by using more than one recommendation system.

4. Challenges in Recommender Systems

Even though recommendation systems are successfully used today, they still face many challenges that researchers have been trying to solve in the past and continue to address.

Scalability is one such challenge, especially when using real-world datasets. With the rapid growth in the number of users and items, recommendation systems have now become a subject within the big data domain. Even though model-based methods apply dimensionality reduction to the data, the ever-growing number of dimensions suggests that these methods may eventually become insufficient.

Another issue is the sparsity problem. In systems with such a vast array of item options, users can interact with only a very small fraction of the item space. This not only makes generating recommendations more difficult but also causes only a select group of popular items to be continuously highlighted. Consequently, the likelihood of recommending items that receive no interaction is very low. This sparsity problem is particularly pronounced in applications using collaborative filtering methods, as it becomes very challenging to find similar users and items when only a very small subset of the data has any interaction.

An indirect consequence of this issue is the serendipity problem. This occurs when recommendation systems present users with items that they might not otherwise discover or access on their own. This possibility is nearly nonexistent, especially in content-based systems, and it can also be referred to as the novelty or diversity problem (Hurley & Zhang, 2011). Some systems complement personalized recommendations with popular items. While this might be appealing to some users, most prefer personalized suggestions over repeatedly seeing the same familiar items. As recommendations become more personalized, user loyalty to the system increases. In collaborative filtering methods, popular items are more likely to be rated, which in turn increases their likelihood of being recommended. To ensure diversity, various hybrid methods can be developed.

At the opposite end of the spectrum is the problem of over-specialization, where recommendations are too closely aligned with a user's past preferences. This is particularly evident in content-based collaborative filtering methods. Since recommendations are generated based on the user profile—built from historical preferences—and item similarities, the system is unable to break out of that narrow scope, ultimately reducing the overall quality of the recommendations.

One of the most encountered and researched challenges in recommendation systems is the cold start problem. This issue can be categorized into three groups: new user, new item, and new system. The new user cold start problem arises because users who have just joined the system have no historical interaction data, making it almost impossible to generate recommendations for the system. A similar issue occurs in memory-based systems when new items are added; this is known as the new item cold start problem. Since these items have not been rated or visited before, their likelihood of being recommended is very low. In content-based systems, however, if the item features are fully known, the similarity mechanism can function without issue, and the system is less affected by this problem. The new system cold start problem arises when a system has just been set up, and there is no historical data for either users or items meaning that high-quality recommendations cannot be produced until sufficient data is collected.

Another problem arises when users intentionally provide incorrect ratings. Such ratings need to be detected by the system and removed.

In addition to the challenges encountered in generating recommendations, there are also user-related issues. For example, privacy concerns emerge from the system tracking users and collecting all data about them. To address this issue, systems must obtain user consent and develop applications that comply with each country's specific data protection laws. If a user does not permit the use of their personal data, the system cannot utilize that information.

5. Evaluation Methods

How to measure and evaluate the effectiveness of recommender systems and the quality of the recommendations produced has been an important topic of discussion since the early days. There are two types of approaches here. The first is to estimate the ratings in the rating matrix of the system and measure the estimation error statistically. For this, the mean absolute error (MAE), mean square error (MSE) or root mean square error (RMSE) error criteria are used (Silveira et al., 2019):

$$MAE = \frac{\sum_{i \in R_u} p(i) - r(i)}{|R_u|}$$

$$RMSE = \sqrt{\frac{\sum_{i \in R_u} (p(i) - r(i))^2}{|R_u|}}$$

Another approach is to consider the recommendation process as a decision support or classification process and use criteria such as precision, recall, F1 or

ROC curve to measure the recommendation quality. Precision is the ratio of the number of suitable recommendations to the total number of recommendations made. Recall is the ratio of the recommended objects to the total number of objects that can be recommended. F1 is a combination of these two measures (Bobadilla et al., 2013). Where I_u is the set of objects rated or viewed by user u , $w(r)$ is the object in row r and $I[\cdot]$ is the indicator function, precision and recall are defined as follows (Liang et al., 2018):

$$Precision@R(u, \omega) = \frac{\sum_{r=1}^R \mathbb{I}[\omega(r) \in I_u]}{M}$$

$$Recall@R(u, \omega) = \frac{\sum_{r=1}^R \mathbb{I}[\omega(r) \in I_u]}{\min(M, |I_u|)}$$

In addition, the concept of coverage, also called prediction capacity, is defined as the percentage of objects that each user has not yet evaluated by at least one of its neighbors (Bobadilla et al., 2013). This concept can also be expressed as the ratio of objects that can be recommended to the user to the total number of items.

6. Conclusion

Recommendation systems have become an integral part of our lives, especially with their rapid rise over the past two decades. The \$1 million prize, awarded to the winning team of a competition held in 2006, has since multiplied, has turned into an industry worth billions of dollars today. Researchers continue to work on developing smoother, higher-quality, and more effective recommendation systems. Future studies will not only refine fundamental methods but also leverage deep learning techniques, which have already demonstrated outstanding results, as well as emerging quantum computing approaches. As the number of users, data volume, and data diversity continue to grow, academic research in the field of recommendation systems will undoubtedly expand further.

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CHAPTER X

DATA-DRIVEN HEALTHCARE: THE ROLE OF ARTIFICIAL INTELLIGENCE IN PERSONALIZED MEDICINE

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1. Introduction

The healthcare sector has been undergoing a major transformation in recent years with advances in artificial intelligence (AI) and data analytics. While traditional medical models generally develop treatment approaches based on the average effects of certain diseases on the general population, personalized medicine aims to optimize treatments by taking into account individuals' genetic structure, lifestyle and other personal data. This approach enables earlier detection of diseases, creation of more effective treatment plans and reduction of side effect risks.

1.1. Personalized Medicine and Data-Driven Health

One of the most important components of personalized medicine is the data-driven health approach. This approach involves supporting personal health decisions by analyzing a wide range of data obtained from individuals. Big data sources such as electronic health records (EHR), genetic data, biomarkers and instant data from wearable devices form the basis of this process. However, processing and making meaningful such large volumes and complex data is quite difficult with classical statistical methods. Traditional statistical approaches may not be able to adequately model complex relationships. Therefore, the need for AI and machine learning techniques to effectively process data and optimize personal health decisions is increasing.

1.2. The role of Artificial Intelligence

AI techniques such as machine learning and deep learning can analyze big data to create predictive models, enable earlier detection of diseases, and help create personalized treatment plans. Deep learning models are used in a wide range of areas from medical imaging to genetic data analysis, supporting doctors and researchers to make faster and more accurate decisions.

Some critical applications are:

- **Medical image analysis:** Thanks to AI-supported analysis on radiological and pathological images, diseases such as cancer can be detected at an early stage.
- **Genetic data analysis:** Predicting disease risk based on individual genetic profiles and creating customized treatment plans.
- **Drug discovery:** AI plays a critical role in the discovery of new drug compounds and the optimization of clinical trials.
- **Personalized treatment approaches:** Determining the most appropriate drug combinations according to patients' genetic and biometric data.

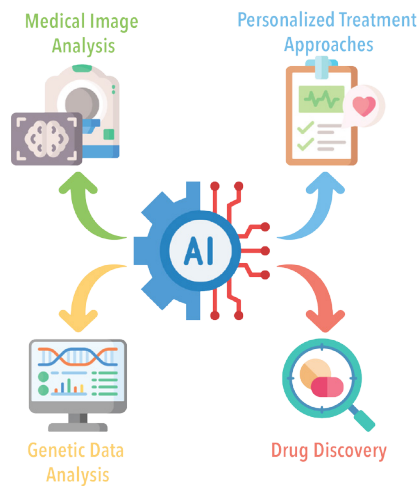


Figure 1. Some AI applications in medicine

In addition, personalized medicine aims not only to treat diseases but also to prevent them. Early warning systems can be developed based on individuals' lifestyles, genetic predispositions and environmental factors. In this way, individuals can be informed about risk factors and make more informed health decisions.

1.3. Ethical and Legal Dimensions

AI-supported personalized medicine applications are an issue that should be carefully addressed from an ethical and legal perspective. The most discussed issues are as follows:

- **Data privacy and patient privacy:** Anonymization of patient data, cybersecurity measures and data sharing should be carried out within the framework of ethical rules.
- **Algorithmic bias and fairness:** It is a great risk for AI models to be biased against certain ethnic groups or socioeconomic levels. Therefore, it is critical that models are transparent and auditable.
- **Regulations and legal liability:** It is not yet fully clear which legal regulations AI-supported health applications will be subject to and who will be responsible for erroneous decisions.

In conclusion, data-driven personalized medicine has the potential to fundamentally transform health systems. However, for this transformation to be successful, technological developments must be addressed together with ethical, legal and social dimensions.

2. Data-driven Health: Core Components

Data-driven health has become one of the cornerstones of modern medicine, with the potential to enhance the efficiency and personalization of patient care processes. This approach integrates diverse data sources and employs advanced analysis methods to optimize healthcare systems. This section will address the core components of data-driven healthcare systems.

2.1. Data Sources

Data generated in the healthcare sector originates from various sources and exhibits significant variation. The primary data sources are:

- **Electronic health records (EHR):** Electronic health records are defined as digital databases that contain patients' past health information, diagnoses, laboratory results, drug prescriptions and treatment plans. The utilization of these systems facilitates data sharing between in-hospital and out-of-hospital healthcare providers, thus increasing the continuity of patient care (Tongsiri, 2013).

- **Genetic and omic data:** Biological data types such as genome sequencing, transcriptomics, proteomics and metabolomics are utilised to ascertain an individual's genetic predispositions and disease risks and are of particular significance in the early diagnosis of cancer and neurodegenerative diseases (Berghoff et al., 2013; Davis et al., 2022).

- **Wearable devices and sensors:** These are devices that continuously collect biometric data such as heart rate, blood pressure, blood oxygen level, sleep patterns and physical activity. This data supports individual health management and also enables the monitoring of chronic diseases (Bianchi & Parke, n.d.).

- **Medical imaging data:** Data obtained from advanced medical imaging techniques such as computerized tomography (CT), magnetic resonance imaging (MRI), and positron emission tomography (PET) are used in AI-supported diagnostic systems. Image processing techniques support the decision-making mechanisms of radiologists (Alongi et al., 2024; Zhang et al., 2024).

- **Patient-generated data:** Data generated by individuals through mobile health applications, online surveys, and personal health diaries contribute to the development of individualized health recommendations (Akram et al., 2017).

2.2. Data Processing and Analysis Techniques

The collected health data is transformed into meaningful information and integrated into clinical decision support systems (CDSS). The main methods used at this stage are:

- **Machine learning and deep learning:** Machine learning, a sub-branch of big data analytics, is widely used in health prediction and diagnosis models. Deep learning has achieved great success, especially in image analysis and NLP (Lakshmi & Devi, 2021; Mishra, 2024).

- **Big data analytics:** Healthcare services generate large amounts of structured and unstructured data. Big data analytics offers innovative solutions in the healthcare sector by using distributed computing techniques to process and make meaningful this data (Kumar & Singh, 2019).

- **Natural language processing (NLP):** NLP techniques used in the analysis of unstructured data such as clinical notes, physician reports, and patient comments are playing an increasingly important role in medical information extraction (Taira, 2010).

2.3. Data Security and Privacy

Health data is extremely sensitive and appropriate security measures must be taken. In this context, the basic elements to be considered are as follows:

- **Protection of personal health data:** Regulations such as GDPR (General Data Protection Regulation) in Europe and HIPAA (Health Insurance Portability and Accountability Act) in the USA require the protection of patient data (Shah, 2023).

- **Anonymization and encryption techniques:** Anonymizing health data is critical to ensuring privacy. Data encryption algorithms are widely used to prevent unauthorized access (Olatunji et al., 2022).

- **Data sharing and access authorization:** Sharing of health data should be regulated in a way that it can only be accessed by authorized healthcare professionals and researchers. Blockchain technology offers new opportunities for secure data sharing (C. Wang et al., 2024).

2.4. Clinical Decision Support Systems

AI and data analytics play an important role in the development of CDSS. The main functions of these systems are as follows:

- **Automatic diagnostic and predictive systems:** Deep learning models assist doctors in the analysis of radiological and histopathological images (W. Wei et al., 2024).

- **Treatment recommendation systems:** Systems that recommend individualized treatment plans based on patients' genetic profiles and health history have made great progress, especially in the field of oncology (Lin, 2020).

- **Clinical risk analysis:** Machine learning-based models help take preventive measures by predicting patients' future health risks (Sun et al., 2022).

The integration of these components forms the basis of data-driven healthcare systems, enabling healthcare services to become faster, more accurate and more personalized.

3. Applications of AI in Personalized Medicine

Personalized medicine is an approach that aims to create customized treatment plans by taking into account patients' genetic, environmental, and

lifestyle factors. AI and big data analytics are among the main drivers of personalized medicine and are revolutionizing healthcare. In this section, the main application areas of AI in personalized medicine will be discussed.

3.1. Genetic and Pharmacogenomic Analysis

Predicting the effectiveness and side effects of drugs based on an individual's genetic profile is one of the most important components of personalized medicine. AI algorithms provide the following main advantages by analyzing large-scale genetic data:

- **Disease risk prediction:** Deep learning models that analyze genetic variations can predict the susceptibility of individuals to certain diseases. For example, the analysis of BRCA1 and BRCA2 gene mutations that carry the risk of breast cancer contributes to the development of early diagnosis strategies (Chow et al., 2024).

- **Pharmacogenetic applications:** AI-supported pharmacogenomic models allow drug selection and dosage adjustments based on individual genetic profiles. These methods are widely used in personalized optimization of chemotherapy drugs (Igwama et al., 2024).

3.2. Disease Diagnosis and Prediction

AI plays an important role in personalized medicine by providing faster and more accurate results compared to traditional diagnostic processes. The main application areas are:

- **Radiological and pathological image analysis:** Deep learning-based image processing systems can detect abnormalities in medical images with high accuracy rates. Convolutional neural networks (CNNs), especially used in cancer diagnosis, can automatically distinguish tumor cells in histopathological tissue samples (Shweikeh et al., 2021).

- **Prediction of disease course:** Machine learning models that analyze patients' past medical records can predict the progression of a particular disease and help determine personalized treatment strategies (Felix et al., 2024).

- **Anomaly detection:** AI is used to diagnose abnormal biological signals and rare diseases. For example, deep learning analysis of electrocardiography (ECG) data makes significant contributions to the detection of cardiac abnormalities (Z. Wang et al., n.d.).

3.3. Personalized Treatment Plans

Personalized treatment applications aim to determine the most effective treatment methods for a patient. AI offers the following contributions in this process:

- **Personalized treatment prescriptions:** Algorithms that evaluate patients' genetic and clinical data suggest the most appropriate treatment plans for specific diseases. Personalized approaches based on patient immune response are being developed, especially in immunotherapies (Shams, 2024).
- **Robotic and autonomous surgical systems:** Robotic surgical systems supported by AI increase the precision of surgical interventions by creating individualized operation plans. The Da Vinci robotic surgical system is one of the most common examples of this technology (Bokhari, 2023).

3.4. Digital Health and Remote Monitoring Systems

AI is making healthcare personalized with digital health platforms and remote patient monitoring systems. The main innovations in this field are as follows:

- **Wearable technologies and sensors:** Wearable devices that instantly monitor physiological data such as heart rhythm, blood pressure, and blood sugar play a critical role in individual health monitoring (Chen et al., 2021).
- **Chatbots and virtual assistants:** Health chatbots supported by NLP can answer patients' health questions, perform symptom analysis, and manage doctor appointments (John et al., 2022).

3.5. Clinical Decision Support Systems

CDSS help doctors make more informed decisions based on patient data. The main functions of these systems are as follows:

- **Preventive health recommendations:** Machine learning models provide personalized health recommendations based on individuals' health history and lifestyle (Sodhi et al., 2024).
- **Disease prediction and risk analysis:** AI-based systems offer early intervention opportunities by predicting the health risks that patients may face in the future (Thakur et al., 2024).

The role of AI in personalized medicine is expanding and making healthcare more individualized, faster and more effective. The development of these technologies will enable more successful diagnostic and treatment processes in the future.

4. The future of Personalized Medicine and the Role of AI

AI and big data analytics play a critical role in the future of personalized medicine. While developing technologies enable healthcare services to become more accessible, faster, and more accurate, they also lead to the adoption of customized treatment approaches that take into account individual patient needs. This section will discuss the future directions of personalized medicine and the role of AI in this context.

4.1. Advanced Genetics and Omics Technologies

The advancement of genetic research and omics technologies is one of the cornerstones of personalized medicine. AI analyzes these large and complex biological data sets to better understand diseases.

- **Holistic omics analyses:** By integrating genomic, proteomic, metabolomic, and epigenomic data, AI can identify disease risk factors and support personalized preventive healthcare (Ozaki et al., 2024).
- **CRISPR and genetic editing:** AI-supported algorithms contribute to the more sensitive and effective use of genetic editing technologies. AI is used in the identification and editing of target genes in CRISPR-based treatments (Dixit et al., 2023).

4.2. AI-supported Biomarker Discovery

AI has become a powerful tool for the discovery and validation of biomarkers. In particular, deep learning algorithms help identify new biomarkers in disease diagnosis and prognosis by analyzing large-scale data obtained from clinical studies.

- **Cancer biomarkers:** AI provides significant progress in identifying new biomarkers that can be used in the early diagnosis of cancer. For example, machine learning algorithms allow cancer to be diagnosed in its early stages by analyzing proteomic data obtained from blood samples (Kong et al., 2014).

- **Neurodegenerative diseases:** AI-based biomarker analysis is becoming increasingly common in the early diagnosis of diseases such as Alzheimer's and Parkinson's. Deep learning models can predict disease progression by detecting subtle changes in brain images (Suganya & Aarthy, 2023).

4.3. Digital Twins and Personalized Simulations

Digital twin technology refers to virtual models created from individuals' biological and health data. Digital twins supported by AI offer significant advantages in personalized treatment simulations and disease management processes.

- **Patient-specific treatment simulations:** AI-supported digital twins can help predict the effects of drug treatments or surgical interventions on patients (Sai et al., 2024).

- **Use of digital twins in clinical trials:** The use of digital twins in clinical trials enables the testing of new treatment protocols on a patient basis and the optimization of personalized medicine strategies (Vidovszky et al., 2024).

4.4. AI-supported Automation and Robotics Technologies

The use of automation and robotics technologies in healthcare is rapidly spreading. AI is improving the quality of healthcare services in many areas, from patient care to surgical procedures.

- **Autonomous surgical robots:** AI-supported surgical robots enable more precise and safer operations. For example, the Da Vinci surgical system performs high-precision procedures in minimally invasive surgeries (Bokhari, 2023).

- **Patient monitoring and care automation:** AI-based systems can monitor patients' routine health checks, provide medical alerts, and reduce the workload of healthcare personnel (Jayant et al., 2024).

4.5 Ethical and Regulatory Challenges

The widespread use of AI in personalized medicine raises some important ethical and regulatory questions. Addressing these issues is critical to ensuring the safe and fair use of technology.

- **Data privacy and security:** Protection of personal health data is one of the biggest concerns in the use of AI-enabled systems. Technologies such as data

anonymization and blockchain play an important role in reducing these risks (Yechuri, 2024).

- **Intervention in algorithmic bias:** Algorithmic bias that may occur in AI systems can lead to unfair outcomes for certain patient groups. Ethical frameworks and regulatory mechanisms need to be developed to overcome this problem (Chinta et al., n.d.)

4.6. Conclusion and Future Perspectives

The future of personalized medicine will continue to be shaped by the development of AI and big data technologies. In the coming years, more sensitive diagnostic methods, personalized treatment protocols, and AI-enabled healthcare services will enable disease management.

5. Future Perspective and Conclusion

Today, AI applications in the field of personalized medicine are rapidly developing and are used in a wide range of areas from medical decision support systems to genetic analyses. It is expected that this field will progress even further in the future, and new generation AI models and technological innovations will enable healthcare services to become more sensitive, predictable and individual-specific. In this section, the future of personalized medicine, the role of AI in this field, and the opportunities and risks that may be encountered will be discussed.

5.1. The Future of Personalized Medicine

One of the most important factors in the development of personalized medicine is the increasing strength of the integration of big data analytics and AI. By integrating genetic, environmental and clinical data, it becomes possible to diagnose diseases earlier and create treatment protocols specific to the individual. In particular, genetic sequence analyses and biomarker-based diagnostic methods will be used more widely in medical applications in the future. However, more effective analysis of data obtained from digital health records, mobile health applications and wearable devices will make personalized healthcare more accessible.

In addition, thanks to pharmacogenomic applications, drug treatments suitable for the individual's genetic structure are being developed. With this approach, the risk of side effects is minimized and the most appropriate dosage regimens for patients can be determined. At the same time, genetic mutations are targeted in cancer treatments called precision oncology, and more effective and less side-effect treatment options are provided.

5.2. *New Generation AI Models*

In addition to traditional machine learning and deep learning approaches, new generation techniques such as large language models (LLMs), graph neural networks (GNNs), federated learning and quantum machine learning are creating a major revolution in personalized medicine.

- LLMs enable the interpretation of patient data and the development of intelligent health assistants that guide doctors. In particular, chatbot systems that analyze patients' symptoms and provide rapid diagnostic suggestions increase access to healthcare services.
- GNNs are used to discover complex relationships on biomedical datasets, providing a better understanding of genetic diseases.
- Federated learning allows training AI models in different hospitals or devices without transferring patient data to a central server, which offers a great advantage in terms of data privacy.
- Quantum machine learning contributes to the acceleration of large-scale calculations in medical imaging and biomedical modeling processes by analyzing large biological datasets.

The integration of these new generation AI models makes diagnosis and treatment processes faster and more reliable. These developments will increase the quality of patient care by making personalized medicine more precise and effective.

5.3. *Potential Opportunities and Risks*

The opportunities offered by AI in personalized medicine include more accurate diagnosis and treatment recommendations, increased efficiency in patient care processes, and reduced healthcare costs. In particular, deep learning-based medical imaging systems provide high accuracy rates in cancer screenings. AI-supported biomarker discovery processes allow for more precise diagnoses and molecular understanding of diseases. In drug discovery and development processes, AI-based simulations accelerate the development of new drugs and optimize clinical trial processes.

However, these technologies also have some risks:

- **Data privacy and security:** Subjecting patient data to large-scale analysis increases the risk of privacy violations.
- **Bias in AI algorithms:** Imbalances in training data can lead to certain patient groups being misdiagnosed or deprived of appropriate treatment options.

- **Transparency and explainability issues:** Not fully understanding how AI decisions are made can create a situation where reliability is questionable for healthcare professionals.
- **Regulatory deficiencies:** The lack of an adequate regulatory framework for AI-based medical applications can limit the effective and safe use of the technology.

Therefore, it is of great importance to integrate ethical rules and legal frameworks into AI-based healthcare systems.

6. Conclusion

As a result, personalized medicine supported by artificial intelligence makes healthcare more efficient and personalized to the individual. Thanks to big data analytics, new generation machine learning techniques and biomedical innovations, it has become possible to diagnose diseases earlier and optimize treatment processes.

In order for these technologies to be used effectively, ethical standards must be established, algorithmic bias must be prevented and data security must be ensured. Strengthening regulatory policies is of critical importance for the safe and widespread adoption of AI-based healthcare systems.

In the future, personalized medicine applications are expected to lead to radical changes in the healthcare sector together with AI-supported healthcare systems. Scientific developments and ethical frameworks will determine the direction of this transformation and continue to shape healthcare services in every field.

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CHAPTER XI

DIAGNOSIS OF PARKINSON'S DISEASE WITH DEEP LEARNING METHODS

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1. Introduction

Parkinson's disorder is extensively appeared as the second one most universal neurological disease globally, following Alzheimer's disorder in phrases of its enormous impact (Singh, Pillay, & Choonara, 2007). According to records launched with the aid of using the World Health Organization, an expected 10 million people global are presently dwelling with Parkinson's disease (Thuy, Nutt, & Holford, 2012). A major issue associated with this condition is the frequent delay in its diagnosis during the early stages. This diagnostic lag often allows the disease to progress, leading to irreversible neurological damage that is largely resistant to treatment. In numerous cases, these severe complications eventually result in mortality

The disease mainly affects people over the age of 60. Although the symptoms are well known, there are many new cases of Parkinson's with different and new symptoms. The prognosis of Parkinson's disorder is a totally complicated trouble and there's no suitable scale to estimate the severity of Parkinson's disorder. It is a neurodegenerative sickness that impacts motor feature because of a lower in dopamine degrees withinside the brain, which in flip has physical consequences at the body (Pagano, Ferrara, Brooks, & Pavese, 2016).

Parkinson's disease is mainly caused by the inability of neurons to regenerate, leading to a loss of critical brain functions. As a person gets older,

neurones begin to die and become irreplaceable. Neurons produce a chemical known as dopamine, which plays a crucial role in controlling body movements and facilitating communication between nerve cells. As dopamine stages start to decline with age, the neurological kingdom additionally begins to gradual down and, as a result, numerous modes of conversation in the mind are affected. These situations extensively have an effect on the patient's first-class of day by day life. The patient may experience decreased motor control abilities, walking and balance problems. Apart from these conditions, impairments in skills such as speech and hand movements may occur.

The signs and symptoms of Parkinson's sickness are classified into most important groups: motor signs and symptoms and non-motor signs and symptoms.

Non-motor symptoms alone are not predictive of the disease. However, when used with markers from cerebrospinal fluid and dopamine transporter imaging, the disease can be predicted (Challa, Pagolu, Panda, & Majhi, 2016).

Diagnosis of Parkinson's disorder appears tremendously easy, however in a few sufferers it is able to be pretty difficult even for knowledgeable neurologists (Hess & Okun, 2016). Many strategies were proposed to diagnose Parkinson's disease (PD). Selecting an appropriate and effective diagnostic method for Parkinson's disease is crucial, as it enables early intervention and significantly enhances the patient's quality of life. The methods used in the diagnostic phase generally include clinical examination, neuroimaging techniques, biochemical tests and neurophysiological evaluations.

In the clinical examination, the physician assesses the patient's motor skills, movements, tremors and balance. Clinical findings focus on the symptoms of the disease and are usually performed by a specialist. There is not yet a reliable and applicable diagnosis or marker for the diagnosis of Parkinson's disease (Lau & Breteler, 2006). The early diagnosis of Parkinson's Disease relies heavily on the clinical expertise and experience of healthcare professionals. Doctors talk to patients, carefully observe their movement characteristics and go through a series of procedures to determine whether the patient has Parkinson's disease. Although the American Academy of Neurology (ANN) does not officially endorse any clinical diagnosis, the most common clinical criteria used to diagnose The clinical diagnostic criteria for Parkinson's disease are established by the United Kingdom Parkinson's Disease Society Brain Bank (UKPDSBB), as referenced by Hess and Okun (2016).

Neuroimaging Techniques Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), unmarried photon emission computed tomography (SPECT), purposeful magnetic resonance imaging (fMRI) and 123I-Inoflupane

(DATASCAN) techniques are used (Pagano, Niccolini, & Politis, Imaging in Parkinson's disease, 2016).

Magnetic resonance (MR) images, especially the visible evaluation of T2- and T1-weighted sequences in Parkinson's disorder patients, function a precious device to supplement traditional diagnostic methods (Mahlknecht et al., 2010).

PET imaging techniques are used to measure the activity of dopamine and its receptors in the brain and to assess the degree to which the dopaminergic system is affected in Parkinson's disease. This diagnostic method aids in identifying Parkinson's disease by detecting the degeneration of dopaminergic neurons, as highlighted by Pavese and Brooks (2009).

SPECT imaging method is used to visualise dopamine activity. It is an effective method to evaluate dopaminergic losses in Parkinson's disease. SPECT is also used to assess blood flow in the brain. It is used to distinguish the consequences of Parkinson's ailment and different neurodegenerative diseases (Pavese & Brooks, 2009).

Functional magnetic resonance imaging (fMRI) is a essential device withinside the prognosis and knowledge of Parkinson's disease. Frequency analysis of BOLD signals in the listening state (without mental activity) reveals patterns of spontaneous activity in the brain (Kwak, et al., 2012).

Studies using mind financial institution statistics from the United Kingdom and Canada have discovered that almost 25% of clinicians inaccurately diagnose Parkinson's disease, as suggested via way of means of Tolosa, Wenning, and Poewe (2006). Therefore, locating the perfect prognosis of Parkinson's disorder remains a hard assignment to be solved. Physicians diagnose Parkinson's disease by carefully analyzing the patient's medical history and conducting a thorough clinical examination, as noted by Joseph (2008).

Specialised radiologists often take a long time to review medical images and their speed of decision-making is affected. Due to the confined quantity of correctly educated radiologists, delays in analysis and decision-making extensively effect the high-satisfactory of existence for patients. Therefore, it's far vital to automate photograph analyses to conquer the postpone withinside the diagnostic procedure and boom the accuracy (Ker, Wang, Rao, & Lim, 2017).

The layout and improvement of modern technology in clinical diagnostics relies upon at the accuracy of virtual techniques, the use of AI technologies is required to reduce diagnostic time and achieve better accuracy than expert doctors (Singh, Singh, & Singh, 2019).

Nowadays, there is a tremendous progress in artificial intelligence-based techniques, especially in the field of health. Artificial intelligence permits the

early detection of Parkinson's disease, imparting each well timed intervention and cost-effective solutions. Especially with the rapid increase in neuroimaging data in the world, research on Parkinson's disease has accelerated. The maximum critical purpose for researchers to recognition on Parkinson's ailment is that there may be nonetheless no modern-day strategy to this problem.

Today, Parkinson's disease is on the rise, especially in the elderly population, and its early diagnosis is becoming increasingly difficult for doctors. Research in the literature draws attention to the limitations of diagnostic methods and emphasises that artificial intelligence has an important potential in this process. The advantages of artificial intelligence techniques in evaluating the symptoms of Parkinson's disease, extracting meaningful results from large data sets and personalised treatment are noteworthy. Studies in the literature show how artificial intelligence applications can be used in different fields in the diagnosis of Parkinson's disease.

In this context, a summary of the literature is given below by examining the studies involving the use of artificial intelligence in the detection of Parkinson's disease.

Convolutional Neural Networks (CNN) had been hired to distinguish among people with Parkinson's sickness and wholesome controls, as established with the aid of using Alissa et al (2022). With data collected from 87 subjects from Leeds teaching hospitals, CNN models were trained and information augmentation techniques, different data representation methods and image resolutions were tested. These techniques positively affected the classification performance and an accuracy rate of 93.5% was obtained.

A system is proposed to analyse the movement of normal and Parkinson's patients by printing text on a computer. For this system, the data set prepared by Kaggle and MIT is used. The data set consists of 55 healthy and 162 Parkinson's patients. Wavelet transform is utilised in the study. A smaller but more powerful CNN network is used than SqueezeNet and AlexNet networks. (Bernardo, Damasevicius, Ling, Albuquerque, & Tavares, 2022), which redesigned the SqueezeNet architecture to increase its success, 90% accuracy was achieved.

Medical imaging techniques offer the potential to elucidate Parkinson's disease and optimise treatment strategies. In this context, current literature studies provide important information on how imaging techniques can be improved and used more effectively. Imaging techniques, combined with artificial intelligence, are poised to play a pivotal role in enhancing the diagnosis and treatment of Parkinson's disease. Below, research with artificial intelligence

packages withinside the detection of Parkinson's sickness with the assist of scientific snap shots are mentioned in detail.

A examine making use of a dataset comprising eighty two healthy people and a hundred Parkinson's sufferers from the PPMI public database established big findings. The statistics have been normalised the usage of diverse photograph preprocessing techniques. The normalized statistics have been in the end processed the usage of a 2D Gaussian filter, using a smoothing kernel of length 5x5 and a standard deviation of 0.8. AlexNet, a convolutional neural network (CNN) architecture, became applied for the category of Parkinson's disease. In this examine, the very last 3 layers of the pre-trained AlexNet version have been changed the usage of transfer learning, achieving an accuracy price of 88.90%, as suggested through Sivaranjini and Sujatha (2020).

(Shah, Zeb, Shafi, Zaidi, & Shah, 2018) used a dataset of 250 healthy and 250 Parkinson's patients from the PPMI public database. CNN model was considered for the proposed system. In this proposed architecture, MR images are taken as input and labelled as Parkinson's and healthy. The model consists of eight main layers in total. The structure incorporates 3 convolutional layers, max-pooling layers, dense layers, and an output layer. When the results of the model are analysed in all experiments, the classification accuracy is between 95% and 98%.

FMRI dataset obtained from PPMI platform was used. There are 604 fMRI images in the dataset. After distinguishing between people affected and unaffected by Parkinson's disease, 3D images were converted into 2D images. Minimum-maximum scaling was used to normalise the data to a specific dimension. Transfer learning was used to perform image classification. VGG-16, VGG-19 and INCEPTIONV3 are used. In this study by (Sajeeb, et al., 2021), the best accuracy was achieved with VGG-19.

(Dehghan, Naderan, & Alavi, 2022) performed classification with SPECT images obtained from the PPMI platform. 650 SPECT images were used. The CNN architecture and SPECT image augmentation processes used are emphasised as the novelty of this study. The model consists of three stages: data preprocessing, training-testing and evaluation. As a result of the tests, the best accuracy rate was achieved with VGG-19.

(Magesh, Myloth, & Tom, 2020) used images of 430 Parkinson's patients and 212 healthy individuals from the PPMI database. The data were analysed on phantoms through attenuation and correction processes. In the study, the predictions of complex artificial intelligence models were interpreted using

superpixels and LIME annotations. By implementing the optimal threshold value in the system utilizing the CNN VGG16 model, the accuracy rate improved from 92% to 95.2%.

2.Methodology

2.1 Artificial Neural Networks

Artificial neural networks are structures designed to imitate the functioning of the human brain's neural structure, permitting them to carry out unique tasks. These networks are widely utilized in various fields today, including image processing, speech recognition, finance, healthcare, automation, manufacturing, sports, and gaming.

The first study in this field was carried out in 1943 by mathematician Walter Pitts and neuro-physiologist Warren McCulloch. They made the first neural network modelling on this date. They explained how neurons work in an article called 'The Logical Calculus of the Ideas Immanent in Nervous Activity' (McCulloch & Pitts, 1943).

2.2 Deep Learning

Today, the rapid increase in the amount of data necessitates the development of more complex model structures and new algorithms. In this context, deep learning techniques enhance the ability of computers to effectively address and solve more complex problems. During the late 1980s, artificial neural networks emerged as a significant area of interest within the fields of Machine Learning and Artificial Intelligence. Although artificial neural networks have been successfully used on many topics, interest in this technology has declined. In 2006, Hinton et al. introduced 'Deep Learning' (DL), which is defined as 'next generation neural networks' based on artificial neural networks (ANN) (Hinton, Osindero, & Teh, 2006).

2.3. Convolutional Neural Networks and Architecture

Convolutional neural networks are grounded within the pioneering studies carried out with the aid of using Hubel and Wiesel at the visible cortex of monkeys and birds. Later, inspired by this paper, Kunihiko Fukushima introduced the convolution process in the field of ESA. These studies have contributed significantly to the current success of ESA. In particular, Yan Le Cunn's ESA model called LeNet-5 was developed using back propagation and

adaptive weights and has an important place in the field of ESA (Ajit, Acharya, & Samanta, 2020). This structure consists of layers arranged one after the other. The major architectures used today are different modifications of LeNet-5.

2.3.1. Convolutional Layer

The convolution layer is one of the maximum vital factors withinside the structure of Convolutional Neural Networks (CNNs). An N-dimensional image as input is processed by convolution filters and an output feature map is generated. The filter weights, defined as kernels, are randomly assigned at the beginning of the training process and learn the important features by adjusting with each training period. Since the CNN input is a multi-channel image, the convolution process is performed in a multi-channel image format, not in a vector format compared to the traditional neural network (Alzubaidi, et al., 2021).

2.3.2. Pooling Layer

Pooling layer is the process of reducing the size by sampling to reduce the complexity of more layers. This process is similar to reducing resolution in image processing. The pooling layer does not alter the number of filters. Among the most widely used pooling techniques is max pooling. This method divides the image into sub-regions, returning only the maximum value within the sub-region.

In maximum pooling, the process is usually applied in 2x2 dimensions. When pooling is applied to the upper left 2x2 blocks of the image, the image shifts by 2 units, focusing on the upper right section. This indicates that step 2 is used in the pooling layer (Albawi, Mohammed, & Al-Zawi, 2018).

2.3.3. Fully Connected layer

In the fully connected layer, all nodes of the neurons are interconnected, similar to the structure found in traditional neural networks. Each node from the pooling layer is hooked up as a vector to the primary layer of the completely related layer. These layers may require long periods of time in the training process because they contain the most parameters.

One of the major drawbacks of the fully connected layer is its high number of parameters, which demand extensive computational resources during training. To address this, reducing the number of nodes and connections is crucial. The dropout technique is often employed to compensate for the removed nodes and links (Albawi, Mohammed, & Al-Zawi, 2018).

2.4. Transfer Learning and Ready CNN Architectures

Transfer learning is a method of utilising prior knowledge from different environments to solve a problem. This learning approach is inspired by the way people learn, people do not learn from scratch for any problem, learning is built on previous knowledge about the subject. For example, learning to play an instrument is considered easier for a person who knows how to play another instrument.

Training convolutional neural community fashions on very huge datasets can take numerous days or maybe weeks. The way to shorten this process is to use the weights of pre-trained ImageNet models. Utilizing the best-performing models directly can significantly accelerate the training process.

2.4.1. VGG-16

The Vgg-16 and 19 had been delivered via way of means of K. Simonyan and A. Zisserman at Oxford University in 2014 in the paper 'Very Deep Convolutional Networks for Large Scale Image Recognition'. They are named after the Visual Geometry Group. The VGG-16 structure accommodates a complete of sixteen layers, including thirteen convolutional layers and three completely linked layers. During the 2014 ImageNet competition, this model achieved an error rate of just 7.3%.

2.4.2. ResNet

Developed in 2015 by Microsoft Research, the 'Residual Network' or ResNet architecture offers a unique approach to eliminate the problem of overlearning. Although it is deeper than VGG networks, it has lower complexity. In ResNet architecture, some layers in the network bypass the incoming data and connect it directly to the output. This 'residual connection' structure is effective in reducing the overlearning problem. Connecting the incoming data directly to the output makes the connections that the network needs to learn easier and increases the learning capacity (He, Zhang, Ren, & Sun, 2016).

2.4.3. MobileNetV2

MobileNet was developed by Google in 2017 and designed as a low memory usage and lightweight neural network for mobile devices (Sandler, Howard, Zhu, Zhmoginov, & Chen, 2018). It offers advantages in embedded systems and applications requiring low computational cost.

MobileNetV2 has a complete of fifty three layers, inclusive of fifty two convolutional and 1 completely related layer. The architecture consists of 16 inverted blocks linked to 3 convolutional layers, bottleneck blocks, a convolutional layer, and a fully connected layer. Its fundamental components are the depthwise separable convolution and pointwise separable convolution layers (Sandler, Howard, Zhu, Zhmoginov, & Chen, 2018).

2.4.4. InceptionV3

In the article describing GoogLeNet, InceptionV3 network is proposed by taking the structure of InceptionV2 network as an example. The main feature of InceptionV3 that distinguishes it from InceptionV2 is the use of not only convolution layers but also cluster normalisation and fully connected layer as auxiliary classifiers (Kızrak, 2018). The architecture of the network is based on the principle of dividing the traditional 7x7 convolution operation into three separate 3x3 convolution operations. In the Inception part, there are 3 Inception modules of 35x35 size, each containing 288 filters. With the grid reduction technique, this size was reduced to 17x17 grids and 768 filters. After this process, an 8 x 8 x 1280 grid was generated using 5 different factorised Inception modules. The computational cost of the network is about 2.5 times higher than GoogLeNet and more efficient than VGGNet (Szegedy, Vanhoucke, Loffe, Shlens, & Wojna, 2016).

2.4.5. DenseNet

DenseNet architecture was introduced in 2016 by Gao Huang et al. in the paper 'Densely Connected Convolutional Networks'. This architecture uses dense connections to connect feature maps to each other, which provides more connectivity and learning capacity. Thanks to its dense connections, better results are obtained by using fewer parameters.

DenseNet networks are designed to solve the gradient problem caused by network depth. Each layer withinside the structure takes enter from all previous layers and passes its characteristic maps to all following layers. This dense connectivity ensures optimal information flow across layers (Mahdianpari, Salehi, Rezaee, Mohammadimanesh, & Zhang, 2018).

3. Application

The use of hand drawings and MR images in the early diagnosis of Parkinson's disease has recently been frequently examined, especially with

artificial intelligence applications. In this study, applications were carried out using deep learning architectures on drawing and MR image datasets obtained from the Kaggle platform.

In this study, datasets have been utilized: the Spirals and Waves dataset, comprising hand-drawn photographs from each wholesome people and Parkinson's sufferers, and the MRI dataset, which incorporates MRI scans of Parkinson's sufferers and wholesome controls. The data, sourced from the Kaggle platform, have been meticulously organized the use of diverse pre-processing and evaluation techniques.

3.1. Drawing Data Set

The statistics set includes each wholesome and Parkinson's patients' drawings withinside the shape of spirals and waves. The dataset incorporates a complete of 204 drawings, with 102 from wholesome people and 102 from Parkinson's patients. As a result of the analysis of the data set, it was determined that the average size of the drawings of healthy individuals was 320 x 320 pixels and the average brightness value was 210. The average size of the drawings of Parkinson's patients is 326 x 326 pixels and the average brightness is 208.

3.2. MR Images Data Set

The dataset utilized in this study was sourced from the Kaggle platform. It includes magnetic resonance images (MRIs) of both individuals with Parkinson's disease and those without the condition. The dataset consists of MRI scans from 221 people recognized with Parkinson's disorder and 607 people with out the condition. Analyses were performed to determine whether the data had been preprocessed. Most of the images range between 216 x 216 and 256 x 256 pixels. In the brightness analysis, the pictures of Parkinson's patients are generally darker and the average brightness value is in the range of 35-39. In normal images, this value is in the range of 39-44.

3.3. ResNet50 Model

This study focuses on developing an efficient approach for the early detection of Parkinson's disease. To achieve this, a pre-trained ResNet-50 architecture was adapted through transfer learning techniques, specifically tailored for the early diagnosis of Parkinson's disease. Below, the stages of the model are explained step by step:

- Data preparation and preprocessing: Images were converted to 224 x 224 pixels.
- Data augmentation: To enhance the model's ability to generalize and ensure greater diversity in the training data, the dataset was scaled up fourfold.
- Loading a pre-educated ResNet-50 Model: The version is primarily based totally on a pre-educated ResNet-50 version. This version turned into educated at the ImageNet dataset and used to extract fashionable photo features. The model is loaded with its weights.
- Customisation of the model: For the ResNet-50 model, the upper layers were removed so that dataset-specific features would not be used for learning. A flattening layer was added to smooth the feature maps, followed by densely connected layers containing 256 neurons and ReLU activation.
- Optimisation and loss function settings: The version become optimised the usage of the Adam optimisation algorithm. The learning rate of the Adam optimisation algorithm become set to 0.0001 and binary cross-entropy become used because the loss function.
- Training the model: The performance of the model was evaluated for 10 epochs and accuracy rates were calculated. With this model, 98.14% accuracy rate was achieved for MR data.

The same model was applied to our second dataset, Parkinson's disease hand drawings dataset. With this model, 99% accuracy rate was achieved for hand drawings data.

3.4. Vgg-16 Model

The VGG-16 version, some other approach used within the early prognosis of Parkinson's disease, is primarily based totally on the switch gaining knowledge of approach. Below, the implementation tiers of the VGG-sixteen version are defined step through step:

- Data preparation and preprocessing: To ensure compatibility with the model's input requirements, all images were resized to dimensions of 224 x 224 pixels.
- Data augmentation: To enhance the diversity of the training data and improve the model's generalization performance, the dataset was enlarged four times.

- Utilizing a pre-trained VGG-16 architecture: The VGG-16 model, previously trained on the ImageNet dataset, was employed to extract high-level image features.

- Customisation of the model: The pre-trained classification layers are removed and a customised classification layer suitable for our dataset is added. Since Parkinson's disorder is a binary category process, the very last output layer contains one neuron and the sigmoid activation feature is used. Instead of training all layers of the VGG-16 model, the model was trained more efficiently by focusing on the added special classification layer.

- Optimisation and training: Adam optimisation algorithm was used to train the model and binary cross-entropy was used as the loss function. With the designed model, 98.38% accuracy rate was achieved for MR data.

The same model was applied to our second dataset, Parkinson's disease hand drawings dataset. With this model, 98% accuracy rate was achieved for hand drawings data.

3.5. MobileNet Model

Another approach created the use of transfer learning withinside the early prognosis of Parkinson's ailment is the MobileNet version. Below, the tiers of the version are defined step through step:

- Data preparation and preprocessing: All photographs had been adjusted to a decision of 224 x 224 pixels to make sure compatibility with the model's enter requirements.

- Feature extraction : A pre-trained MobileNetV2 model was used using transfer learning. The MobileNetV2 model uses feature maps from the lower layers.

- Model customisation: The model starts by processing the feature maps obtained by transfer learning. In the first layer 512 neurons and ReLU, in the second layer 256 neurons and ReLU activation are used. In the last layer, classification is performed with one neuron and sigmoid activation. Optimisation and training: The model was built with the Adam optimisation algorithm and a binary cross-entropy loss function was used. It was trained on MR images for 10 steps and achieved 99% accuracy.

The same model was applied to our second dataset, the Parkinson's disease hand drawings dataset. After 10 training iterations on the hand-drawn images dataset, the model achieved a perfect accuracy score of 100%.

3.6. InceptionV3 Model

InceptionV3 model is another method created using transfer learning for early diagnosis of Parkinson's disease. Below, the stages of the model are explained step by step:

- Data preparation and preprocessing: : Images were converted to 224 x 224 pixels.
- Data augmentation: To enhance the diversity of the training data and improve the model's generalization performance, the dataset was scaled up fourfold.
- Customisation of the model: Since the InceptionV3 model classification layers were not suitable for the dataset, the layers were removed. A layer that smooths the feature maps was added and the average value of each map was used. A dense layer comprising 512 neurons and ReLU activation was incorporated, succeeded by another dense layer with 256 neurons. The architecture was finalized with an output layer utilizing sigmoid activation to perform binary classification.
- Optimisation and training: Training the model involved the Adam optimizer and the binary cross-entropy loss function, which is well-suited for binary classification tasks. The model was trained with Parkinson's MRI and hand drawing datasets and achieved 94.15% accuracy in 10 steps in MR images.

The same model was applied to our second dataset, the Parkinson's disease hand drawings dataset. The model was trained on the hand drawings data for 10 steps and an accuracy rate of 94.5% was achieved.

3.7. DenseNet201 Model

Another method created using transfer learning for early diagnosis of Parkinson's disease is the DenseNet201 model. Below, the stages of the model are explained step by step:

- Data preparation and preprocessing: The dataset was resized to a suitable size for processing by the model. The images were resized to 224 x 224 pixels.
- Data augmentation: To enhance the diversity of the training data and improve the model's ability to generalize, the dataset was enlarged four times its original size.
- Customisation of the model: The pre-trained DenseNet201 model was loaded and the classification layers were extracted. The feature maps were smoothed and averaged. The first dense layer has 512 neurons and the

second dense layer has 256 neurons to learn less complex features and prevent overlearning. In the last layer, binary classification is performed using a neuron with a sigmoid activation function.

- **Optimisation and training:** Using the Adam optimizer and binary cross-entropy loss function, the model underwent 10 training iterations, resulting in an accuracy of 95%.

The same model was applied to our second dataset, the Parkinson's disease hand drawings dataset. The model was trained for 10 steps and an accuracy rate of 95% was achieved for the hand drawing data.

4. Results

In this study, the problem of early detection of Parkinson's disease, a degenerative disorder of the nervous system that causes difficulty in movement control, tremors, muscle stiffness and loss of balance, is analysed. Parkinson's disorder is a tough hassle to diagnose and there's no acknowledged remedy technique yet. The most important contribution for this disease is early diagnosis. The difficulty of the problem is also recognised by doctors; so much so that the misdiagnosis rate of the disease is as high as 25%. In order to solve this problem, researchers have focused on this subject in recent years.

Especially today, artificial intelligence and deep learning strategies are an increasing number of getting used to diagnose and predict complicated neurological situations inclusive of Parkinson's disease. Artificial intelligence-based systems can detect Parkinson's disease in the early stages, slowing the progression of the disease and enabling early intervention. This plays a critical role in enhancing patients' quality of life and reducing treatment expenses.

This study is known as the first classification study with deep learning architectures over two different datasets to predict Parkinson's disease. The innovative approach of this study focuses on predicting Parkinson's disease not only from MRI images, but also from alternative data sources such as hand drawings. The reason for adopting this approach is the possibility that Parkinson's disease cannot be predicted from MR images alone. This research seeks to address a gap in the current literature by conducting a comparative analysis of Parkinson's disease across multiple datasets, rather than relying on a single dataset for evaluation.

For this purpose, in the first part of the study, customised models were created using deep learning models ResNet-50, VGG-16, MobileNet,

InceptionV3 and DenseNet201 transfer learning. For each of these customized models, the top layers were eliminated, and a binary classification layer was integrated into the final layer. These models were applied to both MR dataset and hand drawing dataset.

The performance outcomes of the deep learning models trained on the MR dataset are presented in Table 1. In all models, results were obtained at levels that can be called high success. MobileNet achieved the highest score with 99% accuracy. This result is expected when we consider the structure of the models. Because MobileNet has a lightweight structure compared to other models, contains fewer parameters and requires less computational power, which enables it to give good results especially in limited resources. MobileNet uses Depthwise separable convolution instead of conventional convolution. This feature allows MobileNet to learn more efficient features using fewer computational resources.

Table 1. Results of applying deep learning models to the MR dataset

Data set	Deep Learning Models	Accuracy Rates
MR Data Set	ResNet50	%98,14
	VGG-16	%98,38
	MobileNet	%99
	InceptionV3	%94,14
	DenseNet201	%95

Table 2 shows the results of the deep learning models trained on the hand drawing dataset. In general, more successful results were obtained than the results obtained from MRI images. A slight performance increase was observed in all models except the VGG-16 model. There may be various reasons why the performance of hand drawings is better than the datasets where the same models are applied one-to-one. Especially data quality may be an important factor here. Data set limits may also be a factor. The most important reason for the better results of the drawing images is the readability of the images. MR images are often low in contrast and can be complex, which is a factor affecting the analysis of the images. On the other hand, hand drawing images are more distinct and have higher contrast. This allows the model to obtain more accurate results. In the hand drawing dataset, MobilNet provided the best accuracy rate with 100%.

Table 2. Results of applying deep learning models to hand drawing dataset

Data set	Deep Learning Models	Accuracy Rates
Hand Drawings Data Set	ResNet50	%99
	VGG-16	%98
	MobileNet	%100
	InceptionV3	%94,1
	DenseNet201	%95

In contrast to previous research, this study assesses the effectiveness of deep learning algorithms using both MRI images and hand-drawn datasets.

In particular, the achievement of 100% accuracy rate in the hand drawings dataset shows how effective this method is in the diagnosis of Parkinson’s disease.

Consequently, deep learning models have demonstrated high accuracy in the early detection of Parkinson’s disease. This study confirms that artificial intelligence systems and classification methods can be used successfully in disease diagnosis. In particular, it has been observed that the image-based diagnosis method, in which the MobileNet model stands out with its high accuracy rate, can be used effectively. Future studies with more patient data will provide hope for people living with this disease. Unfortunately, there is no detailed dataset on Parkinson’s disease. In this study, different data sets were used, and such studies may be more effective for this disease, whose symptoms vary from person to person. In future studies, a hybrid research can be carried out with different architectures over different data sets. In future studies, the classification of images can be performed with the help of attributes to be obtained by using HOG, LBP, GLMC, SIFT, SURF and deep learning feature extraction methods, which are not included in our study. These features can be combined with various methods and comparisons can be made with new features. In addition, the advantages of ensemble learning techniques can be utilised by using models with different attributes together. Another suggestion is that if the data set can be obtained, studies showing the level of the disease can be carried out. In addition, biological biomarkers such as blood, saliva or cerebrospinal fluid of the patient, as well as data such as dopamine activity and specific gene mutation analysis can be included in the model.

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CHAPTER XII

DEEP LEARNING CLASSIFICATION OF NEURODEGENERATIVE DISEASES USING MAGNETIC RESONANCE IMAGING DATA

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1. Introduction

Neurodegenerative diseases encompass a range of disorders characterized by progressive degeneration or loss of neurons, leading to structural and functional impairment in the brain. These conditions result in significant cognitive deficits, motor dysfunction, and decreased overall quality of life. Alzheimer's Disease (AD), Parkinson's disease, Mild Cognitive Impairment (MCI), Huntington's disease, Amyotrophic Lateral Sclerosis (ALS), and Multiple Sclerosis (MS) are among these neurodegenerative disorders whose prevalence and mortality rates have dramatically increased in recent years. Early diagnosis of these diseases is crucial since late diagnosis severely limits the options available for halting or reversing disease progression, thus placing significant socioeconomic burdens on societies and national economies (Alsubaie et al., 2024; Huang et al., 2018). Among these diseases, AD and

MCI are particularly prevalent, often manifesting through gradual cognitive decline and eventual loss of independence. MCI, considered a precursor stage to Alzheimer's and other dementia-related conditions, is characterized by subtle yet measurable cognitive deficits. Although individuals with MCI may still perform daily tasks, approximately 30–40% will progress to AD within five years, highlighting the critical importance of early diagnosis and accurate classification of MCI stages.

Progress in medical research and healthcare has led to better health outcomes and extended life expectancy. As a result, the global population is projected to reach approximately 11.2 billion by 2100. Consequently, the elderly population is expected to constitute around 21% of the global population by 2050, representing nearly two billion individuals. This demographic shift has led to an increased prevalence of age-related diseases, particularly AD, which currently accounts for 60–80% of dementia cases globally. With increasing longevity, it is projected that AD will affect approximately 131.5 million people worldwide by 2050, becoming a significant socioeconomic burden across both developed and developing nations. Although the precise etiology and pathogenesis of AD and related disorders remain uncertain, advancements over the past three decades have substantially enhanced our understanding of the mechanisms underlying dementia and led to the development of novel diagnostic and therapeutic methods (Dong et al., 2019; Marzban et al., 2020). The global prevalence of AD is rapidly increasing due to an aging population, making it a major public health challenge. Recent data indicates that AD is the sixth leading cause of death in the United States, with its mortality rates rising substantially over the past two decades (Marzban et al., 2020). At a pathological level, AD is marked by the buildup of neurofibrillary tangles and amyloid plaques, which interfere with neuronal signaling, lead to nerve cell loss, and cause structural brain atrophy.

Neuroimaging, a discipline focused on examining the nervous system's structure, function, and pathological changes, utilizes non-invasive methods for brain imaging. Recent rapid advancements in imaging technologies have rendered neuroimaging techniques powerful tools in medical research and diagnosis. The increased prevalence of neurological diseases has fostered the development of new neuroimaging technologies and novel data analysis techniques to evaluate obtained images effectively (Zhang et al., 2020). While Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) are optimal for revealing structural brain changes, additional neuroimaging techniques such as functional MRI (fMRI), Positron Emission Tomography

(PET), and Diffusion Tensor Imaging (DTI) provide further insights into detailed disease assessment. In clinical settings, combining multiple imaging modalities is a common practice aimed at capturing richer features and thus enhancing diagnostic accuracy (Amini et al., 2021; Chen et al., 2021). This approach enables a more comprehensive understanding of structural, functional, and metabolic alterations in the brain, improving diagnostic accuracy and facilitating early intervention. MRI, in particular, plays a crucial role in neurodegenerative disease assessment by providing detailed structural and volumetric information, allowing clinicians to detect subtle brain atrophy and abnormalities associated with early disease stages. Its non-invasive nature, high resolution, and sensitivity to microstructural changes make it a fundamental tool in both clinical and research settings. However, despite the advantages of multimodal neuroimaging, integrating and analyzing data from multiple modalities presents significant challenges. Differences in resolution, acquisition protocols, and imaging artifacts can complicate data alignment and interpretation, while the high costs and logistical difficulties of multimodal data collection limit its accessibility in routine clinical practice.

Neuroimaging data is analyzed through a series of structured steps to extract meaningful information for clinical and research applications. This process typically includes preprocessing, where raw images undergo noise reduction, skull stripping, spatial normalization, co-registration, and segmentation to ensure consistency across datasets. Next, feature extraction techniques such as volumetric measurements, texture analysis, functional connectivity mapping, and tractography are applied to quantify brain structure and function. Finally, statistical analysis and classification methods, including voxel-based morphometry (VBM) and region-of-interest (ROI) analysis, are used to compare brain structures and identify disease-related patterns. Traditionally, neuroimaging analysis relies on manual feature selection and expert-driven decisions, making the process labor-intensive, time-consuming, and prone to human error. Variability in feature selection and differences in expertise can lead to inconsistencies, potentially affecting diagnostic reliability and reproducibility (Liu et al., 2018; Zaharchuk et al., 2018). To address these challenges, machine learning (ML)-based automated and semi-automated decision-support systems have been increasingly adopted. These approaches streamline neuroimaging analysis by automating feature extraction, improving classification accuracy, and reducing dependence on manual assessments, ultimately enhancing the consistency and efficiency of neuroimaging-based disease evaluation.

ML, a powerful subfield of artificial intelligence, enables automated analysis and interpretation of complex datasets. A typical ML pipeline consists of data preprocessing, feature extraction or selection, classification, and testing for evaluation. In traditional ML methods, domain experts manually determine relevant features to represent specific disorders, introducing potential biases and limiting reproducibility. While ML has shown success in detecting and classifying neurological disorders such as AD, MCI, epilepsy, schizophrenia, stroke, and autism, its dependence on handcrafted features remains a limitation (Liu et al., 2018; Zaharchuk et al., 2018).

Deep Learning (DL), a subset of ML, overcomes this limitation by automatically extracting disease-specific features directly from raw data, eliminating the need for manual feature engineering. Convolutional Neural Networks (CNNs), one of the most widely used DL architectures for medical image analysis, achieve high diagnostic accuracy by identifying complex spatial patterns within neuroimaging data. CNNs apply successive convolutional layers, pooling operations, and nonlinear activations to extract hierarchical representations of imaging data, improving diagnostic performance and reducing the need for manual intervention (Krizhevsky et al., 2012; LeCun et al., 2015; Yapici et al., 2022; Karakis et al., 2023). Recent studies have demonstrated that CNN-based models achieve superior classification accuracy in identifying neurological diseases, effectively capturing structural and functional alterations in the brain (Liu et al., 2018; Zaharchuk et al., 2018). Consequently, integrating DL into neuroimaging analysis enhances diagnostic efficiency, consistency, and clinical decision-making processes.

In this chapter, we aim to explore DL-based analysis methods for classifying AD and MCI using neuroimaging data. Given the limitations of traditional neuroimaging analysis and ML approaches, DL techniques, particularly CNNs and their variants, have demonstrated significant potential in improving classification accuracy and enabling automated feature extraction. This chapter is structured as follows: Section 2 provides an overview of DL techniques for AD analysis, emphasizing their role in neuroimaging-based classification. Section 3 describes the dataset, detailing the construction of the imaging dataset from the Alzheimer's Disease Neuroimaging Initiative (ADNI), along with preprocessing steps, and outlines the proposed methodology and experimental framework used in this chapter, including model architecture, training strategies, and evaluation metrics.

Section 4 presents the obtained results, including classification performance and comparative analyses. Finally, Section 5 critically discusses the findings, highlighting the advantages, limitations, and potential directions for DL-based AD and MCI classification.

2. Deep Learning Approaches for Alzheimer’s Disease Classification

The classification of AD and MCI using DL techniques follows a structured pipeline that involves several key stages, as given in Figure 1. First, neuroimaging data is acquired from various modalities, including MRI, DTI, PET, and fMRI, providing complementary structural, functional, and metabolic information about the brain. Second, preprocessing steps vary depending on the data type; 3D and multimodal imaging require skull stripping, intensity normalization, co-registration, and segmentation, while 2D image-based approaches typically involve filtering, contrast enhancement, and resizing to optimize input for DL models. Third, a DL model is developed, often based on CNNs, which are particularly effective for analyzing medical images. Fourth, the dataset is split into training, validation, and test sets, and the model is trained using labeled neuroimaging data, where optimization techniques such as backpropagation and gradient descent are used to fine-tune network parameters. Fifth, a validation process is conducted to ensure that the model generalizes well to unseen data, addressing potential issues like overfitting by using techniques such as cross-validation and hyperparameter tuning. If necessary, the model is retrained with adjusted parameters to improve performance. Finally, once the model achieves satisfactory accuracy and robustness, it is tested on independent datasets before being deployed in clinical settings, where it serves as a decision-support tool for medical experts in diagnosing and monitoring AD and MCI.

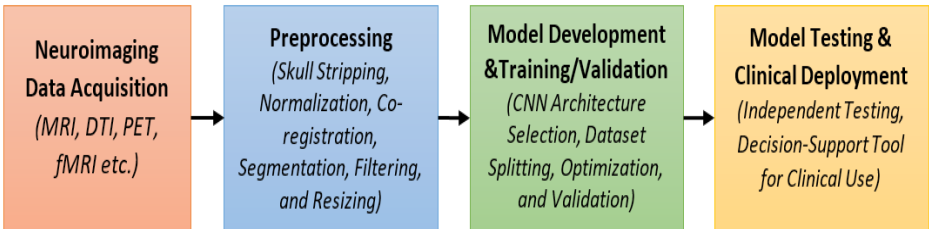


Figure 1. Pipeline for the classification of AD and MCI using the DL techniques.

2.1. CNN Analysis of Single-Modality Data and Transfer Learning Approaches

Single-modality CNN-based approaches utilize neuroimaging data from a single imaging technique, predominantly MRI, to classify AD and MCI (Ashraf et al. 2021; Savaş et al. 2022). These methods often leverage transfer learning by adapting pre-trained CNN architectures to address challenges such as limited dataset sizes, computational complexity, and overfitting, thus enhancing performance and generalizability. Recent studies have demonstrated the effectiveness of transfer learning-based CNN architectures in distinguishing AD stages.

In AD classification tasks, these CNN architectures process MRI images through multiple convolutional layers, where filters extract key spatial patterns, such as brain atrophy, ventricular enlargement, and cortical thinning—hallmarks of neurodegenerative diseases. The extracted feature maps undergo pooling operations to reduce dimensionality while retaining essential information. The final feature maps are then flattened and passed through fully connected layers to generate classification probabilities. When transfer learning is applied, the pre-trained CNN weights from ImageNet are fine-tuned on AD-specific MRI data, allowing the model to learn disease-specific features while retaining the general feature extraction capabilities learned from large-scale datasets.

Pre-trained CNN architectures such as VGG (Simonyan and Zisserman, 2014), ResNet (He et al., 2016), DenseNet (Huang et al., 2017), Inception (Szegedy et al., 2016), and EfficientNet (Tan and Le, 2019) have been widely utilized for AD and MCI classification as they can effectively learn hierarchical features from neuroimaging data. These models follow a structured pipeline where input MRI scans are first processed through convolutional layers to extract spatial and textural features, followed by pooling operations to reduce dimensionality and computational cost. The extracted features are then passed through fully connected layers for classification. When combined with transfer learning, these architectures leverage weights pre-trained on large-scale datasets like ImageNet, which enables them to generalize better, even with limited neuroimaging data.

Among these architectures, VGG16 and VGG19 employ small 3×3 convolutional filters stacked sequentially, preserving spatial details while using max pooling layers to progressively reduce feature map dimensions (Simonyan and Zisserman, 2014). However, their deep architecture results in a high number of parameters, increasing computational cost. In contrast, ResNet, particularly ResNet50, introduces residual connections that help mitigate the vanishing gradient problem, making it more effective for training deep networks. These

residual connections improve gradient flow during backpropagation, allowing ResNet to learn complex spatial relationships in neuroimaging data more efficiently (He et al., 2016).

Another widely used model, DenseNet, improves feature propagation by introducing dense connectivity, where each layer receives input from all preceding layers (Huang et al., 2017). This approach enhances feature reuse and reduces the number of parameters required, making it an efficient choice for AD classification. However, as the network depth increases, computational cost and memory usage can become challenging. To address these limitations, EfficientNet introduces compound scaling, which optimally balances the depth, width, and resolution of the network. This strategy allows EfficientNet to achieve high accuracy while maintaining lower computational demands compared to traditional deep CNN architectures (Tan and Le, 2019). However, the performance of transfer learning-based CNN architectures depends on dataset characteristics, imaging modalities, and classification tasks. Model selection should be guided by experimental results rather than predefined architectural advantages.

For instance, Savaş et al. (2022) collected a dataset comprising 4306 T1-weighted sagittal MRI images from the ADNI database, including AD, MCI, and cognitively normal (CN) cases, and reported that EfficientNet models, particularly EfficientNetB0 and EfficientNetB3, yielded the highest classification accuracies, highlighting their practical applicability in clinical diagnosis. Similarly, a recent review emphasized that CNN models combined with transfer learning methods consistently outperform traditional ML approaches by automatically capturing hierarchical features from neuroimaging data, thereby reducing dependence on manual feature selection and improving diagnostic accuracy and reliability, especially in scenarios with limited data (Suresha et al., 2023).

Building on these findings, Arafa et al. (2024) applied transfer learning to classify AD and CN using MRI scans, further reinforcing the effectiveness of pre-trained CNN architectures. Their study utilized VGG16 fine-tuned on the Kaggle Alzheimer's classification dataset, which contained 5121 training and 1279 testing MRI scans, achieving 97.44% accuracy in AD classification. To enhance model generalization, data augmentation techniques, including rotation, width and height shifts, shearing, horizontal flip, and zooming, were applied. The results aligned with Savaş et al. (2022) in demonstrating that transfer learning significantly enhances classification performance, particularly in datasets with a limited number of labeled MRI scans.

Mahmud et al. (2024) and Sertkaya et al. (2024) both explored DL-based approaches for AD classification using transfer learning, leveraging pre-trained VGG architectures on a publicly available OASIS-2 (Open Access Series of Imaging Studies-2) dataset of 6400 MRI images across four diagnostic categories: Non-Dementia (3200), Very Mild Dementia (2240), Mild Dementia (896), and Moderate Dementia (64). Sertkaya et al. (2024) improved VGG16 by freezing its learned weights and extracting multichannel features, which were then refined using the Minimum Redundancy Maximum Relevance (MRMR) algorithm for feature selection. This enhancement led to an improved classification accuracy of 98.6%, outperforming the baseline VGG16 model. Similarly, Mahmud et al. (2024) applied transfer learning with ensemble models, combining VGG16, VGG19, DenseNet169, and DenseNet201, and achieved 96% accuracy. Both studies employed data augmentation (e.g., rotation, translation, contrast adjustment, brightness modification, and cropping in Sertkaya et al, 2024; horizontal/vertical shifting and flipping, rotation, zooming, brightness modification, etc. in Mahmud et al, 2024) to compensate for class imbalance, particularly addressing the severe underrepresentation of the Moderate Dementia (MD) class. However, while data augmentation increases sample count, it does not introduce true pathological diversity, which may limit the validity and clinical reliability of these models, particularly in real-world AD classification scenarios.

Ahmet et al. (2024) developed a custom CNN model for AD classification using MRI scans from the ADNI dataset. Their model aimed to classify three diagnostic categories: AD, MCI, and CN. The dataset included 199 subjects, with 42 AD, 97 MCI, and 60 CN cases. They evaluated different train-test splits, achieving high accuracy rates in binary classifications (100% for AD vs. CN, 92.93% for CN vs. MCI, and 99.21% for AD vs. MCI). However, when applied to a three-class classification (AD vs. MCI vs. CN), the accuracy dropped to 93.86%, highlighting the increased complexity of multi-class classification. Notably, this performance was achieved without transfer learning or data augmentation.

Ashraf et al. (2021) further reinforced the effectiveness of transfer learning-based CNN architectures in AD classification by comparing multiple pre-trained models, including DenseNet, ResNet, and Inception, on MRI scans from the ADNI dataset comprising 379 subjects (94 AD, 138 MCI, and 146 CN). Their study demonstrated that DenseNet achieved the highest three-class classification accuracy (99.05%), followed by ResNet50 (94.09%) and Inception (94.78%).

Data augmentation techniques (flipping, rotation, illumination, and zoom in/out) were employed to expand the limited original dataset, emphasizing again that transfer learning and data augmentation significantly enhance classification performance, even in complex multi-class scenarios.

In summary, recent single-modality CNN studies predominantly utilize MRI datasets (e.g., ADNI and OASIS) to classify AD, MCI, and CN cases. To mitigate class imbalance, which is common in publicly available datasets, data augmentation techniques such as rotation, translation, flipping, illumination adjustment, and zooming have frequently been employed. Moreover, leveraging transfer learning-based CNN architectures, particularly DenseNet, EfficientNet, and ResNet models, has consistently improved classification accuracy by effectively addressing overfitting issues and capturing meaningful hierarchical features. Therefore, employing transfer learning methods combined with appropriate data augmentation techniques is recommended to achieve robust and clinically applicable performance in multi-class AD and MCI classification tasks.

2.2. CNN Analysis of Multimodal Imaging Data

Several studies have explored ML and DL approaches for AD and MCI classification using neuroimaging data. Research findings suggest that models incorporating multimodal imaging data (MRI, DTI, PET) tend to achieve higher classification accuracy compared to single-modality approaches, as they provide complementary structural, functional, and metabolic insights into brain degeneration (Liu et al., 2018; Marzban et al., 2020; Agostinho et al., 2021). For instance, Agostinho et al. (2021) reported classification accuracies of 0.98 and 0.97 when combining MRI, DTI, and PET features, reinforcing the importance of multimodal integration. However, integrating multimodal data poses several challenges. First, aligning, normalizing, and preprocessing images from different modalities is complex, requiring sophisticated data harmonization techniques to ensure compatibility. Second, collecting multimodal datasets is difficult in clinical practice, as not all patients undergo every imaging modality, leading to missing data and reduced dataset uniformity. Third, the computational cost is significantly higher, especially for 3D CNN-based models, which require larger memory and longer training times due to the increased number of parameters. Despite these challenges, multimodal and 3D imaging-based approaches remain promising, particularly for early-stage MCI detection, where single-modality models often struggle with borderline cases (Marzban et al., 2020; Wen et al., 2018).

Traditional ML models, such as Support Vector Machines (SVM) and Naïve Bayes (NB), have been widely used for AD classification. Dyrba et al. (2013) utilized Fractional Anisotropy (FA) and Mean Diffusivity (MD) maps extracted from DTI images as input for SVM and NB models, achieving accuracy scores ranging from 0.68 to 0.83. Similarly, Douaud et al. (2013) demonstrated that early MCI-related anomalies could be detected using volumetric MRI features and cerebrospinal fluid-based regions of interest (ROIs). However, these models struggle to capture complex spatial relationships in neuroimaging data, as they depend on manually selected features, which can introduce bias and limit generalizability, making them less effective compared to deep learning approaches that learn hierarchical representations directly from raw images.

More recent approaches have leveraged DL techniques, particularly CNN-based architectures, to extract hierarchical features directly from neuroimaging data. Payan et al. (2015) compared 2D and 3D CNN models, showing that 3D CNNs outperform their 2D counterparts in AD/MCI classification tasks due to their ability to preserve volumetric spatial information. Khvostikov et al. (2018) trained CNNs using MRI and DTI-derived ROIs, achieving accuracy scores of 0.97, 0.80, and 0.66 for AD vs. CN, AD vs. MCI, and MCI vs. CN classification, respectively.

A major challenge in AD classification is the presence of difficult cases, particularly in distinguishing MCI from AD and CN. Binary classification models (AD vs. CN) tend to achieve higher accuracy as they exclude ambiguous intermediate cases. However, when MCI is included, classification performance decreases, as early-stage MCI cases may resemble healthy controls, while late-stage MCI may appear similar to AD (Liu et al., 2018; Wen et al., 2018). This challenge has driven the adoption of multimodal learning approaches, where integrating MRI and DTI has been shown to reduce misclassification errors and improve generalization (Marzban et al., 2020).

In conclusion, multimodal imaging methods that integrate MRI, DTI, and PET aim to provide complementary information, potentially enhancing classification accuracy for AD and MCI compared to single-modality techniques. However, these methods present several challenges, including increased computational complexity, data harmonization difficulties, and the practical constraints of obtaining complete multimodal datasets from patients. Despite these limitations, multimodal deep learning approaches, particularly those employing 3D CNN architectures, have been explored for their potential to capture complex neurodegenerative patterns, improve diagnostic reliability, and better distinguish subtle differences among AD, MCI, and CN cases.

3. Experimental Study on Deep Learning

This section presents the dataset used in the study, the CNN-based transfer learning methods applied for classification, and the performance metrics used to evaluate the models. The dataset preparation process, including preprocessing steps and multi-view data generation, is described in detail. Additionally, the performance of different CNN architectures is analyzed to assess their effectiveness in the multi-class classification of neurodegenerative disease stages.

3.1. Dataset

In this chapter, an experimental study was conducted using a five-class dataset obtained from the ADNI (<https://adni.loni.usc.edu/>). ADNI is a large-scale research initiative aimed at improving clinical trials for the prevention and treatment of AD. It compiles neuroimaging, biochemical, and genetic biomarkers from individuals aged 55-90, covering AD, Early MCI (EMCI), MCI, Late MCI (LMCI), and CN cases. The dataset is collected from 63 research centers across Canada and the United States and is accessible to registered researchers for academic purposes, with attribution to ADNI in published works. The dataset details are provided in Table 1. The images were acquired using T1-weighted MPRAGE MRI scans, with individuals being followed from 2010 to 2024, meaning multiple scans were available for the same subjects. As part of preprocessing, the downloaded MRI scans were first stored in 3D NIfTI (*.nii) format to preserve volumetric information. Following this, 2D slice extraction was performed for each subject across different anatomical views. Specifically, for axial slices, between 90 and 120 slices were extracted per subject, while for sagittal and coronal views, 60-120 and 90-170 slices per subject were extracted, respectively. This step ensured a standardized input format for the DL models while retaining critical anatomical details from different orientations. As shown in Table 1, this process resulted in a large and diverse dataset, incorporating multiple slices per subject across different anatomical planes. Consequently, three separate datasets were created, each corresponding to one of the three imaging views (axial, sagittal, and coronal), allowing for a comprehensive evaluation of different perspectives in AD and MCI classification. In addition, Figure 2 presents representative MRI scans from the dataset across sagittal, coronal, and axial views. In this study, separate datasets were prepared for each view, with all three containing images from the five diagnostic categories, including CN, EMCI, MCI, LMCI, and AD.

Table 1. Dataset Properties

Image View	AD	EMCI	MCI	LMCI	CN	Overall
Axial	18476	19220	19840	19530	18383	95449
Coronel	48276	50220	51840	51030	48033	249399
Sagittal	36356	37820	39040	38430	36173	187819

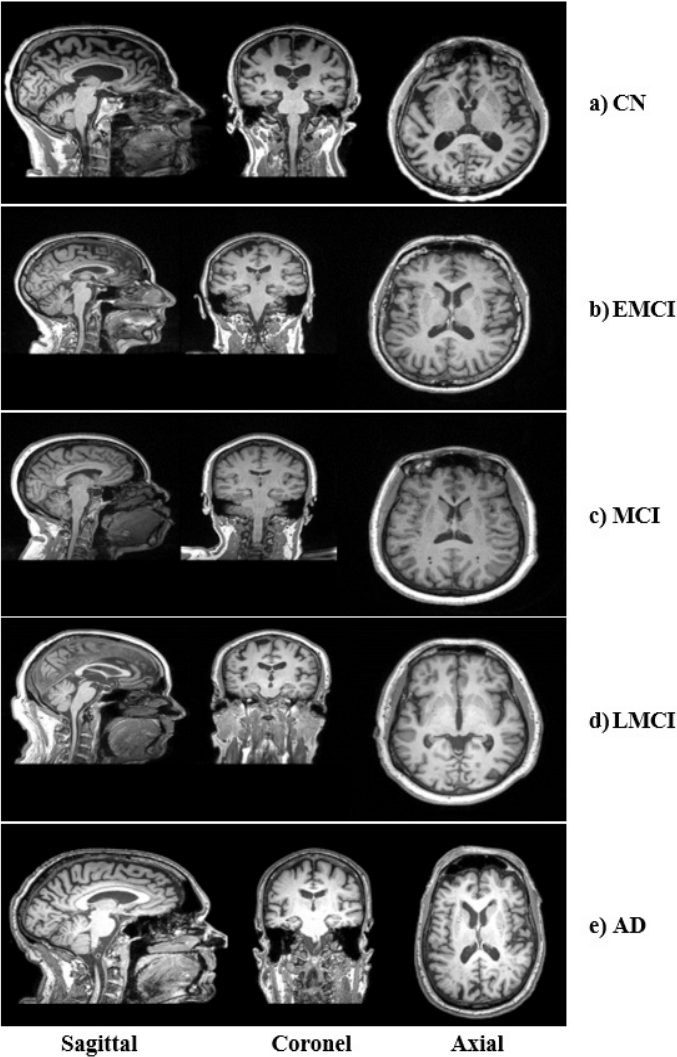


Figure 2. Representative MRI scans from ADNI, showing CN, EMCI, MCI, LMCI, and AD across sagittal, coronal, and axial views.

3.2. Deep Learning Model

In this study, ResNet50, DenseNet121, and InceptionV3 were selected for CN, EMCI, LMCI and AD classification based on preliminary analyses, which demonstrated their effectiveness in handling neuroimaging data. ResNet50 was chosen for its residual connections, which mitigate the vanishing gradient problem and improve training efficiency in deep networks. DenseNet121 was utilized for its dense connectivity, which enhances feature reuse and reduces the number of parameters required, making it an efficient choice for complex classification tasks. InceptionV3 was included due to its multi-scale feature extraction capability, allowing it to capture both fine and coarse-grained structural patterns within MRI scans, improving classification robustness.

The DL models were implemented using the Keras and TensorFlow libraries in Python 3.9. The training and evaluation were conducted on an Intel i9-12900KS @ 3.40 GHz CPU, NVIDIA RTX A6000 48 GB GPU, and 64 GB RAM. The dataset was split into training, validation, and test sets with ratios of 70%, 10%, and 20%. The input images were resized to 256×256 pixels before being fed into the models. The feature extraction process involved Global Average Pooling (GAP), followed by two fully connected layers (256 and 128 neurons) with ReLU activation. The final classification layer used the softmax activation function to predict one of the five diagnostic categories. The models were optimized using the ADAM optimizer with categorical cross-entropy as the loss function. Dropout and L2 regularization were applied to mitigate overfitting. The momentum coefficient, learning rate, weight decay and dropout were set to 0.9, 0.0001, 0.00001, and 40%, respectively. The training was performed for 50 epochs with a batch size of 128.

3.3. Performance Metrics

The classification performance of the CNN models was evaluated using accuracy, specificity, precision, recall, and F1-score metrics, calculated based on the confusion matrix. The matrix categorizes predictions into True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) to assess model reliability.

Accuracy represents the overall correctness of the model by measuring the proportion of correctly classified cases (Eq.1):

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (1)$$

Specificity evaluates how well the model identifies non-positive cases (Eq.2):

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) \quad (2)$$

Precision calculates the proportion of correctly identified positives among all predicted positives (Eq.3):

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (3)$$

Recall (sensitivity) measures how well the model detects actual positive cases (Eq.4):

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (4)$$

F1-score balances precision and recall, making it useful for handling class imbalances (Eq.5):

$$\text{F1} = 2(\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (5)$$

These metrics provide a comprehensive evaluation of the model's ability to classify different MCI and AD stages effectively.

3.4. Results & Discussion

Table 2 demonstrates that the classification performance of DenseNet121, InceptionV3, and ResNet50 is consistently high across Axial, Coronal, and Sagittal views, with accuracy values ranging from 0.9158 to 0.9246. Among the models, InceptionV3 and ResNet50 achieve the highest accuracy, reaching 0.9246 in the Sagittal view, while DenseNet121 maintains stable performance across all views. Specificity, recall, precision, and F1-scores remain above 0.91, indicating reliable classification. The results suggest that all models perform similarly, with slight variations in different views, and no single architecture or view significantly outperforms the others. The sagittal view tends to yield slightly better accuracy, which may indicate that certain anatomical features are better captured in this orientation.

Figure 3 presents the normalized confusion matrices for DenseNet121, InceptionV3, and ResNet50 across axial, coronal, and sagittal MRI views. The classification results for the five diagnostic categories reveal that while the models perform well in distinguishing MCI, LMCI, and AD, they consistently struggle to differentiate CN and EMCI, leading to higher misclassification rates between these two classes. Among the tested models, InceptionV3 achieves the highest classification performance in the coronal view, with accuracy exceeding

0.72 for all classes. However, the overall results indicate that no single model or MRI view consistently outperforms the others across all diagnostic categories. This suggests that integrating multiple views or leveraging multimodal data may further enhance classification robustness, particularly in distinguishing early-stage neurodegenerative changes.

Table 2. Classification Performance Metrics of the DL Models across Three Views.

View	DL Model	Accuracy	Specificity	Recall	Precision	F1-score
Axial	Densenet121	0.9206	0.9801	0.9178	0.9388	0.9160
	InceptionV3	0.9212	0.9805	0.9216	0.9382	0.9178
	ResNet50	0.9158	0.9789	0.9131	0.9287	0.9116
Coronel	Densenet121	0.9206	0.9801	0.9178	0.9388	0.9160
	InceptionV3	0.9219	0.9805	0.9200	0.9238	0.9199
	ResNet50	0.9220	0.9805	0.9192	0.9386	0.9177
Sagittal	Densenet121	0.9236	0.9811	0.9241	0.9423	0.9204
	InceptionV3	0.9246	0.9811	0.9218	0.9431	0.9201
	ResNet50	0.9239	0.9811	0.9244	0.9423	0.9206

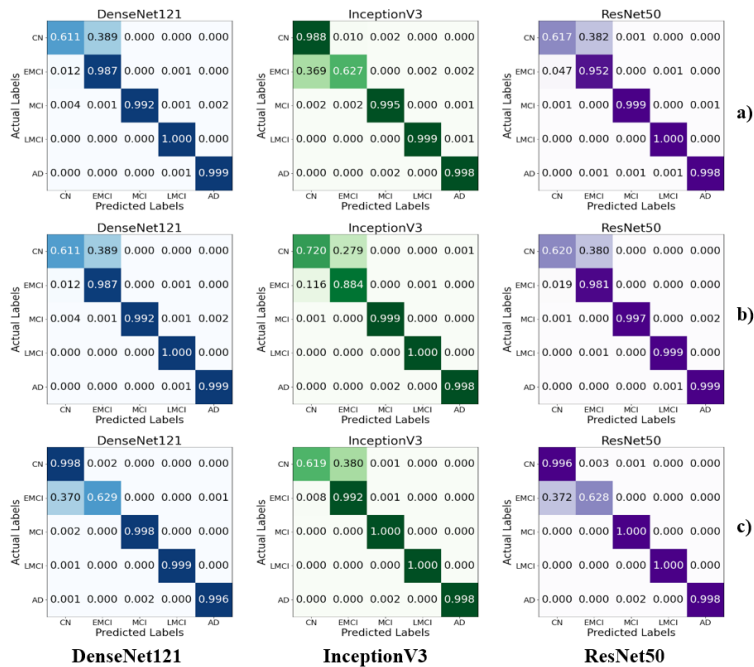


Figure 3. Normalized Confusion Matrices for DL Models across Different MRI Views: (a) Axial, (b) Coronal, and (c) Sagittal.

Our experimental analysis aligns with previous studies, demonstrating the effectiveness of transfer learning-based CNN models in AD and MCI classification. Similar to prior research, our results confirm that pre-trained architectures such as DenseNet121, ResNet50, and InceptionV3 provide reliable classification performance, achieving an overall accuracy of approximately 0.92. While data augmentation has been widely used in previous works to enhance generalization, our study did not employ augmentation techniques due to the large sample size and balanced class distribution in our dataset. However, given its benefits in improving robustness, data augmentation remains a valuable technique for future studies.

A key challenge observed in our findings, as in previous studies, is the misclassification between CN and EMCI, which suggests that additional strategies may be needed to improve the distinction between early disease stages. Addressing this issue could involve integrating multimodal neuroimaging data (e.g., combining MRI with DTI or PET) to capture complementary features. Furthermore, ensemble learning approaches that combine multiple CNN architectures or feature fusion techniques may enhance classification robustness. Another potential direction is the exploration of 3D CNN models, which could better preserve spatial information across slices and provide improved feature representation. Future studies should consider these advanced methodologies to further refine classification accuracy, particularly in distinguishing closely related diagnostic categories.

4. Conclusion

This chapter presents deep learning-based approaches for the multi-class classification of AD and MCI using neuroimaging data. Both single-modality and multimodal imaging strategies are discussed, highlighting their advantages and challenges in AD diagnosis. Transfer learning with pre-trained CNN architectures is examined as an effective method for improving classification performance, particularly when using single-modality MRI scans. A review of existing studies provides insight into how different architectures and imaging modalities contribute to AD/MCI classification.

In addition to the literature review, an experimental study presents a comparative analysis of three widely used CNN models—ResNet50, DenseNet121, and InceptionV3—across different MRI views. The results provide an evaluation of model performance, demonstrating the impact of transfer learning in neuroimaging-based classification. While classification

accuracy remains high, challenges persist in differentiating early-stage MCI from CN cases. Future studies may explore multimodal data integration, ensemble models, or 3D CNN architectures to enhance classification robustness. This chapter contributes to the understanding of the DL applications in neurodegenerative disease diagnosis.

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CHAPTER XIII

A HYBRID FEATURE SELECTION APPROACH FOR IDENTIFYING OBESITY- RELATED TAXONOMIC BIOMARKERS

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1. Introduction

In the past four decades, a rapid increase in the number of obese individuals has been observed worldwide (Şahin T. et al., 2022). Global research indicates that obesity is expected to affect more than 20% of men and 18% of women in the coming years (Lobstein T. et al., 2022). Nowadays, the increasing prevalence of obesity poses a significant global challenge to healthcare systems (Puljiz Z. et al., 2023).

Unhealthy eating habits, genetic predisposition, and insufficient physical activity are among the primary etiological factors of obesity (Yetkin İ. et al., 2017). Moreover, environmental factors such as stress, inadequate sleep duration, and gut microbiota are also noted to play a role in the development of obesity (Durmaz B., 2019). In particular, studies conducted after 2010 have demonstrated that microorganisms in the human body exist in a specific balance,

and this balance has significant effects on the individual's health (Yetkin İ. et al., 2017). In addition to obesity, many other diseases have been reported to be associated with the disruption of this microbial balance (Vamanu E. et al., 2021). Studies on the microbiota have revealed that gut permeability, bile acids, and the production of bioactive compounds such as short-chain fatty acids are regulated by the microbiota, and these mechanisms could trigger the development of obesity (Overby HB et al., 2021). Furthermore, there is an increasing body of research on the potential role of gut microbiota in the treatment of obesity (Angelidi AM et al., 2022). In recent years, despite the use of metagenomic data in various studies aimed at identifying obesity-specific biomarkers, the effects of gut microbiota on obesity are still considered an active area of research. Metagenomic data obtained from patients and controls can be downloaded from published studies, and the analysis of these datasets can facilitate the identification of potential obesity biomarkers (Xu C. et al., 2023).

Advancements in next-generation sequencing technologies have significantly contributed to the digitalization of biology by enabling large-scale production of biological data. In parallel with these developments, innovations in bioinformatics have facilitated the analysis of large biological datasets, allowing for detailed characterization of the human gut microbiome and the identification of potential biomarkers for disease diagnosis (Xu C. et al., 2023). Research has revealed that microorganisms in the gut play critical roles in immune response, DNA damage, and cell proliferation (Şahin T. et al., 2022). The active interaction of the gut microbiota with human cells has emphasized the importance of microbiota analysis in the context of obesity (Puljiz Z. et al., 2023). In this regard, the gut microbiome is considered a guiding factor and a potential source of biomarkers for obesity diagnosis (Napolitano M. et al., 2020). Studies based on biological experiments in this field have suggested that certain species belonging to genera such as *Faecalibacterium*, *Prevotella*, *Ruminococcus*, *Fusobacterium*, *Blautia*, *Dialister*, *Bacteroides*, *Parvimonas*, and *Akkermansia* may play a role in the development of obesity (Xu Z. et al., 2022). However, the same review study points out inconsistencies in the findings of previous research and states that the key microorganisms associated with obesity have not yet been clearly defined. The complex nature of metagenomic studies has made the use of artificial intelligence-based analysis methods increasingly important in this field (Liu Y., 2022). In this context, several studies noted that by selecting taxonomic features, meaningful relationships can be established between microbiota profiles and disease conditions, enabling more effective identification of biomarkers (Chanda D. et al., 2022).

Various features can be derived for metagenomic disease prediction, where one of the most commonly used features is the measurement of bacterial abundance values in samples (Liu Y. et al., 2022) (Bakir-Gungor et al., 2023). Sequences obtained from metagenomic analyses provide fundamental information about which organisms are present in the sample and the extent of representation of each organism. These data can be used as features in machine learning models. Feature selection plays a critical role in disease prediction processes, particularly in identifying significant bacterial species and eliminating features that do not contain meaningful information for classification, which can finally improve the accuracy of the model. A recent review study (Marcos-Zambrano LJ. et al., 2021) reported that different feature selection techniques have strengthened the disease diagnosis models generated from human gut metagenomic data. However, the study does not provide a definitive recommendation on which method should be preferred. Developing models capable of precise classification could enable their effective use in disease diagnosis. Recently, Marcos-Zambrano LJ. et al. (2021) analyzed 89 studies and examined different machine learning techniques used in microbiome data classification. This review reported that supervised learning algorithms are widely used in microbiota analysis, with Random Forest (RF), Support Vector Machines (SVM), Logistic Regression (LR), and k-Nearest Neighbor (kNN) methods standing out. In machine learning-based metagenomic analyses, multiple factors, such as class imbalance, feature diversity, and data quality, must be considered. To improve model performance, it is crucial to comparatively test different methods and identify the most suitable algorithm.

To identify disease-causing microorganisms associated with obesity, this study proposes to use different feature selection methods including two Recursive Cluster Elimination based algorithms (SVM-RCE and RCE-IFE), and three traditional feature selection algorithms including Select K Best, XGBoost and Information Gain, along with different classification algorithms. The objective of this study is to achieve higher machine learning performance using fewer metagenomic features and to provide superior classification performance by reducing computational costs. In this study, the performance of different approaches is compared using metagenomic data related to obesity. The model developed using feature selection methods aims to contribute to diagnosis by minimizing the microbiota components responsible for obesity. For this purpose, disease prediction is performed using common features identified by feature selection methods, and the obtained performance values are investigated. This study, which is crucial for the diagnosis and early detection of obesity, is expected to serve as a guide for obesity researchers. With the goal of

further improving the performance values for obesity diagnosis, future studies can focus on the use of deep learning algorithms applied on the metagenomic data gathered from the literature.

2. Materials and Methods

The dataset used in this study was created by Gupta et al. (2020) and consists of 903 different microorganisms measured for 2978 samples. Initially, microorganism data associated with obesity was obtained from various data sources and subjected to preprocessing. During this process, missing or erroneous data were corrected, the data was normalized, and it was made suitable for analysis. In this study, to identify obesity-related disease-causing microorganisms, we propose to use two Recursive Cluster Elimination based algorithms (SVM-RCE and RCE-IFE), and three traditional feature selection algorithms including Select K Best, XGBoost and Information Gain, combined with different classifiers.

2.1. SVM-RCE (*Support Vector Machine – Recursive Cluster Elimination*)

Yousef et al. introduced the “Recursive Cluster Elimination (RCE)” technique to the literature via combining it with a Support Vector Machine (SVM) classifier (Yousef et al., 2007). In the SVM-RCE method, the data is initially split into training and test sets. Then, the first 1000 features are selected using the t-test method and clustered using the K-means algorithm. Studies have shown that the K-means algorithm performs better in the SVM-RCE method compared to other clustering algorithms (Kuzudisli et al., 2023). After clustering, these clusters are scored based on their classification performance using SVM, and low-scoring clusters are eliminated at a predetermined rate. The remaining features within the clusters are combined, and the model’s performance is evaluated using the test data initially set aside. The clustering and subsequent steps are repeated until the desired number of clusters is reached, providing performance evaluations for different cluster sizes.

2.2. RCE-IFE (*Recursive Cluster Elimination with Intra-cluster Feature Elimination*)

RCE-IFE (Kuzudisli et al., 2025) is an extension of the SVM-RCE method, where feature elimination is performed after cluster elimination. In other words, RCE-IFE simultaneously and recursively applies both cluster

elimination and feature elimination. In this method, the Random Forest algorithm is used to score the clusters and the features within the remaining clusters. Similar to the SVM-RCE method, the samples are initially split into training and test data, and the features are clustered using the K-means algorithm. However, after scoring the clusters and eliminating low-scoring clusters, low-scoring features within each cluster are removed at a certain rate. This feature extraction process is performed after each clustering step. This additional step ensures that not only are the clusters eliminated, but noisy or irrelevant features within the remaining clusters are also cleaned, enhancing the quality of the model's input data.

2.3. The Proposed Method

Firstly, the performances of the SVM-RCE and RCE-IFE algorithms were examined. Secondly, to diagnose obesity and identify bacterial biomarkers that could contribute to obesity, three different feature selection methods (Information Gain, Select K Best, and XGBoost) and five different classification methods (AdaBoost, LogitBoost, Decision Tree, Random Forest, and XGBoost) were combined and the performance of the proposed approach is measured.

The proposed approach focuses on the features having a scaled importance value ≥ 0.5 , computed by the feature selection methods. This approach reduces the number of features and aims to achieve superior performance with fewer features. Machine learning models are retrained and tested using the intersection features. The results obtained were thoroughly evaluated from both biological and statistical perspectives, and the compatibility of the findings with existing literature was discussed. In this way, a model was developed for identifying bacterial biomarkers associated with obesity and contributing to the diagnostic process. The methods used in this study and the proposed approach are shown in Figure 1.

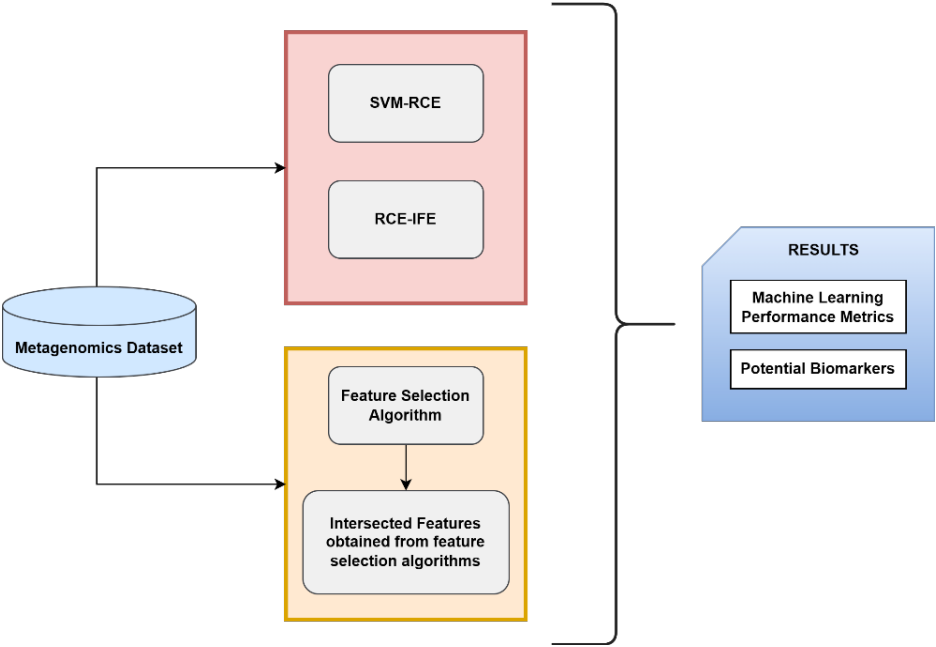


Figure 1. Workflow

3. Results

Recently the imbalance of gut microbiota has been associated with obesity (Erkul C. et al., 2020). Therefore, the biological species found in the human gut microbiome could serve as biomarkers for obesity diagnosis. This study analyzes an obesity-related metagenomic dataset containing relative abundance values for 903 different species and 2978 samples, aiming to identify microorganisms with predictive capabilities for obesity. Two RCE based algorithms (SVM-RCE, RCE-IFE) and three feature selection algorithms were used to separate irrelevant features (microorganisms) and to identify discriminative features (microorganisms). The dataset was split into 80%-20% for training and testing, respectively. The algorithm was run 10 times, and the average results and standard deviation values were recorded. The Information Gain, Select K Best, and XGBoost feature selection algorithms were run in conjunction with the LogitBoost, Decision Tree, AdaBoost, Random Forest, and XGBoost classification algorithms to identify the top 100 important features. In this phase, when each feature selection algorithm was applied with different classifiers, the importance value for each biological species was calculated. Features (biological species) with an importance value greater than the 0.5 threshold were selected. As displayed in Figure 2, when such

a threshold is applied, SKB, XGBoost, and IG algorithms detected 10, 15, and 32 features, respectively. As shown in Figure 2, 11 species represent the total of binary intersections of feature selection methods. The performance of these 11 features are also assessed for diagnosing obesity.

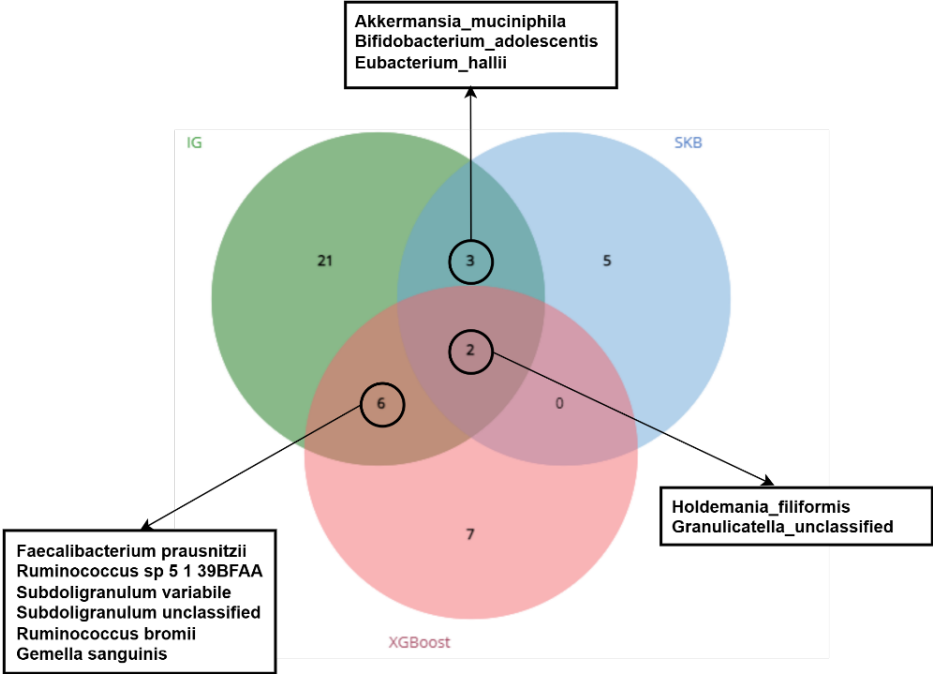


Figure 2. The identified microorganisms having scaled importance score greater than 0.5

Figure 3 and Table 1 shows the AUC metrics obtained using all features (903 species), the top 100 features using SKB, IG, and XGBoost methods, 11 features that are selected by these feature selection methods with an importance value higher than the 0.5 threshold and identified by at least two methods. Upon reviewing these results, it is evident that the performance results obtained using the XGBoost feature selection method outperform those from other approaches. Combining the XGBoost feature selection method with the XGBoost classification algorithm resulted in the highest AUC metric (0.89) using 100 features. A careful examination of the metrics in Figure 3 and Table 1 indicates that the performance results obtained using all features are approximately similar to those obtained using intersection features (except for the Random Forest and XGBoost algorithms). While intersection features consist of only 11 features,

all features include 903 distinct species. The similarity in machine learning performance between the results obtained using all features and those using only the 11 intersection features emphasizes the significance of these 11 species.

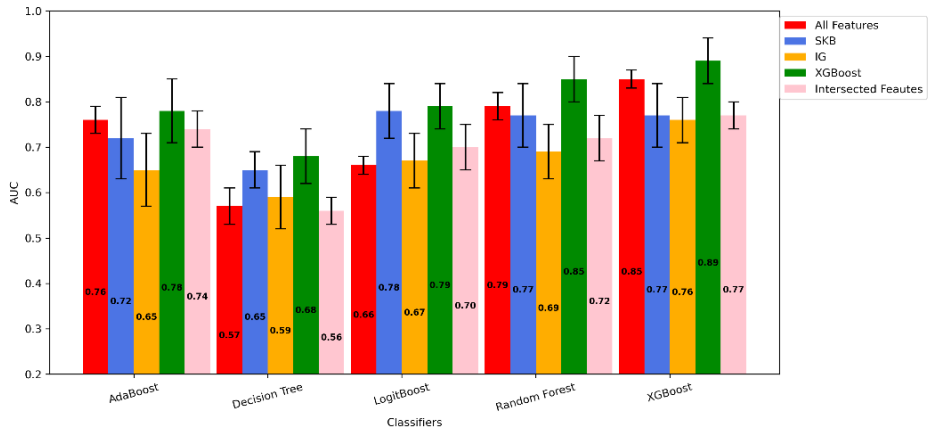


Figure 3. AUC metrics obtained using different feature selection algorithms and different classifiers applied on obesity-associated metagenomics dataset.

Additionally, various metrics such as accuracy, sensitivity, specificity, and F1-measure are also used for performance evaluation. In this context, Table 1 presents the machine learning performance of different feature selection methods and different classification algorithms on the obesity-related metagenomic dataset. Since 10-fold cross-validation was used in our experiments, the table shows the average values and standard deviations of the performance metrics across 10 separate tests. Upon reviewing these values, the best results were achieved with the XGBoost method. When examining the sensitivity and specificity values, the strength of the prediction performance for both positive and negative classes is noteworthy. As a feature selection algorithm, XGBoost outperforms the other feature selection algorithms. Regarding classification algorithms, the best results were observed with XGBoost, followed by Random Forest and LogitBoost algorithms.

Table 1. Performance metrics obtained by different feature selection methods and classifiers on the obesity-related metagenomic dataset. Acc represents Accuracy, Sens represents Sensitivity, Spe represents Specificity, F-meas represents F-measure, AUC represents Area Under the Curve.

Table 1. Performance metrics obtained by different feature selection methods and classifiers on the obesity-related metagenomic dataset. Acc represents Accuracy, Sens represents Sensitivity, Spe represents Specificity, F-measure represents F-measure, AUC represents Area Under the Curve.

XGBoost					
Model	Acc	Sens	Spe	F-measure	AUC
AdaBoost	0,73 ± 0.06	0,74 ± 0.08	0,72 ± 0.09	0,73 ± 0.06	0,78 ± 0.07
DT	0,68 ± 0.05	0,70 ± 0.07	0,67 ± 0.08	0,68 ± 0.05	0,68 ± 0.06
LogitBoost	0,71 ± 0.05	0,74 ± 0.06	0,68 ± 0.06	0,72 ± 0.05	0,79 ± 0.05
RF	0,76 ± 0.08	0,79 ± 0.07	0,74 ± 0.11	0,77 ± 0.08	0,85 ± 0.05
XGBoost	0,81 ± 0.06	0,84 ± 0.08	0,78 ± 0.06	0,81 ± 0.06	0,89 ± 0.04
Select k Best					
Model	Acc	Sens	Spe	F-measure	AUC
AdaBoost	0,66 ± 0.05	0,67 ± 0.05	0,66 ± 0.08	0,67 ± 0.04	0,72 ± 0.08
DT	0,65 ± 0.04	0,65 ± 0.04	0,65 ± 0.07	0,64 ± 0.03	0,65 ± 0.03
LogitBoost	0,71 ± 0.03	0,70 ± 0.06	0,73 ± 0.07	0,71 ± 0.02	0,78 ± 0.05
RF	0,69 ± 0.06	0,70 ± 0.06	0,68 ± 0.12	0,69 ± 0.05	0,77 ± 0.07
XGBoost	0,71 ± 0.06	0,70 ± 0.07	0,71 ± 0.07	0,70 ± 0.06	0,77 ± 0.07
Information Gain					
Model	Acc	Sens	Spe	F-measure	AUC
AdaBoost	0,59 ± 0.08	0,62 ± 0.12	0,56 ± 0.1	0,60 ± 0.08	0,65 ± 0.08
DT	0,58 ± 0.06	0,61 ± 0.07	0,55 ± 0.09	0,59 ± 0.06	0,59 ± 0.06
LogitBoost	0,62 ± 0.05	0,63 ± 0.09	0,60 ± 0.05	0,62 ± 0.06	0,67 ± 0.06
RF	0,66 ± 0.04	0,68 ± 0.03	0,63 ± 0.07	0,66 ± 0.03	0,69 ± 0.05
XGBoost	0,68 ± 0.05	0,71 ± 0.04	0,65 ± 0.1	0,69 ± 0.04	0,76 ± 0.05

The performance metrics obtained by the SVM-RCE and RCE-IFE algorithms on obesity-associated metagenomic dataset are shown in Table 2. Both algorithms were executed with default parameters. In the SVM-RCE method, after the data is split into training and test sets, the most important 1000 features are determined using the t-test and clustered using the K-means algorithm. Then, the clusters are scored with SVM based on their classification performance, and the low-scoring ones are eliminated at a predetermined rate. The remaining features are combined, and the model's performance is tested. This

process is repeated iteratively until the desired number of clusters is reached. In the RCE-IFE method, the Random Forest algorithm is used to score the clusters and the remaining features within them. Similar to the SVM-RCE method, the dataset is split into training and test sets, and the features are clustered using the K-means algorithm. However, in this method, after the low-scoring clusters are eliminated, the low-scoring features within each cluster are also removed at a certain rate. This iterative process is performed after each cluster elimination step, ensuring that noisy or unnecessary features are cleaned.

Table 2. Performance metrics of SVM-RCE and RCE-IFE methods on obesity-associated metagenomic dataset. Acc represents Accuracy, Sens represents Sensitivity, Spe represents Specificity, F-mea represents F-measure, AUC represents Area Under the Curve.

SVM-RCE						
# of Cluster	# of features	Acc	Sens	Spe	F-mea	AUC
30	630,16	0,677 ± 0.03	0,202 ± 0.07	0,911 ± 0.04	0,288 ± 0.08	0,637 ± 0.06
20	550,21	0,679 ± 0.03	0,199 ± 0.07	0,915 ± 0.03	0,286 ± 0.09	0,638 ± 0.06
10	393,43	0,677 ± 0.03	0,182 ± 0.07	0,920 ± 0.03	0,267 ± 0.09	0,640 ± 0.05
5	250,23	0,676 ± 0.03	0,158 ± 0.07	0,931 ± 0.03	0,237 ± 0.09	0,644 ± 0.05
2	145,29	0,679 ± 0.02	0,133 ± 0.06	0,947 ± 0.03	0,210 ± 0.09	0,627 ± 0.06
1	93,25	0,676 ± 0.02	0,102 ± 0.06	0,959 ± 0.02	0,180 ± 0.08	0,614 ± 0.06
RCE-IFE						
# of Cluster	# of features	Acc	Sens	Spe	F-mea	AUC
30	299,05	0,705 ± 0.04	0,476 ± 0.08	0,818 ± 0.05	0,515 ± 0.07	0,755 ± 0.05
20	233	0,706 ± 0.04	0,482 ± 0.08	0,816 ± 0.05	0,518 ± 0.07	0,752 ± 0.05
10	158,17	0,707 ± 0.04	0,467 ± 0.07	0,825 ± 0.05	0,512 ± 0.06	0,741 ± 0.05
5	93,86	0,700 ± 0.04	0,422 ± 0.10	0,836 ± 0.04	0,476 ± 0.09	0,726 ± 0.06
2	50,51	0,700 ± 0.03	0,361 ± 0.11	0,867 ± 0.05	0,435 ± 0.10	0,691 ± 0.07
1	27,43	0,696 ± 0.03	0,303 ± 0.12	0,890 ± 0.04	0,388 ± 0.11	0,663 ± 0.07

3.1. Biological Evaluation of The Findings

Figure 2 shows the commonalities between the selected 10, 15, and 32 features having an importance score greater than 0.5, using the SKB, XGBoost, and IG algorithms, respectively. As seen in Figure 2, all three feature selection methods commonly identified the following two biological species:

Granulicatella unclassified and *Holdemania filiformis*. When examining the pairwise intersections, 11 different species come into the scene. These species include *Akkermansia muciniphila*, *Bifidobacterium adolescentis*, *Eubacterium hallii*, *Holdemania filiformis*, *Granulicatella unclassified*, *Faecalibacterium prausnitzii*, *Ruminococcus* sp. 5 1 39BFAA, *Subdoligranulum variabile*, *Subdoligranulum unclassified*, *Ruminococcus bromii*, and *Gemella sanguinis*.

Additionally, as shown in Figure 4, five different species were commonly identified among the top 10 features determined by the SVM-RCE and RCE-IFE methods. The species include *Lachnospiraceae bacterium* 9 1 43BFAA, *Oribacterium sinus*, *Candidate division TM7 single cell isolate TM7c*, *Lachnospiraceae bacterium* 4 1 37FAA, and *Gemella sanguinis*.

The effect of *Lachnospiraceae bacterium* on obesity is previously reported in (Burakova et.al. 2022). Among the top 10 features identified by the SVM-RCE and RCE-IFE methods and the feature selection algorithms, *Gemella sanguinis* was commonly identified (as shown in Figure 4). There is no research directly linking the microorganism *Gemella sanguinis* to obesity; however, numerous studies have established its association with diabetes mellitus (Mancini et. al., 2024). Additionally, the microorganism *Granulicatella unclassified* was commonly identified between the RCE-IFE method and the 11 different intersection feature species (shown in Figure 4). Several studies highlight the positive association between *Granulicatella unclassified* and obesity (Chen et. al., 2020). The features discussed in detail above are highlighted as potential taxonomic biomarkers for obesity.

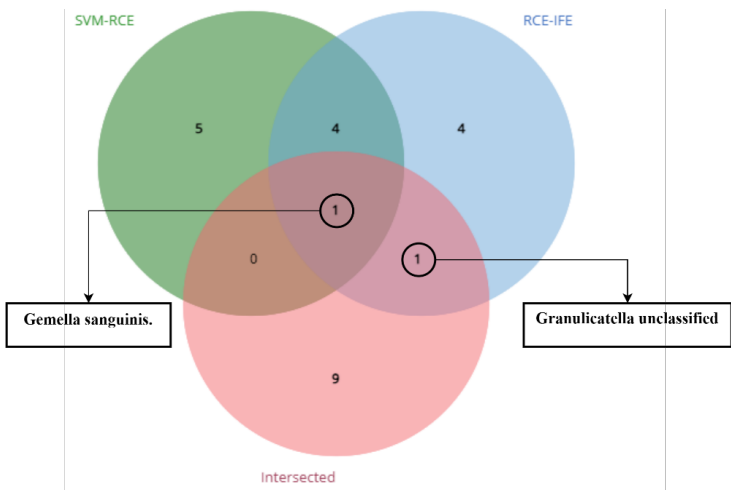


Figure 4. Commonalities among the microorganisms identified by SVM-RCE, RCE-IFE methods and intersected features identified by feature selection algorithms

4. Conclusion

Studies conducted in the past decade have frequently emphasized that the gut microbiota plays an essential role in the onset of obesity, in addition to the genetic and environmental factors (Cheng Z. et al., 2022). In this study, to detect taxonomic obesity biomarkers, two different recursive cluster elimination methods (SVM-RCE and RCE-IFE) and different feature selection techniques are employed. Obesity is predicted via developing a classification model on the metagenomic data obtained from the human gut microbiota, and potential biomarkers for obesity are identified. While the SVM-RCE and RCE-IFE methods define feature clusters and perform classification, the proposed approach reduces the number of features by using various feature selection methods, and the selected features are re-employed during the classification phase. SVM-RCE and RCE-IFE methods did not achieve high accuracy metrics. When evaluating the performance of the model created by different feature selection methods, the highest AUC value (0.89) is obtained using the top 100 features of the 903 species. Firstly, we have selected the features having scaled importance scores higher than 0.5 and secondly, among these features we focused on the features commonly detected among two feature selection methods. To this end, it has been shown that a classification model solely based on 11 biological species can be used for diagnosing obesity patients with an AUC score of 0.77. In other words, based on the relative abundance data of the 11 biological species found in individuals' feces, obesity can be predicted with 77% AUC. Compared with the literature, these findings suggest that the identified biological species can be considered as potential taxonomic biomarkers for obesity. Future studies include obesity diagnosis and detection not only using metagenomic data but also through the incorporation of different omics data types. Additionally, deep learning methods will be integrated to further improve the success rate.

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CHAPTER XIV

APPLICATION OF ADAPTIVE NEURO-FUZZY INFERENCE SYSTEMS (ANFIS) FOR THE PREDICTIVE MODELING OF DEBINDING BEHAVIOR IN PRESSURELESS HOT FORMED ZIRCONIA CERAMICS

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1. Introduction

There are three most studied forms of zirconia. Monoclinic (m), tetragonal (t) and cubic (k). Which polyform ZrO₂ has depends on temperature and pressure (Mohd Foudzi et al., 2013). Tetragonal zirconia polycrystalline is a popular engineering ceramic material. It has good mechanical properties such as high fracture toughness. In heating and cooling cycles, the starting and ending temperatures of the transformation are not a fixed temperature but a transformation range. Stabilizers such as MgO, CaO, Y₂O₃, CeO₂ are used to stabilize zirconia ceramics due to the negativities that occur in the structure during phase transformations (Casellas et al., 2001; Chiou & Lin, 1997; Green et al., 2018; Hafizoğlu et al., 2021). Zirconia is commonly used as a high-temperature material. Zirconia is an effective refractor, due to its high melting point, low thermal conductivity, a coefficient of thermal expansion closely resembles that

of iron-based materials, and good resistance to slag attacks (Boyras, 2018; Çitak & Boyraz, 2014; Israfıl Kucuk & Tahsin Boyraz, 2019; Wei & Lin, 1998).

The process of powder injection moulding combines powder technology and injection moulding, and involves multiple stages such as mixing, separation, and sintering. A homogeneous compound is formed by blending ceramic powder with binders during mixing. Filling the raw material into moulds during injection moulding can be made easier by using binder to provide viscosity to the powder. Moreover, binders ensure that the ceramic powder retains its original shape throughout decomposition and until the start of the sintering process. Powder technology and injection moulding are combined in ceramic injection moulding. By using the moulding process, ceramic components with complex shapes can be produced at a low cost. The ceramic injection moulding process includes several stages such as mixing, injection moulding, segregation and sintering. In this study, the pressureless hot forming method was studied as an alternative method. The non-pressure hot forming method is based on the principle of mixing ceramic powders prepared with various powder preparation methods with a suitable binder system and forming them by hot casting. The binder system consists of two components: base binder and surface processor. It has a lower cost and easier production process compared to other forming methods. The use of binder and the removal of binder in this method are disadvantages compared to some methods. Binder removal in samples shaped by this method is carried out in two stages: the powder bed method, in which the capillary attraction mechanism is applied, and thermal pyrolysis (decomposition). Usually up to 40% of the binder is removed in the first stage. This is a desired situation. Because after removing the bed dust, the parts must have sufficient strength to clean and transport their surfaces; The binder remaining in the body serves this function. The remaining binder evaporates and disappears from the body by increasing the temperature in the second stage (Liu et al., 2001; Md Ani et al., 2014).

Materials science is a broad research area that requires powerful analysis methods to understand, model, and predict the properties of different materials. Experimental studies are often difficult due to high cost, time requirements, and complex data analysis processes. In this context, fuzzy logic-based methods offer an effective solution, especially in modelling nonlinear systems. Adaptive Neuro-Fuzzy Inference System (ANFIS) has become a powerful approach in the prediction and optimization of complex systems in materials science by combining the learning capacity of artificial neural networks and the ability of fuzzy logic to model uncertainties. Artificial intelligence-based ANFIS methods

offer a hybrid learning system that provides flexibility and high accuracy rates in modelling and predicting material properties. This method uses existing information in the best way possible and allows estimating unknown parameters by discovering new patterns, thanks to its ability to optimize membership functions and parameters by learning from experimental data. In addition, while traditional modelling methods require certain assumptions, ANFIS works with the principle of data-driven learning, allowing the nonlinear and uncertain structure of the system to be analyzed more effectively. In this study, the modelling of material properties using ANFIS is discussed, the performance of different membership function combinations is evaluated and the structure that produces the best results is determined. The accuracy of the model is evaluated by testing it with error metrics. This study examines the contribution of ANFIS to the modelling and prediction processes in materials science. Within the scope of the study, the modelling capabilities of ANFIS are discussed in detail and the advantages of the method in terms of accuracy, generalization and reliability are emphasized. In this context, it is shown that the developed model provides a fast, flexible and computationally efficient alternative compared to traditional methods by supporting experimental processes in materials science (Yüksek, Boyraz, & Akkus, 2024; Yüksek, Boyraz, & Akkuş, 2024a) .

2. Materials and Methods

2.1. Materials Production

In this study, the shaping and debinding properties of ZrO_2 -CaO-MgO based ceramics by the pressureless hot forming method were examined, then the experimental debinding results were analysed and modelled with machine learning using the obtained experimental data. Adaptive Neuro-Fuzzy Inference Systems (ANFIS) was used to predict the debinding behaviour of zirconia based ceramics using MATLAB's neural network tool panel.

Two types of raw materials were used in experimental studies: ceramic powders and binder system. ZrO_2 powders (Serp, France), CaO and MgO (Merck, Germany) with submicron grain size were used. A two-component binder system was chosen for this study. The binder contains 95% paraffin (Merck, Germany) and 5% oleic acid (Merck, Germany) by weight. Paraffin is the main binding component. Oleic acid was used to improve the wetting properties between the powder and binder. In this study, zirconia ball and acetone medium were used in the wet preparation of powders by mechanical alloying method. The powders were dried in an oven at 110 °C for 24 hours before and after mixture preparation.

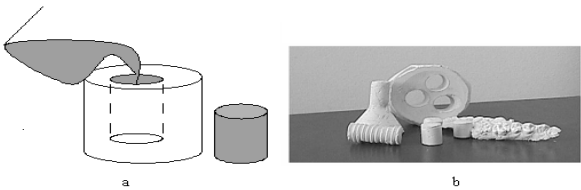
Three different $\text{ZrO}_2\text{-MgO}$ (ZM), $\text{ZrO}_2\text{-MgO-CaO}$ (ZMC) and $\text{ZrO}_2\text{-CaO}$ (ZC) compositions given in Table 1 were prepared. Then, the samples were shaped by the pressureless hot forming method in the proportions given in Table 2. After this, binder removal studies were carried out. Figure 1 shows the schematic view of the pressureless hot forming method and some shaped parts. Then, binder removal studies were carried out on the prepared samples and the experimental results were modelled and compared with Adaptive Neuro-Fuzzy Inference Systems (ANFIS).

Table 1. Composition of the prepared powder mixtures.

Code	Chemical Composition (% mole)		
	ZrO_2	CaO	MgO
ZM	90.5	-	9.50
ZMC	90.5	4.75	4.75
ZC	90.5	9.50	-

Table 2. Compositions of mixtures prepared for non-pressure hot forming.

Materials	Weight, %	Volume, %	Amount, gr
Powder Mixture	79	40	197.5
Paraffin	19.95	57	49.875
oleic acid	1.05	3	2.625



Şekil 4.2. a) Basınçsız Sıcak Şekillendirme yöntemi, b) Basınçsız Sıcak şekillendirilmiş parçalar

Figure 1. a) Pressureless Hot Forming method, b) Formed parts.

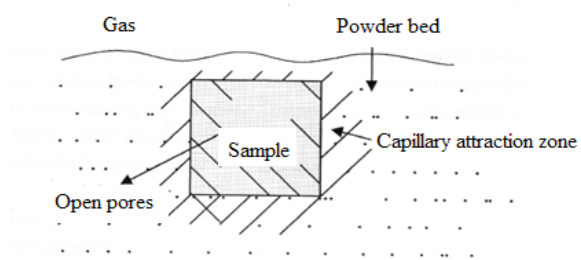


Figure 2. Schematic representation of the binder removal mechanism by capillary force

Debinding studies on shaped samples were carried out with a two-stage method. 1. Capillary attraction mechanism (powder bed method) and 2. Thermal pyrolysis (decomposition). The first step in debinding is embedding the samples in absorbent bed powder. Kaolin powder was used as bedding powder. The moisture of the bed powder was removed before the debinding process. The bedding dust is placed in a container and the samples are 5 cm wide. They are lined up at intervals and covered with bed dust. In this study, which is shown schematically in Figure 2, 1 °C/min. A heating rate of was used. The binders were partially removed by keeping the samples in the oven at temperatures of 100-180 °C for 1-10 hours (Boyras, 2008 and 2010; Lenk, 1994).

2.2. Adaptive Neuro-Fuzzy Inference System (ANFIS)

ANFIS was developed by J.S.R. Jang (Jang, 1993) in 1993 and offers the advantageous features of artificial neural networks and fuzzy logic systems in a single framework. The basis of the proposed hybrid structure is the combination of the intuitive decision-making capabilities of fuzzy inference systems and the learning capabilities of artificial neural network approaches (Abbas et al., 2022). ANFIS architecture provides high-performance modelling of nonlinear functions with Takagi-Sugeno type fuzzy models. In the learning process, system parameters are optimized by using both forward propagation and back propagation algorithms (Rodrigues et al., 2010). Artificial neural networks (ANN) are computational models that learn complex relationships and make predictions by copying biological neural networks (Rumelhart et al., 1986). It processes data through structures such as multilayer perceptron (MLP) and optimizes network weights by using the back propagation algorithm in the learning process.

ANNs are programming models inspired by biological neural systems and capable of learning, generalizing and predicting complex relationships between details and data (Fausett, 1994) (Haykin & Haykin, 2009). ANNs, which have a multi-layer structure, consist of input layers, hidden layers and output layers. The process is carried out by optimizing the weights and one of the most widely used applications is the initiation of error backpropagation. In this process, the model learns the relationships between inputs and outputs by iteratively, increasing the accuracy range. Especially in continuous systems, the ability to separate effective results allows artificial neural networks to have a wide range of use (Graupe, 2013).

Fuzzy logic was developed by Lotfi A. Zadeh in 1965 and emerged as an approach to model uncertainties by softening the sharp boundaries of classical logic (Zadeh, 2009). Since imprecise or ambiguous data is frequently

encountered in real-world problems, fuzzy logic aims to integrate human-like reasoning processes into computer systems (McNeill & Thro, 1994). Fuzzy sets, unlike classical set theory, offer a structure in which elements have a certain degree of membership. The level of belonging of an element is determined by membership functions and takes a value between 0 and 1. This approach enables the modelling of concepts that cannot be defined with sharp boundaries, such as “hot” (Bai & Wang, 2006). Fuzzy logic systems operate based on “If-Then” rules (Goguen, 1973) (Klir & Yuan, 1995). Fuzzy logic is an important component in the ANFIS structure. While transforming input data into fuzzy sets allows the system to model uncertainties, effective solutions are produced for nonlinear problems by making inferences through fuzzy rules. This hybrid structure optimizes the learning process and creates a more flexible and powerful estimation mechanism by incorporating uncertainties into the mode

ANFIS offers a hybrid approach by combining the learning ability of artificial neural networks with the interpretability and uncertainty modelling ability of fuzzy logic (Vargas et al., 2024). The power of ANN in parameter optimization allows ANFIS to dynamically learn fuzzy rules and increase system performance. Thanks to this structure, membership functions and rule bases of fuzzy systems, which are usually determined manually, are optimized with a data-driven approach. Thus, ANFIS becomes a powerful tool that can both reach high accuracy rates and offer an explainable and flexible modelling method (Walia et al., 2015).

One of the biggest advantages of ANFIS is its ability to automatically optimize membership functions and parameters. Artificial neural networks use algorithms such as backpropagation or hybrid learning to speed up this process. During the training process, the system constantly updates the membership functions and the results of fuzzy rules using input-output data. Thus, it can make predictions with minimum error by adapting to the observed data in the most appropriate way. The architecture of ANFIS consists of five layers, each of which performs a specific function (Figure 3). The first layer addresses uncertainty by converting the input data into fuzzy sets. The second layer creates and processes if-then rules based on fuzzy logic. The third layer determines the importance of each rule by normalizing the outputs from the previous layer with the weighted average method. In the fourth layer, the fuzzy results are converted to numerical values, while the last layer produces a single crisp result by summing the outputs from all nodes. This multi-layered structure makes ANFIS an effective tool for modelling and controlling nonlinear complex systems (Vargas et al., 2024) (Salleh et al., 2017) (Yetilmezsoy et al., 2015) (Abdelfattah et al., 2022) (Yüksek et al., 2017).

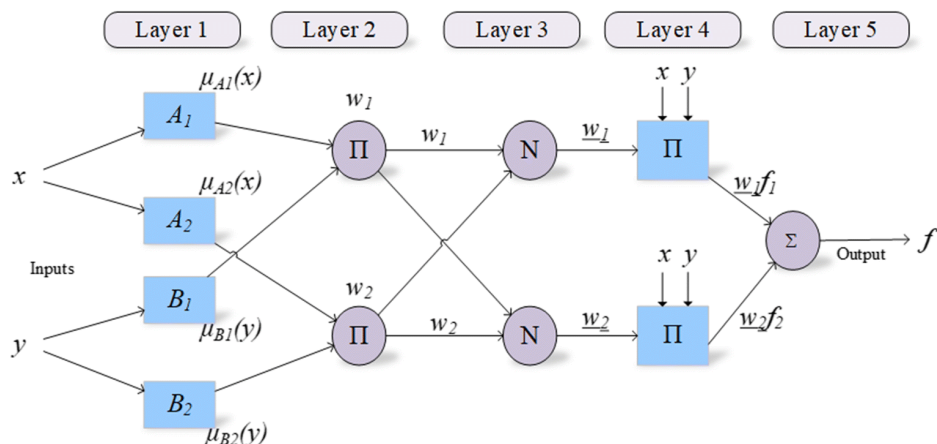


Figure 3- Adaptive network based fuzzy logic inference system
(Haznedar et al., 2021)

2.3. Mathematical Model of ANFIS

ANFIS is a learning system that combines artificial neural networks with the Sugeno fuzzy logic model. The Sugeno type fuzzy model is a fuzzy inference system widely used in system modelling and control applications. This model divides input variables into certain sets based on fuzzy logic rules and determines system outputs through membership functions in the defuzzification layer. In the Sugeno model, output functions are defined as a zero-order constant coefficient or a first-order linear function. A typical fuzzy rule in a Sugeno fuzzy model is expressed as shown in Equation 1:

$$\text{If } A = x \text{ and } y = B \text{ then } f(x, y) = px + qy + r \quad (1)$$

Here, A and B represent the fuzzy sets of input variables, and $f(x, y)$ represents ANFIS automates the optimization process by combining the rule-based structure of the Sugeno model with artificial neural networks. The system uses derivative-based learning algorithms to determine the weights of the rules and the parameters of the membership functions. The error backpropagation algorithm optimizes the parameters of the membership functions, while the least squares method is used to determine the coefficients of the output functions. Thanks to this learning process, ANFIS can effectively model nonlinear systems by learning from the data and increase the prediction performance (Abbas et al., 2022)(Xinqing et al., 1996) .

3. Material Modelling and Data Preparation with ANFIS

3.1. Material Selection and Properties

A correlation table is a statistical tool used to determine the linear relationship between variables. Values range from -1 to +1; a positive correlation (between 0 and +1) indicates that the variables increase together, and a negative correlation (between -1 and 0) indicates that as one variable increases, the other decreases. The magnitude of the value indicates the strength of the relationship.

In the given correlation table (Table 3), the relationship between temperature (°C), time (hours) and eliminated binder (%) was examined in three different conditions (C0, C50, C100). According to the findings, there is a strong and positive relationship between temperature and the amount of binder eliminated (0.765 for C0, 0.758 for C50, 0.791 for C100). This situation shows that as the temperature increases, the rate of binder elimination also increases. The correlation between time and the amount of binder eliminated is at a lower level (0.511 for C0, 0.487 for C50, 0.445 for C100), which indicates that the effect of time on the binder elimination process is weaker compared to temperature. The correlation between temperature and time is 0, indicating that there is no linear relationship between these two variables. This correlation analysis reveals that temperature is the most important factor in the binder elimination process, while time plays a less effective role. Such analyses are important for process optimization and control and help determine which variables are more effective.

Table 3- Input and Output Parameters Correlation

	<i>Temperature</i> C°	<i>Time</i> Hour	<i>Eliminated Binder</i> %
C0			
Temperature	1		
Time	0	1	
Eliminated Binder	0.765659	0.511593	1
C50			
Temperature	1		
Time	0	1	
Eliminated Binder	0.758393	0.487717	1
C100			
Temperature	1		
Time	0	1	
Eliminated Binder	0.791242	0.445473	1

Descriptive statistics table (Table 4) is a method used to summarize and analyse the basic characteristics of the data set. Descriptive statistics help to understand the central tendencies (mean, median), spread (standard deviation, variance) and shape of the distribution (skewness, kurtosis) of the data distribution. While standard deviation and variance indicate how much the data varies, skewness and kurtosis values indicate whether the distribution is symmetrical or the effect of extreme values. The smallest and largest values help to determine the range of the data, while the confidence interval shows the level of reliability of the measurements at a certain confidence level. Such statistics are used to support the decision-making process in data analysis, detect anomalies, evaluate relationships between variables and provide preliminary information in modelling processes (Becerra et al., 2021)(Kaur et al., 2018) .

Table 4. Descriptive statistics

	<i>Temperature</i>	<i>Time</i>	<i>Eliminated Binder</i>		
	C0-C100	C0-C100	C100	C50	C0
Average	140.00	5.50	49.92	47.37	46.30
Standard Error	2.74	0.30	0.97	0.90	0.96
Median	140.00	5.50	51.88	49.79	48.18
Standard Deviation	25.96	2.89	9.24	8.53	9.07
Sample Variance	674.16	8.34	85.31	72.85	82.24
Kurtosis	-1.23	-1.23	1.13	1.22	0.68
Skewness	0.00	0.00	-0.44	-0.83	-0.80
Smallest	100.00	1.00	23.98	21.35	19.16
Biggest	180.00	10.00	75.09	67.78	64.20
Reliability Level(95.0%)	5.44	0.60	1.93	1.79	1.90

The dataset to be used in the creation of ANFIS models was meticulously processed in line with the data preparation processes. First, missing data and noisy observations were determined and data quality was increased with appropriate data processing techniques. Then, normalization and scaling processes were applied and possible corrections were made in order to optimize the learning process of the model and to eliminate the scale differences between different variables. In this way, stability was ensured in the parameter learning

of the ANFIS model and the risk of over-learning was minimized. In the last stage, the dataset was divided into training and test groups in order to evaluate the generalization ability of the model. In line with these stages summarized in Table 1 and Table 2, the dataset created for C0, C50 and C100 was arranged in a way to increase the accuracy of ANFIS models and made ready for the analysis process. This structured data processing process increases the model performance and ensures reliable estimates.

3.2. Establishment and Training of the ANFIS Model

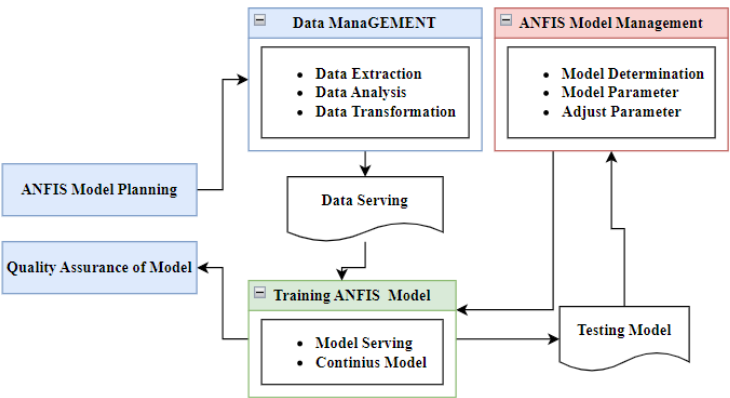


Figure 4. General Process Diagram of the Developed System

In the ANFIS model training process, Figure 4, the focus was primarily on the model planning phase. In this phase, the problem to which the model will be applied was defined, input and output variables were determined, and the necessary preliminary analyses were performed to create appropriate fuzzy rules. Following the determination of the model structure, the data management process was initiated and in this context, data sets were analysed and data transformation processes were applied with appropriate feature engineering methods. In order to process the data reliably, missing and noisy observations were cleaned, and normalization and scaling techniques were used to improve the learning process of the model.

In the model training phase, ANFIS models were trained with the specified parameters and parameter adjustments were made to optimize the performance of the model. In the process of determining the model parameters, the most appropriate weights and membership functions were determined using learning algorithms (Prof.Dr. Adem KALINLI, 2017; Talpur et al., 2017; Tsoukalas & Uhrig, 1997a) The trained model was evaluated with

the principle of continuous learning and improved in line with the specified performance criteria. After the training process of the model, the testing phase was passed and verification procedures were performed. The performance of the model on the test data was analysed, the error rates and success metrics obtained were evaluated and the training process was repeated in order to optimize the model.

This iterative process, which ensures the continuous development of the model, has been continued until the best system performance is reached. In the last stage, the quality assurance of the model has been provided, verification tests have been performed and the general performance of the model has been analysed. In this process, the ANFIS model with the most appropriate parameter combination has been determined by taking into account the generalization ability and accuracy rates of the model and the system has been made applicable reliably.

3.2.1. Selection of ANFIS Membership Functions

While training ANFIS models, membership functions in fuzzy logic systems play a critical role in terms of fuzzifying input parameters and effectively performing the learning process of the system. Membership functions form the basis of the fuzzy inference mechanism by determining the degree to which input variables are related to a certain fuzzy set. Correct determination of these functions increases the accuracy of the system while minimizing errors and ensuring that the most appropriate model is obtained (Dubois & Prade, 2000; Hudec, 2016; Tsoukalas & Uhrig, 1997b). Membership functions are mathematical functions that determine how the input variables will be distributed within the range they are defined. Using different membership functions is necessary to test how the model will adapt to different data sets (Feng & Lu, 2019). In ANFIS models, determining the most appropriate membership function for each input variable increases the generalization ability of the system and provides higher accuracy against unknown data. In this study, different membership functions given in Figure 5 and Table 5 were defined separately for the input variables of the model and the training process was completed using each of them. Thus, the effect of each function on the model performance was evaluated and the error metrics obtained with different combinations were compared. As a result of this evaluation, the most appropriate membership function combination that has the lowest error rate and provides the highest generalization ability was determined and optimized for the system.

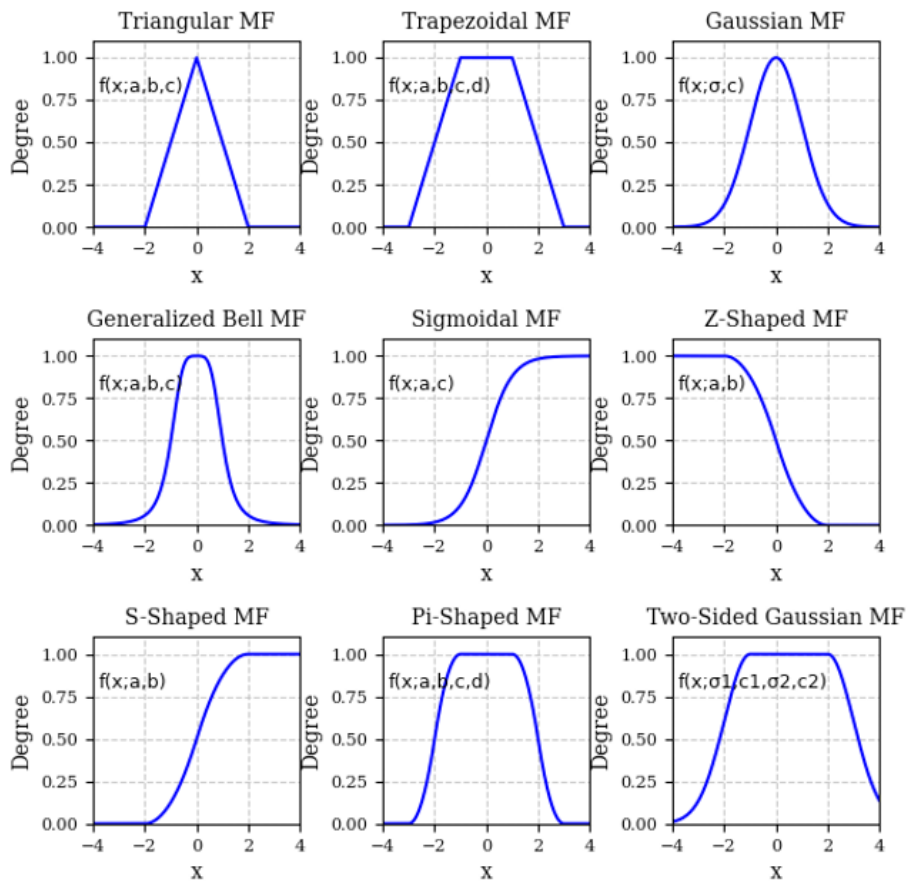


Figure 5- Fuzzy Member Ship Functions

Table 5- Fuzzy Member Ship Functions Explanations

Function	Mathematical Expression	Parameters
Triangular	$0, x \leq a;$ $(x-a)/(b-a), a \leq x \leq b;$ $(c-x)/(c-b), b \leq x \leq c;$ $0, c \leq x$	a=left foot, b=peak, c=right foot
Trapezoidal	$0, x \leq a;$ $(x-a)/(b-a), a \leq x \leq b;$ $1, b \leq x \leq c; (d-x)/(d-c), c \leq x \leq d;$ $0, d \leq x$	a=left foot, b=left shoulder, c=right shoulder, d=right foot
Gaussian	$\exp(-(x-c)^2/(2\sigma^2))$	c=center, σ =standard deviation
Generalized Bell	$1/(1+ x-c/a (2b))$	a=width, b=slope, c=center
Sigmoidal	$1/(1+\exp(-a(x-c)))$	a=slope, c=inflection point
Z-Shaped	$1, x \leq a;$ $1-2((x-a)/(b-a))^2, a \leq x \leq (a+b)/2;$ $2((b-x)/(b-a))^2, (a+b)/2 \leq x \leq b;$ $0, x \geq b$	a=start of descent, b=end of descent
S-Shaped	$0, x \leq a;$ $2((x-a)/(b-a))^2, a \leq x \leq (a+b)/2;$ $1-2((b-x)/(b-a))^2, (a+b)/2 \leq x \leq b;$ $1, x \geq b$	a=start of ascent, b=end of ascent
Pi-Shaped	$S(x;a,b) \cdot Z(x;c,d)$ <i>(Product of S and Z functions)</i>	a,b,c,d=combination parameters
Two-Sided Gaussian	$\exp(-(x-c_1)^2/(2\sigma_1^2)), x \leq c_1;$ $\exp(-(x-c_2)^2/(2\sigma_2^2)), x \geq c_2;$ $1, c_1 < x < c_2$	σ_1, c_1 =left Gaussian; σ_2, c_2 =right Gaussian

3.2.2. Determination of Training Process and Parameters in ANFIS

Model

After the data set was created, the data set was divided into 80% training data and 20% model test data in order to best evaluate the performance of the system. In the process of determining the most appropriate membership function for the input variables of the model, all membership functions shown in Figure 5 and Table 5 were defined separately and training was performed for each combination. In this process, “4” membership functions were determined for each input variable and the model was trained in accordance with this structure. “Linear function” was preferred as the output function. During the training process, performance analysis was performed using certain error metrics to increase the accuracy level of the model. The output values produced for each membership function combination were tested with the error metrics of Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R² (Determination Coefficient) and the ANFIS parameters that gave the best results were determined (Chicco et al., 2021; ‘Stock Prediction System Based on Bi-Directional LSTM Model’, 2024a; ‘Stock Prediction System Based on Bi-Directional LSTM Model’, 2024b; Yüksek, Boyraz, & Akkuş, 2024b; Yüksek et al., 2025). Error backpropagation and Least Squares methods were used together in the training process of the model. The parameters given in Table 6 were used in the training process.

Table 6- ANFIS Model Parameters

Number of Membership Functions	4
Output Membership Function Type	Linear
EpochNumber	500
Initial StepSize	0.1
Learning Rate	0.1
Momentum	0.5

In this way, it is aimed for the model to reach the highest generalization ability without experiencing overfitting problems. As a result, the combination with the lowest error rate among all membership functions tested in the training process was determined and optimized for the system. Thus, the developed ANFIS model was optimized as a system that can make high-accuracy predictions and has high generalization ability.

3.2.3. ANFIS Training Results

Table 7. ANFIS Model Produced Values Metrics

		MAE	MSE	RMSE	MAPE	R ²	MF
C0	Training	0.916	1.266	1.125	2.131	0.984	'trimf' 'trimf'
	Test	1.288	2.984	1.728	2.969	0.944	
	Validation	1.820	5.092	2.257	4.556	0.921	
	Model Test	1.081	1.993	1.412	2.541	0.974	
C50							'trimf' 'pim'
	Training	0.904	1.601	1.265	1.930	0.977	
	Test	0.775	1.235	1.111	1.672	0.982	
	Validation	1.289	2.716	1.648	2.692	0.958	
	Model Test	1.042	1.930	1.389	2.207	0.960	
C100							'trimf' 'trimf'
	Training	1.108	1.938	1.392	2.426	0.977	
	Test	0.997	1.543	1.242	2.203	0.981	
	Validation	1.182	1.987	1.409	2.498	0.975	
	Model Test	1.736	4.602	2.145	3.838	0.951	

As a result of the training, the models that exhibit the best performance were determined and the error metrics of these models are presented in Table 7 for C0, C50 and C100. Here, the MODEL Test data set is used as the data set provided to the model developed as a result of the training and reserved only for the testing phase. This test data set plays a critical role in measuring the performance of the model with new data that it has not learned during the training process. These results support the success of the model with the high accuracy values produced and show that ANFIS can produce reliable predictions with a suitable configuration.

The success of the model is supported not only by the tabular data but also by the graphical analysis. The values produced by the real and ANFIS models of the training data and MODEL Test data sets are compared with four different graphical methods:

- **Regression Comparison Graphs:** Shows the accuracy of the model by examining the regression relationship between the real and predicted values. A high R² value for training and test data is an important indicator that supports the success level of the model.

- **Value Comparison Graphs:** These are graphs that directly compare the real values with the values predicted by the model. The closeness of the model’s predictions to the real values reveals the performance of the model.
- **Residuals Comparison Graphs:** Analyses the model’s prediction errors and examines whether the model produces a systematic error. Low residual values indicate that the model has a reliable prediction mechanism.
- **Difference Comparison Graphs:** Visualizes the differences between the predictions produced by the model and the real values, allowing the analysis of the regions where the model makes the highest errors. The minimum level of the differences obtained is another important criterion that supports the success of the model.

As a result of these analyses, it was determined that the developed ANFIS model produced successful predictions with high accuracy rates. The results supported by the error metrics shown in Table 5 and verified by graphical analyses prove that the model exhibited satisfactory performance in both training and testing stages. Thus, as a result of the study, it was shown that ANFIS provided a reliable and high-performance prediction mechanism when optimized with the correct parameters and appropriate membership functions (Figure 6 for C0, Figure 7 for C50, Figure 8 for C100).

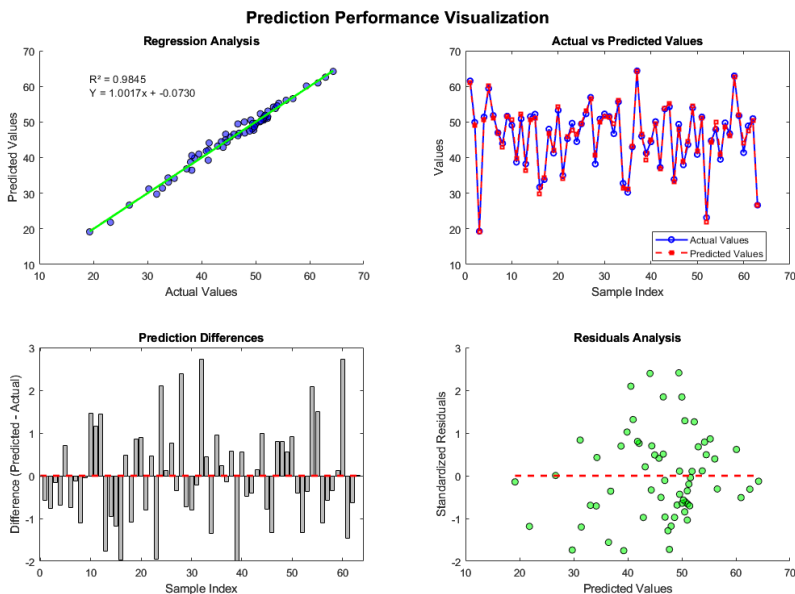


Figure 6a. Training Dataset Result Comparisons for C0.

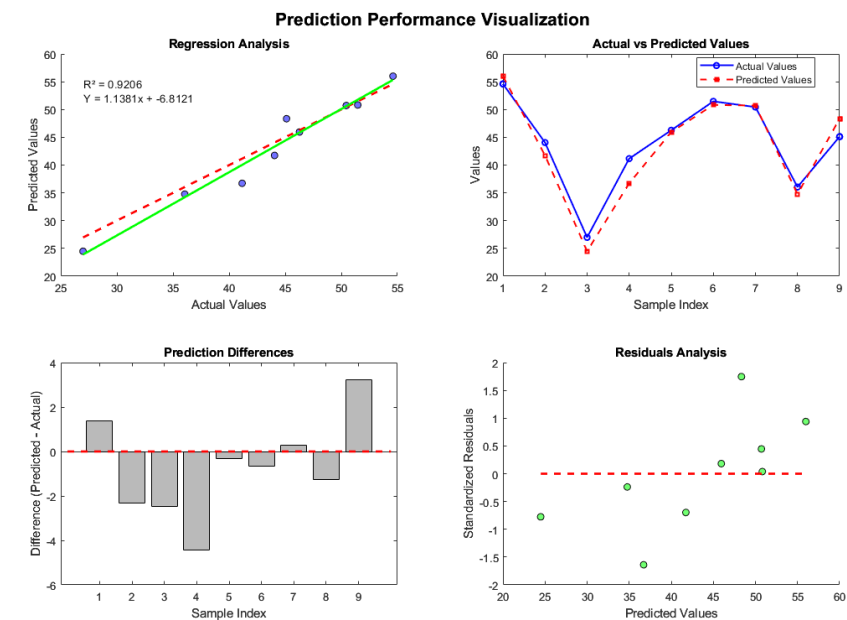


Figure 6b. Model Test Dataset Result Comparisons for C0.

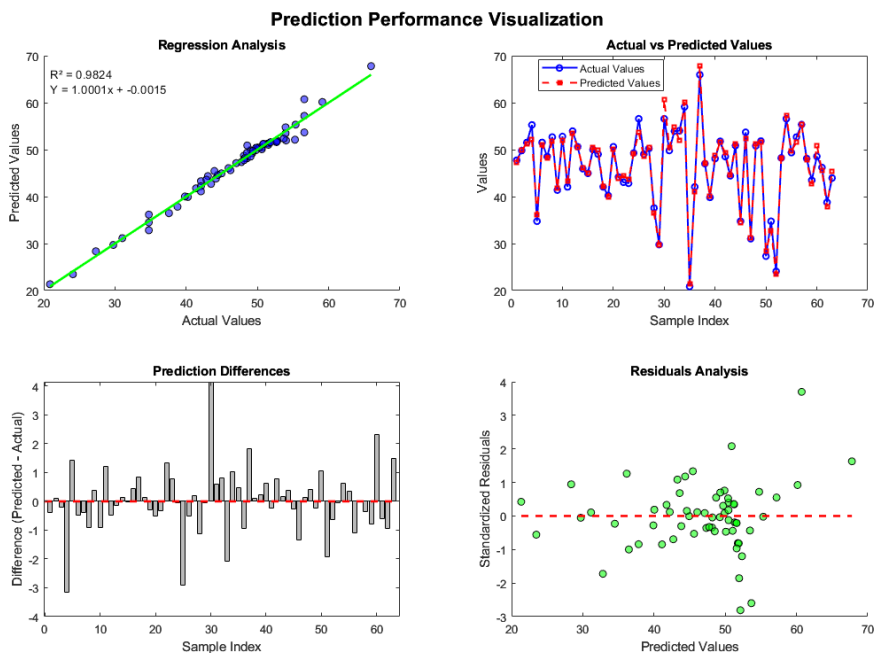


Figure 7a. Training Dataset Result Comparisons for C50.

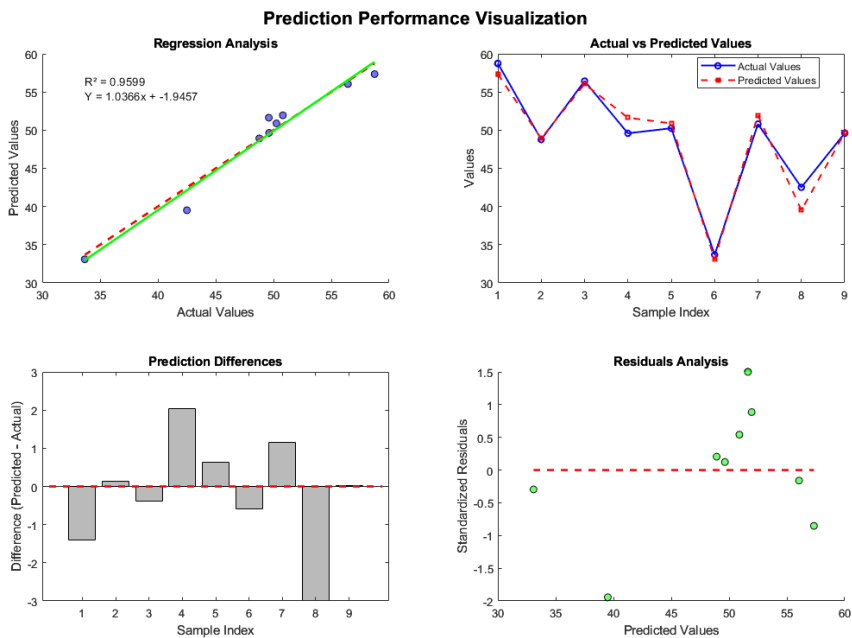


Figure 7b. Model Test Dataset Result Comparisons for C50.

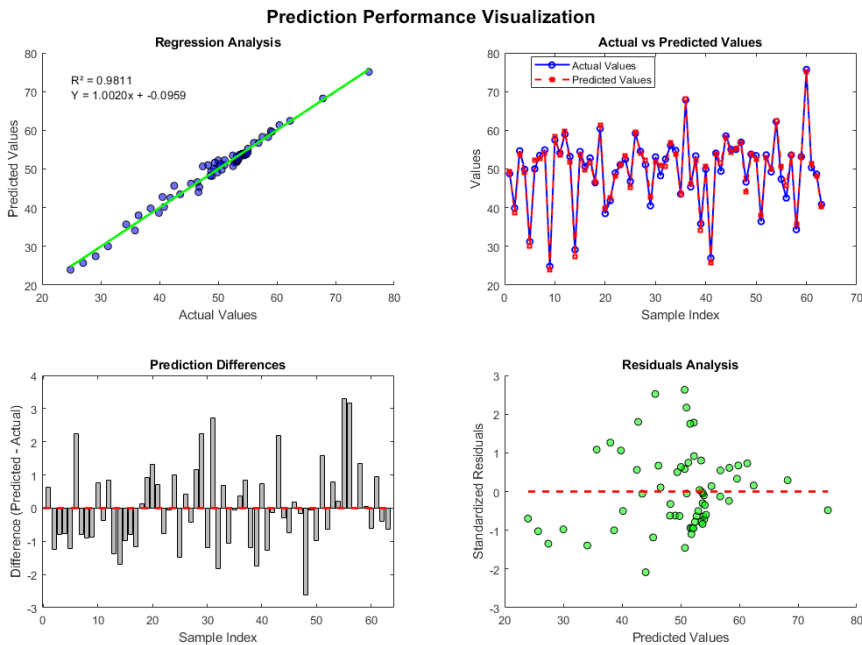


Figure 8a. Training Dataset Result Comparisons for C100.

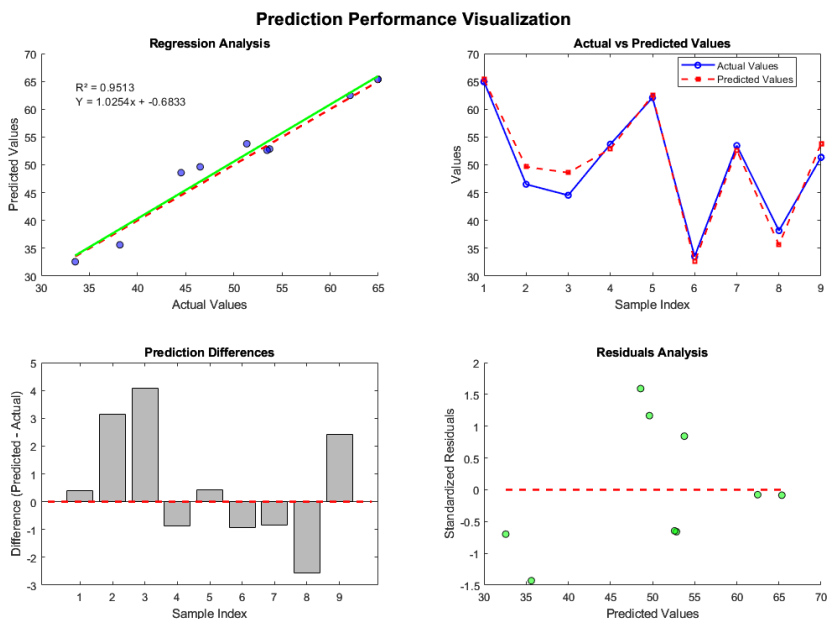


Figure 8b. Model Test Dataset Result Comparisons for C100.

4. Conclusion and Discussion

Materials science requires powerful analysis methods to understand, model and predict the properties of different materials. In this study, ANFIS approach was used to model and predict the physical processes of the forms produced by non-pressurized hot forming method of zirconia material, one of the high temperature engineering ceramics. ANFIS offers a powerful alternative in predicting nonlinear systems by combining the learning capacity of artificial neural networks and the ability of fuzzy logic to model uncertainties. In the modelling process performed using different membership functions, the accuracy of the system was tested with MSE, RMSE, MAE and R^2 error metrics. The results show that ANFIS is a computationally efficient method that supports experimental processes by providing high accuracy in material modelling processes.

In this study, the pressureless hot forming process of zirconia material was modelled using the ANFIS method and the performance of different membership functions was compared. The error metrics in the training, testing, validation and model testing stages were analysed for three different models (C0, C50 and C100) tested in the study.

For the C0 model, the lowest error rates were obtained in the training phase, with a MAE value of 0.916 and an R^2 value of 0.984. However, an increase in error rates was observed in the testing and validation phases, and it was determined that the generalization ability of the model was limited. The C50 model produced the most successful results with a MAE of 0.775 and an R^2 value of 0.982 in the testing phase, and stood out as the model with the lowest error rates. This shows that the model in which the pimf and trimf membership functions were used together had the best estimation performance. In the C100 model, although the error rates were low in the testing phase, the R^2 value dropped to 0.951 in the model testing phase, indicating that the model's adaptation to different data sets was limited.

In general, the modelling results performed with the ANFIS method reached high accuracy levels and it was determined that different membership function combinations had a significant effect on the model performance. The model with the lowest error rates, C50, produced more stable results compared to other models and stood out as the most successful model in terms of reliability by keeping the error rates low during the model testing phase. The success of ANFIS in predicting and modelling data that is difficult to obtain through physical experiments shows that the study makes a significant contribution to the field of materials science. In this context, it is suggested to expand ANFIS-based modelling approaches for different materials and production processes in the future.

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CHAPTER XV

FUZZY LOGIC BASED TEMPERATURE CONTROL SYSTEM DESIGN

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1. Introduction

Temperature control is critical in many sectors such as industry, energy, aerospace, food processing, chemical engineering and home automation (Khan, Adhikary, & Khan, 2019). In modern manufacturing processes, maintaining specific temperature ranges is essential to ensure both product quality and energy efficiency. Precise temperature control is also vital in the biotechnology and pharmaceutical industries (Miralles, Huerre, Malloggi, & Jullien, 2013).

Traditional control methods used to achieve optimum performance and energy efficiency are usually based on specific mathematical models and optimized for linear systems (ATAM, 2016). Proportional-Integral-Derivative (PID) controllers, which are widely used in control engineering, are often preferred in temperature control systems. PID controllers use a feedback mechanism that processes the error signal so that the system reaches the specified reference temperature. However, the effectiveness of PID controllers can be limited due to the dynamics and uncertainties of the system. In particular, for nonlinear and time-varying systems, the classical PID control needs to be continuously re-tuned, which can create practical difficulties (Çetin & İplikci, 2015).

In real-world applications, temperature control systems can exhibit unexpected deviations due to environmental factors, heat losses, changing load conditions and nonlinear dynamics (Jamil, Tu, Ali, Terriche, & Guerrero, 2022).

In such cases, conventional control methods may struggle to adequately handle uncertainties and performance losses may occur.

At this point, fuzzy logic-based control (FLC) emerges as an intelligent and flexible alternative to conventional methods (Yadav, Kumar, Kumar, & Yadav, 2018). This feature allows designing a more flexible and adaptive control system against temperature variations. Moreover, thanks to the FLC's ability to better manage uncertainties, more effective results can be achieved in nonlinear temperature control applications.

Fuzzy logic-based temperature control is becoming an important component of modern control systems, offering a flexible and adaptive approach that requires less tuning in industrial processes (Ramizares, Teves, Arboleda, & Bangeles, 2024). In this study, the design and implementation of fuzzy logic control using MATLAB platform will be investigated and a comparative analysis with conventional PID control will be presented.

2. Material method

2.1. Step response in a first order heater system

A first-order heater system is typically characterized by a time constant (τ) and a constant gain (K). The step response of such a system shows how the output changes when a unit step input is applied to the input of the system. (Campos, Ruzyk, Castaldo, & Reis, 2021) The heater system can be modeled by the differential equation given in Equation 1.

$$\tau \frac{dT(t)}{dt} + T(t) = Ku(t) \quad (1)$$

Where:

T(t) : Variation of temperature with time

u(t) : Input signal (power applied to the heater)

K : Gain of the system (effect of heater power on temperature)

τ : System time constant (shows how fast the system reacts)

This differential equation is transformed into the transfer function in equation 2 using the Laplace transform.

$$G(s) = \frac{K}{\tau s + 1} \quad (2)$$

This expression is a typical first order system transfer function. The step response of a system shows how its output changes when a unit step function

$(U(s)=1/s)$ is applied to the input of the system. The output function of the system is calculated using equation 3:

$$Y(s) = G(s)U(s) = \frac{K}{\tau s + 1} \frac{1}{s} \quad (3)$$

This is found in the time domain given in Equation 4 using the inverse Laplace transform: after solving the equation with the partial fraction's method.

$$y(t) = K \left(1 - e^{-\frac{t}{\tau}} \right) \quad (4)$$

The real-time step response of the temperature control system is obtained using the MATLAB/Simulink application shown in Figure 1. In this system, a heater system is used to reach the specified temperature level and a DC-DC converter is preferred as the power source.

The DC-DC converter used to provide the necessary power to the heater system is driven by the PWM (Pulse Width Modulation) output signal of the Arduino Mega microcontroller. Accordingly, Pin 3, one of the digital outputs of Arduino Mega, was used to send a PWM signal to the DC-DC converter. Thanks to this signal, the amount of energy transferred to the heater system was controlled and the temperature change was observed.

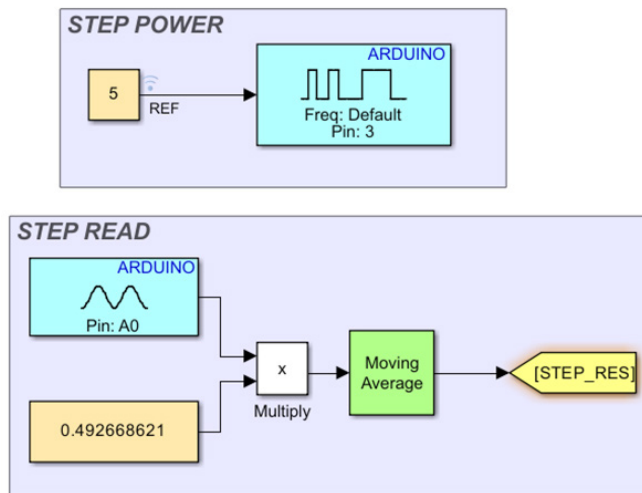


Figure 1. The real-time step response of the temperature control system

In the system, a 2000 W resistance was preferred as the heating element. LM35 temperature sensor was used to monitor the temperature change of

the heating system. The LM35 sensor measures the ambient temperature and transmits this information to the microcontroller via the A0 analog input pin of the Arduino Mega.

Throughout the experimental process, a constant PWM signal was applied to the heater system. In order to observe the dynamic response of the system, the PWM signal was applied continuously until the temperature reached the specified saturation point. Thus, the response of the system to the temperature change was analyzed and the step response was obtained. Figure 2 shows the step response of the heater system. As can be seen in the step response of a temperature control system, the constant system consists of a delay from a certain initial temperature, followed by an increase and saturation of the temperature. This signal is used to find the transfer function of the system.

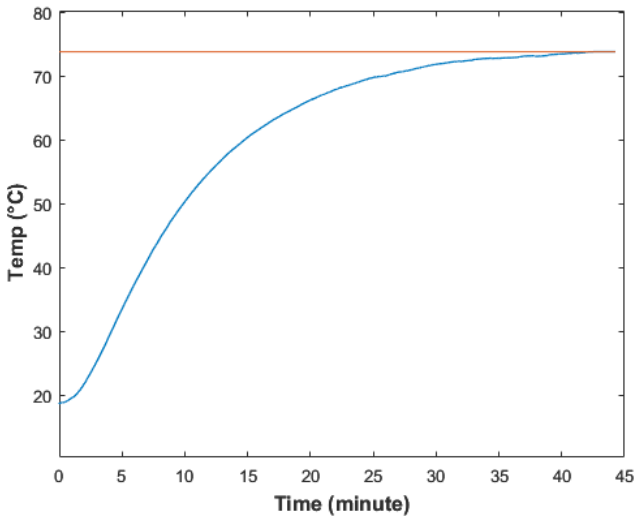


Figure 2. Step response

Methods such as Ziegler-Nichols Open Loop Method (First Order Plus Dead Time - FOPDT Approach) (Pathak, Gautam, & Tripathi, 2022), Least Squares Method and Curve Fitting are used to determine the transfer function of the system using the step response (Lin & Yu, 1977).

In addition, the data obtained from the step response of the temperature control system were analyzed using MATLAB System Identification Toolbox and a dynamic model of the system was created. As a result of this modeling process, the transfer function of the system is obtained and shown in Equation 5.

This transfer function was modeled with an accuracy of 97.57% as a result of the validation analysis.

$$TF = \frac{0,0484}{s - 0.00136} \quad (5)$$

2.2. PI Controller Design for Temperature Control System

Designing a Proportional-Integral (PI) controller for a temperature control system involves determining the appropriate proportional (K_p) and integral (K_i) gains to ensure stable, fast and accurate temperature regulation of the system. By minimizing steady-state error and overshoot, the PI controller ensures that the system reaches and stabilizes at the specified temperature.

The main objective of the PI controller is to ensure that the system reaches the reference temperature value quickly, while at the same time preventing unstable oscillations and unwanted overreaction. The proportional (K_p) gain determines the system's response to the error, while the integral (K_i) gain corrects the error accumulated over time, bringing the steady state error closer to zero. In this way, the temperature control system provides a more precise and stable operation.

Figure 3 shows a simulation performed in MATLAB/Simulink environment. In this simulation, a PI controller is implemented for temperature control and the response of the system is analyzed. The gain parameters (K_p and K_i) of the PI controller are automatically tuned using MATLAB's PID Tuner tool. The PID Tuner analyzed the dynamic model of the system and determined the optimal control parameters, thus helping to tune the system to meet the desired performance criteria.

Thanks to this approach, the temperature control system is fast, stable and has a low margin of error.

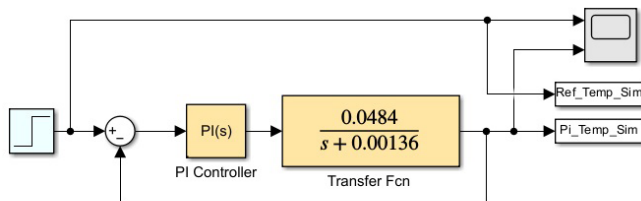


Figure 3. PI controller MATLAB/Simulink application

A screenshot of the PID Tuner interface, where the gain parameters of the PI controller are determined, is shown in Figure 4. This tool analyzes the dynamic characteristics of the system and automatically adjusts the proportional (P) and integral (I) gains of the PI controller to optimize them.

The proportional gain (K_p) value determined by the PID Tuner is 0.4052 and the integral gain (K_i) value is 0.002. These values are optimized to ensure that the system operates fast, stable and with minimum error. While determining the parameters of the PI controller, the PID Tuner performs the optimal balancing process by considering performance criteria such as transient response, overshoot, rise time and settling time.

Thanks to these tuned parameters, when the block response of the PI controller is compared with the tuned system response, it is observed that the controller is designed in accordance with the system dynamics and provides the best temperature regulation. This optimization process made it possible for the temperature control system to operate both precisely and stably. The parameters of the PI controller were set according to the transfer function determined using the step response of the system. Accordingly, the modeling and simulation of the PI-based temperature control system was carried out in MATLAB/Simulink environment. In the Simulink model, the temperature system is modeled as a closed loop and the interaction between system inputs and outputs is visualized. Figure 3 shows the MATLAB/Simulink implementation of the temperature system of the PI controller. In this Simulink application, the reference temperature value desired to be reached by the heating system is selected as 60°C. Figure 5 shows the characteristics of the system performance when the simulation is run under these initial conditions. As shown in Figure 5, the output signal of the system is obtained according to the parameters determined in the PI controller design. The PI controller reached the maximum overshoot value at $t=240s$, raising the temperature to 73°C, and then the system continued to react and stabilized to the reference temperature value at $t=700s$.

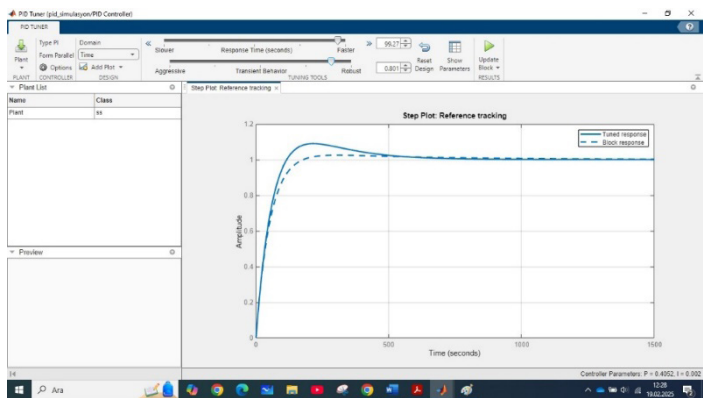


Figure 4. MATLAB PID tuner tool

2.3. Fuzzy Logic Controller Design

Fuzzy logic offers a system that works in a similar way to human decision-making, with the ability to generate meaningful and feasible solutions from specific, uncertain or approximate data (Swathi, Ebienezar, Swathi, & Suruthipriy, 2023). While traditional logic systems usually require precise and unambiguous inputs, fuzzy logic can work successfully with uncertain, noisy or incomplete data. This makes it highly suitable for controlling complex and nonlinear systems.

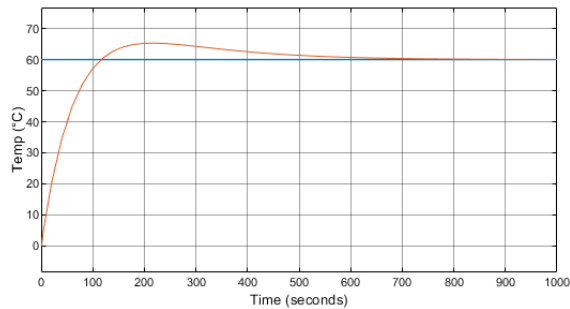


Figure 5. Output characteristics of PI Control with MATLAB/Simulink

Unlike traditional methods based on mathematical modeling, fuzzy logic control systems use an approach that works with linguistic expressions and relies on human intuition. This facilitates the control of processes that are difficult to describe with precise formulas. Fuzzy logic systems produce their outputs as a smooth and continuous function, even when they have a wide range of inputs. That is, even if there are sharp changes in the inputs, the fuzzy logic controller provides a smooth and fluid output without creating sudden jumps and instabilities.

One of the biggest advantages of fuzzy logic is that it can produce more than one output by processing multiple input variables thanks to its rule-based processing. While in classical control methods, it may be necessary to derive a mathematical model of a system, this is not necessary in fuzzy logic. Instead, a rule-based inference mechanism determined by IF - THEN logic is used. In this way, different output scenarios can be generated based on multiple input variables.

The basic parameters of fuzzy logic theory include elements such as fuzzification, rule base, inference mechanism, and stabilization, which determine how a system will work. These parameters are important components

that support the flexibility and wide range of applications of fuzzy logic. Figure 5 shows a diagram of the basic elements of fuzzy logic theory.

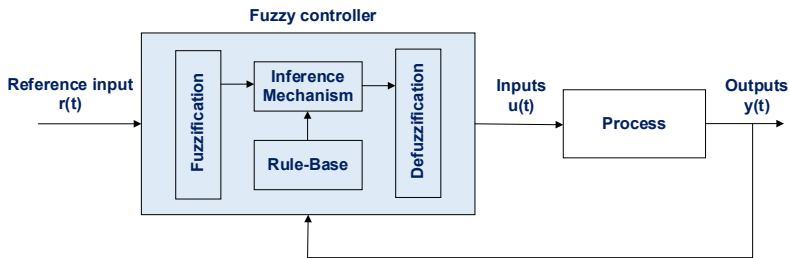


Figure 5. Basic components of a fuzzy logic controller

A fuzzy logic controller can be considered as an artificial decision maker operating in real time in closed-loop systems. By comparing the output of the system ($y(t)$) with the reference input ($r(t)$), it determines the control input ($u(t)$) and aims to achieve the desired performance of the system.

Four Basic Components of a Fuzzy Controller

1. Rule Base: Contains the “If - Then” rules on how to control the system.
2. Inference Mechanism: Calculates the control input by determining the appropriate rules for the current situation.
3. Fuzzification Interface: Transforms the incoming data into fuzzy values suitable for the rule base.
4. Clarification Interface: Translates the fuzzy results into precise control signals.

The design of a fuzzy controller requires understanding the closed-loop dynamics of the system and determining the appropriate control rules. This information can be obtained from human experts based on experience or from mathematical analysis of the system. By testing the designed rules and inference strategy, the performance of the control system is evaluated. Unlike conventional methods, fuzzy control offers the flexibility to manage uncertain and complex systems without the need for precise mathematical models.

A temperature controller using fuzzy logic is used to generate the control signal of a DC-DC converter. The generation of this control signal is determined by the error difference between the reference temperature value and the measured temperature value. The configuration of the fuzzy logic controller system is

shown in Figure 6. The fuzzy logic controller uses two inputs: first, the error (E) and second, the variation of the error (CE). The duty cycle determined by these two inputs allows the controller to generate the appropriate PWM (Pulse Width Modulation) signal to the DC-DC converter.

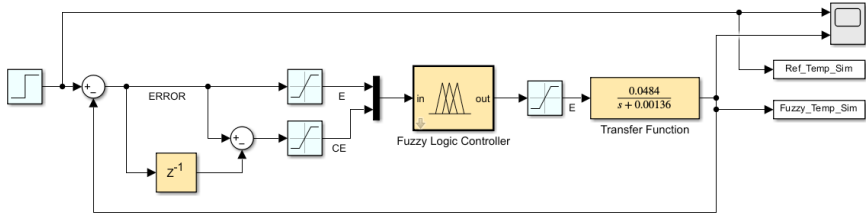


Figure 6. Fuzzy Logic controller MATLAB/Simulink diagram

Membership functions are defined as follows: Negative Large (NB), Negative Small (NS), Zero (ZE), Positive Small (PS) and Positive Large (PB).

A controller designed according to the 25 fuzzy rules defined in Table 1 is used. In this table, rows and columns represent the input variables Error (E) and Error Variation (CE). The values in the table cells represent the output variable (D).

Table 1. Rule table

	NB	NS	ZE	PS	PB
NB	NB	NB	NB	NS	ZE
NS	NB	NB	NS	ZE	PS
ZE	NB	NS	ZE	PS	PB
PS	NS	ZE	PS	PB	PB
PB	ZE	PS	PB	PB	PB

The membership functions used for the error variable (E) are shown in Figure 7. These functions consist of five triangular membership functions defined between -100 and 100. The membership functions for the error change are presented in Figure 8. These functions are modeled with five triangular membership functions ranging from -10 to 10. This structure allows the controller to categorize the input variables into specific fuzzy sets and generate a suitable control signal for the output variable.

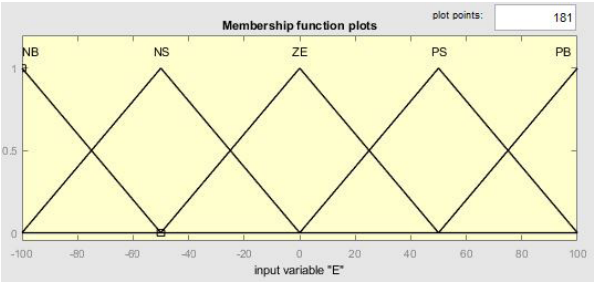


Figure 7. Error (E) membership function

PWM is a technique commonly used in electrical systems to regulate power or output voltage. A PWM signal produces a discrete signal at a specific frequency. This signal is shaped by a parameter called “duty cycle”. Duty cycle refers to the active (i.e. high) time within one cycle of the signal, relative to the total cycle time.

The Duty (D) variable determines the width of the PWM signal. For example:

- When $D = 0$, the signal is not active at all (0% duty cycle).
- When $D = 255$, the signal is fully active (100% duty cycle).

The amplitude of the PWM signal changes in direct proportion to this duty cycle. In this way, the output power of the system can be precisely controlled.

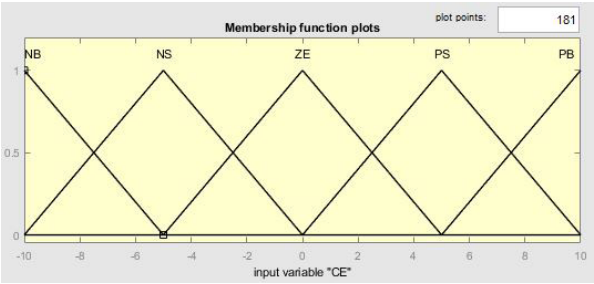


Figure 8 Change of error (CE) membership function

As shown in Figure 9, the five triangular membership functions allow to adjust the output power of the heater system much more accurately by appropriately modeling the duty cycle values between -255 and 255. This allows the system to operate more efficiently and reach the desired target faster and more stably.

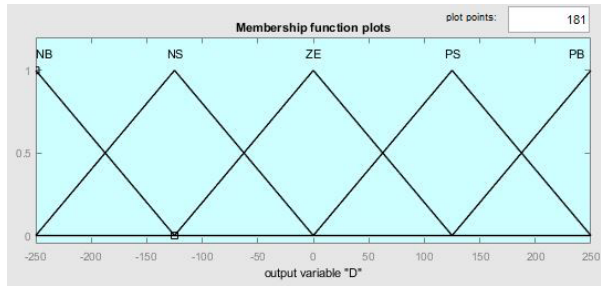


Figure 9 Duty (D) membership function

Figure 10 shows the simulation results of temperature control with fuzzy logic controller. As can be seen in the figure, the fuzzy logic controller reaches the reference temperature value in much less than 100 seconds and fully settles to the reference value after about 400 seconds. Also, no overshoot is observed.

2.4. Power Circuit Design

As shown in Figure 11, the system consists of three main components: the rectifier circuit, the TLP250 opto-coupler circuit and the forward converter. First, the rectifier circuit takes the mains voltage and first converts the AC signal into pulsative DC; then, using capacitors and similar filter elements, this pulsative signal is converted into a constant, smooth DC voltage. This step provides the clean and stable power supply needed by the other components of the system.

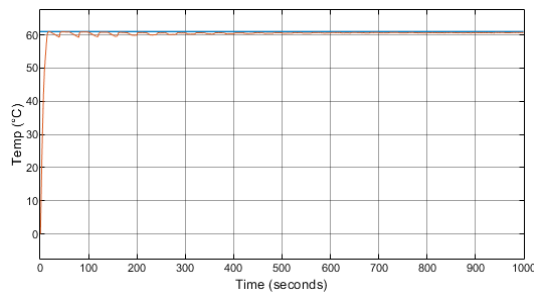


Figure 10. Simulation of temperature control with fuzzy controller

The TLP250 opto-coupler circuit takes the PWM signal from the controller circuit and isolates it via the optical path to the forward converter. This structure not only ensures that the signal is transmitted undistorted but also provides galvanic isolation, preventing high voltage and parasitic noise from reaching the controller circuit. Thus, the control signal required to drive the IGBTs is obtained safely and accurately.

The forward converter processes the DC power from the rectifier in isolation to form the power supply for driving the IGBTs. The PWM signal from the TLP250 switches the IGBTs quickly and reliably, improving the efficiency and performance of the system. As a result, the harmonized operation of these three components ensures both safe power conversion and efficient signal control, enabling the system to deliver high performance in industrial applications.

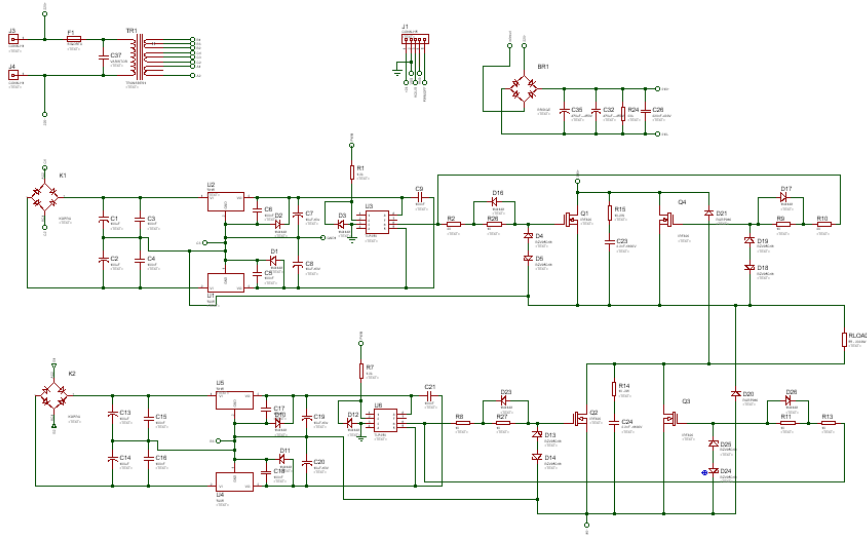


Figure 11. Power circuit

2.5. Experimental Setup

The experimental setup basically consists of three main components. These components are the insulated heating cabinet, the power supply and the control setup. These three components work together to provide the necessary temperature control during the experiment.

2.5.1. Insulated Heating Cabinet

The first component of the experimental setup is the heating chamber, which consists of an insulated furnace body to minimize heat transfer with the external environment. This chamber is specially designed to keep the temperature of the experimental environment under control and to ensure that it is not affected by external factors. The internal volume of the chamber is 35 cm wide, 40 cm deep and 55 cm high. This volume allows for the convenient placement of both the heating element and the sensors used for temperature measurement.

Heating is provided by a 2000Watt nichrome wire heating element located in the ceiling of the cabinet. Nichrome wire is a material that is resistant to high temperatures and has a high electrical resistance, which is why it is often used in such applications.

The LM35 temperature sensor used for temperature measurement is placed far away from the heating element in order to accurately detect the temperature distribution in the cabin and to minimize the direct radiation and convective effects of the heater. Thus, the sensor is able to provide more accurate data about the overall temperature inside the cabin. The structure and layout of the heating cabinet is shown in Figure 12.



Figure 12. Insulated furnace body

2.5.2. Power Supply

The second main component of the experimental setup is the power supply, which provides the necessary energy to the heater. The power supply is equipped with a conventional rectifier and filter circuit, which first converts the AC voltage from the grid into DC and then filters this DC voltage, resulting in a smoother and more stable output. This DC voltage is then transmitted to a power conversion stage called the forward converter. The forward converter acts to adjust the amount of power to be sent to the heater with appropriate control signals.

The physical structure of the power supply circuit is designed as a PCB board on which the components are arranged and the general view of this board is shown in Figure 13. An open circuit diagram of the power board is shown in Figure 11. The detailed working principle of the power supply is explained in

more detail in section 2.4 of the report. In summary, the power supply circuit not only converts the fluctuating energy from the grid into a suitable and stable DC voltage, but also adjusts how much power is transferred to the heater in accordance with the signals set by the control system.

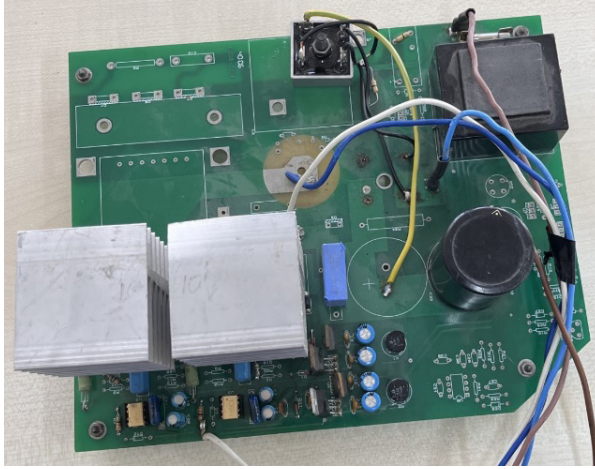


Figure 13. Güç kartı

2.5.3. Control Scheme

The third component of the experimental setup is the control system. This part includes the Arduino Mega 2560 microcontroller board and the sensor and driver circuits connected to this board. This control system works with a control algorithm prepared in MATLAB/Simulink environment. The model designed on Simulink takes the temperature information in the cabin as an input signal and generates a PWM (Pulse Width Modulation) signal to adjust the power of the heater based on this information.

Arduino Mega converts the analog temperature data received from the LM35 temperature sensor into digital form and transmits it to MATLAB. The control algorithm running in MATLAB decides how much power the heater should operate with using this temperature information and sends an appropriate PWM signal to the Arduino. Using this PWM signal, the Arduino drives the forward converter circuit in the power supply and thus precisely adjusts the amount of power to the heater. This closed loop control system ensures stable heating at the desired temperature throughout the experiment. The connections and components of the Arduino and the control system are shown in Figure 14.

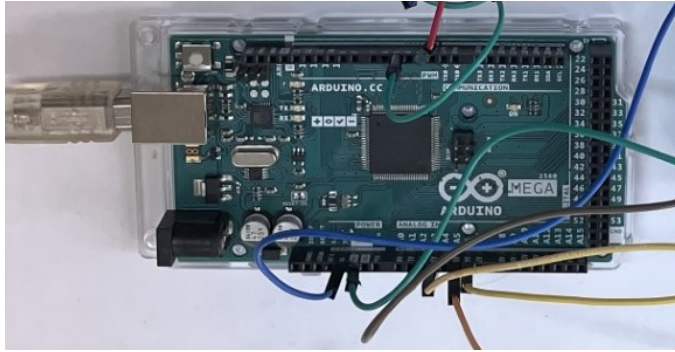


Figure 14. Arduino Mega 2560 microcontroller board

2.6. Real-Time PI Temperature Controller

The MATLAB/Simulink implementation of the PI controller used in real-time temperature control is shown in Figure 15. The hardware infrastructure of this system consists of Arduino Mega 2560 microcontroller. LM35 temperature sensor is used for temperature measurement. The temperature data received from the sensor is read via the analog input port A8 of the Arduino.

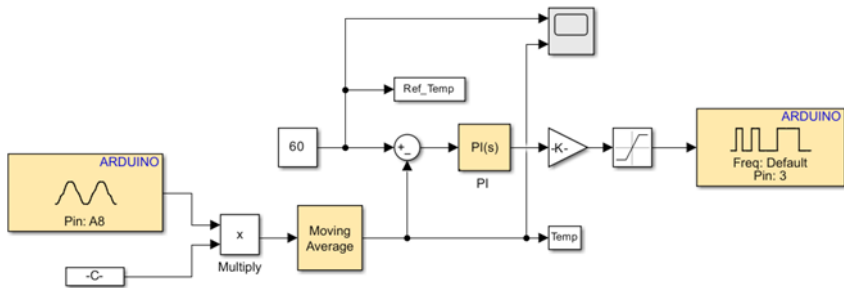


Figure 15. Real-time MATLAB/Simulink application

Arduino's analog ports have 10-bit resolution. Therefore, the temperature-related voltage value from the LM35 is converted into temperature units using a constant multiplier. This multiplier is calculated by dividing the 5V supply voltage of the Arduino by 1023.

In order to make the temperature data more stable and reliable, a moving average filter was applied to the measured temperature value in the Simulink application. Moving average is a statistical method that helps to determine the general trend of the temperature by suppressing short-term fluctuations in the temperature signal.

This filtered temperature value is compared with the reference temperature and applied to the input of the PI controller. The control signal generated by the PI controller, after being normalized, is transferred to the PWM output (Pin 3) of the Arduino and this output signal is fed to the PWM input of the power unit to control the heater element.

During the control process, the temperature values are recorded in the workspace using the “Temp” block in MATLAB/Simulink environment.

The performance of the system is monitored on the graph showing the change in time between the reference temperature and the measured temperature (Figure 16). As can be seen in this graph, the controller causes an overshoot over the reference temperature at around 90 seconds, but at around 500 seconds the system stabilizes and the temperature settles to the reference value.

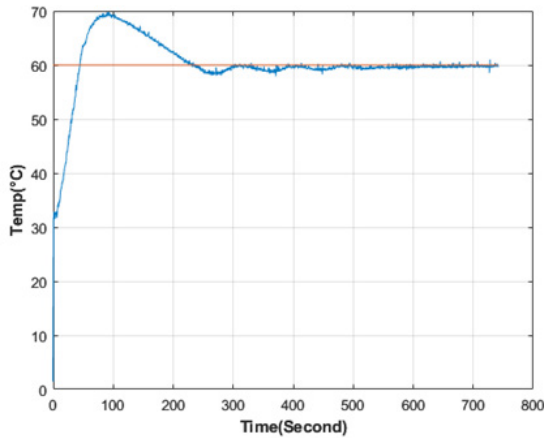


Figure 16. PI controller temperature graph

2.7. Real-Time Fuzzy Logic Temperature Controller

The MATLAB/Simulink implementation of a real-time temperature control system based on a Fuzzy Logic Controller is shown in Figure 17. In the hardware part of this application, the Arduino Mega 2560 microcontroller, which is also used in the PI controller, is in charge. LM35 sensor was preferred for temperature measurements. The temperature information obtained from the LM35 is transmitted to the system through the A8 analog input port of the Arduino.

Arduino’s analog inputs have a resolution of 10 bits. For this reason, the voltage signal obtained from the LM35 sensor is converted into temperature value with the help of a certain coefficient. This conversion coefficient was found by dividing 5V, the supply voltage of the Arduino, by 1023, the total digital value.

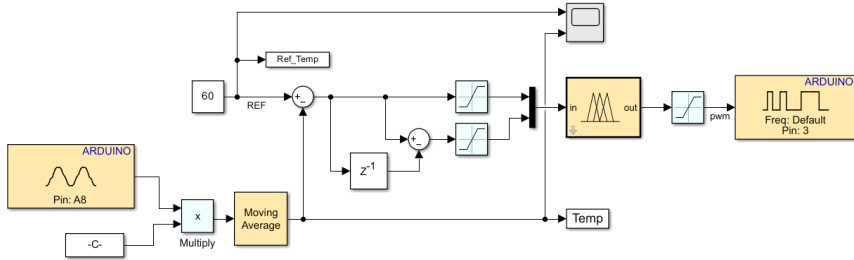


Figure 17. MATLAB/Simulink Real-time fuzzy logic temperature controller

In order to make the temperature information more stable and reliable, a “moving average filter” was applied to the temperature data read in the Simulink model. This filtering method is a statistical technique that reduces short-term fluctuations in the temperature signal while allowing a more accurate tracking of the overall temperature trend.

The filtered temperature value obtained as a result of this process is compared with the targeted reference temperature and transferred to the input of the Fuzzy Logic controller. After the control signal generated by the Fuzzy Logic controller is scaled to an appropriate level, it is directed to the PWM output 3 of the Arduino, which in turn drives the PWM input of the power driver to operate the heater element as desired.

During the control process, the temperature values are recorded in the workspace using the “Temp” block in MATLAB/Simulink environment.

The performance of the system is monitored throughout the control process by means of a graph showing the change over time between the reference temperature value and the actual temperature value measured by the sensor (Figure 18). This graph is an important indicator to visually evaluate both the time-dependent dynamic behavior of the system and the performance of the controller.

Looking at the graph, it is seen that there is an initial temperature when the system is first started and this temperature is lower than the reference temperature. During the control process, thanks to the control signal generated by the Fuzzy Logic controller, the heating element was activated and the temperature was raised in a controlled manner. Meanwhile, the response of the controller was very stable and stable. No overshoot was observed, indicating that the controller has a controlled approach without causing sudden and unnecessary heating while reaching the reference temperature.

Considering the time axis, it was found that the controller reached the reference temperature value at approximately 90 seconds. After this point, the

temperature was kept in a range very close to the target value and it was observed that the system stabilized. When the graph is analyzed, it is clearly seen that the system has completely stabilized by the 100th second. This shows that the Fuzzy Logic controller plays a very effective role in the process of reaching the target temperature and a stable control is achieved without unexpected oscillations or instabilities throughout the process.

In conclusion, the success of the Fuzzy Logic controller on real-time temperature control is clearly seen in this application. Both the ability to reach the reference temperature quickly and the ability to keep the measured temperature constant after reaching the target temperature show that this control approach is a highly effective and reliable method. This type of control structure can be easily preferred in industrial systems or laboratory applications that require temperature control.

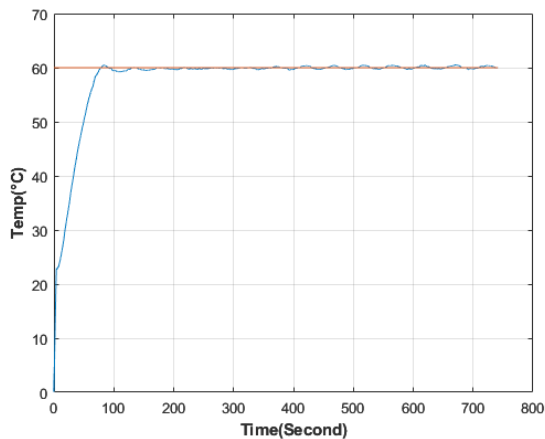


Figure 18. Fuzzy controller temperature graph

3. Conclusion

In this study, the design and performance of a fuzzy logic based temperature control system is investigated. The fuzzy logic controller, which is evaluated in comparison with conventional PI control methods, provides a more precise and stable temperature control by better adapting to the system dynamics.

In the experimental studies, the time for the PI controller to reach the specified reference temperature and the overshoot values were analyzed. It is observed that the PI controller reaches the reference temperature with a certain overshoot and the system stabilizes over time. However, when the fuzzy logic

controller is used, it is found that the system reaches the reference temperature in a shorter time and the overshoot rate is visibly reduced. This result shows that fuzzy logic-based control can perform more effectively especially in nonlinear systems and under variable operating conditions.

The operation of the experimental system with a power supply, Arduino Mega 2560 microcontroller and MATLAB/Simulink based control structure allowed the system to be monitored and optimized in real time. The temperature data obtained with the LM35 temperature sensor was used to evaluate the performance of both PI and fuzzy logic control systems.

The results show that the fuzzy logic controller responds more flexibly, especially to dynamic changes, and makes the system faster and more stable to reach the desired temperature value.

As a result, it is observed that the temperature control system designed with fuzzy logic controller performs better than the conventional PI control method and offers a more stable structure in terms of temperature regulation. Therefore, especially in complex and nonlinear processes, fuzzy logic-based control systems will provide significant advantages in terms of energy efficiency and precise temperature control. Future studies may focus on testing the system at different temperature ranges and in a wider range of industrial applications.

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CHAPTER XVI

ARTIFICIAL INTELLIGENCE-SUPPORTED SOFTWARE QUALITY ASSESSMENT: THE RELATIONSHIP BETWEEN CODE METRICS AND QUALITY WITH ARTIFICIAL INTELLIGENCE

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1. Introduction

Software is a critical component that forms the foundation of modern information systems. Therefore, software quality is a crucial factor that directly determines the overall quality of many systems. Factors such as software accuracy, efficiency, reliability, and maintainability play a significant role in assessing software quality. Traditional software quality assurance methods typically rely on manual code reviews, static and dynamic analyses, test automation, and version management. However, as the complexity of software projects increases and development processes accelerate, the effectiveness of these methods becomes increasingly limited.

In recent years, advancements in artificial intelligence (AI) technologies have the potential to revolutionize software quality assurance processes (Meziane et al., 2010). Artificial intelligence can provide significant benefits in areas such as enhancing software development processes by analyzing big data, detecting and predicting errors, and improving code quality (Amershi et al., 2019).

Additionally, with the help of AI-supported systems, software testing processes can be automated, feedback mechanisms can be provided to developers, and new approaches that enhance quality throughout the software lifecycle can be adopted. As the use of artificial intelligence becomes more widespread, the software industry has started to shift towards more agile and efficient methods in quality assurance. This section will explore the possibilities offered by AI to improve software quality and provide examples from academic research.

2. Software Quality Attributes and Code Metrics

Software quality attributes and code metrics are criteria used to assess the quality and efficiency of software development processes. In the field of software engineering, various metrics are used to enhance software quality and ensure the effectiveness of the development process. These metrics aim to measure the performance, maintainability, flexibility, and security of the software, providing essential data for making informed decisions during the software development process (Pressman & Maxim, 2005).

Software quality attributes and code metrics typically analyze various parameters at different stages of the software lifecycle (design, development, testing, maintenance). These metrics help developers understand how efficient, error-free, and sustainable the software is, while enabling managers to track the progress of projects and set directions for future improvements. Additionally, they can have a significant impact on factors such as software complexity, modularity, testability, and comprehensibility (Concas et al., 2012).

Some of the software quality attributes and code metrics serve as performance indicators based on factors such as code complexity, the amount of code duplication, test coverage, error rates, and user feedback. Software metrics are an indispensable tool for objectively evaluating quality during the software development process and ensuring continuous improvement (Sidhu et al., n.d.).

Software quality attributes can be categorized into four main areas: Complexity, Size, Coupling, and Lack of Cohesion. Software code metrics, as shown in Table 1, are the fundamental characteristics extracted from code that aim to explain these four quality attributes. They are typically obtained using static code analysis tools. Depending on the tool used to obtain these metrics, there may be minor differences in naming conventions or variations in the number of metrics. Generally, all the commonly encountered metrics are provided and presented with their widely accepted names. In Table 1, the quality attribute associated with each code metric is indicated.

Table 1. Source code metrics(Moshin Reza et al., 2021)

Nu	Source Code Metric Name	Related Quality Attributes	Description
1	Class lines of code (CLOC)	Size	The total number of code lines in a class, excluding comments.
2	Weighted method count (WMC)	Complexity, Size	The weighted sum of all methods within a class.
3	Depth of inheritance tree (DIT)	Complexity	The depth level of a class within the inheritance hierarchy.
4	Number of children (NOC)	Coupling	The count of subclasses derived from a class.
5	Coupling between object classes (CBO)	Coupling	The number of other classes a particular class depends on.
6	Response for a class (RFC)	Complexity	The total number of methods that can be invoked in response to a message received by an object of the class.
7	Simple response for a class (SRFC)	Complexity	The count of methods that can be directly called on a specific class instance.
8	Lack of cohesion of methods (LCOM)	Lack of Cohesion	A metric assessing how closely related the methods within a class are.
9	Lack of cohesion among methods (LCAM)	Lack of Cohesion	Evaluates class cohesion based on the parameter types of its methods.
10	Number of fields (NOF)	Size	The total count of attributes (fields) declared within a class.
11	Number of methods (NOM)	Size	The total number of methods defined in a class.
12	Number of static fields (NOSF)	Size	The count of fields declared as static in a class.
13	Number of static methods (NOSM)	Size	The count of methods declared as static in a class.
14	Specialization index (SI)	Complexity	Measures how extensively subclasses override the methods inherited from their parent classes.

15	Class-methods lines of code (CMLOC)	Size	The total number of non-empty, non-commented lines of code inside methods of a class.
16	Number of overridden methods (NORM)	Complexity	The number of inherited methods that have been overridden with the same return type.
17	Lack of tight class cohesion (LTCC)	Lack of Cohesion	A metric that quantifies cohesion between a class's public methods and subtracts it from 1.
18	Access to foreign data (ATFD)	Coupling	The count of attributes from external classes that are directly or indirectly accessed by a given class.

2.1. *Size*

In software development processes, size is one of the fundamental quality attributes used to measure the magnitude of a software system. This attribute is typically evaluated using code metrics such as Lines of Code (LOC), Number of Methods (NOM), and Number of Fields (NOF), and is directly related to the software's complexity, maintainability, and error rates (Srivastava et al., 2019). In software projects, inaccurate size estimations can lead to budget overruns and delays, making precise size measurements a critical factor for project success (Basili et al., 1996). Meta-analysis studies have shown that software size is directly related to quality, with larger projects typically having higher error rates [9]. Additionally, software functional size metrics are used as an important tool in IT governance to better manage development processes and enhance quality (De Castro & Hernandez, 2013). In conclusion, size is an important indicator for predicting the sustainability, maintenance costs, and error probabilities of software.

2.2. *Complexity*

In the software development process, complexity is directly related to a software's understandability, maintainability, and error rates. As complexity increases, the software becomes more difficult to develop, test, and maintain (Yu & Zhou, 2010). Metrics such as Weighted Method Count (WMC) and Response for a Class (RFC) are commonly used to measure software complexity (Nurminen, 2003). Literature studies indicate a strong correlation between software complexity and error rates; higher complexity typically results in more errors and increased maintenance costs (Munson & Khoshgoftaar, 1992).

Especially in large-scale software projects, complexity management is a critical factor in software reliability and cost estimation (Finkbine, 1996). In conclusion, controlling software complexity is a crucial way to enhance software quality and reduce maintenance costs.

2.3. Coupling

In software engineering, coupling measures the level of dependency between software components and has a significant impact on maintainability, error tendency, and reusability (Briand & Daly, 1999). Tightly Coupled means that changes made to one module can affect other modules, which reduces the flexibility of the software (Singh et al., 2012). Coupling is measured using various code metrics such as Coupling Between Objects (CBO) and Number of Children (NOC) (Hitz & Montazeri, 1996). Studies in the literature have shown that the level of coupling directly affects software quality, and high coupling tends to increase error rates (Yang et al., 2005). In conclusion, software designs with low coupling levels lead to the development of more modular, scalable, and sustainable systems.

2.4. Lack of Cohesion

In software engineering, **Lack of Cohesion** measures how related the methods and variables within a class are, providing an assessment of the software's quality. Low lack of cohesion (high cohesion) indicates a modular and well-organized codebase, while a high lack of cohesion value suggests that the class contains multiple responsibilities and may need to be refactored (Dallal & Briand, 2012). Studies in the literature have shown that classes with high cohesion values tend to have negative effects such as increased maintenance difficulty, higher error tendencies, and reduced reusability (Bieman & Kang, 1998). Among the various proposed cohesion metrics, versions such as LCOM1, LCOM2, LCOM3, and LCOM5 exist; it has been found that LCOM5 best evaluates class cohesion (Shih et al., 2001). The LCOM metric is considered an important criterion, especially in object-oriented software development processes, for increasing software modularity and ensuring the sustainability of the code (Curtis, 2002).

Code metrics, used to understand and assess the factors that determine software quality, are a critical tool for analyzing key quality attributes such as software complexity, dependencies, and cohesion. However, manually calculating and interpreting these metrics can be time-consuming and prone to human error. Today, AI-supported methods are being used to make this process more efficient in large-scale software projects. Artificial intelligence can analyze

code metrics to identify weaknesses in software, predict errors, and provide developers with suggestions to enhance quality. The next section will detail the integration of AI into software quality assessment processes and the advantages provided by this technology.

3. The Relationship Between Artificial Intelligence and Software Quality

Artificial intelligence plays a significant role in improving quality throughout the software development process. Software quality is not only directly related to the functionality of the code but also to factors such as sustainability, readability, and scalability. AI-based systems enhance software quality by providing automation and predictive analyses in these areas.

Today, software development processes require new approaches due to rapidly advancing technological innovations and increasing complexity. Continuous improvement of software quality is a critical factor in both reducing development costs and optimizing the user experience. In this context, artificial intelligence technologies play an effective role in enhancing quality by being integrated into software development processes.

AI-supported software development offers innovations in areas such as code automation, error detection, software testing, and performance analysis. Studies showing that AI-based software systems improve instructional quality highlight the successful approach this technology provides across various domains (Krishnamoorthy et al., 2013). One of the biggest advantages of artificial intelligence in enhancing software quality is its ability to analyze large datasets to identify hidden errors and optimization opportunities. This enables developers to make the debugging process faster and more effective.

Academic studies on the impact of artificial intelligence on software quality reveal that this technology can make a significant difference across a wide range of areas, from code analysis to test automation. This section will provide a review of the relationship between artificial intelligence and software quality based on current literature, with a focus on the most important developments in this field.

3.1. Code Quality and Error Detection

AI-based error detection systems automate the evaluation of code quality in software projects. While traditional error detection methods typically involve manual code reviews and static analysis tools, artificial intelligence allows for the rapid and efficient analysis of large-scale codebases. In particular, machine

learning algorithms, when trained on large datasets, have the ability to identify potential errors in the code and provide recommendations to developers.

For example, AI-powered error detection tools developed by Google can analyze millions of lines of code, identify common types of errors, and provide suggestions for fixing them. This enables faster and more effective error management compared to manual debugging processes.

Code quality is one of the most critical components of the software development process. High-quality code not only accelerates the development process but also simplifies the maintenance and sustainability of the software. Artificial intelligence offers significant innovations in evaluating code quality and error detection. Machine learning and artificial neural networks enable the development of algorithms that automatically detect and resolve errors in software (Kaya et al., n.d.).

AI-powered error detection systems can analyze large codebases and recognize error patterns that developers might not manually detect. These systems use methods such as automatic code interpretation and static code analysis to speed up the code review process. In particular, artificial neural networks have shown successful results in identifying potential error points within the code (Taşpınar & Isik, n.d.).

Among current applications, AI-powered error detection and correction systems such as Google's DeepCode and Facebook's SapFix stand out. These systems provide developers with error correction suggestions, reducing the manual debugging process. Additionally, AI-powered code completion tools like GitHub Copilot assist developers in writing faster and error-free code.

As the effectiveness of AI-powered error detection systems increases, the adoption of these technologies in software development processes becomes inevitable. In the future, it is expected that these systems will be further optimized, offering developers more accurate and predictable debugging methods.

3.2. Code Review Processes

Code review processes are a cornerstone of quality assurance in software engineering. Traditional code review methods are typically manual, time-consuming, and prone to errors. Artificial intelligence technologies provide significant contributions in automating and improving this process.

AI-powered systems accelerate code review processes, enabling developers to work more efficiently. Tools like GitHub Copilot optimize software development by offering code completions and providing suggestions regarding

potential errors, thereby enhancing the overall development workflow (Sidhu et al., n.d.). Additionally, AI systems based on Natural Language Processing (NLP) analyze code comments and documentation, providing developers with better documentation and enhancing the overall clarity and understanding of the codebase..

AI-based systems not only automate error detection in code review processes but also provide suggestions to enhance code readability and efficiency, contributing to overall code quality improvement (Dülger, 2021). These systems learn from large datasets to present best coding practices to developers, while also identifying faulty code patterns, thus helping to improve code quality.

Today, AI-powered tools are revolutionizing code review processes. These tools analyze the context of the code to detect errors and security vulnerabilities, while also guiding developers on how to write better code. Additionally, AI-based systems speed up the code review process, providing time savings in software development workflows (Ataseven, n.d.).

The integration of artificial intelligence into code review processes will continue to evolve in the future, offering developers customized error analysis, code optimization suggestions, and automatic correction options. These advancements will make significant contributions to enhancing software quality, marking the beginning of a new era in the field of software engineering.

3.3. Software Updates and Optimization

Software updates are critical processes regularly performed to close security vulnerabilities, improve performance, and add new features. Traditional update management often requires manual code reviews and testing processes, which can be time-consuming and prone to errors. Artificial intelligence optimizes this process, making it more efficient and reliable. (Jhamat et al., 2020).

AI-powered update systems use machine learning algorithms to analyze the current state of the software and can proactively identify potential security vulnerabilities. Additionally, these systems provide optimization suggestions that enhance software performance by utilizing system resources more efficiently. These approaches accelerate automated updates in large-scale software projects and improve system stability.

In large-scale software projects, continuous integration and delivery (CI/CD) processes are critical for maintaining software quality. Artificial intelligence performs data-driven analyses in these processes, predicting potential errors in software updates and offering preventive suggestions. For example, AI-powered

analysis systems used in Microsoft's DevOps processes detect potential errors in new code updates in advance, thereby improving software quality (Borra, 2024).

Today, major technology companies like Google and Microsoft actively utilize AI-based software update and optimization systems. For instance, Google's AI-powered Android updates determine the optimal update frequency for users' devices, thereby enhancing system performance. Similarly, Microsoft's AI-supported Windows updates offer optimizations tailored to each user's device configuration, ensuring that updates are applied more efficiently and seamlessly (Önder & Koç, 2024).

In conclusion, AI-based software update and optimization systems stand out as significant innovations that enhance efficiency in the software industry. In the future, the use of more advanced AI algorithms is expected to make updates more predictable, secure, and performance-focused.

3.4. Software Security

Software security has become a critical issue in the digital age, where cyber threats are increasingly prevalent. Traditional security methods may prove inadequate against complex and evolving attacks. In this context, artificial intelligence plays a significant role in enhancing software security..

Artificial intelligence also plays a crucial role in enhancing software security. While traditional security analyses and penetration tests are typically conducted manually, AI automates these processes, allowing for a broader threat analysis. For example, AI-based systems that analyze attack vectors can detect malicious code fragments and identify security vulnerabilities, thereby improving software quality (Attaallah et al., 2022).

AI-based security systems can perform functions such as anomaly detection, threat prediction, and automatic attack prevention by analyzing large datasets. Machine learning algorithms define attack patterns, enabling the development of more effective defense mechanisms against new and unknown threats (Sahu & Srivastava, 2021).

Especially deep learning-based systems have shown great success in identifying suspicious activities by monitoring network traffic. AI-powered firewalls and threat intelligence systems go beyond traditional signature-based security measures by creating continuously learning and adaptive defense mechanisms (Fang et al., 2020).

Today, large technology companies effectively utilize AI-based software security solutions. For example, Google's AI-powered reCAPTCHA system

continuously employs learning mechanisms to prevent bot attacks. Similarly, Microsoft's AI-supported cybersecurity solutions provide threat intelligence, developing proactive security measures (Almulihi et al., 2022).

In conclusion, AI-based security systems offer revolutionary solutions for detecting and preventing threats. In the future, it is expected that AI will develop more sophisticated security measures and integrate into autonomous defense systems.

3.5. User Experience

Artificial intelligence is also used to optimize the user experience of software. AI-based systems that analyze user feedback can detect issues encountered by users and provide preventive solutions. Particularly in large-scale software projects, user experience data is analyzed by artificial intelligence models to predict future errors and develop proactive solutions. (Kyle et al., 2004)

User experience (UX) plays a central role in modern software development processes. Artificial intelligence significantly contributes to UX optimization by analyzing user behavior and providing personalized experiences. AI techniques such as machine learning and natural language processing help make user interfaces more intuitive, while analyzing user feedback allows for continuous improvements (Aygenç et al., 2020).

AI-based recommendation systems predict user preferences and provide them with the most suitable content. For example, platforms like Netflix and Spotify offer personalized recommendations based on users' viewing and listening history, enhancing user satisfaction. Similarly, e-commerce platforms analyze users' shopping habits to develop more effective marketing strategies (Haoues et al., 2023).

Additionally, chatbots and virtual assistants stand out as another AI application that improves user experience. Through natural language processing techniques, systems have been developed that can understand user queries and provide faster and more accurate responses. For example, systems like Google Assistant and Amazon Alexa enhance the interaction process by understanding users' voice commands, making the experience more seamless (Nagappan & Shihab, 2016).

In conclusion, AI-powered user experience systems enhance user satisfaction by offering personalized and dynamic interactions. In the future, with more advanced AI algorithms, UX processes are expected to become more intuitive, automated, and human-centered.

4. Conclusion

In software engineering, quality is a critical factor for developing sustainable and reliable software systems. Traditional methods used to evaluate and improve software quality, such as manual code reviews and static analysis, may fall short in large and complex projects. In this context, AI-based approaches have the potential to automate software quality assurance processes, making error detection, code optimization, and maintenance processes more efficient.

In this study, the code metrics used to evaluate software quality have been discussed, and their relationship with key quality attributes such as Complexity, Size, Coupling, and Lack of Cohesion has been examined. The integration of AI techniques into software quality evaluation processes enables early detection of errors, acceleration of code review processes, and the creation of quality prediction models. By using AI techniques such as machine learning, natural language processing, and deep learning to analyze code metrics, human error in software development processes is reduced, while the reliability of the software is increased. Today, AI-based error detection systems developed by tech giants like Google, Microsoft, and Facebook have achieved significant success in improving software quality.

In the future, the integration of AI-supported systems into software quality processes is expected to become even more widespread. Specifically, automatic code correction systems, AI-powered test automation, and intelligent error prediction models could make quality management in software projects more efficient. Additionally, the development of AI-based quality measurement systems for large-scale projects will initiate a new era in the discipline of software engineering.

As a result, the combination of artificial intelligence and software quality metrics stands out as a significant innovation that enhances quality in the software industry. With the increase in academic research and the advancement of industrial applications in this field, it will be possible to develop more reliable, scalable, and sustainable software systems.

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CHAPTER XVII

ADDRESSING CLASS IMBALANCE AND MODEL COMPARISON FOR EARTHQUAKE PREDICTION: A STACKING METHOD APPROACH

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1. Introduction

Earthquakes are one of the most devastating natural disasters that affect millions of people worldwide and cause great loss of life and property. Turkey is a country that frequently experiences earthquake disasters due to its location on active fault lines. Therefore, it is vital to develop effective early warning systems in order to minimize earthquake damage. Early warning systems provide warnings before an earthquake occurs by analyzing seismic waves, allowing people to be directed to safe areas and critical infrastructure to be protected.

The effective use of early warning systems in Turkey can play a critical role in reducing loss of life and property, especially in large cities and industrial areas. Early warnings provided depending on the magnitude, epicenter and depth of the earthquake can prevent major disasters by ensuring the protection of schools, hospitals, power plants, transportation systems and other vital facilities. The success of such systems depends on the accuracy of the machine learning techniques used and the effective processing of earthquake data (Kolivand et al., 2024).

Machine learning has an important place in big data analysis and is used as a powerful tool to increase the predictability of natural disasters. While

classifying and analyzing earthquake data allows early warning systems to make more accurate predictions, challenges such as class imbalance can affect the accuracy of this process. Class imbalance can negatively affect the performance of classification models due to the fact that some earthquake types or magnitudes contain more data while others have very few examples (Bhatia, Ahanger, & Manocha, 2023) .

The features used to classify earthquake data are one of the most important factors that directly affect the accuracy of the models. The features used in this study include “magnitude”, which represents the magnitude of the earthquake, “depth”, which indicates the depth of the earthquake in the crust, “cdi” (Community Determined Intensity) indicating the shaking felt by the community, “mmi” (Modified Mercalli Intensity) indicating the acceleration magnitude at the time of the earthquake, and “sig” (Significance) indicating the potential destruction that the earthquake could cause. These features provide critical information to assess the severity and impact of an earthquake, allowing prediction models to produce more accurate results.

Proper processing of these features and presentation of them in a format suitable for classification models directly affects the success of early warning systems. In particular, modeling the alarm levels determined for earthquakes of different magnitudes and depths is an important factor in determining in which cases early warning systems will give a critical alarm. In this context, the accuracy of the predictions can be increased by optimizing and modeling the data using feature engineering processes. In particular, balancing unbalanced data sets, artificial data generation (with methods such as SMOTE) and feature selection methods can positively affect the success of the model.

In this section, we will study a classification problem based on earthquake data and compare the performances of different machine learning models. The models used include RandomForestClassifier (Breiman, 2001) , GradientBoostingClassifier (Friedman, 2001) , XGBClassifier (Chen & Guestrin) , CatBoostClassifier (Prokhorenkova, Gusev, Vorobev, Dorogush, & Gulin, 2018) , and StackingClassifier (Wolpert, 1992) . These models will try to predict alarm situations by analyzing earthquake data, and the effectiveness of each model will be evaluated by performance metrics such as F1 score. In countries with high earthquake risk such as Turkey, the development of such machine learning-supported early warning systems is of great importance in terms of public safety and disaster management.

2. Proposed Method

StackingClassifier model is used to increase the accuracy of earthquake early warning systems . StackingClassifier is an ensemble method that improves the final classification performance by combining the outputs of multiple machine learning models. This model aims to reduce generalization errors by taking advantage of the strengths of different classifiers. In our study, strong classifiers such as Random Forest, Gradient Boosting, XGBoost and CatBoost are used as base learners. The outputs obtained from these base learners are processed by a high-level model based on XGBoost to make the final prediction. This method provides more accurate and reliable predictions by combining the strengths of different classifiers. The flow diagram of the proposed method is given in Figure 1.

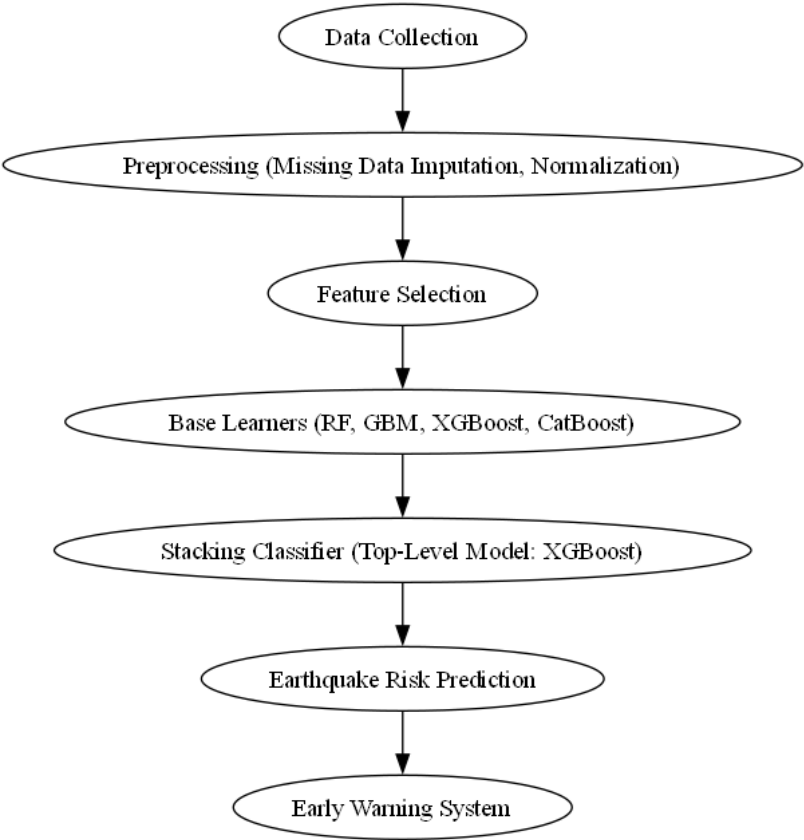


Figure 1 . Flow diagram of the suggested studies

Random Forest creates random subsets from different parts of the data using a collection of decision trees and produces separate predictions for each tree. This model was preferred because it reduces the risk of over-learning and provides high accuracy in large data sets. In addition, it is one of the important advantages of being able to obtain effective results in complex data structures and maintaining the balance of accuracy and stability.

Gradient Boosting is a method that aims to reduce errors by gradually improving weak learners. The model corrects errors made in previous stages with each new learner, allowing for better predictions. In large and variable data sets such as earthquake data, the Gradient Boosting model provides high success by optimizing the learning process.

XGBoost is a boosting algorithm known for its high speed and accuracy. Working with the regularized gradient boosting method, XGBoost has mechanisms that prevent over-learning and can produce results quickly on large data sets. In earthquake early warning systems, the computational efficiency and high accuracy provided by XGBoost are the reasons why the model is preferred.

CatBoost is an algorithm that stands out especially for its ability to process categorical variables. Earthquake data can often contain different categorical information, and it is important to prevent information loss during the processing of such data. CatBoost was preferred because it can show high performance directly with raw data by minimizing feature engineering.

In the proposed method, the predictions produced by the base learners are processed by a meta-model based on XGBoost to make the final prediction. The meta-model combines the information from the base learners to create a more powerful prediction mechanism. This structure minimizes the overall error rate by taking advantage of the strengths of each model.

In the classification phase, earthquake data were analyzed using different machine learning algorithms. RandomForestClassifier, GradientBoostingClassifier, XGBClassifier, CatBoostClassifier and StackingClassifier models were applied and the performance of each model was compared. The effectiveness of the models was analyzed with evaluation criteria such as F1 score and the model with the highest accuracy was determined.

3. Experimental Results

In this section, experimental results of various machine learning models applied on earthquake data are presented. The aim is to estimate the alarm level based on features such as “magnitude” representing the magnitude of

the earthquake, “depth” indicating the depth of the earthquake in the crust, “cdi” (Community Determined Intensity) indicating the shaking felt by the community, “mmi” (Modified Mercalli Intensity) indicating the acceleration magnitude at the time of the earthquake , and “sig” (Significance) indicating the potential destruction that the earthquake could cause . Pre-processing steps were performed on the dataset by applying SMOTE (Synthetic Minority Sampling Method) to handle missing values, scale the features, and eliminate class imbalance. After these steps, the performances of individual classifiers and a stacking model were evaluated.

Figure 2 shows the confusion matrix values for each model.

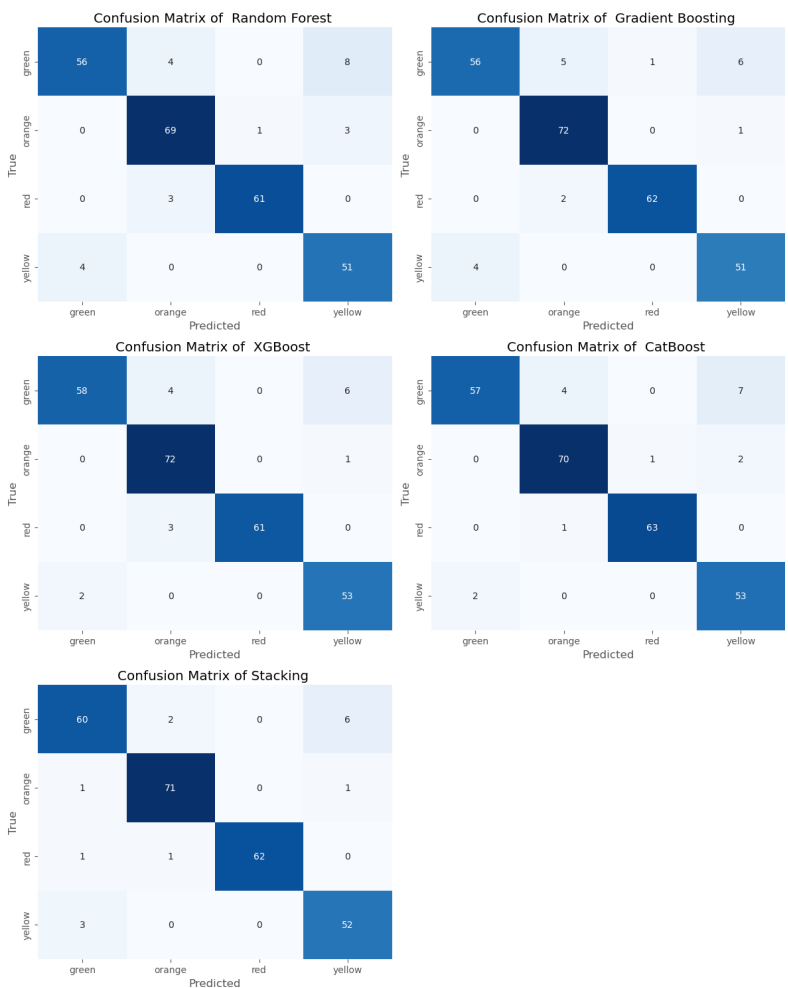


Figure 2 . Confusion matrix of models

The average values of accuracy, F1 score, precision and recall metrics for each model are given in the table below. These metrics were calculated to evaluate the overall performance of the models.

Table 1 . Results of classifiers

Model	Accuracy	Precision	Recall	F1 Scores
RandomForestClassifier	91.15	91.19	91.23	91.03
GradientBoostingClassifier	92.69	92.70	92.65	92.54
XGBoostClassifier	93.85	94.03	93.90	93.78
CatBoostClassifier	93.46	93.47	93.63	93.35
StackingClassifier	94.23	94.10	94.23	94.12

4. Conclusions

This study proposes a StackingClassifier based approach by comparing different machine learning models to increase the accuracy of earthquake early warning systems. The proposed method aims to produce more accurate and reliable predictions by combining powerful classifiers such as Random Forest, Gradient Boosting, XGBoost and CatBoost. In the study, the effectiveness of different models is analyzed with evaluation criteria such as F1 score and the effects of the class imbalance problem on model accuracy are investigated.

Experimental results have shown that the StackingClassifier model provides higher accuracy compared to individual learners. In particular, the success of the proposed method in classifying earthquakes of different magnitudes and depths increases its usability in early warning systems. In addition, it has been observed that techniques such as SMOTE applied to eliminate class imbalance have a positive effect on the model performance.

Future studies may aim to evaluate the overall performance of the model with larger and updated data sets. In addition, it is possible to increase the model accuracy with deep learning-based approaches and more complex time series analyses. This study provides an important contribution to the development of earthquake prediction and early warning systems and forms the basis for data-driven approaches that can be integrated into disaster management processes.

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CHAPTER XVIII

EDGE AND FOG COMPUTING WITH ARTIFICIAL INTELLIGENCE METHODS ON IOT-BASED BIG DATA

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1. Introduction

In the digital age we are in, with the widespread use of Internet of Things (IoT) technologies, there has been a significant increase in the generation of big data in various areas, from manufacturing to healthcare, from transportation to smart cities. When this big data produced by IoT devices is attempted to be processed in traditional cloud computing infrastructures, problems such as increased bandwidth usage, extended latency, and security vulnerabilities are encountered. For this reason, Edge and Fog computing approaches, which aim to increase system efficiency by processing data at points close to where it is generated, are gaining importance.

Edge and Fog layers, unlike the centralized cloud computing model in IoT-based systems, enable data to be processed at nodes by positioning it close to its source, the perception or service layer. While the Edge layer allows data processing at points closest to sensors and IoT devices that produce data, the Fog layer performs this process through servers in social living spaces or regional servers. This approach supports real-time applications by reducing latency while also minimizing network bandwidth consumption. In this context, artificial intelligence (AI) methods play a critical role in increasing the effectiveness and meaningfulness of big data analytics at the Edge and Fog layers. Deep learning, machine learning, and other AI techniques enable more effective analysis of data from sensors and similar IoT devices, accelerating decision-making processes

and enabling systems to operate autonomously. For example, in a smart city application, instant analysis of data from traffic cameras at local nodes to manage traffic is vital to optimize traffic flow and prevent accidents.

This study examines the advantages of integrating Edge and Fog computing models with artificial intelligence techniques in IoT-based big data environments. In particular, the effects of these approaches on critical parameters such as security, energy efficiency, latency, and system performance will be evaluated. The interaction of Edge and Fog computing with artificial intelligence in the developing IoT ecosystem will make significant contributions to the construction of more intelligent and efficient systems in the future.

2. Related Works

In recent years of digital transformation, the rapid spread of Internet of Things (IoT) technologies has made it necessary to process the big data in real time. Traditional cloud computing infrastructures cannot always respond adequately to this need due to high latency and bandwidth limitations. Distributed data processing models such as Edge and Fog Computing have been developed to overcome this problem. Edge Computing provides a low-latency and efficient solution by processing data at the closest point to the source, while Fog Computing optimizes data management by acting as an intermediate layer between the central cloud and edge devices. Supported by artificial intelligence (AI), these systems have the ability to instantly analyze data, make predictions and make smart decisions. This integration provides critical advantages in many areas, especially smart cities, healthcare, industry 4.0 and autonomous systems. In the study of Moosavi et al., the performance of the most advanced end-to-end security schemes of the Internet of Things systems in the field of healthcare is analyzed. The basic requirements for solving security problems for IoT systems in the field of healthcare are determined. A prototype was developed and tested to solve security problems. The prototype was created with a Panda board, a TI SmartRF06 card and WiSMotes. The platform setup created for testing the prototype was supported by Ubuntu OS and simulated with Contiki's cooja network simulator (Moosavi et al., 2018). Mary et al., in their published studies, although there are many protocol studies on energy-efficient protocols, they tried to design a protocol for wireless sensor networks, considering that these protocols are not suitable enough for "WSN" wireless sensor networks. The simulation was done using the Contiki operating system. The Cooja simulator, which is the object-oriented Java simulator of the operating system, was used.

All operations were tested for star topology. As a result of the tests, it is seen that the NullRDC driver consumes too much power and therefore is not suitable for a network that needs to operate on battery power. Contiki MAC offers low power consumption, high delivery rate and reasonable latency with low overhead (Mary et al., 2018). IoT and artificial intelligence studies are also examined under the title of smart cities. Advances in artificial intelligence approaches to automatically extract and classify disturbing sound events have great potential and application in the development of smart cities. In this study, an urban sound event classification system based on deep learning technologies is created using MEL frequency cepstral coefficients as feature extractors and Convolutional Neural Networks as classifiers. The designed system is trained and tested using UrbanSound8K and results in high classification accuracy in 92.67% of the tested results. In addition, verification of unknown sound events is also applied. In addition, the algorithm is implemented on a wireless sensor unit (WSU) that can record and classify sounds in the urban environment and send the classification data to the cloud. This application is suitable for real-time sound classification that can be used in smart cities based on the Internet of Things technology. Based on this situation, an IoT smart city layer using artificial intelligence for urban sound editing is proposed in the study (Domazetovska et al., 2022). In recent years, security in urban areas has become increasingly centralized and increasingly focuses on citizens, institutions, and political powers. Security problems have a different nature; to name a few, we can consider problems arising from the mobility of citizens, then move on to micro crimes, and finally face the ever-present risk of terrorism. It is vital for the safety of citizens that a smart city is equipped with a sensor infrastructure that can warn security managers about a possible risk. The use of unmanned aerial vehicles (IHAs) to manage the needs of citizens in order to prevent possible risks to public safety is becoming widespread. These risks have further increased with the use of these devices to carry out terrorist attacks in various parts of the world. Detecting the presence of drones, their small size, and the presence of only rotating parts. This study presents the results of studies conducted on the detection of the presence of IHAs in open/closed urban sound environments. For the detection of IHAs, sensors that can measure the sound emitted by IHAs and algorithms based on deep neural networks are used. The obtained results suggest the adoption of this methodology to increase the security of smart cities (Ciaburro G. & Iannace G., 2020). Emotional intelligence research, which emerged with the development of spring intelligence, is also examined on the big data produced

by sensor-based IoT systems. Studies in the field of IoT and Emotion Recognition reveal efforts to combine interdisciplinary knowledge and explore the effects of technological advances on the understanding and representation of human emotions. For example, the study by Gyrard et al., developed the concept of Emotional Knowledge Graph (EmoKG), which organizes emotional information from various fields using technologies such as Artificial Intelligence, Knowledge Graph and Semantic Reasoning. This approach shows how IoT technologies can be used to collect physiological data and how EmoKG can be used to recommend foods that can increase emotions within the naturopathy recommendation system. This study offers an important contribution with potential impacts on emotional intelligence and human health (Gyrard et al., 2022). In another study, a new layer is proposed to examine the emotional and physical states of employees in Virtual Reality (VR) environments. This layer offers the opportunity to collect data from simulated virtual environments and develop adaptable models for different contexts. It is built on an architecture focused on sensory perception, natural actions, narrative participation and social features. The authors evaluated the impact of each concept on IVR applications by proposing a set of categories supported by an artificial intelligence module for data analysis and feedback. In addition, physiological parameters were collected using VR glasses with storage and processing capacity. This facilitates integration with control, performance and IoT contexts. The main goal is to identify behavioral patterns that will support decision-making processes and emotional health management of employees (Llins et al., 2023). Another research aims to present a reliable and efficient speech emotion recognition framework that can work in real-time environments. Paralinguistic features of speech are used to train supervised machine learning models for emotion recognition. Experimental analysis and classification are performed using algorithms such as Gaussian Naïve Bayes, Random Forest, k-Nearest Neighbors, Support Vector Machine, and Multilayer Perceptron. SVM and MLP are found to be the best performing models with 77.8% and 79.6% accuracy rates, respectively (Jha et al., 2022). Another area where AI and IoT are integrated is the energy field and studies on sustainable energy management are available in the literature. In a different study, the AFED-EF algorithm is investigated for optimal virtual machine allocation aiming to reduce energy consumption in cloud data centers. The algorithm demonstrates higher energy efficiency by optimizing resource allocation for IoT application demands (Zhou, et al., 2021). Similarly, IoT and artificial intelligence methods act together for patient monitoring systems. Dziak

and his colleagues propose an IoT-based information system for indoor and outdoor use in their studies. A system is being developed to monitor the health problems that elderly people living alone will encounter in their daily lives. For the system design, the accelerometer module AltIMU-10 v4, the heart rhythm monitors sensor Polar's T34 and the Arduino ATmega32u4 module are used and tested with different ML algorithms (Dziak et al., 2017).

3. Internet of Things

The concept of IoT was first put forward in 1999 at the AutoID laboratories of the Massachusetts Institute of Technology. At the World Summit on the Information Society meeting held in 2005, the International Telecommunication Union (ITU) published the ITU Internet Report 2005: Internet of Things and officially proposed a definition for the concept of the Internet of Things (Kaya,2021). There are many intensive studies carried out in many institutions and organizations around the world regarding the concept of the Internet of Things. In line with these studies, different definitions regarding the concept of the Internet of Things are emerging; The basis of the concept of the Internet of Things is based on the ability of every object in our environment to interact and cooperate with each other in order to perform a common function using technologies such as sensors, mobile phones, etc. through a unique addressing system (Turgut, 2018). Marjani and colleagues define the concept of the Internet of Things (IoT) as creating a platform for sensors and digital devices to communicate seamlessly in a smart environment and ensuring that information is shared appropriately between the platforms (Marjani et al., 2017). Al-Fuqaha and colleagues define the Internet of Things (IoT) as a set of systems that control or regulate the seeing, perceiving, thinking and decision-making of physical objects, sharing data and communicating with each other (Al-Fuqaha et al., 2015).

Continuous communication of objects and devices is realized through various wireless technologies such as Bluetooth, WiFi, ZigBee, WSN, LPWAN and cellular network. These communication devices control data and receive commands from remotely controlled devices, directly integrated with the physical world through computer-based systems to improve living standards. More than 50 billion devices, including sensors, smartphones, laptops and game consoles, are expected to be connected to the Internet through various heterogeneous access networks provided by technologies such as radio frequency identification (RFID) and wireless sensor networks (Atzori et al., 2010). In general, an IoT system is a set of digital systems that include a large number of IoT devices, IoT

infrastructures, services, applications, and other applications or services, which can be modeled into four main layers as shown in figure 1 (Li, et al., 2015).

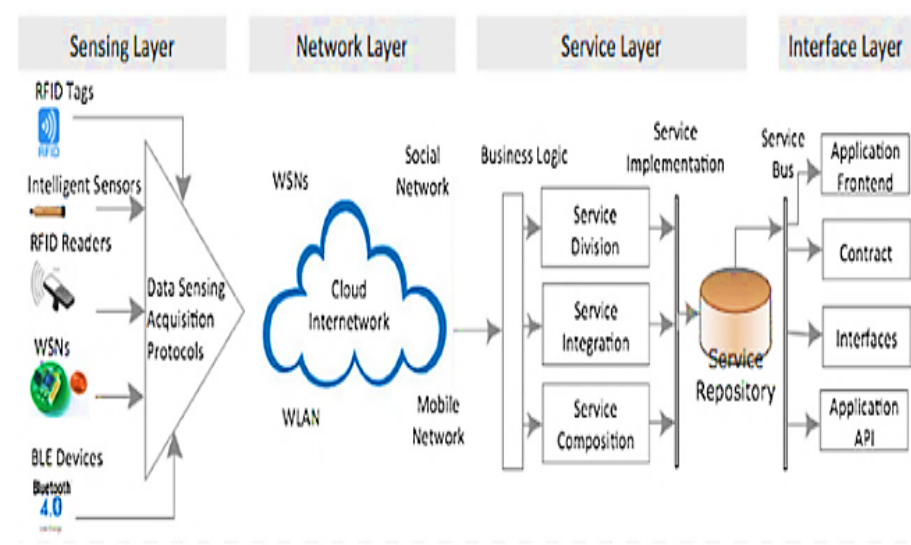


Fig.1: IoT Basic Architecture

Sensing Layer: It includes sensing devices such as smart sensors, radio frequency identification (RFID) and IoT client components to detect and obtain information (Kaya et al., 2023). It is the lowest layer in the IoT architecture and completes the task of collecting data from the physical environment. This layer includes devices such as sensors and actuators and perceives environmental parameters (e.g., temperature, light, motion, humidity, pressure) and converts them into digital data. This layer is the foundation of IoT systems because other layers operate solely based on the data obtained from this layer. The sensing layer is critical in terms of sensor accuracy and data quality (Dui, 2025).

Network Layer: It is the layer that supports the connection infrastructure with the Internet and other devices (Kaya et al., 2022). It is the layer that provides data transmission between IoT devices. Thanks to this layer, it plays a role in the secure transmission of data by using network communication protocols that enable data exchange between sensors and devices. It allows devices to communicate with each other or with a central data processing unit (e.g., cloud or edge computing) and generally uses wireless communication technologies such as Wi-Fi, ZigBee, 5G. The reliability of this layer is critical for the proper operation of IoT systems because devices cannot function effectively without data transmission and network connectivity. (Akhaldi, 2024).

Service Layer: It is the layer where the process of providing and managing services to users or other applications is seen (Bayram & Kaya, 2023). It is a layer that provides functional services in the IoT system. This layer analyses the collected data and draws meaningful conclusions from this data. The service layer provides the infrastructure where various services such as data management, data analysis, data processing and decision making must be performed. The layer usually works integrated with cloud computing or edge computing technologies and processes large data sets and transmits these results to users or applications (Arachchiege, 2024).

Interface Layer: It provides an interface to users or other services (Li et al., 2015). It provides integration between IoT devices and the application layer, allowing devices to communicate with each other and with users. This layer manages device connections, monitors their status, and is responsible for security. It provides data transmission between devices through different protocols and APIs, remotely controls and monitors them. It manages data efficiently by ensuring compatibility between devices, but it can also face challenges such as security and protocol diversity (Donta, 2022)

3.1. Edge Computing

In recent years, with the widespread adoption of sensor-based systems, the basic four-layer IoT architecture has become insufficient to manage the increasing volume of data. To control this data intensity, an intermediary layer is required. The edge computing layer is positioned immediately after the perception layer and acts as this intermediary. Edge computing is a data processing model that enables data to be processed near the source where it is generated, before being sent to a central location (Li and Wang, 2019). This layer reduces data transmission latency, optimizes bandwidth usage, and provides faster response times for real-time applications by processing the data directly from sensors. Local data processing also facilitates privacy protection and reduces the burden on central servers, resulting in a more efficient computing infrastructure (Shi et al., 2016).

Edge computing and the Internet of Things (IoT) are two critical technologies that optimize data management for intelligent systems. IoT is a system where smart devices, such as sensors and cameras, communicate with each other to collect real-time data. The edge computing architecture shown in figure 2 processes this data near its source, reducing dependence on central data centers. Consequently, data transmission latency is minimized, network congestion is alleviated, and real-time decision-making processes are

accelerated (Yousefpour et al., 2019). The integration of edge computing and IoT is particularly beneficial for intelligent applications requiring immediate intervention, such as traffic management, energy optimization, and emergency alerts, enabling the development of more efficient and secure intelligent systems (Satyanarayanan, 2017).

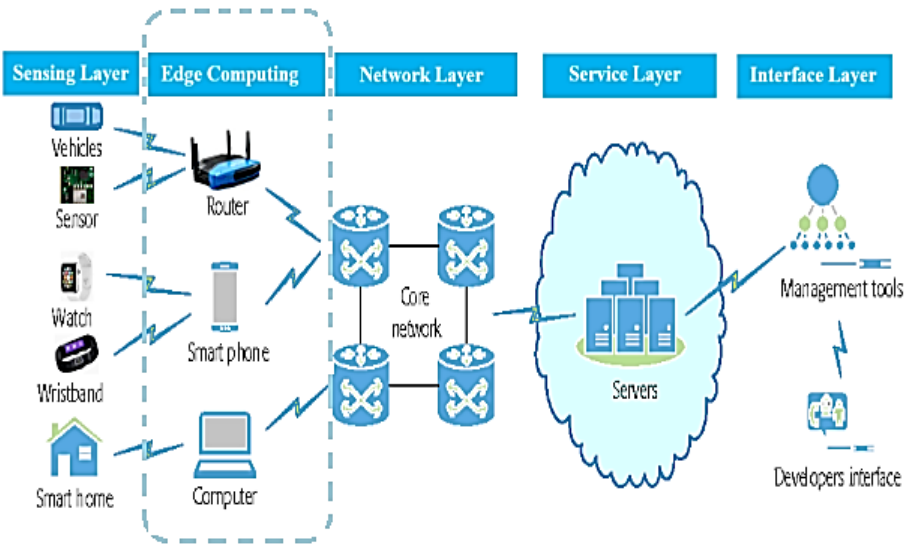


Fig.2: IoT Edge Computing Architecture

The integration of IoT and edge computing technologies provides highly accurate predictions based on sensor data. In intelligent systems, local processing through edge computing near the data source allows data to become more accurate, complete, noise-free, and easier to analyze before being transmitted to storage points (Zhang et al., 2018). This local processing also reduces data transmission latency, enabling real-time traffic density analysis and real-time predictions. These predictions optimize traffic signals and control data flow, reducing data congestion or suggesting alternative methods. Real-time traffic density predictions achieved through this technology contribute to a more responsive and adaptable transportation system in sensor-based environments, making the digital world smarter, more efficient, and sustainable (Katsigiannis, 2024).

3.2. Fog Computing

Fog computing is a new distributed computing model that extends the edge of the cloud computing network, addressing requirements such as low

latency and location awareness, especially for IoT applications. Traditional cloud computing models focus on centralized data centers for data processing and storage. However, the rapid increase in the number of IoT devices and the volume of data they generate causes issues in centralized cloud computing infrastructures, such as latency, bandwidth limitations, and security concerns. In this context, the concept of fog computing has emerged to mitigate these issues by processing data closer to its source (Bonomi et al., 2012). As shown in figure 3, fog computing forms an intermediary layer between edge devices and the central cloud, bringing computation, storage, and networking services closer to the network edge. This approach is characterized by low latency, wide geographic distribution, mobility support, and real-time application capabilities (Bonomi and ark., 2012). The fog architecture consists of three layers: edge devices, fog nodes, and the central cloud. While edge devices generate data, fog nodes process this data locally and forward it to the central cloud only when necessary, minimizing network traffic and latency (NIST, 2018). One of the primary advantages of the fog computing layer is the reduction in latency achieved by processing data near its source, which is crucial for real-time applications requiring high precision (Chiang and Zhang, 2016). Another advantage is that local data processing reduces the need to transfer large datasets to central cloud servers, thereby alleviating bandwidth burdens on the network (Dastjerdi and Buyya, 2016). Furthermore, local data processing enhances security and privacy by preventing the transmission of sensitive personal information to central servers. Despite these advantages, fog computing also presents challenges. Ensuring interoperability across different hardware and software platforms requires the development of standard protocols and interfaces. Additionally, it is crucial to design effective protection mechanisms against new security threats arising from the distributed architecture and resource management complexities (Stojmenovic and Wen, 2014).

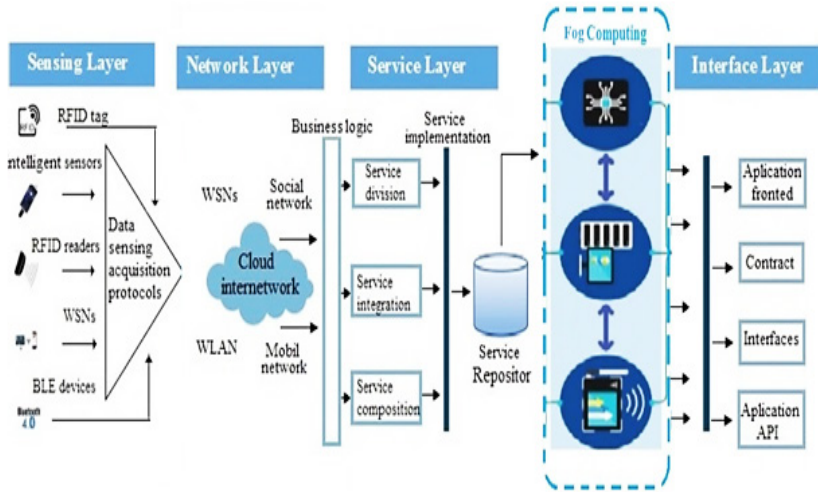


Fig.3: IoT-Fog Computing Architecture

4. The Relationship Between IoT and Big Data

Big data is a term that describes the exponential growth of both structured and unstructured data. It is characterized by high velocity, large volume, and extensive variety, presenting challenges in management, analysis, storage, and processing (Adam et al., 2017). The volume of data generated by sensors, social media, healthcare applications, temperature sensors, and other software applications and digital devices is increasing significantly. As a result of this massive data production, big data emerges (Marjani et al., 2017). The big data process is examined in two main sections: data management and data analysis. The aim of data management is to collect, store, clean, and prepare data for the analysis process (Sushree et al., 2018). The goal of data analysis is to model, analyze, and interpret the data to gain insights (Ge et al., 2018). Paakkonen and Pakkala propose a big data architecture that considers data sources and storage as the input and infrastructure of the big data process (Paakkonen & Pakkala, 2015). The proposed architecture consists of data extraction, data loading and preprocessing, data processing, data analysis, data transformation, and data visualization stages (Sushree et al., 2018).

The components of big data are divided into five subgroups. In a 2001 Gartner research report, these subgroups were defined as a three-dimensional “3V” model: variety, volume, and velocity (Adam et al., 2017). A transformation is occurring as data produced from various sources shifts from analog to digital environments. The concept of big data entered the computing world in the

early 2000s through industry analyst Doug Laney’s 3V model. According to Laney’s widely accepted definition, big data is described by Volume, Velocity, and Variety (Patgiri & Ahmed, 2016). Later, the SAS Institute added two more dimensions: Value and Verification. Figure 4 illustrates the characteristics of big data components, explaining volume, velocity, variety, value, and verification (Šahovska et al., 2019).

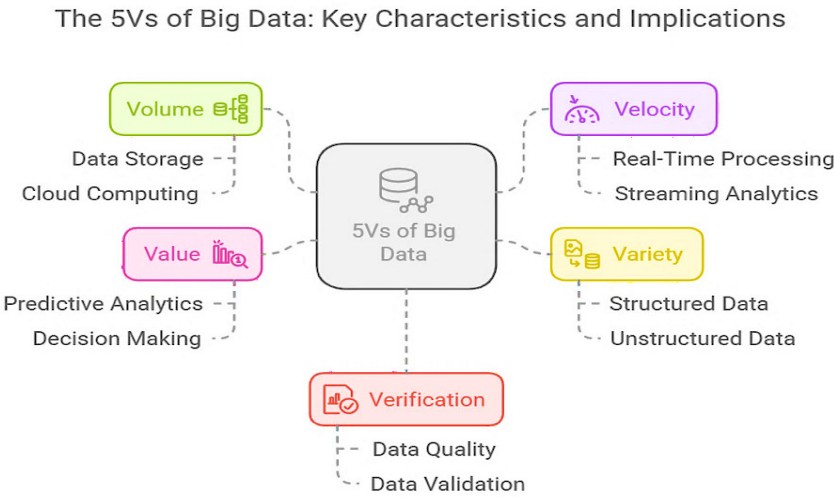


Fig.4: Big Data Components

✓ **Volume**

Volume refers to the size of the data collected. It indicates the amount of data generated (Karaboğa et al., 2022). Big data technology ensures the storage and retrieval of this data using various data storage techniques (Karaboğa et al., 2022).

✓ **Velocity**

Velocity represents the speed at which data is generated, stored, analyzed, and processed (Al-Nuamini et al., 2015). It reflects the rate at which data flows in and out of the system in real time (Sushree et al., 2018). Thus, big data requires high-speed connections and large bandwidth (Tosi et al., 2024).

✓ **Variety**

Variety refers to the different sources and types of data produced. As a result of increased data generation, most data is now unstructured and uncategorized, meaning it cannot be easily tabulated (Al-Nuamini et al., 2015). The term “variety” represents different categories of structured and unstructured data (Das et al., 2018).

✓ Value

Value refers to the importance of the data or the value derived from the information it contains (Tunc-Abubakar et al., 2023). It includes large volumes of diverse data that offer high-quality analyses and facilitate decision-making (Gonçalves S. et al., 2024).

✓ Verification

The accuracy of the information forming the data stream is crucial. To ensure that analyses support accurate decision-making, data must be sourced from reliable sources. Additionally, data accuracy is essential for ensuring the trustworthiness of the information (Ariffin, et al., 2023). In Big Data, the concept of verification addresses biases, noise, and unstructured data (Al-Ateeg, et al., 2022). As the volume of collected data increases, many useless, corrupted, or abnormal data points are also generated. With new Big Data technologies, such unnecessary data can be monitored and cleaned (Ghasemaghahi, et al., 2022). Increasing importance is being given to supporting real-time Big Data analysis (Al-Nuamini, et al., 2015).

Statistics indicate that by 2025, the number of internet users will reach approximately 6 billion, making the production of vast amounts of data an inevitable reality (Lan et al., 2017). IoT generates Big Data for many reasons, with one of the most significant being the transmission of sensor-generated data to network environments (O’Leary, 2013). IoT applications are among the largest sources of Big Data (Ahmed E., et al., 2017). This has led to the need for the integration of IoT and Big Data (Lan et al., 2017).

An IoT Big Data analytics platform must dynamically manage IoT data while ensuring interoperability with various heterogeneous objects (Saldatos, 2017). While cloud storage is the most widely accepted platform for storing large IoT data across all IoT domains, it is not specifically tailored to IoT data (Ge M., et al., 2018). However, in IoT, Big Data processing and analytics can be performed closer to the data source using edge computing or fog computing (Ahmed & Rehmani, 2016). Although the increase in data variety and volume due to IoT may seem like a disadvantage, it accelerates the development of Big Data analysis and applications. Furthermore, applying Big Data technologies to IoT accelerates both research advancements and business model innovations (Marjani et al., 2017). Ultimately, the relationship between IoT and Big Data is mutually reinforcing in terms of value addition, as shown in figure 5.

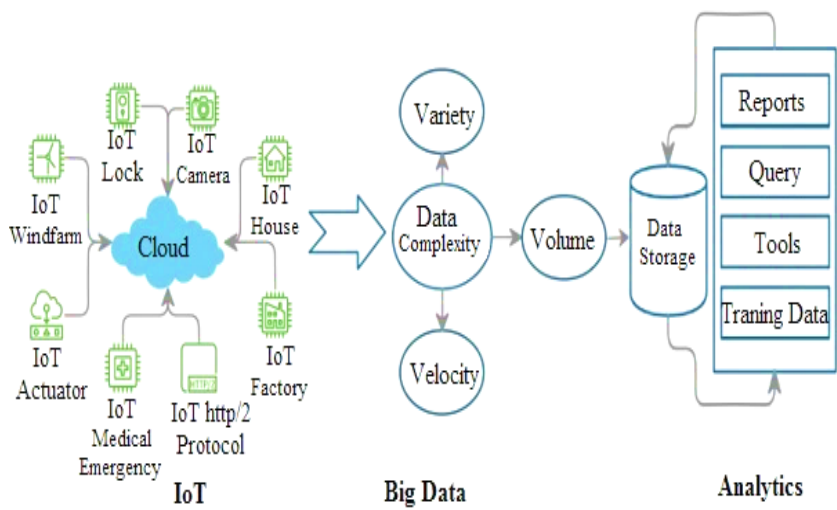


Fig.5: The Relationship Between IoT and Big Data

The need for Big Data technology in IoT applications is inevitable. These two technologies are already recognized in the field of information technology. However, despite this recognition, developments in Big Data technology progress more slowly than advancements in IoT applications. As a result, the rapid advancement of both technologies is interdependent. As shown in figure 5, the relationship between IoT and Big Data is examined in three parts. The first part deals with managing the data sources created by the interaction of sensors in the network environment. The second part is the Big Data domain, which consists of data differing in volume, velocity, and variety. The third part involves the application of data analysis and processing tools such as MapReduce, Spark, Storm, Flink, and Kafka. Statistical and machine learning methods are used for these analyses (Siddiga et al., 2016). Although many mechanisms exist for Big Data management, the evolving nature of Big Data also changes the requirements for data capture, preprocessing, and analysis. Big Data analysis requires the same or faster processing speed at minimal cost compared to traditional analysis methods for handling high-volume, high-velocity, and diverse data (Mukhopadhyay & Bandyopadhyay, 2014).

5.Conclusion

With the rapid proliferation of Internet of Things (IoT) devices in the digital age, the production of big data has increased, making the effective management

of this data a critical issue and the focal point of information management. When processing large-scale data from IoT devices using traditional cloud computing infrastructures, limitations such as latency, the inability to ensure real-time processing, and network bandwidth constraints emerge. Therefore, edge and fog computing approaches, which process data closer to the point of generation to reduce latency, provide significant solutions for big data management and analytics. Integrating artificial intelligence (AI) methods into these processes offers substantial advantages in data analytics and decision-making processes.

Edge and fog computing allow data processing on IoT devices or network gateways, reducing dependence on central systems and enabling real-time analysis. Meanwhile, fog computing acts as a bridge between central cloud and edge computing, allowing data to be processed on a broader and more regional scale. This minimizes issues related to high-latency data transmission while increasing system reliability and efficiency.

AI methods provide effective solutions in big data management, enabling advanced analyses and the development of decision-support systems in IoT-based big data environments. To achieve this, deep learning, machine learning, and optimization techniques play significant roles in big data analytics. As seen in the machine learning literature, these methods are used for tasks such as anomaly detection, prediction, and pattern recognition by processing data from IoT devices. For instance, machine learning algorithms can be used to predict traffic congestion in smart city applications. Deep learning, particularly in complex data analyses such as image and speech recognition, can be integrated with edge and fog computing systems to develop real-time decision-making mechanisms. Optimization algorithms facilitate more efficient operation of IoT devices by improving resource management and energy efficiency. For example, dynamic energy management using artificial neural networks can minimize energy consumption.

Processing IoT-based big data on edge and fog computing platforms alleviates the burden on central cloud systems while enabling real-time analysis. This approach offers benefits across a wide range of applications, from healthcare services to smart agriculture. In the future, more advanced AI algorithms and improved hardware are expected to enhance these systems' effectiveness further. In conclusion, AI-supported edge and fog computing approaches in processing IoT-based big data optimize data processing workflows while enhancing system performance. Increasing research in this field will contribute significantly to the development of smarter and more efficient systems.

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CHAPTER XIX

SMART IRRIGATION SYSTEMS: AN APPLICATION OF ARTIFICIAL INTELLIGENCE TECHNIQUE BASED AUTOMATIC IRRIGATION SYSTEM

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1. Introduction

In addition to its importance as a fundamental need for humans, water is also extremely important for plants. Population growth in the world also increases people's need for agricultural products. Water resources used for growing agricultural products account for 70% of the total water consumption (Taştan, 2019). The use of more efficient and effective irrigation is inevitable due to the decrease in agricultural water resources along with changes in climatic conditions.

In order to make an effective agricultural irrigation, it is necessary to use technological approaches instead of classical methods. In an irrigation performed by classical methods, farmers perform periodic irrigation depending on the type of the product they produce. The amount of irrigation is usually determined based on observation. Such classical methods are prone to human errors, and a failure to perform irrigation properly in certain circumstances negatively affects the agricultural activities (Nguyen et al., 2020). The use of information

technologies in agricultural areas has led to the emergence of smart and effective irrigation systems. The success of smart irrigation systems in terms of water consumption is also demonstrated in many studies that utilize information technologies (Angelopoulos et al., 2020; Campos et al., 2020; Canales-Ide et al., 2019). The common goal of these studies is to reduce water consumption and introduce an efficient irrigation program. In addition, different strategies are also being developed for the management of water resources used in agricultural irrigation (Ismail et al., 2020).

On the other hand, each plant has varying degrees of water requirement. The water requirement of each plant also varies depending on the life cycle of the plant. Given the climatic conditions, it is understood that numerous factors are effective in the irrigation program. Methods such as sprinkler and drip irrigation are often used in agricultural irrigation systems. Studies conducted in the literature have mostly focused on finding the most optimal values for the amount of water used in these methods by taking into account the climatic conditions. At this point, however, some important factors are not taken into account by these methods. The first is that each plant has a different water requirement. Second, water requirement varies depending on the life cycle of each plant. For this reason, studies taking into account the entire life cycle of the plant or studies that consider only the instant irrigation program do not provide effective results. This necessitates developing new approaches and seasonal optimization for today's agricultural irrigation. In addition, advances in intelligent technologies are also indispensable for agricultural irrigation programs. New generation technologies used in irrigation systems also increase the technological demands in agriculture (Işık et al., 2017). As in all areas, agricultural irrigation programs also strive to achieve optimal values in line with the advances in information technologies (De Ocampo & Dadios, 2017)

Numerous methods can be used in smart irrigation systems. Fuzzy logic is one of these methods. The understandable structure of the fuzzy logic, its advantages in fault tolerance, and its use in irrigation systems make this method popular (Li et al., 2019). Irrigation algorithms made by using fuzzy logic have demonstrated the efficiency and success of this method (Alomar & Alazzam, 2019; Fierro-Chacon & Torres-Tello, 2019; Hamouda, 2017; Poyen et al., 2020). However, the systems proposed in existing studies are limited regarding the best performance. In order to obtain a maximum efficiency for the success of the system, it is necessary to use different methods in a hybrid model in the system.

The low-cost smart irrigation system developed in this study was created using a hybrid algorithm that combines genetic algorithm and fuzzy logic. As the sample plant in the study, walnut was chosen since it is grown in many regions in Turkey. Walnut has four different life cycles. A separate irrigation program was developed for each stage. First, the actual water requirement was calculated for each cycle, and these values were used to create fuzzy logic structure. Then, a hybrid structure was created using the genetic algorithm to optimize the performance of the fuzzy logic structure. This structure was implemented separately for each cycle, and a low-cost prototype system was implemented. The proposed irrigation system takes soil moisture and ambient temperature as input data for each cycle separately, and outputs the necessary water requirement. The boundary values of the membership functions created in the fuzzy logic were optimized using the genetic algorithm to maximize the system performance.

2. Material and Methods

2.1. *Fao Penman Monteith*

The United Nations Food and Agriculture Organization (FAO) proposes the FAO Penman-Monteith (FAO-PM) method for determining the water needs of plants (Allen et al., 1998) FAO-PM method aims an irrigation with a more precise calculation by taking into account the geographical location of the agricultural region, soil structure, climatic conditions and the life cycle of the plant to be grown. The equation used to calculate the water requirement of a plant according to this method is given in Equation 1. This equation is used for calculating the reference evapotranspiration (ET₀) value of a plant (Bülbül et al., 2021).

$$ET_0 = \frac{0,408 * \Delta * (R_n - G) + \gamma * (\frac{900}{T + 273}) * u_2 * (e_s - e_a)}{\Delta + \gamma * (1 + 0,34 * u_2)} \quad (1)$$

In Equation 1, ET₀ represents reference evapotranspiration (mm day⁻¹). R_n represents net radiation at the crop surface (MJ m⁻² day⁻¹). G represents soil heat flux density (MJ m⁻² day⁻¹). T represents the mean daily air temperature at a height of 2 m (°C). U₂ represents wind speed at a height of 2 m (m/s⁻¹). e_s represents saturation vapor pressure (kPa). e_a represents actual vapor pressure (kPa). Δ stands for the slope of the vapor pressure curve (kPa/°C⁻¹). Γ stands for the psychrometric constant (kPa/°C⁻¹).

The reference evapotranspiration (ET_0) value of the plant calculated according to the FAO-PM method is used to determine the water requirement of that plant depending on its life cycle. The water requirement of a plant according to its life cycle is determined by calculating the ET_c (crop evapotranspiration) value (Bülbül & Öztürk, 2022) Calculation of the ET_c (crop evapotranspiration) value is given Equation 2.

$$ET_c = k_c * ET_0 \tag{2}$$

In Equation 2;

ET_c : Crop evapotranspiration

k_c : Average effects of evaporation in soil and plant characteristics

ET_0 : Reference evapotranspiration.

The k_c value varies according to the type and life cycle of the plant. FAO Penman Moteith method is more successful in terms of water consumption compared to classical irrigation systems (Mannan J et al., 2020).

The life cycle of the walnut plant, taken as an example in this study, consists of 4 cycles and lasts an average of 173 days. Actual measurements were made for 173 days and the amount of water needed by the walnut plant was calculated according to the FAO-PM method. Figure 1 shows the plant water requirement determined by the FAO-PM method.

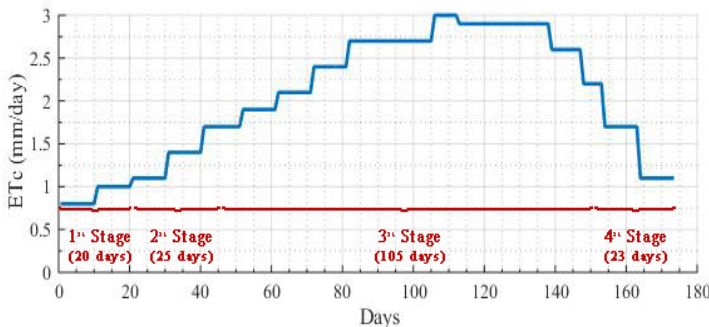


Figure 1: Actual plant water consumption calculated according to the FAO-PM method (ET_c)

The developed genetic-fuzzy-based model is expected to make an optimal prediction based on water requirements. The optimal irrigation program for each cycle is determined depending on these estimates. At this point, data on climatic conditions should be known to estimate optimal values. These data are the basis for creating estimates at each cycle of the plant.

2.2. Fuzzy Logic

Fuzzy logic is a technique employed to handle complex, uncertain, indefinite, hard to model attributes in order to clarify information by supporting linguistic expressions (Pospíchal, 1996). Fuzzy controller consists of four basic elements (Zadeh, 1965):

- i. Fuzzification
- ii. Defuzzification
- iii. Rule-base
- iv. Inference mechanism

Fuzzification is the process of converting a real numeric input values into linguistic terms by the membership functions.

Defuzzification is the process of converting a fuzzy set transferred from a fuzzy inference engine unit to truth values.

Rule-base refers to linguistic expression of the control rules that characterize the expert's knowledge, skills, and control strategy.

Inference mechanism executes fuzzy logic using rules and establishes a connection between the input and output space using the fuzzy rule base (Karataş et al., 2020). The general structure of the fuzzy logic is shown in Figure 2.

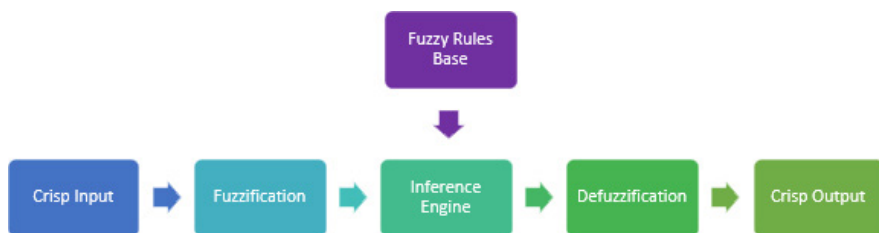


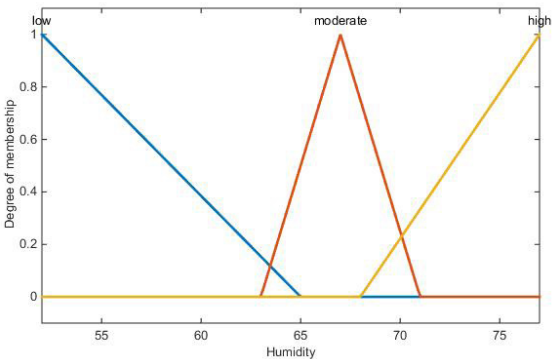
Figure 2: Fuzzy logic structure

Fuzzy logic based decision making processes are successfully implemented in agricultural irrigation applications (Mendes et al., 2019). Fuzzy logic is used to estimate the plant water requirement in the smart irrigation system. In the prototype model developed, ambient temperature and soil moisture values were used as input data for the fuzzy logic. The membership functions and boundary values necessary for the hybrid system were determined according to the measured values. Ambient temperature, soil moisture and plant water requirement boundary values measured by the direct observations are presented in Table 1.

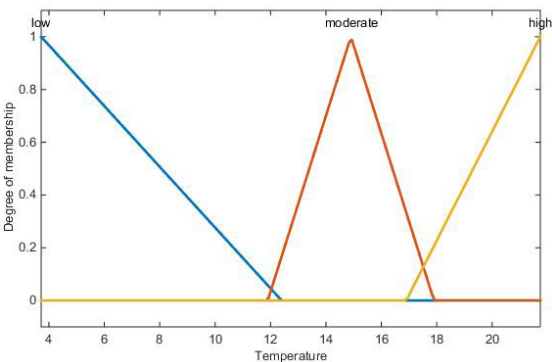
Table 1: Ambient temperature, soil moisture and plant water requirement limit values

Parameters	Minimum	Maximum
Temperature	3,72 C°	21,74 C°
Soil Moisture	% 52	% 77
Plant Water Requirement	0,8 mm day ⁻¹	3 mm day ⁻¹

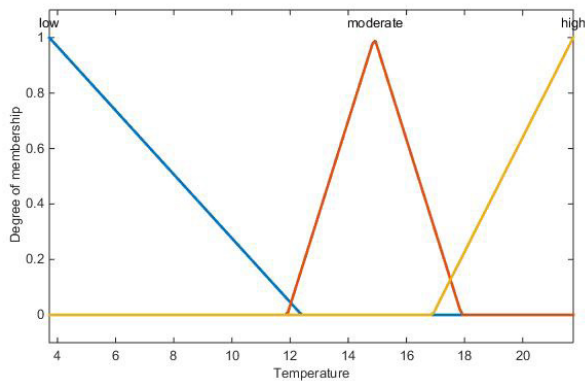
The values of the membership functions created in the fuzzy logic according to the boundary values of the input and output parameters are shown in Figure 3. Since they are commonly applied in the literature with successful outcomes, the triangular membership function, Mamdani inference method, and centroid defuzzification method were preferred. (Mohapatra & Lenka, 2016; Mushtaq et al., 2016). The membership function data were divided into three groups: low, medium and high.



a) Membership function for humidity [Input]



b) Membership function for temprature [Input]



c) Membership function for the amount of water [Output]

Figure 3: Fuzzy logic input-output membership functions

Nine rules were set for the fuzzy logic, consisting of two inputs and one output. The rule base of the system is presented in Table 2.

Table 2: Fuzzy logic rule base

Rules	Soil Moisture	Temperature	The amount of water
1	Low	Low	Medium
2	Low	Medium	Medium
3	Low	High	High
4	Medium	Low	Low
5	Medium	Medium	High
6	Medium	High	High
7	High	Low	Low
8	High	Medium	High
9	High	High	Medium

2.3. Genetic Algorithm

The goal in optimization problems is to find unknown parameter values within certain boundaries (Murty, 2003). Artificial intelligence optimization algorithms are often used to solve optimization problems. In evolutionary artificial intelligence algorithms, the best known algorithm is the genetic algorithm, which is based on the principle of finding the best individuals (Martins & Neves, 2020). The steps of the genetic algorithm are as follows (Jaiswal et al., 2019):

Step 1: A random initial population is created, consisting of individuals (chromosomes) that are candidates for the solution.

Step 2: The fitness value of each individual is calculated according to the fitness function defined for the problem. This value indicates the quality of the solution.

Step 3: Individuals with higher fitness values are selected to be included in the next generation.

Step 4: A crossover operation is applied between the selected parents with a certain probability, resulting in the creation of new individuals.

Step 5: Small modifications are made to the genes of individuals with a certain probability.

Step 6: Repeat step 2 until the condition is true.

A genetic algorithm uses an initial population consisting of some solutions found in the solution space. In each generation, new individuals are created by individuals of the previous generation. These individuals are formed by crossover and mutation events. In the genetic algorithm that uses natural selection, dominant individuals (good solutions) are likely to survive (Karaboğa, 2011).

In order for the proposed system to use the water resource most effectively, the boundary values of the fuzzy logic membership functions were optimized by a genetic algorithm. The “MATLAB” programming language was used to create the program necessary for the optimization of boundary values. Figure 4 shows the flow chart of the genetic-fuzzy-based system developed.



Figure 4: Genetic-fuzzy model flowchart

As a result of the tests, parameters were determined as shown in Table 3.

Table 3: Genetic algorithm parameters and values

Genetic Algorithm Parameters	Values
Population Number (n)	100
Solution Space (D)	15
Selection Rate (c)	0,9
Mutation Rate (m)	0,03
Iteration Number (T)	1500

In a hybrid system, each gene in the population falls within the boundary values of the membership functions of the genetic algorithm. There are 15 parameters to be optimized, excluding the minimum and maximum boundary values, to create triangular membership functions. These parameters are shown in Figure 5.

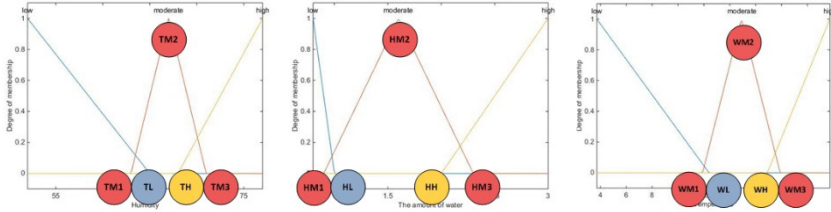


Figure 5: Fuzzy logic parameters to optimize

Each gene consists of these 15 parameters. Example gene sequence is presented in Table 4. where, T is temperature, H is soil moisture and W is water requirement. L denotes low, M middle, and H high triangular membership functions.

Table 4: Gen sequence

TL	TM1	TM2	TM3	TH	HL	HM1	HM2	HM3	HH	WL	WM1	WM2	WM3	WH
----	-----	-----	-----	----	----	-----	-----	-----	----	----	-----	-----	-----	----

In the Initialize Population step, the boundary values in the Table 1 are taken into account for creating each gene, while the 15 parameters found in each gene are determined by considering the boundary functions in Equations 3-8.

$$P = \{T(\text{Temperature}), H(\text{Humidity}), W(\text{the amount of water})\} \quad (3)$$

$$P_k L_i(x) = \begin{cases} P_k L_{\min}, & P_k L_i < P_k L_{\min} \\ P_k L_i, & P_k L_{\min} < P_k L_i < P_k L_{\max} \\ P_k L_{\max}, & P_k L_i > P_k L_{\max} \end{cases} \quad (4)$$

$$P_k M1_i(x) = \begin{cases} P_k L_{\min}, & P_k M1_i < P_k L_{\min} \\ P_k M1_i, & P_k M1_i < P_k M2_i \wedge P_k M3_i \\ P_k L_{\max}, & P_k M1_i > P_k L_{\max} \end{cases} \quad (5)$$

$$P_k M2_i(x) = \begin{cases} P_k L_{\min}, & P_k TM2_i < TL_{\min} \\ P_k M1_i, & P_k M1_i < P_k M2_i < P_k M3_i \\ P_k L_{\max}, & P_k M2_i > P_k L_{\max} \end{cases} \quad (6)$$

$$P_k M3_i(x) = \begin{cases} P_k L_{\min}, & P_k M3_i < P_k L_{\min} \\ P_k M3_i, & P_k M3_i > P_k M1_i \wedge P_k M2_i \\ P_k L_{\max}, & P_k M3_i > P_k L_{\max} \end{cases} \quad (7)$$

$$P_k H_i(x) = \begin{cases} P_k L_{\min}, & P_k H_i < P_k L_{\min} \\ P_k H_i, & P_k L_{\min} < P_k H_i < P_k L_{\max} \\ P_k L_{\max}, & P_k H_i > P_k L_{\max} \end{cases} \quad (8)$$

The fitness function used in the genetic algorithm for the boundary values of the triangular membership functions to be optimized in the hybrid system is given in Equation 9. A fuzzy logic is created according to the values specific for each gene found in the population. All the actual input data for each gene are used in the fuzzy logic created for that gene to estimate the amount of water the plant needs. The fitness function gives the success of the fuzzy logic created for each gene.

$$f(X_i) = \sum_{j=1}^{173} |W_j - evalfis_i(T_j, H_j)| \quad (9)$$

In the Equation 9;

$f(X_i)$: i. gen's fitness value

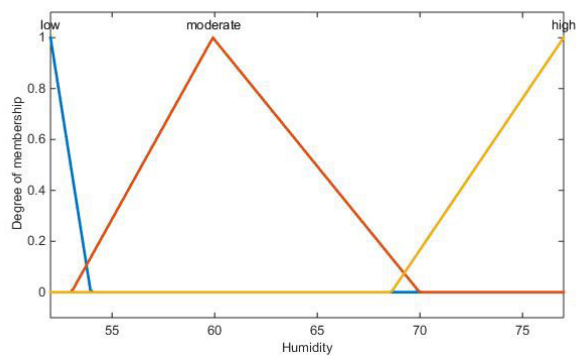
W_j : The actual water requirement of the plant for the j. day

T_j : j. day temperature value

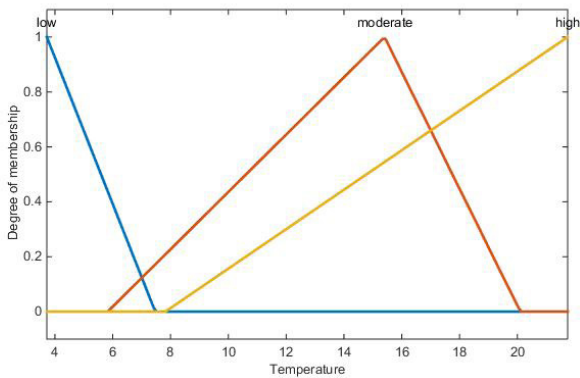
H_j : j. day soil moisture

$evalfis_i$: Plant water requirement prediction of fuzzy logic created for the i. gene

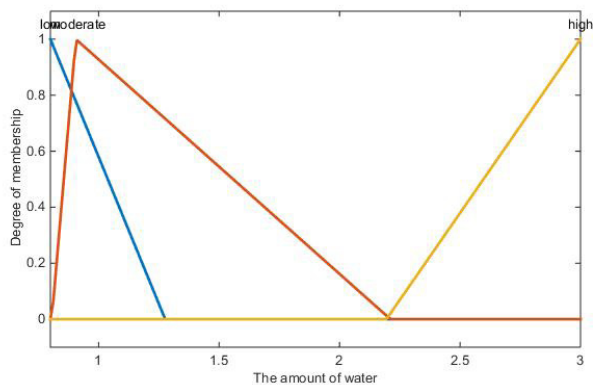
In the genetic-fuzzy model algorithm developed, the roulette wheel selection was used for gene selection. In the crossover step, single-point crossover was used as the gene crossover method. The hybrid system was executed by performing 1500 iterations. The best gene selection was made according to the calculated fitness values as stated in Equation 9. The membership functions of soil moisture, temperature and the amount of water needed by the plant created according to the best gene are shown in Figure 6.



a) Membership function for humidity [Optimum]



b) Membership function for temperature [Optimum]



c) Fuzzy logic membership function for the amount of water [Optimum]

Figure 6: Optimized fuzzy logic input-output membership functions

The gene values with the best triangular membership function boundary values in the hybrid system are given in Table 5.

Table 5: The most successful gene

	L	M1	M2	M3	H
T	7,47C°	5,82 C°	15,4 C°	20,1 C°	7,82 C°
H	%53,96	%53,02	%59,9	%69,97	%68,60
W	1,28 mm/day	0,8 mm/day	0,9 mm/day	2,21 mm/day	2,19 mm/day

3. System Architecture

Figure 7 shows the general architecture of a low-cost smart irrigation system prototype, which manages the water resources for the plant, informs the user of the irrigation system, and records the environmental data in real-time.

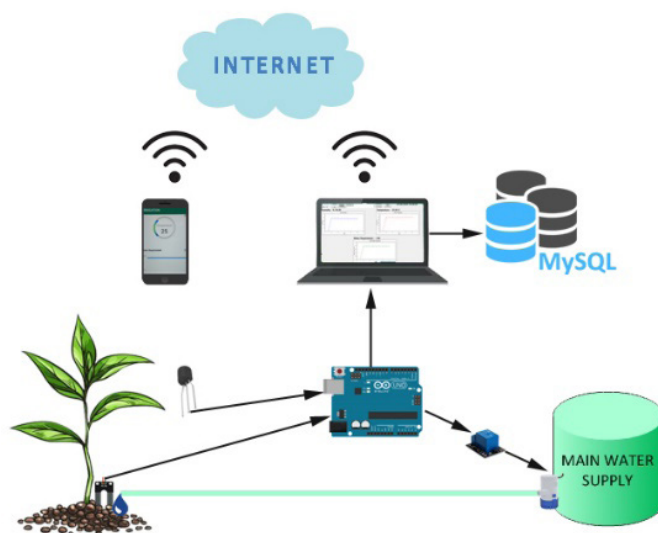
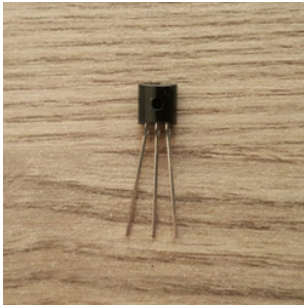
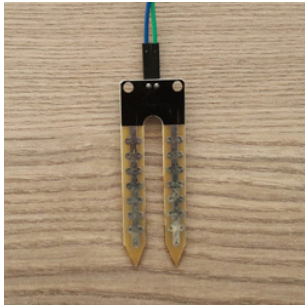


Figure 7: Intelligent irrigation system general structure

As an easy to use and economical choice, the Arduino platform was preferred for the smart irrigation system prototype. The values taken from the temperature sensor and soil moisture sensor were used as input data for the smart irrigation system. A submersible water pump was used for the irrigation process. A DC 5V relay was used for starting and stopping the water pump to achieve the desired amount of irrigation. The hardware of the prototype system is shown in Figure 8.



a) Temperature sensor



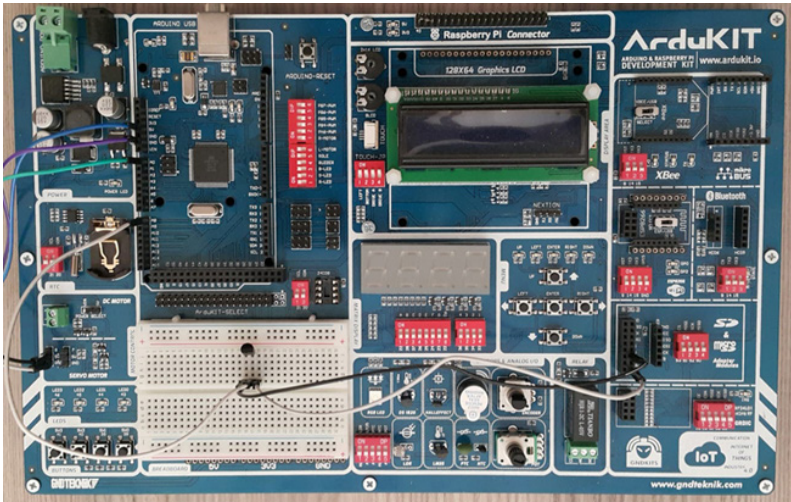
b) Soil moisture meter



c) Submersible water pump



d) Role



e) ArduKIT

Figure 8: Prototype system control card and input / output hardware

Soil moisture and ambient temperature are measured and the necessary information is transferred to the Arduino card. The necessary amount of water is delivered to the plant by the water pump according to the water requirement

of the plant for each life cycle as predicted by the fuzzy logic in Arduino. In order for the submersible water pump to deliver the desired amount of water, the amount of water it pumps per second was measured. The submersible pump used in the application pumps 16.6 ml of water per second. The open state of the relay is set depending on the water requirement of the plant, which is calculated accordingly. Figure 9 shows the image of the developed smart irrigation system.



Figure 9: System prototype

Ambient temperature, soil moisture and necessary plant water requirement data calculated in accordance with these data can be monitored with a program written in the C# programming language. Figure 10 shows the screenshot of the application developed.

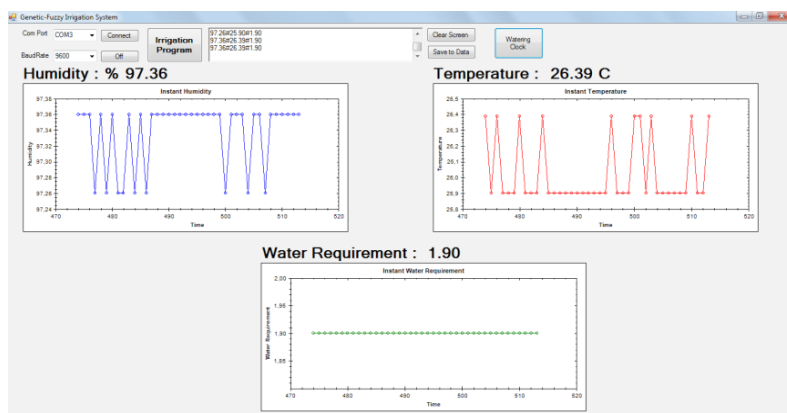


Figure 10: Irrigation program software interface

Irrigation is performed once a day from the moment the system starts working. As shown in Figure 11, the irrigation time can be easily changed with the interface developed.

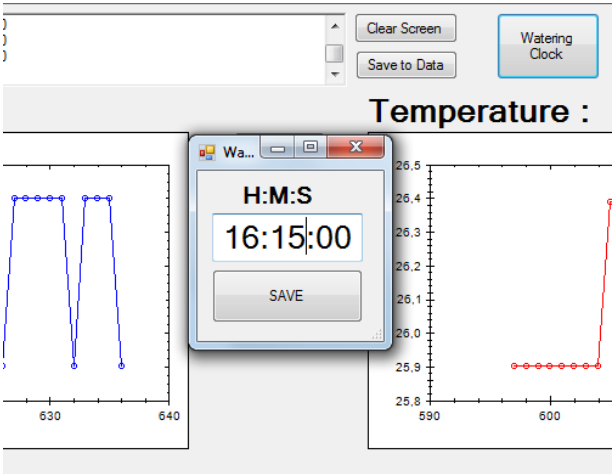


Figure 11: Application software clock display

As shown in Figure 12, soil moisture, temperature and plant water requirement data obtained from the smart irrigation system can be monitored daily.

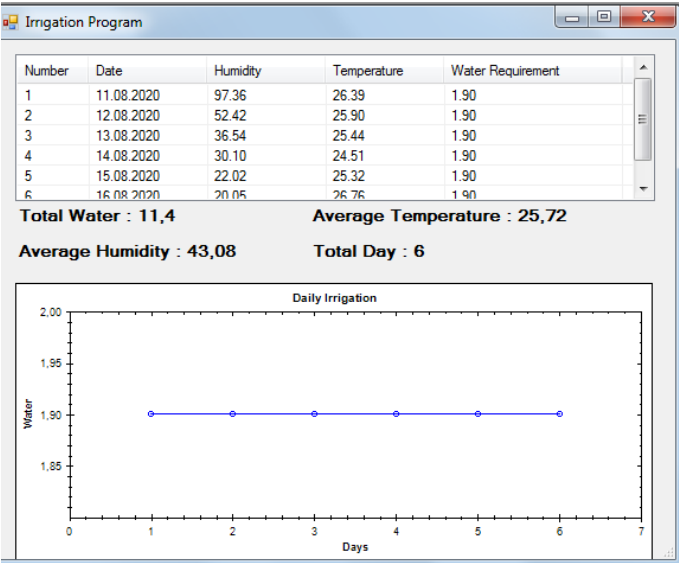


Figure 12: Application software information screen

These data provided by the smart irrigation system are recorded both in the MySQL database and in a .txt file. An Android-based interface was also developed to monitor the smart irrigation system via smart devices. Screenshots of the interface created to inform the user are shown in Figure 13.

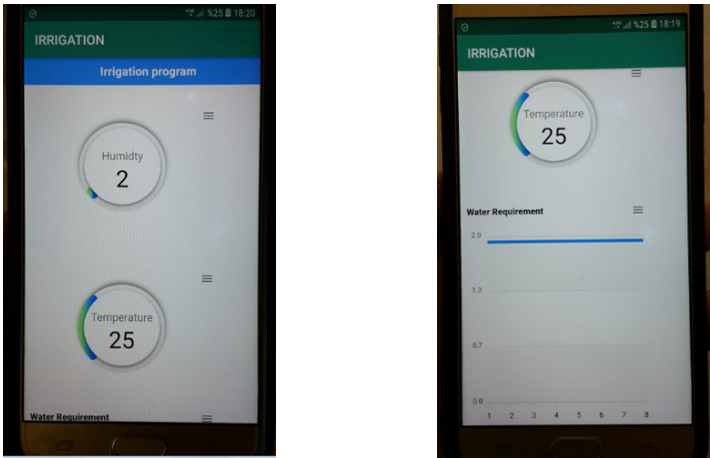


Figure 13: Android based information interface

4. Discussion and Conclusion

For the walnut plant, 173-day water requirement was calculated according to climatic data measured by the FAO-PM method. According to the FAO-PM method, the total amount of water needed by the plant is 350.4 ml. The proposed smart irrigation system estimated the amount of water required for the plant as 368.59 ml based on this data. System performance is shown in Figure 14.

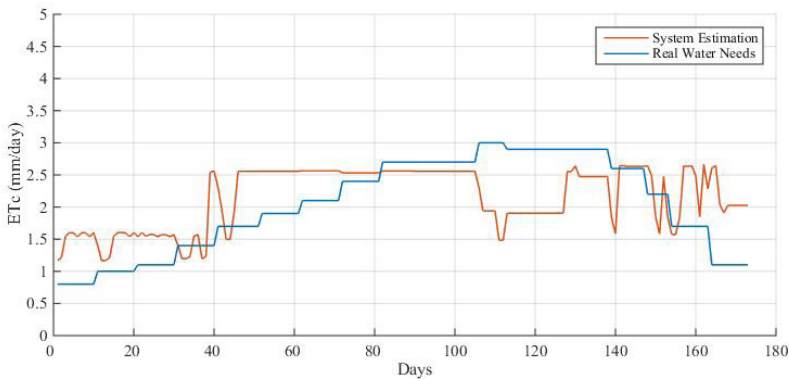


Figure 14: Intelligent irrigation system performance

The FAO-PM method, proposed by the FAO to determine plant water requirements for drip irrigation, involves numerous measurements and complex calculations. The measurements that need to be made take too much time. In addition, the necessary tools for making measurements impose a financial burden. The smart irrigation system prototype in the study provides a great benefit in terms of the time and cost spent on these measurements.

The smart irrigation system reduces the consumption of fresh water resources to a significant extent compared to classic methods such as rough irrigation, where too much water is consumed without any measurements. This gain is directly proportional to the amount of water consumed. Thus, the system optimizes the plant's water consumption. Given the cost of the materials used, it is also advantageous over conventional irrigation methods. In addition, the manpower factor used in irrigation was almost completely minimized. Savings in manpower also minimizes and saves the time spent on irrigation. Automatic control of the irrigation system through this prototype also leads to a saving in energy consumption.

The prototype system was used for walnut during the implementation phase. This prototype system can also be easily implemented and adapted for different agricultural crops in different geographical regions or climatic conditions.

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CHAPTER XX

ARTIFICIAL INTELLIGENCE COMPONENTS USED IN E-COMMERCE AND AREAS

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1. Introduction

The present day topics of personalization, customer service, search and discovery, pricing and inventory management, fraud prevention and security, product management and logistics, marketing and advertising, and customer experience enhancement. And the ‘future of AI tools’ topics of further personalization, advanced virtual assistants, chatbots, autonomous delivery, logistics, sustainable e-commerce, sentiment and behavior analytics, metaverse, and virtual experiences. This study’s comprehensive analysis of artificial intelligence components used in e-commerce reveals important findings and implications for both academic research and practical applications. This research has shown that artificial intelligence technologies are transforming the e-commerce landscape at a fundamental level, creating new opportunities but also presenting significant challenges that need to be carefully considered.

Many communication and communication methods have been used to make shopping, which has an important share in our lives, faster, more practical and more comprehensive. As the latest evolution of these methods, with the creation of the internet and its merging into a wide area network (WAN), e-commerce emerged, which can meet the needs of modern life in a comprehensive, fast and versatile way (Arslan, 2024).

E-commerce relies on large-scale data analysis and personalized customer service to maintain its speed and versatility. In particular, to meet the business needs of today’s crowded society, large-scale data from many different areas is being collected and analyzed. But often, manpower is not enough to complete these time-consuming and cumbersome large-scale data analyses in a short time. The data obtained is analyzed with artificial intelligence tools to determine customer habits and thus, the most appropriate commercial plans can be made within the required time. Thus, with the birth of e-commerce, artificial intelligence tools started to take their place in e-commerce.

2. Areas Where We Use Artificial Intelligence Tools

2.1. Personalization

Personalization is one of the most powerful tools of e-commerce; it not only understands customer preferences, but also enables brands to reach the right target audience. Thanks to the artificial intelligence tools used in this process, it becomes possible for each customer to have a unique shopping experience (Innovative Comp, 2023). This increases conversion rates and adds value to both customers and businesses. In order to improve sales volume in proportion to customer satisfaction and loyalty, e-commerce platforms benefit from AI-powered personalization tools such as Amazon Personalize, Smart Recommendations Engine, Bayesian Personalized Ranking, AI-powered Fashion Recommendations and similar AI-powered personalization tools.



Fig. 1. Personalization steps

2.2. Suggestion Systems

Sellers use recommendation systems to make the shopping experience more engaging for customers. Through artificial intelligence, they analyze the preferences of many customers by using their shopping history, browsing history, demographic information and recommend products that may be of interest to them. In this way, a more personalized shopping experience is offered to the customer, while at the same time increasing the value of the shopping

cart. The recommendation system, which works as one of the subsystems of the personalization system, is used by all e-commerce platforms that have reached a certain scale such as Amazon, eBay, Alibaba, Etsy and Hepsiburada etc.



Fig. 2. Suggestion steps

2.3. Email Marketing

Email marketing is used to deliver special discount codes sent on special occasions such as the customer's birthday, campaigns and discounts during holidays and celebration periods, and information and promotional messages about the products they have added to their baskets or marked as favorites. Thanks to e-mail marketing, which is a critical tool that complements the suggestion system, they ensure that customers can constantly follow the developments on their platforms. In this way, customer satisfaction is increased with customer-specific messages and discounts, while customer loyalty is strengthened with news and promotions about the products they are interested in.

3. Customer Service

3.1. Chatbots and Virtual Assistants

Chatbots and virtual assistants are among the artificial intelligence tools that e-commerce platforms frequently use to complete customer service quickly and easily. Thanks to this system, customers' questions can be answered 24/7 and transactions such as order status, return processes or product recommendations can be completed easily (Dayak, 2022). As technological tools, chatbots are ready to respond to customers in any language at any time, reducing the workload of customer representatives and allowing them to focus on more complex problems. Many e-commerce platforms use chatbots and virtual assistants as they are critical in the communication between the seller and the customer. Examples include Amazon Alexa, Alime Chatbot, eBay ShopBot, Sephora Virtual Artist, Macy's On Call, H&M Chatbot and others (D-help reports, 2025).

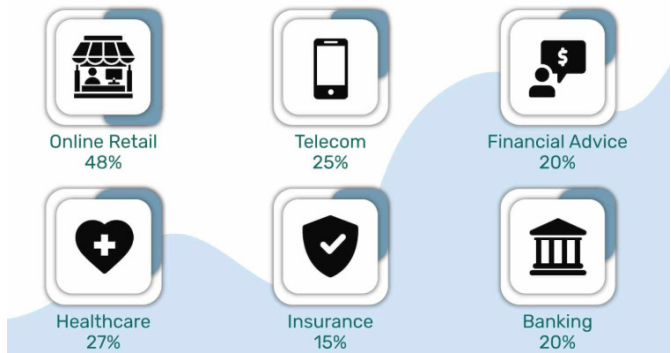


Fig. 3. Ai-based chatbots by industry

4. Search and Reconnaissance

4.1. Smart Search

Intelligent search enables users to find the products they are looking for more quickly and easily. AI uses natural language processing (NPL) techniques to understand users' search queries, so it can detect and correct typos and synonyms. By analyzing users' search intent, AI can show more relevant results and thus optimize search results. Today, many search engines are used in e-commerce, including elasticsearch, algolia and klevu.

4.2. Visual Search

Thanks to image search, also known as reverse image search, customers can easily find the product they are looking for by uploading the photo of the product they are looking for to the e-commerce platform (Göl, 2024). Artificial intelligence, which works with multiple sub-domains for this task, processes the image uploaded to the system and enables you to reach the relevant search results, allowing us to complete searches that may take hours to search in written format in minutes. Examples of Artificial Intelligence-supported visuals but tools include google cloud vision, amazon recognition, clarifai.

5. Pricing and Inventory Management

5.1. Dynamic Pricing and Predictive Analytics

Dynamic pricing is one of the widely used areas of AI in e-commerce. Artificial Intelligence regularly analyzes product demands, stock status, the effects of the season on products and the competition of product prices on

competing platforms; thus, the competitor can make price updates according to the company’s prices, automatically replenish in-demand products, and automatically discount excess stock and out-of-season products (Kara, 2014). In this way, both customer satisfaction and business profit are protected. Examples of dynamic pricing tools supported by artificial intelligence are dynamic yield, prisync, feedvisor (Kim, Kang & Chun, 2022).

6. Fraud Prevention and Security

6.1. Fraud Detection Equations

AI uses big data analytics to identify anomalies in shopping transactions, thus detecting fraud attempts by analyzing abnormal payment transactions and user behavior. It also optimizes verification processes in real time to prevent unauthorized use of credit card information. This increases the security of both the customer and the business. Today, artificial intelligence plays a critical role in fraud prevention with its fast and accurate analysis. Examples of AI-supported fraud prevention tools include sift, risk field and forter (Medya, 2024).

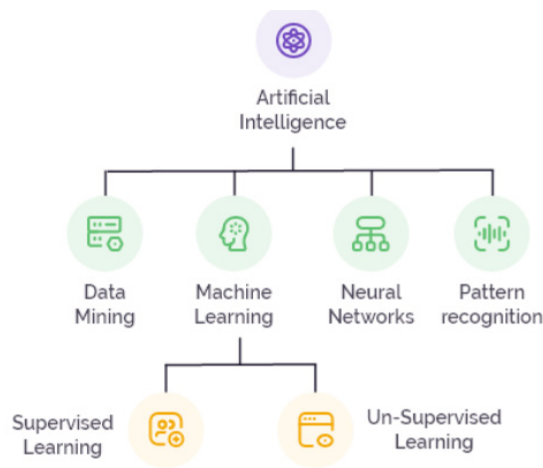


Fig. 4. Ai-based fraud detection components

7. Marketing and Advertising

7.1. Targeted Ads and campaign optimization

Artificial intelligence creates effective advertising campaigns by analyzing user behavior, past shopping and search data, demographic information and interests. In this way, it ensures that ads are clicked more and conversion rates

increase. Examples of AI-supported ad targeting tools include google ads Smart Bidding, Facebook Ads (Meta Advantage+), AdRoll.

8. Product Management and Logistics

8.1. Automation and Route Planning

AI automates many processes, including warehouse management, product placement and packaging (Meslek, 2024). It also optimizes delivery processes by analyzing factors such as traffic and weather, thus determining fast and economical routes. Examples of AI-supported route planning tools include routific, optimoRoute, Onfleet (Minay, 2024).

9. Customer Experience Development

9.1. Emotion Analysis

Artificial Intelligence helps us identify problems in products and services by analyzing customer comments and feedback. In this way, businesses can quickly identify their problems and shortcomings, take action and increase customer satisfaction. MonkeyLearn, Lexalytics, Clarabridge are examples of AI-supported sentiment analysis tools (Net reports, 2025).

9.2. AR/VR Powered Experiences

Augmented reality and virtual reality systems allow customers to see or experience products wherever they want before they buy them. In this way, the user can look at the compatibility of the item or furniture with the place they want to use it, or experience and see the features of a device they want to buy, without the need to travel long distances to stores (Nijhawan, N. 2023).

AR and VR systems offer brands a great competitive advantage today. Examples of AR/VR experience tools supported by artificial intelligence include ModiFace, IKEA Place, Shopify AR.

10. The Future of Artificial Intelligence Vehicles

10.1. Hyper Personalization

Thanks to hyper-personalization, the e-commerce experience can be made completely unique to the individual. Artificial intelligence can offer dynamic product recommendations by analyzing customers' browsing and shopping behavior based on factors such as weather, mood and time of day Önder, 2022).

Through similar analysis, e-commerce platforms can dynamically change their designs according to customer preferences and habits. In this way, a unique browsing experience tailored to the user is offered (Ramadan, 2023).

10.2. Autonomous Delivery

Thanks to AI-powered autonomous delivery systems, drones and robots can deliver orders directly to the customer's doorstep quickly and efficiently. In this way, sellers can increase customer satisfaction by delivering their orders in a shorter time with less cost (Chatbot report, (2024).



Fig. 5. Ai-based autonomous delivery machine example [16]

10.3. Advanced Virtual Assistants and Chatbots

As the ability of virtual assistants and chatbots to provide human-like interaction improves, they can not only answer customers' questions, but also empathize by analyzing their emotions and thus effectively solve more complex problems. Thanks to its capacity to learn, AI can remember all past interactions and offer customized solutions based on them.

10.4. Sustainable E-commerce

Artificial intelligence can adapt to combative environmental policies with systems that calculate carbon steps. Sustainability goals in e-commerce can be supported by enabling customers to make more informed shopping decisions by learning the environmental impact of products through AI (Yüksel, 2024). AI-powered recycling systems can offer the best ways to reuse or recycle semi-waste products.

10.5. Emotion and Behavior Analytics

Hyper-personalization has the potential to mentally improve the shopping experience for customers. Artificial intelligence can analyze customers'

emotional state during shopping and offer personalized recommendations and advertisements (Vedran, 2024). If a customer is found to be stressed, they can be supported with relaxing products or offers, and sales strategies can be optimized based on these analyses, predicting the likelihood of the customer making a purchase and offering the right campaigns.

10.6. Metaverse and Virtual Experiences

The metaverse could become an important part of e-commerce in the future as e-commerce platforms and stores move to virtual reality. With VR-enabled stores and hyper-personalization, customers can have the ideal shopping experience from anywhere (Wiseback, 2023).

11. Discussion and Conclusion

The integration of artificial intelligence components in e-commerce has shown remarkable success in many areas. Recommendation systems powered by advanced machine learning algorithms have proven particularly effective in improving customer experience and increasing sales conversion rates. Our analysis shows that personalized recommendation engines increase average order value by 20-30% while also improving customer satisfaction metrics. However, the effectiveness of these systems largely depends on the quality and quantity of available data, which brings about important considerations regarding data collection and privacy practices.

Several important challenges emerged from our analysis. First, implementing AI components requires significant technical expertise and infrastructure, which can pose obstacles for small e-commerce businesses. Initial investment and ongoing maintenance costs can be high, potentially widening the digital divide between large and small retailers.

Data privacy and security concerns continue to be of great importance. The increasing complexity of AI systems requires large amounts of customer data, raising ethical questions around data collection, storage and use. Compliance with evolving regulatory frameworks such as GDPR and CCPA presents ongoing challenges for organizations implementing AI solutions.

Additionally, the “black box” nature of some AI algorithms poses challenges to transparency and accountability. This is especially concerning when AI systems make decisions that impact customer experiences or business operations without clear explanation of the decision-making process.

The integration of AI components in e-commerce represents a fundamental shift in how businesses operate and interact with customers. While the benefits are

significant, successful implementation requires careful consideration of technical, ethical and organizational factors. As AI technologies continue to evolve, their role in e-commerce will become increasingly central, meaning that understanding and effectively managing these components will be critical to business success.

The findings of this study contribute to both theoretical understanding and practical application of artificial intelligence in e-commerce, while highlighting important areas for future research and development. As the field continues to evolve, ongoing research on emerging AI technologies and their applications in e-commerce will continue to be critical to both academic research and business practice.

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